COLLISION RECORDS, ARITHMETIC, AND RAW LOCAL INVERSION IN A TRIANGULAR LORENTZ GAS

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Abstract. For the original triangular Lorentz gas with disk radii in $[9/20, 47/100]$, we prove uniform localization of every global disk-word length minimum relative to its actual physical reference record. A variational integration-by-parts inequality and the physical turning anchors yield an explicit contraction of the full relative contact and flight energy after each layer of 94 collisions. The estimate includes inactive comparison contacts and fixed-word singular reference closures, and requires no physical continuation to a minimum. A transverse strip theorem retains both incident disks and every third lattice obstacle. It gives, together with the contraction and the exact collision measure, an exponentially small complete positive source outside which all central minimum records are physical and have precisely the original section observations. We retain the central physical clock's exact endpoint work and all four Fourier marks. Under the separately identified inherited pressure and optical input, the small central comparison costs vanish in collision-normalized original-roof essential height, with an explicit exponential rate. Every predecessor variational, arithmetic, induced-return, optical, clearance and analytic proof body is preserved in the native source chain. The exceptional source height, shallow comparison capacity and singular paired budgets are still required for the original raw pointwise local limit and four-coordinate inversion; those final theorems are not claimed complete.

Introduction

A physical collision record carries more information than the displacement of a particle at a large time. In the triangular Lorentz gas considered here, the observations are the two integer lattice coordinates, the total unit-speed physical time, and the integer occupation of a specified collision section. Conditioning on the original initial and terminal section tests identifies the actual return event. The problem is to obtain its raw pointwise local limit, with all collision labels and microscopic roof windows retained, and then the four-coordinate local inverse theorem.

The fixed family consists of equal circular scatterers on the triangular lattice, with $R \in [9/20, 47/100]$. The collision probability is $d\alpha\, dp/(4\pi)$ and its original section normalization is $c_* = 91/(10000\pi)$. The event (1.1), arithmetic reference (1.4) and target (1.5) are unchanged. In particular occupation counts the initial state and excludes the terminal state, whose membership is tested separately. All reflections are actual first collisions. The physical flight bounds $\tau_- = 3/50$ and $\tau_+ \le \sqrt{3847/124} < 28/5$ and the finite lattice budgets are proved in Section 2.

The complete comparison problem and its inherited gap. For an actual regular reference $Q_0, \dots, Q_m$ with ordered disk centers O_i, minimize

$$A(Y) = \sum_{i=0}^{m-1} |Y_{i+1} - Y_i| \quad \text{over} \quad \mathcal{K} = \prod_{i=0}^{m} \overline{B}(O_i, R).$$

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A minimum of this compact convex problem need not be a physical billiard record. Its contacts can be inactive and its flights can meet unlisted obstacles. Those possibilities are retained throughout the comparison argument.

The inherited dual certificate D_Q and positive gap $G = A(Y) - D_Q$ satisfy, on the original length sublevel,

$$0 \le G \le E_R(Q), \qquad E_R(Q) = R(2 - c_0 - c_m).$$

With $\delta_i = Y_i - Q_i$, e_i the difference of the unit edge directions, $F_i = \ell_i |e_i|^2$ and $H_i = c_i |\delta_i|^2$, the flight and contact parts consume one shared positive gap. The preceding complete sublevel theorem, Theorem 43.3, already proves $\sum_i |\delta_i|^4 \le 10^8 E_R(Q)$, including fixed-word closed physical references. The exact all-center clearance and observation transport of module 303 is also inherited. Neither result is reclassified as new here.

Localization of every minimum without a physical path. The new Lemma 45.1 uses the Cartesian minimum condition in every disk. It bounds the relative contact and direction energy inside a cutoff by work in its two transition layers. All comparison edges remain separated from zero length, so this argument applies at inactive contacts as well as at physical minima. The physical turning theorem supplies an interior contact with $c_i \ge a_0 = 1/416640$ in every complete $N = 94$-flight reference block. It follows from the explicitly proved lattice horizon and bending bound, not an abstract hyperbolic comparison model.

The first consequence, Theorem 45.4, is the uniform algebraic estimate

$$\sum_{i=d}^{m-d} |\delta_i|^4 \le \min\{10^8 E_R(Q), K_{\mathrm{tail}}(d+1)^{-3}\}.$$

Its proof pays the cutoff work, obtains a nonlinear recurrence and checks every depth including an isolated central contact. It also provides localized contact and flight energies. The full proof is retained because its constants and its exact mechanism give independent useful comparison bounds.

The anchor interval identity permits a stronger conclusion. Define the full energy at depth d by

$$\mathcal{E}_d = \frac{1}{R} \sum_{\substack{0<i<m \\ d\le i\le m-d}} H_i + \frac{1}{2} \sum_{i=d}^{m-d-1} F_i.$$

Theorem 48.3 proves

$$\mathcal{E}_{d+94} \le q_{\mathrm{a}} \mathcal{E}_d, \qquad \mathcal{E}_d \le E_R(Q) q_{\mathrm{a}}^{\lfloor d/94 \rfloor}, \qquad q_{\mathrm{a}} = \frac{3263680}{3263681} < 1. \tag{0.1}$$

The two chosen anchors pay the exact work at both cuts from the removed layer. Their contact energy is disjoint from the interior energy being estimated. Young's inequality retains both cut flights, including the case when the cuts share a single flight. The initial energy is at most G, hence at most $E_R(Q)$. This proves a contraction of the complete energy with no collision-count multiplier.

The constants in (0.1) are conservative. Their mathematical significance is that they are fixed uniformly before all word and collision limits. The deterministic q_{a} does not replace the original guard constant $q_0 = 47/53$. Lemma 48.1 further proves that all global minima have the same unit directions, active index set and active vertices. Their possible differences lie along inactive straight chains. No uniqueness of the full minimum, physical continuation to it, or billiard flow through a singular reference is asserted.

Clearance of all actual obstacles and an exact source. A change parallel to an original flight should be accounted for using its two actual incident disks. Lemma 46.1 does this with $\beta_i = \max(|v_i^{\perp} \cdot \delta_i|, |v_i^{\perp} \cdot \delta_{i+1}|)$:

$$\Gamma_i(Y) \geq \min\{\Gamma_i(Q), \tau_-\} - \beta_i \quad (2\beta_i < \tau_-).$$

Here Γ_i is the clearance of the complete closed segment from every third lattice disk, including a disk visited elsewhere in the same word. The proof treats the strip projections of the incident disks, the longitudinal extensions, endpoints and changes of the minimizing center. Its quadratic transverse budget is a new refinement of the inherited fourth-power Hausdorff budget.

Together with (0.1), the strip theorem provides exponentially small true clearance and incidence witnesses for any deep physical failure of any minimum. A separate pointwise phase estimate supplies the witness for disagreement with the exact original section. The original section's collar satisfies the explicit bound $\nu\{\mathrm{dist}_\infty(x, \partial Y_R^*) \leq \rho\} \leq \rho/(2\pi)$ for $0 < \rho \leq 1/200$.

Theorem 50.3 defines a Borel source $\mathcal{E}^{\mathrm{obs}}_{D,R,m}$ using only these original physical observables and proves

$$\sup_{m,R,n,k,\,0\leq h\leq 1} \left\| (L_{m,R})_*(\mathbf{1}_{\mathcal{E}^{\mathrm{obs}}_{D,R,m}} \mathbf{1}_{\Pi_{n,k,m,R}} h\, d\nu_R^*) \right\|_{\mathrm{TV}} \leq \epsilon_D \longrightarrow 0 \quad (D \to \infty).$$

The displayed ϵ_D is an explicit sum of three geometric tails. This is a complete positive-source estimate: the countable physical word sum precedes the uniform bound, and no independence between flight times is assumed. It uses the exact one-flight clearance flux, invariant collision probability and the original section conditioning once.

On its complement, every minimum has an actual first-hit central record $Y_D, \ldots, Y_{m-D}$ with exactly the original section indicator at each of its contacts. Its displacement, collision count, occupation and two endpoint tests are therefore exact. No assertion of physicality for the omitted end portions is needed or made. This comparison class is an explicitly measured part of the original source, and is not substituted for the original conditional event.

The physical clock, its endpoint work, and roof height. For $d \leq b = m - d$, on a proved physical central comparison, Theorem 47.1 gives the exact identity

$$L_d(Y) - L_d(Q) = v_{b-1} \cdot \delta_b - v_d \cdot \delta_d + \frac{1}{R} \sum_{i=d+1}^{b-1} H_i + \frac{1}{2} \sum_{i=d}^{b-1} F_i.$$

The first two terms form the boundary work W_d. They are retained in the central Fourier phase; the nonnegative remainder is the small physical clock cost. The section proof pays all contact-state changes, duplicates only the terminal incoming edge as required, and keeps the terminal indicator separate.

The unconditional comparison statements are positive inequalities of measures pushed forward by $L_{m,R}(Q)$, the original *full* physical roof. Theorem 49.1 and Corollary 50.4 identify exactly the additional input for a roof density: Input 11.1, with the specified physical stable arclength, optical coordinate factors and uniform word-pressure constant. Under that input the small costs have bounded density on the whole roof line. For $d_m = \lceil \theta m \rceil$, $0 < \theta < 1/2$, the exponential costs satisfy an $O(m^2 e^{-\lambda_0 m})$ bound after collision normalization, with $\lambda_0 = \theta |\log q_{\mathrm{a}}|/94$. Their central clock Fourier error is uniformly small on $|s| \leq e^{\lambda m}$ for every $\lambda < \lambda_0$, after its exact boundary phase, and uniformly in both lattice twists and the return twist.

These conditional density estimates concern small comparison costs on the exact central class. Total variation of its exceptional source does not estimate that source's roof essential height. Neither statement by itself supplies all physical multipliers, a full unbounded Fourier integral or roofwise capacity.

The retained architecture and the remaining limit. All forty-four complete predecessor core proof bodies 260–303, the model and the analytic interfaces are preserved byte for byte and included exactly once in the current native manuscript. They contain the optical Jacobi and endpoint trace completion, marked incidence pressure, persistent low-load shortage, one- and multiple-mark physical clearance, cross-flight control, homogeneous growth, complete critical-sector summation, radial coarea, endpoint jets and resolvents, physical tubes, bending, and the global variational and summable sublevel theorems. In particular the physical curve-growth hypothesis is discharged by the native chain 281–283, and the complete regular critical coefficient and common physical-roof unscaled summations are inherited results. Their completion is not presented as a new assumption or a new contribution.

The preserved historical native assembly carries the complete earlier geometry, periodic orbit, arithmetic, inducing, spectral, conditional-measure and inverse chain. It contains 390 exact source files and is rebuilt separately with its original assembler. The archive also retains all 355 distinct numbered source variants covering 1–303, every original predecessor file and all 90 retrieved A2-DYN reports. The proof ledger distinguishes a directly proved finite geometric assertion, a consequence at an inherited input, and the global analytic inputs still requiring specialist reconstruction.

The governing remaining condition stays exactly

$$\lim_{B\to\infty}\limsup_{m\to\infty}\sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} \left(\mathsf{U}^{\mathrm{low}}_{B,m}(t) - K_B P^{\mathrm{ng}}_{r_B,m}(t)\right)_+ = 0. \tag{0.2}$$

Here M is fixed, the complete positive physical sums precede all suprema, each outer width is fixed before the collision limsup, and B is outermost. The exceptional-source height, positive shallow source, critical and singular pairing budgets, full physical normalization and comparison-pool capacity remain explicit obligations. The new depth schedule is not inserted into the original microscopic guard. The raw pointwise target and four-coordinate inverse consequence are not asserted proved.

Source succession and the scope of external results. The actual complete source parent of this author revision is commit `fb99e4d67ef74912f6c69b05c9abff3207572e7a`, with article tree `5575389060a6ba0ddaa0079be066ae41b107a49c`. It contains the complete native chain through 303 and has been actually rebuilt. Its failed remote source qualification and its incomplete archival manifest remain recorded; neither invalidates the readable native proof bytes nor counts as successful CI. Higher-numbered source-free wrappers and skipped qualification steps do not define a scientific successor.

The controlling retrieved report, commit `f01482035a0b945b50f1f9c30f30ef35077558dc`, explicitly examines the author source `31792379408c5ace2c09284c2eb22d2331023433` through 276. Its mathematical objections are applied here with that scope; it did not review 277–309. It is an author-requested AI-assisted report, not a journal decision or an independent specialist reconstruction. The response locates each of its six major and nine secondary requests. Favorable assessments of the separate A2 geometry paper supply no A2-DYN review evidence.

The new localization and source proofs use their displayed geometric and measure arguments. The perturbative anisotropic framework of Demers–Zhang [6], the geometric-potential pressure theory of Baladi–Demers [7], and the tested flow local limit theorems of Dolgopyat–Nándori [8] are retained with their actual dynamical, function-space, pressure-gap, arithmetic and observable hypotheses. They are not identified with (0.2). General convex duality [9] is background to the explicit disk-word proof. Actual source, mathematical-program and native compilation receipts accompany the manuscript; finite diagnostics and CI are auxiliary reproducibility checks.

1. The unchanged physical problem

Put $R_- = 9/20$ and $R_+ = 47/100$, and let $R \in I = [R_-, R_+]$. Circular scatterers have centers in $\Lambda = \mathbb{Z}(1,0) + \mathbb{Z}(1/2, \sqrt{3}/2)$. Write T_R for the regular collision map and $(\alpha, p) \in (\mathbb{R}/2\pi\mathbb{Z}) \times (-1,1)$ for its coordinates; p is the outgoing tangential velocity and $\mathrm{d}\nu = \mathrm{d}\alpha\, \mathrm{d}p/(4\pi)$. Physical flights satisfy $\tau_- \le \tau_R \le \tau_+$, where $\tau_- = 3/50$ and τ_+ is the fixed finite-horizon bound.

The section is specified here to fix every event in the argument. Put $h_0 = 1/200$. Let Y_R be the union of the four open coordinate squares of half-side h_0 centered at

$$(0,0), \quad (\pi/6, -1/2), \quad (-\theta_R, \sin\theta_R), \quad (\pi/3 - \theta_R, \sin\theta_R).$$

Here $\theta_R = 2\arctan q_R$, and q_R is the unique root in $(49/100, 53/100)$ of

$$8Rq(1+q^2) - 4q(1-q^2) + \sqrt{3}(1 - 6q^2 + q^4) = 0.$$

Set

$$U = \{|\alpha - \pi/6| < 3/50,\ |p| < 3/20\}, \qquad Y_R^* = Y_R \cup U, \qquad c_* = \frac{91}{10000\pi}.$$

The rectangles are disjoint and $\nu(Y_R^*) = c_*$. We always use $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$, not a renormalization of a smaller source.

At m flights, let $L = L_{m,R}$ be their total length, $K^{\mathrm{c}}_{m,R}$ their lattice displacement, and $A_{m,R} = \sum_{j=0}^{m-1} \mathbf{1}_{Y_R^*} \circ T_R^j$. The event is

$$\Pi_{n,k,m,R} = Y_R^* \cap \{K^{\mathrm{c}}_{m,R} = k,\ A_{m,R} = n\} \cap T_R^{-m} Y_R^*. \tag{1.1}$$

The initial visit is counted; the terminal visit is tested separately. On this event the actual nth return has collision count m and roof L. Denote its density by $p_{n,R}(k,m,t)$.

Write $c_j = \sqrt{1 - p_j^2}$ and let Δ_j be the least clearance of flight j from any third disk, excluding its two incident disks. Keep the original smooth function $0 \le \Theta \le 1$, zero on $(-\infty, 1]$ and one on $[2, \infty)$. With $q_0 = 47/53$, $d_j = \min(j, m-j)$ and $d_j' = \min(j, m-1-j)$, set

$$H^\varepsilon = \prod_{j=0}^{m} \Theta\!\left(\frac{c_j}{\varepsilon q_0^{d_j/4}}\right) \prod_{j=0}^{m-1} \Theta\!\left(\frac{\Delta_j}{\varepsilon q_0^{d_j'/4}}\right), \qquad W^\varepsilon = 1 - H^\varepsilon, \tag{1.2}$$

$$b^\varepsilon(t)\,\mathrm{d}t = L_\#(W^\varepsilon \mathbf{1}_{\Pi_{n,k,m,R}} \nu_R^*). \tag{1.3}$$

In particular first incidence and first clearance outside any selected critical neighborhoods are still present. No section-distance factor has been inserted into this guard.

1.1. The arithmetic reference and the target.

For $y = (k_1, k_2, t, n)$ put

$$\mu_R = (0, 0, \bar\tau_R, c_*), \quad Z = (y - m\mu_R)/\sqrt{m}, \quad \mathcal{C}_{m,R,n,k}(M) = \{t : |Z| \le M\}.$$

We retain the finite spectral-peak reference, including peaks close to but not on the unit circle. To specify its meaning, the collision record observable is $(\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$. Let $\lambda_j(R,\vartheta)$ and $\mathsf{P}_j(R,\vartheta)$ be its continued simple twisted eigenvalue and projection in each of the fixed peak charts, and let χ_j be their frequency partition. At the real maximum a_j of $\log|\lambda_j|$ put

$$z_j = \lambda_j(R, a_j), \quad \mu_j = i^{-1}\nabla \log\lambda_j(R, a_j), \quad H_j = -\nabla^2 \log\lambda_j(R, a_j),$$

with $\operatorname{Re} H_j \geq aI_4$ for a common $a > 0$. If $\eta_R = \mathbf{1}_{Y_R^*}$ and ℓ denotes mass evaluation, put $B_j = \chi_j(R, a_j)\ell(M_{\eta_R}\mathsf{P}_j(R, a_j)(\eta_R\nu))$. Define

$$G_H(z) = (2\pi)^{-4} \int_{\mathbb{R}^4} e^{-iv\cdot z - v^{\mathsf{T}} Hv/2}\, \mathrm{d}v,$$

$$\mathcal{L}_{m,R}(y) = \operatorname{Re}\left[c_*^{-1} \sum_j e^{-ia_j\cdot y} z_j^m B_j G_{H_j}\left((y - m\mu_j)/\sqrt{m}\right)\right]. \tag{1.4}$$

Only the charts active at R are summed. This is the same transition reference as in the retained spectral theorem. The existence and uniform estimates for these scalar data belong to that analytic input, not to the transport argument below. No frequency branch or arithmetic zero class is deleted.

The governing assertion remains

$$\sup_{R,\, 1\leq n\leq m,\, k\in\mathbb{Z}^2} \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} |m^2 p_{n,R}(k, m, t) - \mathcal{L}_{m,R}(k_1, k_2, t, n)| \longrightarrow 0. \tag{1.5}$$

Let $\mathcal{E}_M$ denote this error's collision limsup. An empty central target contributes zero. The order of the uniformity is $\sup_R \operatorname{ess\,sup}_t$, so exceptional density sets may depend on R.

2. An explicit all-direction physical horizon and lattice budget

We give a quantitative proof of finite horizon for the unchanged round scatterers. It is independent of a simulated trajectory, a symbolic model, a pressure theorem and the compactness argument retained in Section 24. It makes the finite geometric constants in the physical source construction directly auditable.

Lemma 2.1 (A row rectangle for a primitive lattice direction)**.** *Let v be a primitive vector of the triangular lattice, $l = |v|$, and $h = \sqrt{3}/(2l)$. For every $q \in \mathbb{R}^2$ there is a lattice center z whose components relative to q, along v/l and its unit normal, satisfy $|x| \leq l/2$ and $|y| \leq h/2$. Thus for a line direction making angle δ with v,*

$$|(z-q)\cdot e^{\perp}| \leq \tfrac{h}{2}|\cos\delta| + \tfrac{l}{2}|\sin\delta|, \qquad |(z-q)\cdot e| \leq \tfrac{1}{2}\sqrt{l^2+h^2}. \tag{2.1}$$

Proof. Primitivity permits v to be completed to a lattice basis. The parallel lattice rows are consequently spaced by the covolume $\sqrt{3}/2$ divided by l. Choose a nearest row to q, and then a nearest lattice center along that row. Its two coordinates obey the stated bounds, regardless of the longitudinal offset of the row. Projection and Cauchy–Schwarz give (2.1). All choices are in the original Euclidean lattice; no linear map deforms the circular disks. □

Theorem 2.2 (A uniform, explicit physical free-flight bound)**.** *For every $R \in [9/20, 47/100]$, every straight segment avoiding the interiors of all radius-R lattice disks has length at most*

$$\tau_{\max} \leq \sqrt{\frac{3847}{124}} < \frac{28}{5} < 6. \tag{2.2}$$

In particular the same bound holds for every genuine free flight and all its grazing limits. The statement is simultaneous over all initial points and all directions; no exceptional direction is discarded.

Proof. The dihedral symmetries of the triangular lattice reduce an unoriented line direction to $0 \leq \vartheta \leq \pi/6$. Put $t = \tan\vartheta/\sqrt{3} \in [0, 1/3]$. We use four primitive lattice vectors, with the

indicated ranges:

range of t	v	$\lvert v\rvert^2$
$[0, 1/60]$	$(1, 0)$	1
$[1/60, 1/12]$	$(11/2, \sqrt{3}/2)$	31
$[1/12, 1/5]$	$(5/2, \sqrt{3}/2)$	7
$[1/5, 1/3]$	$(3/2, \sqrt{3}/2)$	3

The last three vectors have lattice coordinates $(5, 1), (2, 1), (1, 1)$, so they are primitive. For the horizontal vector the squared normal width in (2.1) is

$$W_0(t)^2 = \frac{3}{16}\frac{(1+2t)^2}{1+3t^2}.$$

This is increasing on $[0, 1/60]$, since the logarithmic derivative of $(1+2t)/\sqrt{1+3t^2}$ has numerator $2-3t > 0$. For $v = ((2a+1)/2, \sqrt{3}/2)$, write $s = 2a+1$ and $N = a^2+a+1$. The normal width is

$$W_a(t) = \frac{\sqrt{3}}{4\sqrt{1+3t^2}}\left\{\frac{s+3t}{2N} + |st-1|\right\}.$$

On each chosen interval for $a = 5, 2, 1$ one has $t \le 1/s$. The expression in braces is positive and has derivative $3/(2N) - s < 0$. Since the denominator increases, W_a is decreasing. It therefore suffices to check its left endpoint. The four exact squared widths and their positive gaps below $R_-^2 = 81/400$ are

v type	$\max W^2$	$81/400 - \max W^2$
horizontal	961/4804	1181/480400
$a = 5$	13697401/73866304	31513139/1846657600
$a = 2$	529/3136	2651/78400
$a = 1$	75/448	393/11200

Thus, for every direction and midpoint q, Lemma 2.1 provides a lattice center at *strictly* less than R_- normal distance from the line. The longitudinal distance is at most $\sqrt{l^2 + 3/(4l^2)}/2$. Among the four lengths its largest value is $\sqrt{31 + 3/124}/2 = \sqrt{3847/124}/2$. If a free segment were longer than twice this number, the perpendicular foot of the chosen center would lie inside the segment and inside an open radius-R_- disk, a contradiction. This proves the first inequality in (2.2). The comparison with $28/5$ follows by squaring positive quantities: $3847 \cdot 25 = 96175 < 97216 = 784 \cdot 124$. Since the disk radius is only increased for $R \ge R_-$, the entire argument is uniform over the original family. Strict normal clearance in the covering argument includes limiting directions and grazing free segments; no collision derivative at grazing is used. □

Corollary 2.3 (Explicit enclosing candidate counts, not word counts)**.** *Keep the original constants $R_+ = 47/100$, $\delta_* = 3/200$, the intrinsic τ_+ of the retained source papers, and every original event. Then*

$$N_f \le 150, \qquad N_\circ = N_* \le 150, \qquad N_{\mathrm{pair}} \le 162, \qquad J_f = 2N_f + 1 \le 301.$$

Here N_f is the first-hit candidate count of (24.1)*, $N_\circ$ is the one-mark/cross-flight count of* (19.7)*, and N_{pair} is the difference count in* (18.1)*. In particular the angular first-hit partition in Lemma 21.2 has at most* 1201 *intervals. These are upper bounds for the intrinsic counts, not assertions that all candidate centers can occur or that the smallest physical horizon is* $28/5$*.*

Proof. Theorem 2.2 allows the enclosing constant $\overline{\tau} = 28/5$. The corresponding three enclosing radii are

$$D_f^{\mathrm{enc}} = 327/50, \qquad D_\circ^{\mathrm{enc}} = 1311/200, \qquad D_{\mathrm{pair}}^{\mathrm{enc}} = 657/100.$$

Their squares are respectively 106929/2500, 1718721/40000 and 431649/10000. The first two lie in $[42, 43)$ and the last in $[43, 44)$. A lattice point with integer coordinates (a, b) has squared length $Q(a, b) = a^2 + ab + b^2$, which is integral. The enclosing sets are therefore exactly $0 < Q \le 42$ and $0 < Q \le 43$.

For completeness this finite count does not depend on a floating-point circle. The identity $Q = (a + b/2)^2 + 3b^2/4$ implies $|b| \le 7$ when $Q \le 43$, and by symmetry $|a| \le 7$. Direct solution of the quadratic inequality gives the following inclusive integer ranges for $b \ge 0$:

b	a for $Q \le 42$	#	a for $Q \le 43$	#
0	$[-6, 6]$	13	$[-6, 6]$	13
1	$[-6, 5]$	12	$[-7, 6]$	14
2	$[-7, 5]$	13	$[-7, 5]$	13
3	$[-7, 4]$	12	$[-7, 4]$	12
4	$[-7, 3]$	11	$[-7, 3]$	11
5	$[-7, 2]$	10	$[-7, 2]$	10
6	$[-6, 0]$	7	$[-7, 1]$	9
7	$[-5, -2]$	4	$[-6, -1]$	6

Negative rows have the same cardinalities under $(a, b) \mapsto (-a, -b)$. The totals, excluding the origin, are $12+2(12+13+12+11+10+7+4) = 150$ and $12+2(14+13+12+11+10+9+6) = 162$. The intrinsic candidate sets are subsets of these enclosing sets, which proves all three inequalities. The stated branch bounds now follow from the already proved formulas $2N_f + 1$ and $8N_* + 1$. No collision word, pairing sheet, homogeneous grazing strip or probability normalization is counted by these finite numbers. □

This quantitative result replaces no dynamical hypothesis by a finite simulation. It supplies the geometric finite-horizon input used in the retained source and curve-growth proofs. It does not supply the long-time marked pressure, the sum at a common physical roof, or the collision-normalized positive-source estimate.

3. The physical source and the inherited analytic inputs

We specify the inputs to the new geometric argument, separating the existence of the original physical source from the endpoint regularity that will be proved below. The complete historical proofs and their qualifications are retained in the accompanying archive.

For fixed m, R, n, k, let Θ_{phys} be the original disjoint incident-component pairing-sheet space. It is the countable Borel sum of the maximal regular components of the original exact event, parametrized by

$$(u, \theta) = (F, \arg \nabla_q F), \qquad q = (\alpha_0, \alpha_m), \qquad a(\theta) < u < b(\theta), \quad \ell = b - a > 0.$$

The sheet identity, and not just its angular coordinate, is included in θ. Integration $d\theta$ means the original one-copy physical sum with angular Lebesgue measure. The regular components, their Borel endpoint maps, their physical multiplicity convention, and their common coarea versions are the retained pairing-space input. This article does not replace them by symbolic collision words. Every component carries the same exact event (1.1). Put

$$\begin{aligned} &g = \nabla_q F = R(-p_0, p_m), \quad z = |g|, \quad I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i}/c_i, \\ &\omega = |\partial_{p_0} \alpha_m|^{-1}, \quad \chi = |\det D_q(F, \arg g)|, \quad A = \frac{\omega}{4\pi c_* \chi}, \quad f^\varepsilon = AW^\varepsilon. \end{aligned} \tag{3.1}$$

The density formula is used initially on the regular interior. Once its endpoint traces have been established, the original pairing is

$$f_-^\varepsilon = f^\varepsilon(a+,\theta), \quad f_+^\varepsilon = f^\varepsilon(b-,\theta), \quad c^\varepsilon = \min(f_-^\varepsilon, f_+^\varepsilon), \quad d\pi^{\varepsilon,+} = c^\varepsilon \mathbf{1}_{\{a<u<b\}} d\theta\, du. \tag{3.2}$$

Write $p_\vee = \max(f_-^\varepsilon, f_+^\varepsilon)$ and $e_\partial = |f_-^\varepsilon - f_+^\varepsilon|$. Thus $p_\vee = c^\varepsilon + e_\partial$.

Input statement 3.1 (Regular variation and completed physical averages)**.** The inherited regular-point optical calculation gives, almost everywhere on each regular segment,

$$|\partial_u f^\varepsilon| \le C(1+\varepsilon^{-1}) A I_m / z. \tag{3.3}$$

Let $\mathsf{T}_\Delta^\varepsilon$ be the complete paired roof density restricted to $0 < \ell < \Delta$, with no additional selector, and let

$$v^0(t)\, dt = F_\#(I_m z^{-1} \mathbf{1}_\Pi \nu_R^*), \qquad A_r f(t) = (2r)^{-1} \int_{t-r}^{t+r} f(u)\, du.$$

For every previously fixed $\varepsilon, r, \Delta, \rho > 0$,

$$\limsup_m \sup_{R,n,k} m^2 \|A_r \mathsf{T}_\Delta^\varepsilon\|_\infty \le C\{\varepsilon^{1/16} + (1+\varepsilon^{-1})\Delta\}, \tag{3.4}$$

$$\limsup_m \sup_{R,n,k} m^2 \|A_\rho v^0\|_\infty \le C. \tag{3.5}$$

The constants are independent of the previously fixed widths. The coarea identity identifies the full physical sum of AI_m/z with v^0. Fixed-count local finiteness and common roof versions are retained. These are averaged, not pointwise, bounds.

These are the positive pairing, uncut volume and weighted optical variation statements in Section 2 of the frozen article [5]. Their exact historical source labels are recorded in the proof ledger. The new proof uses (3.3) only on regular interiors and supplies its integrable passage to the physical endpoints.

3.1. **The original microscopic source.** The outer schedule is

$$\varepsilon(B) = A_0 B^{-1/12}, \qquad r_0 = c_0^{\mathrm{out}} \varepsilon(B)^4, \qquad \delta(B) = \tfrac{3}{4} r_0, \qquad \Delta_B = 8\delta(B). \tag{3.6}$$

The fixed constants A_0, c_0^{out} are the inherited constants; c_0^{out} is written this way to avoid confusion with the initial incidence c_0. Keep the original remainder selector H_B, the paid critical-arm family $\mathscr{C}_B$, and the adaptive paid family $\mathscr{D}_B$ of the historical microscopic reduction. Their memberships are Borel, disjointized by first physical membership, and their total outer costs are at most $C\varepsilon(B)^{-1/32}$. For a previously fixed $0 < h_B < \delta(B)$ put

$$\begin{aligned} \widetilde{H}_B &= H_B \mathbf{1}_{(\mathscr{C}_B \cup \mathscr{D}_B)^c} \mathbf{1}_{\{0<\ell<h_B\}}, \\ \mathsf{U}_B^{\mathrm{micro}}(t) &= \int c^{\varepsilon(B)} \mathbf{1}_{\{a<t<b\}} \widetilde{H}_B(\Psi(\theta,t))\, d\theta. \end{aligned} \tag{3.7}$$

No sign cancellation, deletion of an endpoint class, or change of section occurs in this definition. Here Ψ denotes the original physical parametrization.

Input statement 3.2 (The scalar microscopic reduction)**.** For every fixed M, define

$$\mathcal{H}_M(T) = \limsup_m \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t \in \mathcal{C}_{m,R,n,k}(M)} T(t).$$

The historical exact positive reduction gives

$$\mathcal{E}_M \le \mathcal{H}_M(\mathsf{U}_B^{\mathrm{micro}}) + C_M B^{-1/384}. \tag{3.8}$$

The assertion (1.5) is equivalent to $\mathcal{H}_M(\mathsf{U}_B^{\mathrm{micro}}) \to 0$ for every fixed M. All widths and costs are fixed at the outer stage before the complete physical sum, the essential suprema, and the collision limsup; B is outermost.

This is the unchanged theorem in historical module 255. The fixed-positive-residence estimate in module 253 and the earlier paid-arm estimates are retained as its inputs. A bound uniform in a previously fixed positive residence is not used at a collision-dependent residence.

4. The reduced optical action

Fix an actual collision itinerary with $m \geq 1$ flights. Its visited centres are $O_0, \dots, O_m$, its collision points are $x_i = O_i + Rn(\alpha_i)$, and $n(\alpha) = (\cos\alpha, \sin\alpha)$. Write $\tau_i = |x_{i+1} - x_i|$. All variations in this section are made on this actual itinerary before a physical boundary is reached. They neither continue an event through a forbidden collision nor change an occupation label.

Lemma 4.1 (The original section stays away from endpoint grazing)**.** *On the closure of the original initial and terminal section tests,*

$$|p_0|, |p_m| \leq \frac{9}{10}, \qquad c_0, c_m \geq c_\circ := \frac{\sqrt{19}}{10}. \tag{4.1}$$

Proof. The only section centres with tangential coordinate exceeding $1/2$ in absolute value have coordinate $2q_R/(1+q_R^2)$. This function is increasing on $(0,1)$, and

$$\frac{2(53/100)}{1+(53/100)^2} + \frac{1}{200} < \frac{9}{10}.$$

The other squares and the rectangle U satisfy the same strict bound. Passing to their closures proves the assertion. □

Let $S(\alpha_0, \dots, \alpha_m) = \sum_{i=0}^{m-1} \tau_i$ be the broken-path action. At a regular specular trajectory its interior variables are stationary. Eliminating them locally gives the actual endpoint action $F(q)$, where $q = (\alpha_0, \alpha_m)$.

Lemma 4.2 (Second variation and its Schur complement)**.** *For a real variation $\xi = (\xi_0, \dots, \xi_m)$ at an actual regular trajectory,*

$$\begin{aligned} D^2S[\xi,\xi] = R\left(c_0\xi_0^2 + c_m\xi_m^2 + 2\sum_{i=1}^{m-1} c_i\xi_i^2\right) \\ + R^2 \sum_{i=0}^{m-1} \frac{(c_i\xi_i + c_{i+1}\xi_{i+1})^2}{\tau_i}. \end{aligned} \tag{4.2}$$

The reduced Hessian $G = D_q^2 F$ satisfies

$$R\,\mathrm{diag}(c_0, c_m) \leq G, \qquad \lambda_\circ I_2 \leq G \leq \Lambda_\circ I_2, \quad \lambda_\circ = \frac{9\sqrt{19}}{200}, \quad \Lambda_\circ = \frac{47}{6}. \tag{4.3}$$

These inequalities are uniform in the radius, the collision count, and the original exact labels.

Proof. For a flight with unit direction v, the Hessian of its length in its endpoints is the squared perpendicular displacement divided by the flight length. The acceleration of $Rn(\alpha)$ is $-Rn(\alpha)$. At the departure and arrival points, respectively, $v\cdot n_i = c_i$ and $v\cdot n_{i+1} = -c_{i+1}$. Also $(Rn_i')\cdot v^\perp = Rc_i$ and $(Rn_{i+1}')\cdot v^\perp = -Rc_{i+1}$. Thus one flight contributes $R(c_i\xi_i^2 + c_{i+1}\xi_{i+1}^2)$ and the corresponding square in (4.2). Summing proves that formula.

Let $H = D^2S$, $E = \{0, m\}$ and $I = \{1, \dots, m-1\}$. For $m > 1$, the interior block is positive definite. The implicit function theorem and differentiation of the stationary action give

$$G = H_{EE} - H_{EI}H_{II}^{-1}H_{IE}, \qquad \eta^{\mathsf{T}} G \eta = \min_{\zeta} (\eta, \zeta)^{\mathsf{T}} H (\eta, \zeta). \tag{4.4}$$

The order in the last expression places endpoint coordinates first. The diagonal endpoint terms in (4.2) prove the first lower bound. Choosing the interior variation to be zero gives $G \leq H_{EE}$.

The latter block is bounded above by $\{R_+ + 2R_+^2/\tau_-\}I_2 = (47/6)I_2$; the same bound covers the single-flight case, where there is no Schur complement. Lemma 4.1 proves the remaining lower bound. □

4.1. **A positive optical denominator.** Put

$$\mathsf{S}(s) = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}, \qquad \mathsf{C}(\kappa) = \begin{pmatrix} 1 & 0 \\ \kappa & 1 \end{pmatrix}, \qquad \kappa_i = \frac{2}{Rc_i}. \tag{4.5}$$

Let M_m be the product from the initial collision to the terminal collision, with the rightmost factor acting first:

$$\mathsf{M}_m = \mathsf{S}(\tau_{m-1})\mathsf{C}(\kappa_{m-1})\cdots\mathsf{S}(\tau_1)\mathsf{C}(\kappa_1)\mathsf{S}(\tau_0), \qquad \mathfrak{b}_m = (\mathsf{M}_m)_{12}. \tag{4.6}$$

For $m = 1$ the product is just $\mathsf{S}(\tau_0)$.

Lemma 4.3 (Exact denominator and a marked incidence bound)**.** *The actual endpoint-coordinate density is*

$$\omega = |\partial_{p_0}\alpha_m|^{-1} = \frac{|G_{0m}|}{R} = \frac{Rc_0c_m}{\mathfrak{b}_m}. \tag{4.7}$$

With $\varrho = 47/59$, one has

$$\omega \le \frac{47}{6}\varrho^{m-1}, \qquad \frac{\omega}{c_j} \le \frac{2209}{72}\varrho^{m-2} \quad (1 \le j \le m-1). \tag{4.8}$$

The second inequality is asserted when $m \ge 2$.

Proof. The first equality with G follows by differentiating $p_0 = -\partial_{\alpha_0}F/R$ with α_0 fixed. For the denominator, the interior block of H is $R^2C_IJC_I$, where $C_I = \mathrm{diag}(c_1, \dots, c_{m-1})$ and J is tridiagonal with

$$J_{ii} = \kappa_i + \tau_{i-1}^{-1} + \tau_i^{-1}, \qquad J_{i,i+1} = \tau_i^{-1}.$$

The corner cofactor in (4.4) gives $|G_{0m}| = R^2c_0c_m/(\prod_i \tau_i \det J)$. Expansion of successive tridiagonal determinants gives $\prod_i \tau_i \det J = \mathfrak{b}_m$. This identity also follows by multiplying the second column in (4.6); both recursions start with $\mathfrak{b}_1 = \tau_0$. The empty determinant convention handles $m = 1$.

All entries in the optical factors are nonnegative. Multiplication of a positive second column $(b, d)^{\mathsf{T}}$ by $\mathsf{S}(\tau_i)\mathsf{C}(\kappa_i)$ changes its top entry to $(1+\tau_i\kappa_i)b+\tau_i d$. Since $1+\tau_i\kappa_i \ge 1+2\tau_-/R_+ = 59/47$,

$$\mathfrak{b}_m \ge \tau_-\varrho^{-(m-1)}. \tag{4.9}$$

For a fixed interior collision j, split the product immediately before and after $\mathsf{C}(\kappa_j)$. Its $(1,2)$ entry is an expression with nonnegative coefficients in κ_j, whose linear coefficient is $\mathfrak{b}_{\mathrm{right}}\mathfrak{b}_{\mathrm{left}}$. The two subproducts have respectively $m-j$ and j flights. Hence

$$\mathfrak{b}_m \ge \kappa_j\tau_-^2\varrho^{-(m-2)}.$$

Substitution into (4.7), and $c_0, c_m \le 1$, proves (4.8). □

Corollary 4.4 (The full inherited incidence insertion)**.** *For the unchanged insertion $I_m = 1 + \sum_{i=1}^{m-1} q_0^{d_i}/c_i$, with $q_0 = 47/53$,*

$$\omega I_m \le 500\varrho^{(m-2)_+}. \tag{4.10}$$

This is an estimate on each actual physical trajectory; it involves no lower cutoff on any interior incidence.

Proof. For $m \ge 2$, $\sum_{i=1}^{m-1} q_0^{d_i} \le 2q_0/(1-q_0) = 47/3$. Use (4.8) and $(47/6)\varrho + (2209/72)(47/3) < 500$. For $m = 1$ the sum is empty and the first optical bound suffices. □

No estimate for a sum over collision words is obtained by simply summing the right side of (4.10). In particular, this per-trajectory exponential estimate is not being substituted for a collision-uniform local limit theorem.

5. Completion of the original roofwise endpoint traces

We use the original incident-component pairing sheets and their constant-angle coordinates $(u,\theta)=(F,\arg\nabla F)$. On one of their positive-length maximal regular segments, write $a<u<b$, $g=\nabla F=ze(\theta)$ and $e^{\perp}(\theta)=(-\sin\theta,\cos\theta)$. The physical section and all original exact-event tests are retained.

Lemma 5.1 (Finite Gauss-coordinate length)**.** *Along a fixed-θ segment,*

$$\frac{dq}{dz}=G^{-1}e,\qquad \frac{du}{dz}=ze^{\mathsf{T}}G^{-1}e,\qquad 2\lambda_\circ\le\frac{d(z^2)}{du}\le 2\Lambda_\circ. \tag{5.1}$$

Consequently z is strictly increasing, both endpoint limits of q exist, and

$$\int_a^b\frac{du}{z(u,\theta)}\le\lambda_\circ^{-1}(z_b-z_a)\le\lambda_\circ^{-1}\min\{\sqrt{2}R_+,\sqrt{2\Lambda_\circ(b-a)}\}. \tag{5.2}$$

A critical endpoint of a positive-length segment can only be its incoming endpoint.

Proof. Differentiate $\nabla F(q)=ze$ and then $F(q)=u$. The bounds in (4.3) give (5.1). They also give $|dq/dz|\le\lambda_\circ^{-1}$. Since $0<z\le\sqrt{2}R_+$, the two endpoint limits of q exist in the chosen angular lifts. Integrating $du/z=e^{\mathsf{T}}G^{-1}e\,dz$ proves the first inequality in (5.2). The second uses $z_b^2-z_a^2\le 2\Lambda_\circ(b-a)$ and $(z_b-z_a)^2\le z_b^2-z_a^2$. If $z_b=0$, strict increase would contradict $z>0$ in the interior. □

Lemma 5.2 (A continuous reduced Hessian at visited grazing)**.** *The reduced Hessian has a continuous extension along a fixed itinerary when any collection of its interior c_i tends to zero, provided $\tau_i\ge\tau_-$ and* (4.1) *hold. Its extension still satisfies* (4.3)*. If at least one interior c_j vanishes, the extended mixed entry G_{0m} vanishes.*

Proof. Let $L=\sum_{i=0}^{m-1}\tau_i^{-1}(e_i+e_{i+1})(e_i+e_{i+1})^{\mathsf{T}}$, and set $C_E=\operatorname{diag}(c_0,c_m)$. For $m>1$ put

$$Z=2I+RC_I^{1/2}L_{II}C_I^{1/2}\ge 2I.$$

Formula (4.4) can be written without inverse incidences:

$$\begin{aligned}G=RC_E+R^2C_EL_{EE}C_E\\ -R^3C_EL_{EI}C_I^{1/2}Z^{-1}C_I^{1/2}L_{IE}C_E.\end{aligned} \tag{5.3}$$

Every term is continuous for $c_i\ge 0$, since Z^{-1} remains bounded. The order inequalities pass to this limit. The marked bound in (4.8), together with $|G_{0m}|=R\omega$, proves the last assertion. There is no interior extension to make when $m=1$. □

At an endpoint of a fixed-θ segment the complete finite trajectory also has a limit. Indeed Lemma 5.1 gives the initial angle and $p_0=-z\cos\theta/R$. For each designated next disk, the entry root of the flight quadratic is continuous up to a tangency. The flight length cannot collapse to zero, and the reflection $v\mapsto v-2(v\cdot n)n$ is continuous at tangency. Induction over the fixed finite itinerary gives the limits of all visited points, velocities, τ_i and c_i. The third-disk clearances have limits as well: the finite horizon restricts the possible nearby disk centres to a finite set, and the minimum distance to each of them is continuous. This argument takes only one-sided limits from the actual event; it asserts no continuation through its boundary.

Theorem 5.3 (Trace completion on the original physical sheets)**.** *On every positive-length original regular pairing segment the coarea factor and weighted density are*

$$\chi = \det D_q(F, \arg g) = e^{\perp\mathsf{T}} G e^{\perp} \geq \lambda_\circ, \qquad f^\varepsilon = \frac{\omega W^\varepsilon}{4\pi c_* \chi}. \tag{5.4}$$

They have finite one-sided endpoint limits. The regular-point variation input (3.3) *implies that* f^ε *extends absolutely continuously to the closed segment and satisfies*

$$|f^\varepsilon(b-) - f^\varepsilon(a+)| \leq C(1+\varepsilon^{-1}) \int_a^b A I_m / z \, du. \tag{5.5}$$

In fact, on each physical sheet,

$$|f^\varepsilon(v) - f^\varepsilon(u)| \leq C(1+\varepsilon^{-1}) \varrho^{(m-2)_+} |v-u|^{1/2} \quad (a \leq u < v \leq b). \tag{5.6}$$

If a limiting trajectory at either endpoint contains an actually visited interior grazing collision, its trace at that endpoint is zero. The original paired minimum on that entire segment is then zero, for every original selector.

Proof. The determinant identity in (5.4) follows by expressing $D \arg g$ as $e^{\perp\mathsf{T}} G / z$. The preceding endpoint-limit argument and Lemma 5.2 give limits for G, ω and χ, the last with a positive lower bound. Every factor in the finite product defining W^ε is continuous in its incidence or clearance argument. Thus f^ε has the asserted finite limits, including at a critical incoming endpoint, where the direction θ is fixed.

Corollary 4.4 and (5.4) bound AI_m by $C\varrho^{(m-2)_+}$. The integral in (5.5) is finite by Lemma 5.1. On compact regular subsegments the flight charts are smooth and the finite clearance minima are locally Lipschitz, so f^ε is locally absolutely continuous. Integrating the inherited regular-point derivative inequality on these subsegments and then passing to the proved endpoint limits establishes absolute continuity and (5.5). The same argument between u and v, using (5.2), gives (5.6).

Finally, if $c_j \to 0$ at a visited interior collision, the marked optical bound gives $\omega \to 0$, while $0 \leq W^\varepsilon \leq 1$ and $\chi \geq \lambda_\circ$. Hence that endpoint trace is zero. The paired density is its nonnegative minimum with the other trace. □

Remark 5.4 (Two different tangency mechanisms)*.* The last assertion concerns a collision in the visited itinerary. A flight which merely becomes tangent to an *unvisited* third disk has zero clearance but need not have any $c_j = 0$. If all visited incidences remain positive, then (4.7) has a positive limit. A zero third-disk clearance makes $W^\varepsilon = 1$ at the limit; consequently the trace can be positive. Such sheets are not removed. Neither are nongrazing critical entrances or positive section-boundary traces. For a nongrazing critical entrance the exact directional limit is $\omega_* W_*^\varepsilon / (4\pi c_* e^{\perp\mathsf{T}} G_* e^{\perp})$. No roof-independent value at the critical point is asserted.

6. The complete physical comparison space

The comparison space is now specified without an unverified endpoint-trace exclusion:

$$\Theta_{\mathrm{ad}} = \Theta_{\mathrm{phys}}. \tag{6.1}$$

More explicitly, it consists of all positive-length maximal regular pairing segments of the original exact event, with their original incident-component identities. Their endpoints may be critical, visited-grazing, third-disk-tangent, or section-boundary limits. Theorem 5.3 supplies finite traces and the endpoint variation inequality on this space. The underlying Borel pairing construction and the averaged analytic estimates remain the inputs in Section 3. No singular trajectory is continued past its physical boundary, and no new source supported at a zero-length segment is introduced.

For fixed $0 < r < \Delta$, define

$$q_+(t,\theta) = \mathbf{1}_{\{0<\ell<\Delta,\ a<t,\ t+r<b\}},$$
$$q_-(t,\theta) = (1-q_+)\mathbf{1}_{\{0<\ell<\Delta,\ a<t-r,\ t<b\}}, \qquad Q_r = q_+ + q_-. \tag{6.2}$$

Thus Q_r is Borel and at most one. Write

$$P_r(t) = \int Q_r p_\vee \, d\theta, \qquad D_r(t) = \int Q_r c^\varepsilon \, d\theta, \qquad E_r(t) = \int Q_r e_\partial \, d\theta.$$

The full pool is retained even on a sheet whose paired source is zero.

Theorem 6.1 (Capacity on the full original comparison space)**.** *At finite collision count, on the original common roof versions,*

$$P_r(t) \le \frac{1}{r}\int_{t-r}^{t+r} \mathsf{T}^\varepsilon_\Delta(u)\, du + C(1+\varepsilon^{-1})\int_{t-\Delta}^{t+\Delta} v^0(u)\, du. \tag{6.3}$$

Consequently, for every previously fixed $K > 0$,

$$\mathcal{H}_M(KP_r) \le CK\{\varepsilon^{1/16} + (1+\varepsilon^{-1})\Delta\}. \tag{6.4}$$

Proof. Every positively oriented selected segment contains $[t, t+r]$; every negatively oriented one contains $[t-r, t]$. Integrating its constant paired minimum over the appropriate window gives r times that minimum. Tonelli and $Q_r \le 1$ therefore give $D_r(t) \le r^{-1}\int_{t-r}^{t+r} \mathsf{T}^\varepsilon_\Delta$. Each selected segment lies inside $(t-\Delta, t+\Delta)$. The completed variation inequality (5.5) and the original one-copy physical coarea sum give $E_r(t) \le C(1+\varepsilon^{-1})\int_{t-\Delta}^{t+\Delta} v^0$. Since $P_r = D_r + E_r$, this proves (6.3). Use (3.4) and (3.5), only on these previously fixed windows, to obtain (6.4). The weights are held at the original roof t throughout; none is differentiated or evaluated at the integration roof u. □

6.1. **The physical grazing partition.** For each interior collision let c_i^- and c_i^+ be its incidence limits at the two ends of a pairing segment. Define the Borel sets

$$\Gamma = \bigcup_{i=1}^{m-1} \{c_i^- = 0 \text{ or } c_i^+ = 0\}, \qquad \mathcal{N} = \Theta_{\text{phys}} \setminus \Gamma. \tag{6.5}$$

For $m = 1$ the union is empty. The limits are Borel, for example by evaluating the interior functions at $a + \ell/j$ and $b - \ell/j$ and letting the integer j tend to infinity. Set

$$P_r^{\text{gr}} = \int_\Gamma Q_r p_\vee \, d\theta, \qquad P_r^{\text{ng}} = \int_{\mathcal{N}} Q_r p_\vee \, d\theta, \qquad P_r = P_r^{\text{ng}} + P_r^{\text{gr}}.$$

The superscripts refer to visited collision incidences in the limiting finite itinerary, not to a lower incidence cutoff.

Theorem 6.2 (Zero original grazing source and paid grazing capacity)**.** *For every original Borel selector $0 \le H \le 1$, at finite count,*

$$\int_\Gamma c^\varepsilon \mathbf{1}_{\{a<t<b\}} H(\Psi(\theta,t))\, d\theta = 0. \tag{6.6}$$

Nevertheless P_r^{gr} need not vanish. It satisfies

$$P_r^{\text{gr}}(t) \le C(1+\varepsilon^{-1})\int_{t-\Delta}^{t+\Delta} v^0(u)\, du, \tag{6.7}$$

$$\mathcal{H}_M(KP_r^{\text{gr}}) \le CK(1+\varepsilon^{-1})\Delta. \tag{6.8}$$

These estimates include the complete positive capacity on this physical class, with no count of angular blocks or collision words.

Proof. Theorem 5.3 gives $c^{\varepsilon} = 0$ on Γ, proving (6.6) before the roof pushforward. On that same class $p_{\vee} = e_{\partial}$, not necessarily zero. Apply (5.5), use the localization $(a, b) \subset (t - \Delta, t + \Delta)$ on $\{Q_r = 1\}$, and complete the one-copy positive physical sum. This proves (6.7). Equation (3.5) at the fixed width Δ proves (6.8). □

7. The original microscopic shortage after grazing payment

All allocations below are scalar allocations at the original roof; they are not couplings of individual source and comparison sheets. For any original source $U(t)$ define

$$V_{r,K}(t) = \min\{U(t), KP_r(t)\}, \qquad R_{r,K}(t) = (U(t) - KP_r(t))_+.$$

The multiplier is $\beta_{r,K} = \min\{1, KP_r/U\}$ on $U > 0$ and zero on $U = 0$. Then $U = V_{r,K} + R_{r,K}$ at finite count and $V_{r,K}$ is the original source multiplied by $\beta_{r,K}(t)$. When a restricted comparison space is used, source outside it is still in U and may receive this same global fraction. It is not therefore retained sheetwise with coefficient one.

Use the exact source (3.7) and, for a concrete outer choice, put

$$h_B = \delta(B)/2, \qquad r_B = \delta(B)/8, \qquad K_B = \max\{1, \varepsilon(B)^{-1/32}\}. \tag{7.1}$$

For all sufficiently large B these satisfy the inherited cost budget. Define

$$\begin{aligned} \mathsf{R}_B^{\mathrm{full}} &= (\mathsf{U}_B^{\mathrm{micro}} - K_B P_{r_B})_+, \\ \mathsf{R}_B^{\mathrm{ng}} &= (\mathsf{U}_B^{\mathrm{micro}} - K_B P_{r_B}^{\mathrm{ng}})_+. \end{aligned} \tag{7.2}$$

The full capacity pool includes any sheets previously excluded solely for lack of a trace verification. On the complete comparison space it is exactly the canonical pool; on a previously restricted space it is an enlargement, not an identification of different capacities.

Theorem 7.1 (Quantitative payment of the visited-grazing class)**.** *Under the retained physical-source and analytic inputs, for the original event and every fixed M,*

$$0 \le \mathsf{R}_B^{\mathrm{ng}} - \mathsf{R}_B^{\mathrm{full}} \le K_B P_{r_B}^{\mathrm{gr}}, \tag{7.3}$$

$$\mathcal{H}_M(\mathsf{R}_B^{\mathrm{ng}} - \mathsf{R}_B^{\mathrm{full}}) \le C_M B^{-95/384}, \tag{7.4}$$

$$\mathcal{E}_M \le \mathcal{H}_M(\mathsf{R}_B^{\mathrm{ng}}) + C_M B^{-1/384}. \tag{7.5}$$

The unchanged pointwise assertion (1.5) *is equivalent to $\mathcal{H}_M(\mathsf{R}_B^{\mathrm{ng}}) \to 0$ for every fixed M. The original microscopic source in these formulas is supported on $\mathcal{N}$ without a change of its positive weight.*

Proof. For $u, a, b \ge 0$, the positive-part function gives $0 \le (u - a)_+ - (u - a - b)_+ \le b$. Apply this with $a = K_B P_{r_B}^{\mathrm{ng}}$ and $b = K_B P_{r_B}^{\mathrm{gr}}$ to prove (7.3) at finite count. Theorem 6.2, $\Delta_B = O(\varepsilon^4)$ and $K_B = O(\varepsilon^{-1/32})$ give

$$\mathcal{H}_M(K_B P_{r_B}^{\mathrm{gr}}) \le C\varepsilon^{-1/32}(1 + \varepsilon^{-1})\varepsilon^4 = O(\varepsilon^{95/32}) = O(B^{-95/384}).$$

This proves (7.4) in the prescribed order. By Theorem 6.1, the accepted full-pool source has height at most

$$C\varepsilon^{-1/32}\{\varepsilon^{1/16} + O(\varepsilon^3)\} = O(B^{-1/384}).$$

Insert its exact positive decomposition into (3.8). Since $\mathsf{R}_B^{\mathrm{full}} \le \mathsf{R}_B^{\mathrm{ng}}$, this proves (7.5) and sufficiency of the displayed residual limit. Necessity follows from Input statement 3.2 and $0 \le \mathsf{R}_B^{\mathrm{ng}} \le \mathsf{U}_B^{\mathrm{micro}}$. Equation (6.6) proves the final assertion. □

This theorem supplies a physical component of the required scarcity estimate, rather than a new certificate class. It removes an entire visited-grazing source class exactly and pays the loss of all its comparison capacity by a faster-decaying original optical volume. It does not yet bound $\mathcal{H}_M(\mathsf{R}_B^{\mathrm{ng}})$ by a quantity tending to zero. In $\mathcal{N}$ every visited endpoint incidence is positive at each fixed count, but its infimum may tend to zero with the count. Third-disk tangencies, nongrazing critical entrances, section-boundary cuts, flat and short-reach segments, and the original comparison shortages all remain with their full weight. A collision-uniform estimate on that remaining source is still required for (1.5). Neither finite-count exhaustion nor the per-itinerary exponential bound is used in its place.

8. A positive payment from the variation on a physical segment

The comparison pool of Section 6 is left unchanged. We first make a different use of the completed endpoint estimate: it pays part of the *original source*, whether or not that source has a comparison window. All statements in this section are made on the inherited one-copy physical pairing space.

Write C_{reg} for a fixed constant in (3.3). For fixed $\varepsilon > 0$ define

$$\begin{aligned} \mathcal{V}^{\varepsilon}(\theta) &= \mathrm{Var}_{[a(\theta),b(\theta)]}\, f^{\varepsilon}(\,\cdot\,,\theta), \\ \mathcal{J}(\theta) &= \int_{a(\theta)}^{b(\theta)} \frac{A(u,\theta) I_m(u,\theta)}{z(u,\theta)}\, du. \end{aligned} \tag{8.1}$$

Endpoint values in the variation are the one-sided physical traces, not values assigned on a continuation beyond the original event.

Lemma 8.1 (Measurability and the full variation bound)**.** *Both functions in* (8.1) *are Borel and finite on every positive-length physical segment. Moreover,*

$$\mathcal{V}^{\varepsilon} = \int_a^b |\partial_u f^{\varepsilon}|\, du \le C_{\mathrm{reg}}(1+\varepsilon^{-1})\mathcal{J}. \tag{8.2}$$

This is a bound for total variation, not merely for the difference of the two endpoint values.

Proof. Pull the closed segment back to $[0,1]$ by $u = a + s(b-a)$, using the endpoint traces at $s = 0, 1$. The interior evaluations are jointly Borel. At $s = 0$ the completed value is the limit of evaluations at $a + \ell/j$, $j \ge 2$; at $s = 1$ it is the limit of evaluations at $b - \ell/j$. The completed endpoint values, and hence the function on the closed product, are jointly Borel. The resulting function is continuous in s. Its variation is the supremum of the sums of absolute increments over finite rational partitions containing 0 and 1. Continuity permits this countable family of partitions, so its supremum is Borel. The integral defining $\mathcal{J}$ is Borel by the same change of variables and nonnegative integration. Theorem 5.3 gives absolute continuity on the closed segment. The variation of an absolutely continuous real function is the integral of the absolute value of its derivative. Integrate (3.3); its right side is integrable by Corollary 4.4 and Lemma 5.1. This proves the equality, inequality and finiteness asserted above. □

For $\gamma \ge 1$, split the original paired coefficient into

$$c_{\mathrm{var}}^{\varepsilon,\gamma} = \min\{c^{\varepsilon}, \gamma\mathcal{V}^{\varepsilon}\}, \qquad c_{\mathrm{per}}^{\varepsilon,\gamma} = (c^{\varepsilon} - \gamma\mathcal{V}^{\varepsilon})_+. \tag{8.3}$$

Thus $c^{\varepsilon} = c_{\mathrm{var}}^{\varepsilon,\gamma} + c_{\mathrm{per}}^{\varepsilon,\gamma}$, with both terms nonnegative. The subscript per will be justified by a physical persistence statement in the next section. It denotes a positive *submeasure* of the original source, not a replacement of its coefficient by a different normalization.

For a jointly Borel selector $0 \le H \le 1$ supported on $0 < \ell < h$, set

$$T_{H,j}^{\varepsilon,\gamma}(t) = \int c_j^{\varepsilon,\gamma}(\theta)\mathbf{1}_{\{a<t<b\}} H(\Psi(\theta,t))\, d\theta, \qquad j \in \{\mathrm{var}, \mathrm{per}\}. \tag{8.4}$$

The original selected paired source is their exact sum. The selector is evaluated at t and is not moved to another roof in the proof.

Theorem 8.2 (A summed payment of the positive short source)**.** *At every finite collision count, on the original common roof versions,*

$$0 \le T^{\varepsilon,\gamma}_{H,\mathrm{var}}(t) \le C_{\mathrm{reg}}\gamma(1+\varepsilon^{-1}) \int_{t-h}^{t+h} v^0(u)\,du. \tag{8.5}$$

Consequently, for every $\varepsilon, h > 0$ and $\gamma \ge 1$ fixed before the collision limit,

$$\mathcal{H}_M(T^{\varepsilon,\gamma}_{H,\mathrm{var}}) \le C\gamma(1+\varepsilon^{-1})h. \tag{8.6}$$

The constants are uniform in the collision count, radius, exact labels, segment length and selector. No positive lower bound for ℓ or for an interior incidence is required.

Proof. For a segment meeting t and satisfying $\ell < h$, one has $(a,b) \subset (t-h, t+h)$. By (8.3) and (8.2), its selected contribution is at most

$$C_{\mathrm{reg}}\gamma(1+\varepsilon^{-1})\mathbf{1}_{\{a<t<b,\,\ell<h\}} H(\Psi(\theta,t)) \int_a^b AI_m/z\,du.$$

First sum these nonnegative quantities over the physical sheets. Tonelli's theorem applies, without an estimate for the number of sheets. At an integration roof u, each retained pair (θ, u) is still a point of the same original exact event. Dropping its selector, which is between zero and one, and using physical multiplicity one therefore bounds the sum by

$$C_{\mathrm{reg}}\gamma(1+\varepsilon^{-1}) \int_{t-h}^{t+h} \int_{\{a(\theta)<u<b(\theta)\}} A(u,\theta)I_m(u,\theta)/z(u,\theta)\,d\theta\,du.$$

The coarea identity in Input statement 3.1 identifies the inner integral with $v^0(u)$. This proves the finite-count inequality. Apply (3.5) at the previously fixed width h, using $\int_{t-h}^{t+h} v^0 = 2hA_h v^0$. Only after the complete sum and this inequality are established are the uniform norms and collision limsup taken. This proves (8.6). □

Corollary 8.3 (The original microscopic schedule)**.** *Keep $\widetilde{H}_B, h_B, r_B, K_B$ exactly as in* (3.7) *and* (7.1)*, and choose*

$$\gamma_B = \max\{2, \varepsilon(B)^{-1/32}\}. \tag{8.7}$$

Let $\mathsf{U}^{\mathrm{var}}_B$ and $\mathsf{U}^{\mathrm{per}}_B$ denote the two sources in (8.4) *with $H = \widetilde{H}_B$, $\varepsilon = \varepsilon(B)$ and $\gamma = \gamma_B$. Then*

$$\mathsf{U}^{\mathrm{micro}}_B = \mathsf{U}^{\mathrm{var}}_B + \mathsf{U}^{\mathrm{per}}_B, \qquad \mathcal{H}_M(\mathsf{U}^{\mathrm{var}}_B) \le C_M B^{-95/384}. \tag{8.8}$$

Proof. The identity holds before pushforward, by (8.3). For sufficiently large B, $h_B = O(\varepsilon^4)$ and $\gamma_B = O(\varepsilon^{-1/32})$. Theorem 8.2 gives the bound

$$C\gamma_B(1+\varepsilon^{-1})h_B = O(\varepsilon^{95/32}) = O(B^{-95/384}).$$

All three quantities $h_B, \varepsilon(B), \gamma_B$ are fixed before m tends to infinity. No collision-dependent residence is inserted into an estimate asserted only for fixed residence. □

Unlike (6.6), this payment does not rely on a zero paired coefficient. If $c^\varepsilon > 0$ and $\mathcal{V}^\varepsilon > 0$, its coefficient in (8.3) is positive. If $c^\varepsilon \le \gamma\mathcal{V}^\varepsilon$, the complete original coefficient is paid. These statements do not assert a positive uniform proportion of paid mass in every exact event.

9. Persistent guarded segments and the same pointwise target

The positive decomposition of the preceding section has a geometric consequence on every segment which retains source. It uses the original incidence and clearance guard, without adding a distance to the section boundary.

Theorem 9.1 (Persistence of the guard on the remaining source)**.** *Let $\gamma > 1$. If $c^{\varepsilon,\gamma}_{\mathrm{per}}(\theta) > 0$, then $c^{\varepsilon} > 0$ and, at every point of the closed physical segment,*

$$|f^{\varepsilon}(u,\theta) - c^{\varepsilon}(\theta)| \le \mathcal{V}^{\varepsilon}(\theta) < \gamma^{-1} c^{\varepsilon}(\theta). \tag{9.1}$$

In particular $f^{\varepsilon} > 0$ and $W^{\varepsilon} > 0$ throughout that segment, including its one-sided endpoints. For each u at least one of the original tests satisfies

$$c_j(u) < 2\varepsilon q_0^{d_j/4} \quad \text{for some } 0 \le j \le m, \qquad \text{or} \qquad \Delta_j(u) < 2\varepsilon q_0^{d'_j/4} \quad \text{for some } 0 \le j < m. \tag{9.2}$$

The witnessing test may change with u.

Conversely, if W^{ε} vanishes anywhere on the closed segment, its complete original paired coefficient is paid: $c^{\varepsilon,\gamma}_{\mathrm{var}} = c^{\varepsilon}$ and $c^{\varepsilon,\gamma}_{\mathrm{per}} = 0$.

Proof. The endpoint minimum is attained at at least one endpoint, where $f^{\varepsilon} = c^{\varepsilon}$. Every increment from that endpoint is bounded by the total variation. Positivity of the remaining coefficient is precisely $c^{\varepsilon} > \gamma \mathcal{V}^{\varepsilon}$, which proves (9.1). If any visited interior incidence vanished at an endpoint, Theorem 5.3 would give $c^{\varepsilon} = 0$. Thus the segment belongs to $\mathcal{N}$. Its optical factor A is finite and positive on the closed segment: the exact formula $\omega = Rc_0c_m/\mathfrak{b}_m$ in (4.7) has a finite positive denominator when all visited incidences are positive, while (5.4) gives $\chi \ge \lambda_\circ$. This also applies at a critical endpoint with its fixed Gauss direction. Since $f^{\varepsilon} = AW^{\varepsilon} > 0$, the guard is positive there. If all inequalities in (9.2) failed, every factor defining H^{ε} would equal one, contradicting $W^{\varepsilon} = 1 - H^{\varepsilon} > 0$.

For the converse, let u_0 be a point where $W^{\varepsilon} = 0$. The finite optical factor gives $f^{\varepsilon}(u_0) = 0$, so the increment from an endpoint attaining c^{ε} gives $\mathcal{V}^{\varepsilon} \ge c^{\varepsilon}$. Apply (8.3). This argument also covers $u_0 = a$ or $u_0 = b$, with the proved physical one-sided values. □

The surviving class

$$\mathcal{P}_B = \{\theta \in \mathcal{N} : c^{\varepsilon(B)}(\theta) > \gamma_B \mathcal{V}^{\varepsilon(B)}(\theta)\} \tag{9.3}$$

is Borel. The source $\mathsf{U}^{\mathrm{per}}_B$ is supported on it; its coefficient is still the positive difference in (8.3), not the full original coefficient. The sum (8.8) retains every original coefficient exactly. No assumption that one fixed bad test covers the entire segment is made. A Borel witness at a specified point can be chosen as the first successful test in the finite ordered list in (9.2); this selection is not needed to prove the payment.

Remark 9.2 (Boundary mechanisms and what is retained). An unvisited third-disk tangency may give $W^{\varepsilon} = 1$ at an endpoint without forcing variation relative to the endpoint minimum. It is therefore not declared paid merely on account of its name. The same applies to a nongrazing critical entrance, an original section cut and a short-reach segment. Each receives precisely the two coefficients in (8.3). When the guard vanishes elsewhere on that segment the entire coefficient is paid by Theorem 9.1; otherwise a positive remaining coefficient is possible. At fixed collision count the optical data on a surviving segment have positive minima, but no such minimum is asserted uniformly in the collision count or the sheet.

9.1. **Comparison with the original nongrazing shortage.** Keep the *full original* nongrazing pool $P^{\mathrm{ng}}_{r_B}$, including its capacity on segments whose source has already been paid. Define only a new source-side remainder:

$$\mathsf{R}^{\mathrm{per}}_B(t) = \big(\mathsf{U}^{\mathrm{per}}_B(t) - K_B P^{\mathrm{ng}}_{r_B}(t)\big)_+. \tag{9.4}$$

The allocation by this pool is scalar at the roof, just as in Section 7. It is not a coupling of individual persistent sheets to individual comparison sheets.

Theorem 9.3 (Positive-source reduction with a paid error)**.** *On the original exact event, at every finite collision count,*

$$\mathsf{U}_B^{\mathrm{micro}} = \mathsf{U}_B^{\mathrm{var}} + \min\{\mathsf{U}_B^{\mathrm{per}}, K_B P_{r_B}^{\mathrm{ng}}\} + \mathsf{R}_B^{\mathrm{per}}, \tag{9.5}$$

$$0 \le \mathsf{R}_B^{\mathrm{ng}} - \mathsf{R}_B^{\mathrm{per}} \le \mathsf{U}_B^{\mathrm{var}}. \tag{9.6}$$

For every fixed M, in the prescribed order of limits,

$$\mathcal{H}_M(\mathsf{R}_B^{\mathrm{ng}} - \mathsf{R}_B^{\mathrm{per}}) \le C_M B^{-95/384}, \tag{9.7}$$

$$\mathcal{E}_M \le \mathcal{H}_M(\mathsf{R}_B^{\mathrm{per}}) + C_M B^{-1/384}. \tag{9.8}$$

The unchanged pointwise assertion (1.5) *is equivalent to $\mathcal{H}_M(\mathsf{R}_B^{\mathrm{per}}) \to 0$ for every fixed M.*

Proof. Equation (9.5) is the positive scalar identity $x = \min(x, y) + (x - y)_+$ applied to $x = \mathsf{U}_B^{\mathrm{per}}$, combined with (8.8). Monotonicity and the unit Lipschitz bound of the positive-part function give (9.6). Corollary 8.3 then proves (9.7).

For completeness, the height of the middle term in (9.5) is at most

$$\mathcal{H}_M(K_B P_{r_B}) \le C\varepsilon^{-1/32}\{\varepsilon^{1/16} + (1 + \varepsilon^{-1})\Delta_B\} = O(\varepsilon^{1/32}) = O(B^{-1/384}),$$

by Theorem 6.1. Insert the three nonnegative terms in Input statement 3.2, and use subadditivity of the ordered upper height. This proves (9.8) and sufficiency of the displayed limit. Necessity follows from the same inherited scalar equivalence and $0 \le \mathsf{R}_B^{\mathrm{per}} \le \mathsf{U}_B^{\mathrm{micro}}$. No essential supremum is interchanged with an integral, and the comparison pool has not been restricted to $\mathcal{P}_B$. □

The estimate still required to conclude (1.5) is

$$\lim_{B\to\infty} \limsup_{m\to\infty} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} \big(\mathsf{U}_{B,m}^{\mathrm{per}}(t) - K_B P_{r_B,m}^{\mathrm{ng}}(t)\big)_+ = 0. \tag{9.9}$$

Theorems 8.2 and 9.1 prove an actual summed payment and a necessary physical persistence property. They do not prove (9.9). In particular a positive nearly constant density on a shrinking segment need not have small pointwise height merely because its roof integral is small. The original raw pointwise theorem, and not an averaged substitute, remains the organizing target.

10. Global uniqueness on a roof–angle fiber

The estimates in this section concern actual finite trajectories, not a continuation through a forbidden event. In particular the regular domain of a lifted itinerary need not be connected or convex. This distinction is important when the original section and occupation tests divide that domain into many components.

Normalize the initial lifted scatterer to $O_0 = 0$. Let $\mathcal{W}_m(R)$ denote the realizable lifted itineraries of m flights. This set is finite by the finite-horizon bound and the lattice geometry. Fix $w = (O_0, \dots, O_m) \in \mathcal{W}_m(R)$. For a physical trajectory write $Q_j = O_j + Rn_j$, let t_j be the counterclockwise rotation of n_j, and put

$$S(Q) = \sum_{j=0}^{m-1} |Q_{j+1} - Q_j|, \qquad g = R(-p_0, p_m), \qquad z = |g|.$$

As a function of the independent Cartesian contacts, S is convex. It is differentiable at every configuration in question because successive scatterer disks are disjoint. Write $\beta_0 = \beta_m = 1$ and $\beta_j = 2$ for $0 < j < m$.

Lemma 10.1 (Two-trajectory support inequalities)**.** *Let Q and $\widehat{Q}$ be two regular physical trajectories with the same lifted itinerary and radius. Set*

$$b_j = 1 - n_j \cdot \widehat{n}_j \geq 0, \qquad s_j = t_j \cdot \widehat{n}_j, \qquad s_E = (s_0, s_m).$$

Their lengths satisfy

$$S(\widehat{Q}) - S(Q) \geq R \sum_{j=0}^{m} \beta_j c_j b_j + g \cdot s_E, \tag{10.1}$$

$$S(Q) - S(\widehat{Q}) \geq R \sum_{j=0}^{m} \beta_j \widehat{c}_j b_j - \widehat{g} \cdot s_E. \tag{10.2}$$

Consequently

$$(\widehat{g} - g) \cdot s_E \geq R \sum_{j=0}^{m} \beta_j (c_j + \widehat{c}_j) b_j. \tag{10.3}$$

No interpolation by physical trajectories is used.

Proof. The physical reflection law and first variation in Cartesian coordinates give

$$\nabla_0 S = -c_0 n_0 - p_0 t_0, \qquad \nabla_j S = -2 c_j n_j \ (0 < j < m), \qquad \nabla_m S = -c_m n_m + p_m t_m.$$

The convex support inequality is $S(\widehat{Q}) - S(Q) \geq \sum_j \nabla_j S(Q) \cdot (\widehat{Q}_j - Q_j)$. Since $\widehat{Q}_j - Q_j = R(\widehat{n}_j - n_j)$, its normal terms are $R\beta_j c_j b_j$ and its two tangential terms are $g \cdot s_E$. This proves (10.1). Reversing the roles of the two configurations leaves b_j unchanged and replaces s_j by $-s_j$. This proves (10.2). Adding the two inequalities proves (10.3). The support inequality is applied to a convex Cartesian function; it requires neither a convex angular chart nor a physical path joining the configurations. □

Theorem 10.2 (Global roof–angle uniqueness)**.** *For a fixed lifted itinerary, radius and direction $e \in S^1$, the physical length is strictly increasing among regular trajectories whose endpoint gradients are $g = ze$, $z > 0$, ordered by z. At a fixed z there is at most one such trajectory. In particular*

$$Q \longmapsto (F(Q), \arg g(Q)) \tag{10.4}$$

is globally injective on the noncritical regular trajectories of that itinerary. This remains true after any of the original exact event tests, and across all their disconnected regular components.

Proof. If $\widehat{g} = g$, the left side of (10.3) is zero. All c_j and $\widehat{c}_j$ are strictly positive, so every $b_j = 0$. Thus all physical contacts, and then all flight directions, coincide.

Now let $\widehat{g} = \widehat{z}e$, $g = ze$ and $\widehat{z} > z > 0$. The configurations cannot coincide. At least one b_j is therefore positive, and (10.3) gives

$$(\widehat{z} - z) e \cdot s_E \geq R \sum_{j} \beta_j (c_j + \widehat{c}_j) b_j > 0.$$

Hence $e \cdot s_E > 0$. The right side of (10.1) is then strictly positive, proving $S(\widehat{Q}) > S(Q)$. Equal roof and angle would force equal z, and the first part of the proof would force equality of trajectories. Restricting an injective map to the original exact event cannot increase its multiplicity. □

Lemma 10.3 (A bounded roof range for each physical itinerary)**.** *For any two regular physical trajectories with the same itinerary,*

$$|S(Q) - S(\widehat{Q})| \leq |Q_0 - \widehat{Q}_0| + |Q_m - \widehat{Q}_m| \leq 4R. \tag{10.5}$$

Proof. With its endpoint contacts fixed, a regular physical trajectory minimizes S over the product of the closed interior scatterer disks. Indeed, for a competitor Y_j in the jth disk, $n_j \cdot (Y_j - Q_j) \le 0$; the convex support inequality and $\nabla_j S = -2c_j n_j$ therefore give the minimizing property. Use the interior contacts of $\widehat{Q}$ as a competitor for the endpoints of Q. Changing only the first and last endpoint changes the sum of flight lengths by at most the sum of their displacements, by the triangle inequality. This proves one direction of (10.5); interchanging the trajectories proves the other. The same triangle inequality covers $m = 1$. □

This range estimate does not control how many original occupation labels occur in a word domain. In particular it is not used to replace the exact occupation test by an unqualified interval test.

10.1. **The original physical sum at a target roof.** For the original section tests, $\chi \ge \lambda_\circ$ by (5.4). On each regular component, the map (10.4) is a local diffeomorphism. Theorem 10.2 shows that all those components of a fixed itinerary have disjoint images, not merely that each component has a nonsingular Jacobian. The equal-gradient argument also gives at most one regular critical trajectory per word, even when the gradient is zero. There are finitely many physical words at each finite count, so these critical points have zero collision area. Consequently, for any nonnegative Borel trajectory weight a, the original roof density has the representation

$$\frac{dF_{\#}(a\mathbf{1}_{\Pi}\nu_R^*)}{dt}(t) = \int_0^{2\pi} \sum_{w \in \mathcal{W}_m(R)} \mathbf{1}_{\{x_w(t,\vartheta) \text{ exists}\}} \frac{\omega(x_w) a(x_w)}{4\pi c_* \chi(x_w)}\, d\vartheta. \tag{10.6}$$

Here x_w is the unique point of the original exact event with the indicated word, roof and angle. A missing point contributes zero. The representation holds for almost every roof, in the original order $\sup_R \operatorname{ess\,sup}_t$. Angular chart overlap is removed before summation; the incident-component convention remains unchanged.

Proposition 10.4 (Monotone completion of the target-roof formula)**.** *Formula* (10.6) *is an identity of nonnegative measures for every nonnegative Borel weight, with extended values allowed before an integrability bound is proved. In particular it applies to the unbounded products* $\prod_{i \in S} c_i^{-1}$.

Proof. Fix m, R, n, k. Cover the noncritical regular physical domain in endpoint coordinates by countably many relatively compact local charts V_j on which $(F, \arg g)$ is a C^1 diffeomorphism into an unwrapped angular chart. Such a cover exists by the nonvanishing Jacobian and second countability. Subdivide as necessary so that both maps have bounded derivatives on each smaller chart. Disjointize the source charts by $B_j = V_j \setminus \bigcup_{k<j} V_k$. The sets B_j are Borel; no chart overlap is counted twice. Angular representatives are also split at a fixed seam, so their range is $[0, 2\pi)$.

For a nonnegative Borel weight h put

$$h_N = \min\{h, N\} \mathbf{1}_{\bigcup_{j \le N} B_j}.$$

This sequence increases to h on the regular noncritical domain. Apply ordinary change of variables on each of the first N charts, restricted to its Borel piece, and test against a nonnegative Borel function of (t, ϑ). The local inverse Jacobian, multiplied by the original collision density, is exactly $\omega/(4\pi c_* \chi)$. Local inverses are continuous and agree where they represent the same physical point, so their Borel restrictions define the displayed x_w without a selection ambiguity. Global uniqueness prevents different regular components of one word from producing the same target. Distinct words are summed, not identified.

Monotone convergence on the source, followed by Tonelli and monotone convergence on the target, now gives the identity for h as an extended nonnegative measure. Only after this passage is an upper pressure bound used to establish finiteness and an L^∞ density. This avoids assuming the integrability of the unbounded marked weight in order to prove its own change of variables.

The complement of the two-sided regular set has zero collision area. On the countably many charts just chosen, absolute continuity of the coordinate changes and their local Lipschitz property give zero target area for its image. The finitely many regular critical points also have zero source area. Removing these sets therefore does not change the weighted source measure. Fubini yields the claimed roof identity for almost every t. All null sets here are chosen at the fixed radius; no exceptional set common to every R is asserted or needed. For the finitely many marked sets at a fixed m, intersecting the corresponding full-measure roof sets provides a common version for their finite positive expansions. □

11. A complete marked pressure bound at the target roof

We now transfer the retained geometric pressure estimate to the roof fibers in (10.6). The previous momentum-fiber bound alone would not justify this transfer: different words at the same roof can have different endpoint momentum magnitudes. The global injectivity just proved supplies the missing multiplicity statement.

Let $j(x) = J^s T_R(x)$ on the full-area set M'_R regular in both infinite time directions, and let $j_l(x) = \prod_{v=0}^{l-1} j(T_R^v x)$, with $j_0 = 1$. The lifted physical one-flight alphabet has at most D members, uniformly in R. Define

$$P_l(R) = \sum_{w \in \mathcal{W}_l(R)} \sup_{x \in [w]_R \cap M'_R} j_l(x), \qquad P_0 = 1.$$

Input statement 11.1 (Uniform physical pressure and stable optical comparison)**.** On the circular family $R \in [9/20, 47/100]$, use the original finite-horizon flight bounds, positive curvature bounds and the original endpoint section cap. The stable Jacobian j is computed with Euclidean arclength along the true stable line in collision coordinates (s, φ), where $s = R\alpha$ and $p = \sin\varphi$. It is restricted to M'_R, the full-area set regular in both infinite directions. Its cocycle uses this same convention on every block, including the empty block. The endpoint weight ω is still the derivative in the original (α, p) coordinates of (3.1); all coordinate factors are included in the uniform optical constant.

For the realizable lifted physical words just defined, the retained geometric theorems state

$$\sup_{R,l} P_l(R) \le Q_*, \qquad \omega(x) \le C_0 c_m(x) j_m(x), \qquad j(x)/c(x) \le C_1. \tag{11.1}$$

All three constants are independent of R, l and the original exact labels. These statements concern positive, unconditioned physical word sums; they are not a twisted or exact-label pressure theorem.

We take $Q_* \ge 1$ and $C_0, C_1 > 0$. These are the uniform geometric pressure theorem in historical module 241 and the stable optical comparison in module 238, with the physical-word domination in module 239. Their complete original sources are pinned in the proof ledger and copied in the provenance directory. They involve no occupation, displacement or roof test. Such tests are dropped only on the upper side of the estimates below. No new pressure estimate for a twisted or point-conditioned operator is assumed.

Lemma 11.2 (Removal of several marked stable factors)**.** *For $S = \{i_1 < \cdots < i_s\} \subset \{1, \dots, m-1\}$, at every regular trajectory in M'_R,*

$$\frac{\omega(x)}{\prod_{i \in S} c_i(x)} \le C_0 C_1^s \prod_{v \in \{0,\dots,m-1\} \setminus S} j(T_R^v x). \tag{11.2}$$

Consequently, if at most one point x_w is chosen from each physical word domain, then

$$\sum_w \frac{\omega(x_w)}{\prod_{i\in S} c_i(x_w)} \le C_0 C_1^s D^s Q_*^{s+1}. \tag{11.3}$$

The choice of point may depend on the word and need not have the same momentum target for different words.

Proof. Use $c_m \le 1$ in (11.1), factor j_m into its individual steps, and apply $j(T_R^i x)/c_i(x) \le C_1$ at each marked index. This gives (11.2).

Delete the s marked steps of a word. The remaining blocks have lengths

$$i_1, \quad i_2 - i_1 - 1, \ \dots, \ i_s - i_{s-1} - 1, \quad m - i_s - 1.$$

For clarity, steps are numbered $0, \dots, m-1$, whereas the marks belong to $1, \dots, m-1$. Their indexing is

$$\underbrace{0, \dots, i_1 - 1}_{i_1} \mid i_1 \mid \underbrace{i_1 + 1, \dots, i_2 - 1}_{i_2 - i_1 - 1} \mid i_2 \mid \cdots \mid i_s \mid \underbrace{i_s + 1, \dots, m-1}_{m - i_s - 1}.$$

For example, $m = 7$ and $S = \{1, 2, 6\}$ leave blocks of lengths $(1, 0, 3, 0)$ and delete three physical symbols. Thus adjacent marks and a final mark are included, and every empty block contributes $P_0 = 1$. Some lengths can be zero. The blocks, translated to the initial fundamental scatterer, together with the s deleted physical step symbols determine the full lifted word uniquely. There are at most D^s choices for those symbols. Bound the remaining cocycle on each block by its own supremum and sum over all block words. Positivity permits adding combinations which do not produce a realizable full word. The resulting upper bound is $C_0 C_1^s D^s$ times the product of the $s+1$ corresponding $P_l(R)$'s, at most $C_0 C_1^s D^s Q_*^{s+1}$. For $S = \varnothing$ it is simply $C_0 P_m(R)$. This argument counts physical step symbols, never the infinitely many homogeneous strips at a removed incidence. □

Set

$$C_\star = \frac{C_0 Q_*}{2 c_* \lambda_\circ}, \qquad K_\star = C_1 D Q_*. \tag{11.4}$$

Neither constant depends on the collision count, radius, exact labels, roof or marked indices.

Theorem 11.3 (Target-roof multiple-incidence insertion)**.** *For every finite m, every original exact event and every $S \subset \{1, \dots, m-1\}$, the nonnegative measure*

$$F_\# \left(\prod_{i \in S} c_i^{-1} \mathbf{1}_\Pi \nu_R^* \right)$$

has a roof density d_S satisfying

$$\|d_S\|_{L^\infty(dt)} \le C_\star K_\star^{|S|}. \tag{11.5}$$

The bound holds on the whole roof line. For arbitrary numbers $a_1, \dots, a_{m-1} \ge 0$, the density $d_{\boldsymbol{a}}$ with weight $\prod_{i=1}^{m-1}(1 + a_i/c_i)$ satisfies

$$\|d_{\boldsymbol{a}}\|_\infty \le C_\star \prod_{i=1}^{m-1} (1 + K_\star a_i) \le C_\star \exp\left(K_\star \sum_{i=1}^{m-1} a_i \right). \tag{11.6}$$

In particular the complete inherited linear incidence weight obeys

$$\left\| \frac{dF_\#(I_m \mathbf{1}_\Pi \nu_R^*)}{dt} \right\|_\infty \le C_\star \left(1 + K_\star \frac{2q_0}{1 - q_0} \right). \tag{11.7}$$

These are finite-count unscaled height bounds, not m^{-2} local limit estimates.

Proof. At an almost-everywhere target (t, ϑ), Theorem 10.2 supplies at most one x_w per word. Apply (11.3) to precisely those points, use $\chi \geq \lambda_\circ$, and integrate (10.6) over the angle interval of length 2π. This gives (11.5) with the constants (11.4). Proposition 10.4 first establishes the nonnegative change-of-variables identity by truncation and monotone exhaustion; the present bound then proves finiteness and the density. At finite m the roof is confined to a bounded interval.

Expand $\prod_i (1 + a_i/c_i)$ into its finite sum over distinct marked sets. Apply (11.5) to each term. The sum of the upper bounds factors exactly as $C_\star \prod_i (1 + K_\star a_i)$. The exponential inequality uses $1 + x \leq e^x$. Finally expand I_m linearly and use $\sum_i q_0^{d_i} \leq 2q_0/(1 - q_0)$. All original exact labels remain in the source throughout. Removing their tests only enlarges the positive word upper sums in the proof; it does not change the source normalization or infer a Gaussian factor from an unconditioned pressure bound. □

Corollary 11.4 (A factorial bound for changing incidence witnesses)**.** *Let $J \subset \{1, \dots, m-1\}$ and let $\eta_i \geq 0$ be arbitrary. On the original event define*

$$N_J(x) = \sum_{i \in J} \mathbf{1}_{\{c_i(x) < \eta_i\}}, \qquad A_J = \sum_{i \in J} \eta_i.$$

For every integer $s \geq 1$,

$$\left\| \frac{dF_\#(\mathbf{1}_{\{N_J \geq s\}} \mathbf{1}_\Pi \nu_R^*)}{dt} \right\|_\infty \leq C_\star K_\star^s e_s((\eta_i)_{i \in J}) \leq C_\star \frac{(K_\star A_J)^s}{s!}, \tag{11.8}$$

where $e_s = 0$ for $s > |J|$. The witnesses may change with the target roof and angle; no single witness is chosen for an entire segment.

Proof. At every regular state,

$$\mathbf{1}_{\{N_J \geq s\}} \leq \binom{N_J}{s} = \sum_{\substack{S \subset J \\ |S| = s}} \prod_{i \in S} \mathbf{1}_{\{c_i < \eta_i\}} \leq \sum_{\substack{S \subset J \\ |S| = s}} \prod_{i \in S} \eta_i / c_i.$$

Push this positive inequality through the original roof and apply (11.5). In the expansion of A_J^s each product with distinct indices occurs $s!$ times; all remaining terms are nonnegative. This proves the last inequality. The argument is finite-count and already completes the full physical sum, so it introduces no collision-count multiplier for the possible witnesses. □

12. Pointwise removal of deep and numerous incidence witnesses

We apply the marked pressure theorem to the unchanged positive persistent source of Section 9. The new payment concerns its value at the original target roof, rather than an average in that roof. Clearance witnesses require a separate estimate and will not be identified with incidence witnesses.

Lemma 12.1 (Domination of the surviving coefficient at every roof)**.** *For $\gamma \geq 1$, on every original physical segment and every point of its closed interval,*

$$0 \leq c_{\mathrm{per}}^{\varepsilon,\gamma} \leq f^\varepsilon(u, \theta) \leq A(u, \theta). \tag{12.1}$$

Proof. Total variation bounds the decrement from an endpoint attaining c^ε, so $f^\varepsilon(u) \geq c^\varepsilon - \mathcal{V}^\varepsilon$. If $c_{\mathrm{per}} > 0$, its value is $c^\varepsilon - \gamma \mathcal{V}^\varepsilon \leq c^\varepsilon - \mathcal{V}^\varepsilon$. If $c_{\mathrm{per}} = 0$, nonnegativity suffices. The last inequality is the unchanged bound $0 \leq W^\varepsilon \leq 1$. □

Thus a target-dependent Borel restriction of the persistent paired source is bounded by the corresponding restriction of the original uncut roof measure. This assertion concerns the surviving coefficient, not the full endpoint minimum on an arbitrary segment.

For $\varepsilon > 0$, use precisely the incidence witness thresholds already forced by the original guard:

$$E_i^\varepsilon = \{c_i < \eta_i^\varepsilon\}, \qquad \eta_i^\varepsilon = 2\varepsilon q_0^{d_i/4}, \quad d_i = \min(i, m-i), \quad 1 \le i < m. \tag{12.2}$$

Write $N^\varepsilon = \sum_{i=1}^{m-1} \mathbf{1}_{E_i^\varepsilon}$. There are at most two indices at each depth. Hence, for an integer $L \ge 1$,

$$\sum_{d_i \ge L} \eta_i^\varepsilon \le \frac{4\varepsilon q_0^{L/4}}{1 - q_0^{1/4}}, \qquad \sum_{i=1}^{m-1} \eta_i^\varepsilon \le A_\varepsilon := \frac{4\varepsilon q_0^{1/4}}{1 - q_0^{1/4}}. \tag{12.3}$$

For any Borel target event E, let

$$\mathsf{U}_B^{\mathrm{per}}[E](t) = \int c_{\mathrm{per}}^{\varepsilon(B),\gamma_B}(\theta) \mathbf{1}_{\{a<t<b\}} \widetilde{H}_B(\Psi(\theta,t)) \mathbf{1}_E(\Psi(\theta,t))\, d\theta.$$

These are restrictions of the source in (8.8); no new normalization is used.

Theorem 12.2 (Uniform target-roof witness estimates)**.** *For every finite m, every B and integers $L, s \ge 1$,*

$$\|\mathsf{U}_B^{\mathrm{per}}[\bigcup_{d_i \ge L} E_i^{\varepsilon(B)}]\|_\infty \le \frac{4C_\star K_\star \varepsilon(B)}{1 - q_0^{1/4}} q_0^{L/4}, \tag{12.4}$$

$$\|\mathsf{U}_B^{\mathrm{per}}[N^{\varepsilon(B)} \ge s]\|_\infty \le C_\star \frac{(K_\star A_{\varepsilon(B)})^s}{s!}. \tag{12.5}$$

The norms are essential suprema on the whole roof line; constants are uniform in the radius and original exact labels.

Fix any $\eta > 0$ and define, at finite count,

$$\begin{aligned} L_m &= \left\lceil \frac{4(2+\eta)}{|\log q_0|} \log(m+1) \right\rceil, \\ s_m &= \left\lceil \frac{4\log(m+e)}{\log\log(m+e^e)} \right\rceil, \\ \mathcal{D}_{B,m} &= \bigcup_{d_i \ge L_m} E_i^{\varepsilon(B)}, \qquad \mathcal{A}_{B,m} = \{N^{\varepsilon(B)} \ge s_m\}, \qquad \mathcal{Z}_{B,m} = \mathcal{D}_{B,m} \cup \mathcal{A}_{B,m}. \end{aligned} \tag{12.6}$$

An empty union contributes zero. For every fixed outer B and M,

$$\mathcal{H}_M(\mathsf{U}_B^{\mathrm{per}}[\mathcal{Z}_{B,m}]) = 0. \tag{12.7}$$

More precisely the deep part satisfies, at every count,

$$\sup_{R,n,k} m^2 \|\mathsf{U}_B^{\mathrm{per}}[\mathcal{D}_{B,m}]\|_\infty \le C\varepsilon(B)(m+1)^{-\eta}. \tag{12.8}$$

Proof. By Lemma 12.1, each source in the first two assertions is bounded by the original uncut density restricted to the same target event. Apply Corollary 11.4 with $s = 1$ to the depth tail and with general s to all indices, then use (12.3). This proves the two finite-count bounds after the physical sum, with no selected uniform witness.

They hold for every L and s at every m. Substituting L_m in the first gives $q_0^{L_m/4} \le (m+1)^{-(2+\eta)}$, which proves (12.8). For fixed B, put $a = K_\star A_{\varepsilon(B)} < \infty$. The elementary factorial bound $s! \ge (s/e)^s$ gives

$$m^2 \frac{a^{s_m}}{s_m!} \le \exp\{2\log m + s_m(1 + \log a - \log s_m)\} = m^{-2+o(1)} \longrightarrow 0.$$

Here $s_m \log s_m = (4+o(1)) \log m$ and $s_m = o(\log m)$. The case $a = 0$ is immediate. If $s_m > m-1$, the event is empty. Subadditivity and positivity complete the proof of (12.7).

The depth and multiplicity thresholds in this proof are substitutions into finite-count inequalities valid simultaneously for all their values. No fixed-window spectral estimate is evaluated at a collision-dependent window. The original outer quantities $\varepsilon(B), h_B, r_B, \gamma_B, K_B$ remain fixed before the collision limit. Thus the proof does not exchange that limit with any outer width or band. □

12.1. **The retained source after this pointwise payment.** Define

$$\begin{aligned} \mathsf{U}_B^{\mathrm{loc}} &= \mathsf{U}_B^{\mathrm{per}}[\mathcal{Z}_{B,m}^{c}], \\ \mathsf{U}_B^{\mathrm{exc}} &= \mathsf{U}_B^{\mathrm{per}}[\mathcal{Z}_{B,m}], \qquad \mathsf{R}_B^{\mathrm{loc}} = (\mathsf{U}_B^{\mathrm{loc}} - K_B P_{r_B}^{\mathrm{ng}})_+. \end{aligned} \tag{12.9}$$

The superscript loc records localization of all incidence witnesses to the two collision-end layers; it makes no claim that clearance witnesses are localized to those layers. The comparison pool is the same full nongrazing pool, without a source-dependent restriction.

Corollary 12.3 (Exact preservation and collision-zero residual difference)**.** *At finite count the original persistent source satisfies*

$$\mathsf{U}_B^{\mathrm{per}} = \mathsf{U}_B^{\mathrm{exc}} + \mathsf{U}_B^{\mathrm{loc}}, \qquad 0 \le \mathsf{R}_B^{\mathrm{per}} - \mathsf{R}_B^{\mathrm{loc}} \le \mathsf{U}_B^{\mathrm{exc}}. \tag{12.10}$$

For every fixed B, M,

$$\mathcal{H}_M(\mathsf{R}_B^{\mathrm{per}} - \mathsf{R}_B^{\mathrm{loc}}) = 0, \qquad \mathcal{E}_M \le \mathcal{H}_M(\mathsf{R}_B^{\mathrm{loc}}) + C_M B^{-1/384}. \tag{12.11}$$

The unchanged raw pointwise assertion is equivalent to $\mathcal{H}_M(\mathsf{R}_B^{\mathrm{loc}}) \to 0$ for every fixed M.

Proof. The source partition is an identity of nonnegative original submeasures. Monotonicity and the unit Lipschitz bound of the positive-part map give its residual inequality. Apply (12.7) and then Theorem 9.3. For necessity use $0 \le \mathsf{R}_B^{\mathrm{loc}} \le \mathsf{U}_B^{\mathrm{micro}}$ and the retained scalar equivalence. Every removed term has been retained in the identity and paid in the target essential-height topology. □

For all sufficiently large B, $2\varepsilon(B) < c_\circ$, so the original endpoint section cap rules out incidence witnesses at $i = 0, m$. At a target point with positive surviving coefficient the original guard still requires an incidence or a clearance witness, by Theorem 9.1. On the retained part, every incidence witness has depth below L_m, and fewer than s_m such witnesses occur. A clearance witness may occur at any original flight, with its original positive weight. These alternatives allow a witness which changes with the roof; they assert no common witness on an entire physical segment.

The precise remaining estimate is

$$\lim_{B\to\infty} \limsup_{m\to\infty} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} (\mathsf{U}_{B,m}^{\mathrm{loc}}(t) - K_B P_{r_B,m}^{\mathrm{ng}}(t))_+ = 0. \tag{12.12}$$

The target-roof pressure theorem proves a new physical weighted summation and the pointwise removal in Theorem 12.2; it does not prove (12.12). In particular the all-incidence bound $C\varepsilon$ furnished by (12.4) with $L = 1$ is unscaled. At a fixed outer B it cannot be multiplied by m^2 and declared small. Nor does an inverse-incidence insertion estimate an inverse third-disk clearance. The remaining source includes the original shallow incidence witnesses, unvisited tangencies, critical entrances, section cuts and capacity deficiencies which are not paid by the proved estimates. No change of the original theorem or of its arithmetic classes is made in stating this boundary.

13. Joint depth and severity of incidence witnesses

The factorial estimate treats the witness thresholds only through their sum. It therefore forgets that there are at most two collision indices at any depth. It also forgets how much an individual incidence lies below its guard threshold. We retain both pieces of information. All estimates below concern the same target-roof measure as Theorem 11.3, with its original exact labels. They use its inherited geometric inputs, not a new conditional mixing or local-limit assumption.

Put

$$r = q_0^{1/4}, \qquad \alpha = -\log r = \tfrac{1}{4}|\log q_0|, \qquad b_\varepsilon = 4K_\star\varepsilon/\alpha. \tag{13.1}$$

For a regular trajectory define its incidence-witness set and load by

$$\begin{aligned} J_\varepsilon(x) &= \{i : 1 \le i < m,\ c_i(x) < 2\varepsilon r^{d_i}\}, \\ \mathscr{L}_\varepsilon(x) &= \sum_{i\in J_\varepsilon(x)} \log\frac{2\varepsilon}{c_i(x)} \\ &= \alpha \sum_{i\in J_\varepsilon(x)} d_i + \sum_{i\in J_\varepsilon(x)} \log\frac{2\varepsilon r^{d_i}}{c_i(x)}. \end{aligned} \tag{13.2}$$

The empty sum is zero. Both summands in the last line are nonnegative. In particular this is not a signed logarithmic weight, and it introduces no clearance factor. The Borel witness set is allowed to vary with the roof and angle.

For a nonnegative Borel weight h, write

$$D[h](t) = \frac{dF_\#(h\mathbf{1}_\Pi \nu_R^*)}{dt}(t)$$

when this density is defined. The positive identities of the preceding section give a common version at fixed m, R, n, k. The next theorem also proves existence and finiteness for its weights.

Theorem 13.1 (Exponential moment of the original incidence load)**.** *For every finite collision count, every $\varepsilon > 0$ and $0 < \lambda < 1$,*

$$\left\|D[e^{\lambda\mathscr{L}_\varepsilon}]\right\|_\infty \le C_\star \exp\left\{\frac{4K_\star\varepsilon}{e^{\alpha(1-\lambda)}-1}\right\}. \tag{13.3}$$

Consequently, for $u > b_\varepsilon$,

$$\left\|D[\mathbf{1}_{\{\mathscr{L}_\varepsilon \ge u\}}]\right\|_\infty \le C_\star \min\left\{1, \exp[-u + 2\sqrt{b_\varepsilon u}]\right\}. \tag{13.4}$$

Both estimates are uniform in the radius, exact labels and target roof. They remain valid with any further Borel restriction of the original measure, including the persistent source through Lemma 12.1.

Proof. Set $y_i = 2\varepsilon/c_i$. At a witness, $y_i > r^{-d_i}$, whence

$$y_i^\lambda = y_i y_i^{-(1-\lambda)} \le y_i r^{(1-\lambda)d_i}.$$

At every regular trajectory we therefore have the positive inequality

$$\begin{aligned} e^{\lambda\mathscr{L}_\varepsilon} &= \prod_{i=1}^{m-1}\left[1 + (y_i^\lambda - 1)\mathbf{1}_{\{i\in J_\varepsilon\}}\right] \\ &\le \prod_{i=1}^{m-1}\left(1 + \frac{2\varepsilon r^{(1-\lambda)d_i}}{c_i}\right). \end{aligned}$$

Apply the finite product insertion (11.6). There are at most two indices at every positive depth, so

$$\sum_{i=1}^{m-1} 2\varepsilon r^{(1-\lambda)d_i} \le 4\varepsilon \sum_{d\ge 1} r^{(1-\lambda)d} = \frac{4\varepsilon}{e^{\alpha(1-\lambda)}-1}.$$

This proves (13.3) after the complete physical sum. Every product is finite before this last geometric upper bound is used; no independence of collisions is assumed.

Positivity gives $\mathbf{1}_{\{\mathscr{L}_\varepsilon \ge u\}} \le e^{-\lambda u} e^{\lambda \mathscr{L}_\varepsilon}$. Writing $v = 1-\lambda$ and using $e^{\alpha v} - 1 \ge \alpha v$, we obtain, for every $0 < v < 1$,

$$\left\| D[\mathbf{1}_{\{\mathscr{L}_\varepsilon \ge u\}}] \right\|_\infty \le C_\star \exp\{-u + vu + b_\varepsilon / v\}.$$

When $u > b_\varepsilon$, take $v = \sqrt{b_\varepsilon/u}$. The independent bound $\|D[1]\|_\infty \le C_\star$ follows from the empty marked set and supplies the minimum in (13.4). Borel restrictions decrease a positive measure. For the persistent source the pointwise coefficient inequality in Lemma 12.1 supplies this decrease. □

Corollary 13.2 (Accumulated depth and severe shallow incidences)**.** *For $v > 0$, each of the events*

$$\left\{ \alpha \sum_{i\in J_\varepsilon} d_i \ge v \right\}, \qquad \left\{ \sum_{i\in J_\varepsilon} \log \frac{2\varepsilon r^{d_i}}{c_i} \ge v \right\}$$

is contained in $\{\mathscr{L}_\varepsilon \ge v\}$ and has the bound (13.4) *whenever $v > b_\varepsilon$. An incidence witness satisfying $c_i \le 2\varepsilon e^{-v}$ is also in that event, irrespective of its depth.*

Proof. Use the nonnegative decomposition in (13.2). For the last assertion, the single indicated summand is at least v. □

Thus a single sufficiently severe witness in an end layer can be paid. The conclusion does not require either a deep witness or the large witness count of Theorem 12.2. It does not assert a bound for an unvisited third-disk tangency.

14. Quadratic tails for the number of incidence witnesses

There is also an explicit finite-count improvement which retains all threshold locations, rather than passing to their total sum. The calculation is deterministic; it makes no independence claim about the billiard or its symbolic itinerary.

For $0 < r < 1$ write

$$(r;r)_a = \prod_{j=1}^{a}(1-r^j), \quad (r;r)_0 = 1, \quad (r;r)_\infty = \prod_{j=1}^{\infty}(1-r^j) > 0, \quad f(s) = \left\lfloor \frac{(s+1)^2}{4} \right\rfloor.$$

The infinite product is positive because the tail of $\sum_j -\log(1-r^j)$ is bounded by a constant times $\sum_j r^j$.

Lemma 14.1 (The two-end depth polynomial)**.** *For $0 \le s \le m-1$, put $E_{m,s}(r) = e_s((r^{d_i})_{1\le i<m})$. Then*

$$r^{f(s)} \le E_{m,s}(r) \le \sum_{a=0}^{s} \frac{r^{a(a+1)/2+(s-a)(s-a+1)/2}}{(r;r)_a (r;r)_{s-a}} \le \frac{(s+1)r^{f(s)}}{(r;r)_\infty^2}. \tag{14.1}$$

For $s > m-1$ the finite polynomial coefficient is zero and only the upper inequalities are asserted. In particular the coefficient has quadratic logarithmic rate, uniformly over $m \ge s+1$:

$$\log E_{m,s}(r) = -\frac{\alpha}{4}s^2 + O_r(s + \log(s+1)). \tag{14.2}$$

This is sharpness of the depth polynomial, not a lower bound for a physical roof density.

Proof. The multiset of finite collision depths is contained in two copies of $1, 2, 3, \ldots$. Coefficientwise positivity bounds its generating polynomial by $\prod_{d\ge1}(1+zr^d)^2$. To compute one factor, for $a \ge 1$ write $1 \le d_1 < \cdots < d_a$ as $d_j = j + h_j$, where $0 \le h_1 \le \cdots \le h_a$. Setting $g_1 = h_1$ and $g_j = h_j - h_{j-1}$ gives

$$\sum_{1\le d_1<\cdots<d_a} r^{d_1+\cdots+d_a} = r^{a(a+1)/2} \prod_{j=1}^{a} \sum_{g_j\ge0} r^{(a-j+1)g_j} = \frac{r^{a(a+1)/2}}{(r;r)_a}.$$

The empty factor contributes one. Multiplying the two positive series proves the middle sum in (14.1); all infinite passages are monotone limits of nonnegative coefficients. The minimum of its exponent over a is $f(s)$, attained by the two most balanced choices. Each denominator is at least $(r;r)_\infty^2$, giving the last inequality.

For the lower bound take the s smallest available depths. If $s = 2k$ these consist of two copies of $1, \ldots, k$ and have sum $k(k+1)$. If $s = 2k+1$ take these and one copy of $k+1$, with sum $(k+1)^2$. These depths are available whenever $m \ge s+1$, including the single central depth when m is even. Their product is one term of $E_{m,s}$. Taking logarithms of the two bounds proves (14.2). □

Theorem 14.2 (Quadratic witness-count tail at the target roof)**.** *With $N^\varepsilon = |J_\varepsilon|$, for every $s \ge 1$,*

$$\left\|D[\mathbf{1}_{\{N^\varepsilon\ge s\}}]\right\|_\infty \le \frac{C_\star(s+1)(2K_\star\varepsilon)^s}{(r;r)_\infty^2}\, r^{f(s)}. \tag{14.3}$$

The event is empty for $s > m-1$. The same estimate holds for the persistent source. For every fixed B and every $\eta > 0$, take

$$\sigma_m = \left\lceil 2\sqrt{\frac{(2+\eta)\log(m+1)}{\alpha}} \right\rceil. \tag{14.4}$$

Then the source with at least σ_m incidence witnesses has zero m^2-normalized essential height, with finite-count rate $m^{-\eta+o_B(1)}$.

Proof. Use the first, unrelaxed inequality in (11.8), with thresholds $2\varepsilon r^{d_i}$, and apply Lemma 14.1. Since $f(s) \ge s^2/4$, the logarithm of m^2 times the upper bound at $s = \sigma_m$ is at most

$$-\eta\log(m+1) + \sigma_m \log(2K_\star\varepsilon(B)) + \log(\sigma_m+1) + \log\frac{C_\star}{(r;r)_\infty^2}.$$

For fixed B the last three terms are $O_B(\sqrt{\log(m+1)} + \log\log(m+2))$. This proves the claim. No outer width is changed. □

The square-root logarithmic threshold is strictly smaller in order than the logarithmic-over-logarithmic threshold of the earlier factorial bound. The estimate depends on the actual two-end depth geometry, not on additional cancellation or a count of physical words.

15. Positive payment and the remaining low-load source

We now apply the joint-load theorem to the original localized persistent source. All quantities $\varepsilon(B), h_B, r_B, \gamma_B, K_B$ are fixed at the outer stage. A load level used below classifies an original trajectory at finite collision count; it is not a spectral, roof, residence or comparison window.

Fix $\eta > 0$ and set

$$u_m = (2+\eta)\log(m+1), \qquad \mathcal{Q}_{B,m} = \{\mathscr{L}_{\varepsilon(B)} \ge u_m\}. \tag{15.1}$$

Keep the previously paid set $\mathcal{Z}_{B,m}$ from (12.6). Define two restrictions of the same positive source, with no new renormalization:

$$\begin{aligned} \mathsf{U}_B^{\mathrm{load}} &= \mathsf{U}_B^{\mathrm{per}}[\mathcal{Z}_{B,m}^c \cap \mathcal{Q}_{B,m}], \\ \mathsf{U}_B^{\mathrm{low}} &= \mathsf{U}_B^{\mathrm{per}}[\mathcal{Z}_{B,m}^c \cap \mathcal{Q}_{B,m}^c], \\ \mathsf{R}_B^{\mathrm{low}} &= (\mathsf{U}_B^{\mathrm{low}} - K_B P_{r_B}^{\mathrm{ng}})_+. \end{aligned} \tag{15.2}$$

The comparison pool is the full original nongrazing pool, not a source-selected pool. The notation low refers only to incidence load; clearance witnesses remain at arbitrary depth.

Theorem 15.1 (A new collision-normalized positive payment)**.** *At every finite count,*

$$\mathsf{U}_B^{\mathrm{per}} = \mathsf{U}_B^{\mathrm{exc}} + \mathsf{U}_B^{\mathrm{load}} + \mathsf{U}_B^{\mathrm{low}}, \qquad 0 \le \mathsf{R}_B^{\mathrm{loc}} - \mathsf{R}_B^{\mathrm{low}} \le \mathsf{U}_B^{\mathrm{load}}. \tag{15.3}$$

For every fixed B, M, the load term has zero ordered height:

$$\mathcal{H}_M(\mathsf{U}_B^{\mathrm{load}}) = 0. \tag{15.4}$$

More explicitly, whenever $u_m > b_{\varepsilon(B)}$,

$$\sup_{R,n,k} m^2 \|\mathsf{U}_B^{\mathrm{load}}\|_\infty \le C_\star \exp\left\{-\eta \log(m+1) + 2\sqrt{b_{\varepsilon(B)}(2+\eta)\log(m+1)}\right\}. \tag{15.5}$$

Consequently

$$\mathcal{H}_M(\mathsf{R}_B^{\mathrm{loc}} - \mathsf{R}_B^{\mathrm{low}}) = 0, \qquad \mathcal{E}_M \le \mathcal{H}_M(\mathsf{R}_B^{\mathrm{low}}) + C_M B^{-1/384}. \tag{15.6}$$

The original assertion (1.5) *is equivalent to $\mathcal{H}_M(\mathsf{R}_B^{\mathrm{low}}) \to 0$ for every fixed M.*

Proof. The three sets in (15.3) disjointly partition the original persistent source. Thus its first equality holds before pushforward and remains exact afterwards. Monotonicity and the unit Lipschitz bound of the positive-part map prove the residual inequality with the same capacity on both sides.

Lemma 12.1 and Theorem 13.1 bound $\mathsf{U}_B^{\mathrm{load}}$ by $C_\star \exp[-u_m + 2\sqrt{b_{\varepsilon(B)} u_m}]$ once $u_m > b_{\varepsilon(B)}$. Multiplication by m^2, followed by $m^2 \le (m+1)^2$, gives (15.5). At fixed B its exponent is $-\eta \log(m+1) + O_B(\sqrt{\log(m+1)})$, tending to minus infinity. The finitely many smaller counts do not affect the collision limsup. This proves (15.4) uniformly in all the stated inner suprema.

Apply the residual inequality and Corollary 12.3 to obtain (15.6). For necessity use $0 \le \mathsf{R}_B^{\mathrm{low}} \le \mathsf{U}_B^{\mathrm{micro}}$ and the original scalar equivalence. For sufficiency use (15.6) and then send B to infinity. This is the original order of limits. The optimizing parameter in the proof of (13.4) was chosen inside a finite-count inequality valid for every $0 < \lambda < 1$; it is not inserted into a theorem which holds only at a previously fixed outer width. □

Corollary 15.2 (Explicit support restrictions after payment)**.** *At any regular target point contributing to $\mathsf{U}_B^{\mathrm{low}}$, put $J = J_{\varepsilon(B)}$ and $N = |J|$. Then*

$$\alpha \sum_{i \in J} d_i < u_m, \qquad \alpha f(N) < u_m, \qquad \prod_{i \in J} \frac{c_i}{2\varepsilon(B)} > e^{-u_m}. \tag{15.7}$$

In particular $N < 2\sqrt{u_m/\alpha}$ and every individual witness has $c_i > 2\varepsilon(B) e^{-u_m}$. All these restrictions are in addition to the retained v111 incidence localization. They impose no clearance localization.

Proof. The load is strictly less than u_m on the retained set. Its nonnegative decomposition gives the first inequality. The sum of the depths of any N indices is at least $f(N)$, by the construction in Lemma 14.1; this proves the second. Exponentiating minus the definition of the load proves the product inequality. Since $f(N) \ge N^2/4$ and all load summands are nonnegative, the remaining assertions follow, including the empty-witness case. □

For sufficiently large B, the original section cap excludes incidence witnesses at the two endpoint collisions. The persistent guard then still forces an interior incidence witness or an original third-disk clearance witness. A state with $N = 0$ is not removed by this revision: the guard may be witnessed entirely by clearance. Critical entrances and section cuts carrying the original positive coefficient are likewise not deleted by their names.

The exact remaining target is

$$\lim_{B\to\infty} \limsup_{m\to\infty} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} (\mathsf{U}^{\mathrm{low}}_{B,m}(t) - K_B P^{\mathrm{ng}}_{r_B,m}(t))_+ = 0. \tag{15.8}$$

Theorems 13.1–15.1 prove new finite-count concentration and positive payments in the required normalized topology. They do not prove (15.8). Low-load shallow incidences, arbitrary-depth clearance (including unvisited tangency), and capacity-deficient positive source remain in that equation. The original physical section, guard, arithmetic reference and exact-label pointwise assertion have not been weakened. The distinction between this remaining assertion and the proved payments is part of the theorem statement, not an inference from software diagnostics or an editorial change of topic.

16. Clearance-to-roof coarea without a fixed-clearance density bound

We keep the physical parameters and every event in Section 1. In this section u is the clearance coordinate of Section B, not physical time. Physical time is denoted by t throughout. Fix R, m, n, k, a flight $0 \le j < m$, a sign σ, and an actual regular marked itinerary. On a local angular lift put

$$\mathcal{Q}_j(\vartheta, u) = (\alpha_0, \alpha_m)(X_{R,j}(\vartheta, \sigma, u)), \qquad S(\vartheta, u) = F(\mathcal{Q}_j(\vartheta, u)).$$

The point used as a mark is an unvisited near miss, not an additional reflection. Every derivative below is taken on the same designated regular itinerary. In particular m, j and the physical roof are not changed by this parametrization.

Theorem 16.1 (The long-record Jacobian and its critical Hessian)**.** *Let $A = D\mathcal{Q}_j$, let $g = \nabla_q F$, and let $G = D_q^2 F$. On every such chart,*

$$|\det A| = (R\omega)^{-1}, \tag{16.1}$$

$$|\det D_{(\vartheta,u)}(S, \arg g)| = \chi/(R\omega) \quad (g \neq 0), \tag{16.2}$$

$$D^2 S = A^{\mathsf{T}} G A + \sum_{a=1}^{2} g_a D^2(\mathcal{Q}_j)_a. \tag{16.3}$$

At a regular critical roof, $g = 0$, the Hessian D^2S is positive definite and

$$\sqrt{\det D^2 S} = \frac{\sqrt{\det G}}{R\omega}. \tag{16.4}$$

These are identities for each finite physical collision count, with all coordinate and section-normalization factors displayed. In particular the Hessian term containing g cannot be dropped away from a critical roof.

Proof. The collision map preserves $d\alpha\, dp$ at fixed R. The signed line map at collision j has absolute Jacobian R, by (B.4). Composition with T_R^j therefore still has absolute Jacobian R with respect to the initial collision coordinates. The map $(\alpha_0, p_0) \mapsto (\alpha_0, \alpha_m)$ has absolute Jacobian ω^{-1} by the definition of ω. Dividing the two determinants gives (16.1). It uses the actual inverse iterate and is independent of where the mark occurs. On a noncritical chart the determinant of $q \mapsto (F, \arg g)$ is χ; the chain rule gives (16.2). Twice differentiating $F \circ \mathcal{Q}_j$ gives (16.3), component by component. At $g = 0$ its last term vanishes, and the retained positive-definite

optical Hessian and the invertibility of A give the assertion. Taking the determinant and using (16.1) proves (16.4).

As a normalization check, the line-coordinate density is $(4\pi Rc_*)^{-1}$. Dividing it by the determinant in (16.2) gives exactly $\omega/(4\pi c_*\chi)$, the original target-roof/Gauss-angle density. There is no additional factor R, R^{-1} or c_*^{-1}. All chart restrictions and exact labels remain on the source side of this change of variables. □

Corollary 16.2 (Uniform determinant growth on regular marked charts)**.** *With the retained constants $\varrho = 47/59$ and $\lambda_\circ$, all regular marked charts with the original endpoint cap satisfy*

$$|\det D\mathcal{Q}_j| \geq \frac{600}{2209}\varrho^{-(m-1)}, \qquad |\det D(S, \arg g)| \geq \frac{600\lambda_\circ}{2209}\varrho^{-(m-1)} \quad (g \neq 0).$$

At a critical roof the same lower bound as in the second inequality holds for $\sqrt{\det D^2 S}$. The constants are uniform in the original radius interval, the exact labels, the flight index, and the physical word. These determinant bounds do not bound individual singular values, a chart's condition number, or the complete physical sum of inverse determinants.

Proof. The retained optical estimate is $\omega \leq (47/6)\varrho^{m-1}$, and $R \leq 47/100$. Insert these in (16.1). Then use $\chi \geq \lambda_\circ$ and $\sqrt{\det G} \geq \lambda_\circ$ in (16.2) and (16.4). The identity $6/(47(47/100)) = 600/2209$ fixes the common constant. □

The following elementary estimate is recorded with proof to avoid importing an unstated finite-type or regularity theorem.

Lemma 16.3 (A one-dimensional band estimate)**.** *Let $f \in C^\ell(I)$ on an interval, with integer $\ell \geq 1$ and $|f^{(\ell)}| \geq \kappa > 0$. For $h > 0$,*

$$|\{x \in I : a < f(x) < a + h\}| \leq 2\ell(h/\kappa)^{1/\ell}. \tag{16.5}$$

The estimate also bounds any Borel subset of that band.

Proof. First restrict the band to a compact subinterval and let its measure be $L > 0$. Its cumulative Lebesgue measure is continuous and one-Lipschitz. The $\ell+1$ quantiles with successive mass gaps L/ℓ can be chosen in its closed support; their successive coordinate gaps are at least L/ℓ. Equivalently one can use interior approximate quantiles and pass to their limits. Write them as $x_0 < \cdots < x_\ell$. Continuity puts $f(x_i)$ in $[a, a + h]$. The divided-difference mean-value theorem, proved by subtracting the interpolating polynomial and applying Rolle's theorem ℓ times, gives some $\xi \in (x_0, x_\ell)$ for which

$$\frac{f^{(\ell)}(\xi)}{\ell!} = \sum_{i=0}^{\ell} \frac{f(x_i) - a}{\prod_{v\neq i}(x_i - x_v)}.$$

The subtraction of a is legitimate because the divided difference of a constant is zero. The gap bound makes the absolute denominator at least $(L/\ell)^\ell i!(\ell - i)!$. Hence

$$\kappa \leq h(\ell/L)^\ell \sum_{i=0}^{\ell} \binom{\ell}{i} = h(2\ell/L)^\ell.$$

Rearrange, then exhaust the original interval by compact subintervals. This also handles a band not initially known to have finite measure. No statement about the number of physical words is used in the proof. □

Theorem 16.4 (A positive finite-type clearance slice)**.** *Suppose $S \in C^{\ell+1}$ on a neighbourhood of $I \times [0, \delta_0]$, and*

$$0 < b_f \leq |S_u| \leq B_f, \qquad |\partial_\vartheta^\ell S(\vartheta, 0)| \geq \kappa_f > 0.$$

The sign of S_u is fixed on this rectangle. Let E be any Borel subset of it and $0 \le h \le H_f$ a pointwise bounded Borel weight. For $0 < \delta \le \delta_0$, the measure

$$S_{\#}(h\mathbf{1}_E\mathbf{1}_{\{0<u<\delta\}}\,d\vartheta\,du)$$

has a density $D_{E,h,\delta}$ with

$$\|D_{E,h,\delta}\|_\infty \le \frac{2\ell H_f}{b_f}\left(\frac{B_f\delta}{\kappa_f}\right)^{1/\ell}. \tag{16.6}$$

In the physical marked record one takes $h = (4\pi R c_*)^{-1}$, or a bounded multiplier of this weight, and E is the original physical domain with all exact labels. The constants must be established on that actual chart; the theorem does not assert their uniformity in m or their summability over all charts.

Proof. At fixed ϑ, $u \mapsto S(\vartheta, u)$ is strictly monotone. Let $U_t(\vartheta)$ denote its unique inverse when it exists. The ordinary one-dimensional substitution, first against nonnegative roof tests and then by Tonelli, gives

$$D_{E,h,\delta}(t) = \int_I \frac{h(\vartheta, U_t)\mathbf{1}_E(\vartheta, U_t)\mathbf{1}_{\{0<U_t<\delta\}}}{|S_u(\vartheta, U_t)|}\,d\vartheta.$$

Missing inverses contribute zero. This argument proves absolute continuity, not merely an estimate conditional on its existence. If $S_u > 0$ and the inverse lies in $(0, \delta)$, then $t - B_f\delta < S(\vartheta, 0) < t$. For $S_u < 0$ the corresponding band is $t < S(\vartheta, 0) < t + B_f\delta$. Both have width $B_f\delta$. Bound the integrand by H_f/b_f and apply Lemma 16.3. Borel cuts, including section cuts, only restrict the integral and never create a chart-count multiplier. The same proof on bounded compact rectangles followed by monotone exhaustion gives the assertion for open intervals and Borel weights. □

Theorem 16.5 (Integrated simple folds, with a sharp local constant)**.** *Assume that on $I \times (0, \delta_0)$ one has $S_{\vartheta\vartheta} \ge a_f > 0$ and $S_u \ge b_f > 0$, and that for each u the unique angular critical point $\vartheta_0(u)$ exists in I. For the same positive weights and selectors as above,*

$$\|D_{E,h,\delta}\|_\infty \le 2H_f\sqrt{\frac{2\delta}{a_f b_f}}. \tag{16.7}$$

The assertion also applies after reversing the roof sign or the clearance-coordinate orientation when the corresponding derivative signs are reversed. It does not require a bounded roof density at each fixed clearance.

Proof. Put $v(u) = S(\vartheta_0(u), u)$. The implicit function theorem and the critical-point equation give $v'(u) = S_u(\vartheta_0(u), u) \ge b_f$. For fixed u there are at most two angular roots of $S(\vartheta, u) = t$, and

$$|S_\vartheta(\vartheta, u)|^2 \ge 2a_f(S(\vartheta, u) - v(u)).$$

Indeed differentiate S_ϑ^2 along each monotone side and integrate from its zero at ϑ_0. On the left side the integration orientation and the sign of S_ϑ reverse together, giving the same inequality. One-dimensional substitution on those two sides gives a fixed-u density bounded by $H_f\sqrt{2/a_f}(t - v(u))^{-1/2}$ for $t > v(u)$ and zero below. The angular critical point itself has zero angular measure. The bound is allowed to be infinite at $t = v(u)$.

For $0 < u < \delta$, if t is in the range of v, let u_t be its unique inverse. The contributing values satisfy $u < u_t$, and $t - v(u) \ge b_f(u_t - u)$. If $t \ge v(\delta)$, replace u_t by δ for an upper bound; if $t \le v(0)$ there is no contribution. In every case

$$\int_0^\delta \mathbf{1}_{\{t>v(u)\}}(t - v(u))^{-1/2}\,du \le 2\sqrt{\delta/b_f}.$$

Integrate the fixed-u estimate by Tonelli to obtain (16.7). Restricting by E preserves the inequality. For $S(\vartheta, u) = \vartheta^2 + u$, $h = 1$, and the whole real angular line, the integrated density is

$$2\{\sqrt{t_+} - \sqrt{(t-\delta)_+}\},$$

whose maximum is $2\sqrt{\delta}$. Thus the constant is attained when $a_f = 2$ and $b_f = 1$. This elementary normal form checks the local analytic bound only; it is not substituted for a billiard trajectory or for the joint-record distribution. □

16.1. **One-copy summation and the remaining uniformity.** On a countable collection of admissible physical charts, first replace their source sets by disjoint Borel differences. The preceding density formulae then add exactly. Their positive upper bounds add as well, with an extended value permitted until a finite bound has been proved. Two impact signs and distinct unvisited centres are retained with precisely the multiplicity in (E.2); no full-word cardinality is inserted. Borel event cuts do not require a new chart merely because they disconnect the selected set.

In particular the per-clearance condition (E.3) is not needed on a chart controlled by Theorem 16.5. An angular fold may have an unbounded conditional density and a bounded integrated density at the same time. What is still required for the full clearance payment is a bound for the complete, exact-label, positively weighted chart sum, in the original $m^2 L^\infty(dt)$ topology. The local constants in (16.6) or (16.7) have not been proved to provide that bound. Charts with $S_u = 0$, degenerate angular behaviour outside a verified finite type, and singular physical boundaries are not discarded. Regular full critical roofs are treated next by a different exact calculation.

17. Critical roofs, actual reversing orbits, and their exact density coefficient

We now identify a dynamical invariant in the critical-roof density, without averaging the roof or replacing the physical trajectories. Let x be a regular m-flight trajectory with endpoints in the original section cap. Write $q_x = (\alpha_0, \alpha_m)$ and $t_x = F(q_x)$. All statements about derivatives of the actual collision map below concern regular trajectories, not a formal continuation through a tangent collision.

Theorem 17.1 (Critical half-paths are physical reversing periodic orbits)**.** *The endpoint roof is critical precisely when $p_0 = p_m = 0$. Such a trajectory extends by the actual reflection law to a regular $2m$-collision periodic orbit, whose period need not be primitive. The extra m flights traverse the original flights in reverse, with total roof $2t_x$ and total lifted displacement zero. Let $\mathcal{I}(\alpha, p) = (\alpha, -p)$ be collision time reversal, and put*

$$A^{\mathcal{I}}_{m,R}(x) = \sum_{j=0}^{m-1} \mathbf{1}_{Y^*_R}(\mathcal{I} T^j_R x).$$

If the half-path belongs to the original exact event, then

$$A_{2m,R}(x) = A_{m,R}(x) + A^{\mathcal{I}}_{m,R}(x). \tag{17.1}$$

One cannot replace the right side by $2n$ without additionally verifying the time-reversed section tests.

Proof. The first-variation identity already retained in (3.1) is $\nabla_q F = R(-p_0, p_m)$, which proves the criticality assertion. At a normal terminal collision the reflection sends the incoming velocity to its negative. The ray therefore retraces the preceding actual flight. At each intervening regular collision the reversed reflection retraces the next preceding flight. At the normal initial collision it returns to the original outgoing state. Thus every reversed flight is an actual free flight, with

no extra or hidden collision, and the orbit closes after $2m$ collisions. Its reversed displacement is the negative of the forward displacement. Its length is twice the original length. No assertion about a primitive period is used.

For $1 \le j < m$ the reversed state at time $2m - j$ is $\mathcal{I}T_R^j x$, while the states at times 0 and m are fixed by $\mathcal{I}$. In the original event both endpoint section tests equal one. Summing the m forward tests and then the terminal test and the $m - 1$ reversed interior tests gives (17.1), since the initial test in $A_m^{\mathcal{I}}$ also equals one. The original section is not assumed invariant under $\mathcal{I}$. □

Theorem 17.2 (Exact stability determinant at a critical roof)**.** *At a regular critical half-path write*

$$G_x = \begin{pmatrix} a & e \\ e & d \end{pmatrix}, \qquad \Delta_x = ad - e^2 > 0, \qquad \mathcal{M}_x = DT_R^{2m}(x).$$

Then $e \neq 0$, and the actual return derivative satisfies

$$\det \mathcal{M}_x = 1, \qquad \operatorname{tr} \mathcal{M}_x - 2 = |\det(I_2 - \mathcal{M}_x)| = \frac{4\Delta_x}{e^2} > 0. \tag{17.2}$$

Consequently the critical density coefficient

$$\mathfrak{J}(x) := \frac{\omega(x)}{2c_* \sqrt{\det G_x}} = \frac{1}{Rc_* \sqrt{|\det(I_2 - DT_R^{2m}(x))|}} \tag{17.3}$$

is an invariant of the actual reversing periodic orbit with its marked initial state and specified half-length m. The coefficient is not a probability conditioned on that periodic orbit.

Proof. The signed optical corner formula (G.1) makes e nonzero on every regular finite trajectory. Differentiate $p_0 = -F_{\alpha_0}/R$ and $p_m = F_{\alpha_m}/R$. Solving the first identity for $d\alpha_m$ gives, in the original (α, p) coordinates,

$$B_x := DT_R^m(x) = \begin{pmatrix} -a/e & -R/e \\ -\Delta_x/(Re) & -d/e \end{pmatrix}, \qquad \det B_x = 1. \tag{17.4}$$

This keeps the sign of e; replacing it prematurely by $|e|$ in the matrix would change a derivative, although not its final trace. At both normal endpoints the derivative of time reversal is $J = \operatorname{diag}(1, -1)$. The reversibility identity $\mathcal{I}T_R^m\mathcal{I} = T_R^{-m}$ therefore yields $DT_R^m(T_R^m x) = JB_x^{-1}J$ and

$$\mathcal{M}_x = JB_x^{-1}JB_x.$$

For an arbitrary determinant-one matrix $B = \left(\begin{smallmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{smallmatrix}\right)$, $JB^{-1}J = \left(\begin{smallmatrix} b_{22} & b_{12} \\ b_{21} & b_{11} \end{smallmatrix}\right)$, so

$$\operatorname{tr}(JB^{-1}JB) - 2 = 4b_{12}b_{21}.$$

Substitution of (17.4) gives $4\Delta_x/e^2$. For a two-dimensional determinant-one matrix, $\det(I_2 - \mathcal{M}_x) = 2 - \operatorname{tr} \mathcal{M}_x$. This proves (17.2) and also gives positive real hyperbolic multipliers. Finally $\omega = |e|/R$, so elementary simplification proves (17.3). The determinant of $I_2 - \mathcal{M}_x$ is invariant under a smooth change of collision coordinates. No periodic-orbit trace formula or distributional limit theorem has been invoked. □

Theorem 17.3 (Critical-roof coarea with arbitrary exact-event cuts)**.** *Fix a regular critical half-path and a sufficiently small endpoint neighbourhood V on its designated itinerary. Let $E \subset V$ be an arbitrary Borel selector, including the actual physical domain, initial and terminal section tests, exact occupation and displacement labels, and any clearance tests. Put*

$$y = G_x^{1/2}(q - q_x), \quad f(y) = F(q) - t_x, \quad w(y) = \frac{\omega(q)}{4\pi c_* \sqrt{\det G_x}}.$$

Choose $r_0 > 0$ so that the closed y-ball is contained in the coordinate neighbourhood, and constants $K, L_w \geq 0$ such that

$$\|D^2 f(y) - I_2\| \leq K|y|, \quad |w(y) - w(0)| \leq L_w|y|, \quad Kr_0 \leq 1.$$

For $0 < s < r_0^2/3$ there is, for every angle θ, a unique $r_s(\theta) \in (0, r_0)$ with $f(r_s e(\theta)) = s$. Define

$$a_E(s) = \frac{1}{2\pi} \int_0^{2\pi} \mathbf{1}_E\big(q_x + G_x^{-1/2} r_s(\theta) e(\theta)\big)\, d\theta.$$

The original positive collision measure restricted to E and this ball has a density D_E near t_x, no atom at t_x, and

$$|D_E(t_x + s) - \mathfrak{J}(x) a_E(s)| \leq 2\pi\sqrt{3}\{2L_w + Kw(0)\}\sqrt{s}. \tag{17.5}$$

There is no contribution below t_x. The constant is independent of the Borel selector E, but is a constant of this local chart, not a claimed uniform-in-m constant.

If E contains a neighbourhood of q_x, then $a_E(s) = 1$ for small s and the one-sided density limit is exactly $\mathfrak{J}(x)$. If the selected physical side is described by finitely many C^1 inequalities vanishing at q_x with nonzero linear parts, then its limiting coefficient is $\mathfrak{J}(x)$ times the angular fraction of the corresponding linear sector. Without such a hypothesis the exact $a_E(s)$ is retained; no limit for it is asserted.

Proof. The retained reduced action is smooth on a regular finite itinerary; the inverse function theorem and the nonzero optical corner give a smooth positive ω there. Thus the displayed constants exist after decreasing r_0. At a critical point $f(0) = 0$, $\nabla f(0) = 0$ and $D^2 f(0) = I_2$. Integrating the Hessian bound along a radial segment gives

$$|\partial_r f(re) - r| \leq Kr^2/2, \qquad |f(re) - r^2/2| \leq Kr^3/6.$$

For $r \leq r_0$ these imply $r^2/3 \leq f(re) \leq 2r^2/3$, $\partial_r f \geq r/2$, and hence the existence and uniqueness of r_s, with $r_s \leq \sqrt{3s}$. They also give

$$\left| \frac{r_s}{\partial_r f(r_s e)} - 1 \right| \leq K r_s, \qquad 0 < \frac{r_s}{\partial_r f(r_s e)} \leq 2.$$

In endpoint coordinates the original collision density is $\omega(q)/(4\pi c_*)$; the linear substitution to y makes it $w(y)\, dy$. Polar coordinates and the monotone radial substitution $t = t_x + f(re(\theta))$, tested against nonnegative roof functions, give the exact density

$$D_E(t_x + s) = \int_0^{2\pi} \mathbf{1}_E(q_s(\theta)) w(r_s e(\theta)) \frac{r_s}{\partial_r f(r_s e(\theta))}\, d\theta.$$

Here $q_s(\theta) = q_x + G_x^{-1/2} r_s e(\theta)$, and any source outside the selected ball is not part of D_E. The point $r = 0$ has zero two-dimensional measure, so it produces no atom. The strict radial positivity also excludes mass below t_x inside the ball. For each angle the difference between the weight in this integral and $w(0)$ is at most $(2L_w + Kw(0))r_s$. Integrating gives (17.5), since $2\pi w(0) = \mathfrak{J}(x)$. This estimate does not differentiate an indicator or require its boundary to have zero roofwise length.

When E contains a neighbourhood, every sufficiently small radial point belongs to it. For the stated finite collection of C^1 cuts, Taylor expansion of each defining function shows convergence of its sign to its nonzero linear part at every angle except the finitely many zero directions. Dominated convergence gives the sector fraction. It is the fraction after the transformation by $G_x^{-1/2}$, not an unweighted fraction in the original endpoint angles. For a general Borel selector no such argument is used. □

Corollary 17.4 (One-sided unvisited boundaries without a fictitious orbit)**.** *Suppose a designated finite broken path reaches an unvisited-disk tangency or a section cut, all its* visited *incidences remain positive, and its endpoint action is critical. The sector formula* (17.5)*, with the original physical-side selector, continues to hold in the one-sided coordinate neighbourhood. At a singular state the second expression in* (17.3) *is* not *asserted to be a derivative of a defined physical return map; the first expression is the one-sided action coefficient.*

Proof. The stationary equations for the visited interior collision points have positive-definite interior Hessian. At the stated limiting path they therefore have a smooth local solution as functions of the two endpoints. The signed optical corner is nonzero because the visited incidences and flight lengths are positive. This supplies the smooth action and endpoint-coordinate Jacobian on a full auxiliary neighbourhood. The portion representing actual trajectories is retained as a Borel subset E; portions intersecting a forbidden disk are never assigned to the physical event. On the admitted side these functions agree with the actual action and original measure. The proof of Theorem 17.3 used arbitrary Borel restrictions, so it applies to this subset without continuing the physical dynamics across its boundary. Simultaneous cuts are included in the same selector. If visited grazing also occurs, this particular smooth extension is not invoked. Its treatment remains the inherited visited-grazing trace argument. Nothing in this corollary equates an area-null boundary point with a zero positive pairing-endpoint trace. □

Proposition 17.5 (A complete finite-count critical coefficient bound)**.** *Let the sum run over all distinct regular critical half-paths in the original exact event, at fixed m, R, n, k. Then*

$$\mathfrak{J}(x) \le \frac{47}{12c_*\lambda_\circ}\varrho^{m-1}. \tag{17.6}$$

Under the retained, explicitly stated geometric-pressure input 11.1, the full positive sum satisfies

$$\sum_x \mathfrak{J}(x) \le \frac{C_0 Q_*}{2c_*\lambda_\circ}. \tag{17.7}$$

This is unscaled. It does not supply an m^{-2} bound for a clearance-selected critical source or a uniform bound for the local error constants in (17.5)*.*

Proof. Use $\sqrt{\det G_x} \ge \lambda_\circ$ in (17.3) and the optical bound for ω in (4.8) to obtain (17.6). The retained global equal-gradient uniqueness gives at most one regular critical half-path per lifted physical word. Such a path extends periodically without singular collisions, so it belongs to the two-sided regular set M_R'. The selected-word sum in (11.3), with no incidence marks, therefore bounds $\sum_x \omega(x)$ by C_0Q_*. Divide by $2c_*\lambda_\circ$. Missing words contribute zero. No multiplication of a per-orbit exponential estimate by the number of words is made. □

17.1. **What the critical calculation does and does not close.** Formula (17.3) identifies an actual return-stability weight in the original roof density; it is not an ansatz for a Markov model and not a periodic-orbit replacement of the invariant measure. Formula (17.5) also keeps positive source on section-cut and one-sided unvisited boundaries rather than assigning it a zero coefficient by area-nullness. The full paired source, its nongrazing comparison pool, and (15.8) are unchanged. A critical *component* coefficient need not equal a discontinuity of the entire density, since other physical components may be present at the same roof.

For a collision-uniform payment it is still necessary to control the actual weighted sum on fixed outer neighbourhoods, its clearance and exact-label localization, and the chart remainders. Choosing a separate collision-dependent neighbourhood merely to separate critical roof values would not prove that fixed-window statement. Neither (17.6) nor the unscaled sum (17.7) is used in that invalid manner. The new identities instead give the exact geometric and stability coefficients that a positive capacity or transfer-operator estimate must control.

18. Exact two-mark clearance flux on one physical flight

The preceding clearance–roof analysis treats a selected near miss as a *mark*, without adding a collision. Two different unvisited disks can be marked on the same actual flight. Here we quantify their physical multiplicity exactly, rather than assuming independence or multiplying by the number of collision words. The result is a second-order *collision-space* clearance theorem; it is not an m^2-normalized pointwise roof estimate.

Throughout let $R \in [R_-, R_+]$, $R_+ = 47/100$, and keep $\delta_* = 3/200$, $0 < \delta_1, \delta_2 \le \delta_*$, the triangular lattice Λ, the actual flight time $\tau \le \tau_+$, and the one-copy physical mark maps X_R, Z_R of Theorem B.2. Define the strictly positive *physical separation* and the finite comparison set

$$\begin{aligned} c_{\mathrm{pair}} &= \sqrt{1 - [2(R_+ + \delta_*)]^2} = \frac{\sqrt{591}}{100} > 0, \\ \mathcal{W}_* &= \{w \in \Lambda \setminus \{0\} : |w| \le \tau_+ + 2(R_+ + \delta_*)\}, \\ N_{\mathrm{pair}} &= |\mathcal{W}_*|. \end{aligned} \tag{18.1}$$

The constant N_{pair} is finite by the *same* finite-horizon assumption already used in Lemma B.1; it is not the number of admissible words, the number of pairing sheets, or an unproved distortion constant.

Lemma 18.1 (Distinct same-flight clearances are uniformly transverse)**.** *Let $z_1 \ne z_2$ be two unvisited disk centers whose clearance values $u_i \in (0, \delta_i)$ are realized by the* same *regular physical flight with unit direction v_ϑ. Let $b_i = (z_i - q) \cdot v_\vartheta^\perp$ be their signed impact parameters. Then both perpendicular feet lie in the open physical flight, $z_2 - z_1 \in \mathcal{W}_*$, and*

$$|(z_2 - z_1) \cdot v_\vartheta| = \sqrt{|z_2 - z_1|^2 - (b_2 - b_1)^2} \ge c_{\mathrm{pair}}. \tag{18.2}$$

In particular, for the first signed impact parameter held fixed, $\partial_\vartheta b_2 = -(z_2 - z_1) \cdot v_\vartheta$ does not vanish on a simultaneous physical two-mark state.

Proof. Lemma B.1 puts each foot in the open physical flight. Thus its along-flight coordinate lies in an interval of length at most τ_+. Since $|b_i| = R + u_i$,

$$|b_2 - b_1| \le 2R + \delta_1 + \delta_2 \le 2(R_+ + \delta_*) = 97/100 < 1.$$

The lattice spacing $|z_2 - z_1| \ge 1$ and orthogonal decomposition along $v_\vartheta, v_\vartheta^\perp$ imply the equality and lower bound in (18.2). The same decomposition gives $|z_2 - z_1| \le \tau_+ + 2(R_+ + \delta_*)$, proving membership in $\mathcal{W}_*$. The impact derivative follows from $dv_\vartheta^\perp/d\vartheta = -v_\vartheta$ with b_1 fixed. Neither a grazing endpoint nor an unvisited near miss is treated as an extra collision. □

For a first center translated to 0, fix $w \in \mathcal{W}_*$ and $\sigma_1, \sigma_2 \in \{\pm 1\}$. Write

$$d_{\sigma_1\sigma_2}(u_1, u_2) = \sigma_2(R + u_2) - \sigma_1(R + u_1). \tag{18.3}$$

For all $0 < u_i < \delta_i$, $|d_{\sigma_1\sigma_2}| < 97/100 < |w|$. Thus the equation

$$w \cdot v_\vartheta^\perp = d_{\sigma_1\sigma_2}(u_1, u_2) \tag{18.4}$$

has exactly two roots modulo 2π, denoted $\vartheta_{w,\sigma_1,\sigma_2,+}(u_1, u_2)$ and $\vartheta_{w,\sigma_1,\sigma_2,-}(u_1, u_2)$, distinguished by the sign of $w \cdot v_\vartheta$. They are smooth on each parameter rectangle. Their one-dimensional change-of-variables Jacobian is

$$\left|\frac{\partial \vartheta}{\partial u_2}\right| = \frac{1}{\sqrt{|w|^2 - d_{\sigma_1\sigma_2}(u_1, u_2)^2}}, \tag{18.5}$$

with u_1 fixed; the sign of σ_2 has unit magnitude.

Theorem 18.2 (An exact physical two-mark coarea identity)**.** *For every nonnegative Borel weight $a(x, z_1, z_2)$ of a genuine outgoing collision state and two* ordered*, distinct unvisited centers (expressed in the initial incident disk's lattice frame), set at a root of* (18.4)

$$x = X_R(\vartheta, \sigma_1, u_1), \quad z_1 = Z_R(\vartheta, \sigma_1, u_1), \quad z_2 = z_1 + w, \quad \mathfrak{A} = \mathbf{1}_{\{z_2 \in \mathcal{Z}_{\delta_2}(x)\}}.$$

Then the full, one-copy *original collision measure obeys*

$$\int \sum_{z_1 \in \mathcal{Z}_{\delta_1}(x)} \sum_{\substack{z_2 \in \mathcal{Z}_{\delta_2}(x) \\ z_2 \neq z_1}} a(x, z_1, z_2)\, d\nu(x) \tag{18.6}$$

$$= \frac{1}{4\pi R} \sum_{w \in \mathcal{W}_*} \sum_{\sigma_1, \sigma_2, \eta \in \{\pm 1\}} \int_0^{\delta_1} \int_0^{\delta_2} \left. \frac{\mathfrak{A}\, a(x, z_1, z_1 + w)}{\sqrt{|w|^2 - d_{\sigma_1\sigma_2}(u_1, u_2)^2}} \right|_{\vartheta = \vartheta_{w,\sigma_1,\sigma_2,\eta}} du_2\, du_1.$$

The mask $\mathfrak{A}$ retains both actual feet*, the same physical collision flight, and the prohibition on unvisited-disk reflections. It must not be removed in an equality. Any original return, displacement, collision, roof, section, or radius test may be included in a as a Borel selector;* (18.6) *never modifies that test or the physical clock.*

Proof. Apply the proved one-mark flux identity (B.2) to the first mark z_1. This gives the factor $(4\pi R)^{-1}$, the sign σ_1, the u_1 integral and the original angular integral. For each (ϑ, u_1), translating z_1 to the origin associates any distinct second center with exactly one $w = z_2 - z_1 \in \mathcal{W}_*$ by Lemma 18.1. The second center has a definite sign σ_2 of its impact and a unique $u_2 \in (0, \delta_2)$; its physical membership is precisely $\mathfrak{A} = 1$. For fixed w, σ_1, u_1, the function $\vartheta \mapsto b_2 = b_1 + w \cdot v_\vartheta^\perp$ is a sine wave around the fixed b_1. On each of the two signed impact bands $\sigma_2(R, R + \delta_2)$, equation (18.4) has exactly two roots, and its derivative has the nonzero magnitude in (18.5). One-dimensional substitution on the two monotone branches therefore gives the sum over η and the displayed denominator. The total sum over w is finite; Tonelli and monotone approximation first for bounded weights then for arbitrary nonnegative Borel a justify all substitutions. Only regular charts of the genuine flight are used; the inherited one-mark construction assigns a fixed null-set convention on singular lines. The formula counts an ordered pair once because its first mark is unique in the first flux identity and its relative w and sign/angle root are unique thereafter. No physical word count, symbolic renewal process, conditional independence, or fictitious collision is introduced. □

Theorem 18.3 (Uniform quadratic bound for two distinct unvisited marks)**.** *For the original invariant collision measure and every pair of allowed clearances,*

$$\int \sum_{z_1 \in \mathcal{Z}_{\delta_1}(x)} \sum_{\substack{z_2 \in \mathcal{Z}_{\delta_2}(x) \\ z_2 \neq z_1}} 1\, d\nu(x) \leq \frac{2N_{\mathrm{pair}}}{\pi R c_{\mathrm{pair}}}\, \delta_1\delta_2 \leq \frac{2N_{\mathrm{pair}}}{\pi R_- c_{\mathrm{pair}}}\, \delta_1\delta_2. \tag{18.7}$$

This bound includes every physical line/flight and both signs; it does not require independence between the two marks.

Proof. Apply Theorem 18.2 with $a \equiv 1$. Drop only the positive *mask in the upper bound.* The denominator is at least c_{pair} by the lattice-spacing calculation, and there are exactly N_{pair} candidate differences and $2^3 = 8$ sign/root choices. Thus the right side is at most $8N_{\mathrm{pair}}\delta_1\delta_2/(4\pi R c_{\mathrm{pair}})$, which is (18.7). The denominator lower bound is pointwise on the entire parameter rectangle, not only almost everywhere on the physically admitted pairs. □

Corollary 18.4 (A sharp first-order unique-clearance law)**.** *Let* $N_\delta = |\mathcal{Z}_\delta|$ *be the* original *one-flight unvisited-near-miss count and write* $C_{\mathrm{pair}} = N_{\mathrm{pair}}/(\pi R_- c_{\mathrm{pair}})$*. Then*

$$0 \le \frac{\delta}{R} - \nu(N_\delta \ge 1) \le C_{\mathrm{pair}}\delta^2, \tag{18.8}$$

$$0 \le \nu(N_\delta \ge 2) \le C_{\mathrm{pair}}\delta^2, \tag{18.9}$$

$$\nu(N_\delta = 1) \ge \frac{\delta}{R} - 2C_{\mathrm{pair}}\delta^2. \tag{18.10}$$

In particular $\nu(N_\delta \ge 1) = \delta/R + O(\delta^2)$*, uniformly in the original radius interval; the* δ/R *term is exactly the one-copy physical flux, not a proposed Poisson law.*

Proof. For every nonnegative integer N, $0 \le N - \mathbf{1}_{\{N\ge1\}} = (N-1)_+ \le N(N-1)/2$ and $\mathbf{1}_{\{N\ge2\}} \le N(N-1)/2$. Also $N\mathbf{1}_{\{N\ge2\}} \le N(N-1)$. Apply these deterministic inequalities to N_δ, use the exact physical mean (B.3), and substitute Theorem 18.3 with $\delta_1 = \delta_2 = \delta$. No limit in collision count is needed for these one-flight statements. □

Corollary 18.5 (Original exact-label records: uniform weak overcount payment)**.** *Fix* m, R, n, k*, the original physical* m*-flight record, and a nonnegative Borel roof test* $0 \le \psi \le 1$*. Let* $0 < \delta_j \le \delta_*$ *for* $0 \le j < m$*. With the original* $\Pi = \Pi_{n,k,m,R}$ *and unchanged normalized source* ν_R^**,*

$$\begin{aligned} 0 &\le \int \mathbf{1}_\Pi \psi(L) \sum_{j=0}^{m-1} \left[N_{\delta_j}(T_R^j x) - \mathbf{1}_{\{N_{\delta_j}(T_R^j x)\ge1\}}\right] d\nu_R^*(x) \\ &\le \frac{C_{\mathrm{pair}}}{c_*} \sum_{j=0}^{m-1} \delta_j^2. \end{aligned} \tag{18.11}$$

For the original depth schedule $\delta_j = 2\varepsilon r^{d'_j}$*,* $0 < 2\varepsilon \le \delta_*$*,* $r = q_0^{1/4}$*, this is at most*

$$\frac{8C_{\mathrm{pair}}\varepsilon^2}{c_*(1-r^2)}, \tag{18.12}$$

uniformly in m, R, n, k *and the test* ψ*. Thus the positive roof measure of within-flight* duplicate marking *has uniform total variation of order* ε^2*, even with the full original exact labels. The corresponding density need not have controlled* L^∞ *height: multiplying this mass estimate by* m^2 *at fixed outer* B *is* not *justified.*

Proof. The integrand in (18.11) is nonnegative. Since $\Pi \subset Y_R^*$ and $\nu_R^* = c_*^{-1}\mathbf{1}_{Y_R^*}\nu$, its integral is bounded by c_*^{-1} times the sum of unconditioned collision integrals with $\psi \le 1$. By invariance of the original collision measure, each integral equals the same one-flight quantity at $T_R^j x$ and is bounded by $C_{\mathrm{pair}}\delta_j^2$ from Corollary 18.4. The depth list has at most two indices at each nonnegative depth, so $\sum_j \delta_j^2 \le 2(2\varepsilon)^2 \sum_{d\ge0} r^{2d} = 8\varepsilon^2/(1-r^2)$. All source and time tests were retained until applying positive domination; no independent probability factor and no exchange of the original limits occurs. □

The exact second-order formula (18.6) isolates the nontransverse configurations which a two-mark pointwise analysis would otherwise need to pay. The separation constant $c_{\mathrm{pair}} > 0$ rules out that particular simultaneous same-flight angle degeneracy on the original radius range. This leaves *single* unvisited-disk clearance, marks on different physical flights, grazing/section/pairing boundary terms, and the nongrazing capacity shortage. In particular the full $\mathcal{H}_M(\mathsf{R}_B^{\mathrm{low}})$ limit and the original raw pointwise local limit theorem remain open; the new $O(\varepsilon^2)$ total variation gain is never substituted for the required m^2L^∞ estimate.

19. Two clearance marks separated by actual reflections

The same-flight theorem of Section 18 is an inherited result. We now treat marks on *different* flights. The intervening reflections, rather than an independence assumption, supply the new transversality. This section proves finite-itinerary geometric identities on the original physical billiard. The uniform sum in the next section explicitly uses the retained physical pressure input; neither section asserts a normalized roof estimate.

19.1. **Physical feet and the reflection derivative.** Put $\kappa_- = 1/R_+$ and $\kappa_+ = 1/R_-$. All radii, clearances, collision coordinates and measures are those of the preceding sections. The following strengthening of the inherited interior-foot bound is useful because it is independent of every visited incidence.

Lemma 19.1 (A uniform physical foot standoff)**.** *For a genuine mark of clearance* $0 < u \le \delta_* = 3/200$ *on a physical flight, its perpendicular foot has distance at least*

$$a_\circ = \sqrt{(1-R_+)^2 - (R_+ + \delta_*)^2} = \frac{\sqrt{1827}}{200} > \tau_- = \frac{3}{50} \tag{19.1}$$

from each incident collision. In particular the foot is strictly interior, uniformly for $R \in [9/20, 47/100]$.

Proof. The interior assertion is already Lemma B.1; it uses that a third center cannot realize its small clearance at an incident endpoint. If q is either incident contact and z is the marked center, lattice spacing gives $|z-q| \ge 1-R$. Orthogonal decomposition along the actual flight gives $t^2 = |z-q|^2 - (R+u)^2$ for the distance $t>0$ between that endpoint and the foot. The right side is at least $(1-R_+)^2 - (R_+ + \delta_*)^2 = 1827/40000$. Finally $1827 > 144$ proves $a_\circ > 3/50$. No extra collision at the foot has been introduced. □

Consider a first mark on flight 0 and a second on flight h, with $h \ge 1$. Let $q_1, \dots, q_h$ be the actual intervening collision points, $c_i > 0$ their normal incidences, a the distance from the first foot to q_1, and b that from q_h to the second foot. Write $\ell_i = |q_{i+1} - q_i|$ for $1 \le i < h$. Thus $a, b \ge a_\circ$ and $\ell_i \ge \tau_-$. Define

$$P(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad K_i = \begin{pmatrix} 1 & 0 \\ 2/(Rc_i) & 1 \end{pmatrix},$$

$$\mathcal{M}_h = P(b)K_hP(\ell_{h-1})K_{h-1}\cdots P(\ell_1)K_1P(a) = \begin{pmatrix} A_h & B_h \\ C_h & D_h \end{pmatrix}. \tag{19.2}$$

For $h=1$ the middle product is empty: $\mathcal{M}_1 = P(b)K_1P(a)$. The letter B_h here is a matrix entry; it is not the outer localization parameter B.

Theorem 19.2 (The exact cross-flight clearance Jacobian)**.** *Translate the first marked center to* 0 *and parametrize its line by* $(\vartheta, u_1, \sigma_1)$ *as in Theorem B.2. On every regular chart where the intervening physical itinerary and second marked center are fixed, let* b_2 *be the second signed impact parameter. At fixed* u_1*,*

$$|\partial_\vartheta b_2| = B_h > 0, \qquad \left|\det D_{(\vartheta,u_1)}(u_1,u_2)\right| = B_h, \quad u_2 = |b_2| - R. \tag{19.3}$$

Moreover $\det \mathcal{M}_h = 1$*, and the pointwise lower bounds*

$$B_h \ge a + b + \sum_{i=1}^{h-1} \ell_i, \tag{19.4}$$

$$B_h \ge a_\circ(1 + 2\kappa_-\tau_-)^{h-1} \tag{19.5}$$

hold for every such trajectory. They require no lower cutoff on any c_i. A visited incidence tending to zero makes its positive kick larger, not smaller.

Proof. For a differentiable variation of an oriented line let J denote its transverse displacement and J' its direction variation, measured at the reference point on that line. Sliding the reference point along the same line adds only the usual free-flight term. Hence travel by distance t sends (J, J') to $(J + tJ', J')$.

Here is the reflection calculation with its signs. With outward normal n, tangent $\mathcal{J}n$ and curvature $\kappa = 1/R$, write $v^- = -cn + p\mathcal{J}n$ and $v^+ = cn + p\mathcal{J}n$, where $\mathcal{J}$ is rotation by $\pi/2$. A contact variation is $\delta q = \mathcal{J}n\,\delta s$. Its transverse components in the incoming and outgoing direction frames are $-c\delta s$ and $c\delta s$, respectively. Specular reflection gives $\delta\vartheta^+ = 2\kappa\delta s - \delta\vartheta^-$. Reverse the transverse frame after each reflection, applying the same sign reversal to J' as to J. The contact displacement then becomes continuous, while

$$J'^+ = J'^- + \frac{2\kappa}{c} J.$$

Thus the exact linearized reflection in these unfolded transverse frames is K_i. This is a derivative of the genuine reflection rule, not an auxiliary reflecting wall at a marked disk.

The first foot is $-\sigma_1(R + u_1)v_\vartheta^\perp$. Differentiating it at fixed u_1 gives a vector parallel to v_ϑ; its transverse component is zero and its direction variation is one. Consequently the transverse field begins with $(J, J') = (0, 1)$. Propagation to the second foot gives (B_h, D_h). Differentiating the second signed distance gives the negative of this transverse component, up to the h frame reversals. For example, at the last contact, differentiation of $b_2 = (z_2 - q_h) \cdot v_h^\perp$ gives $-\delta q_h \cdot v_h^\perp - b\,\delta\vartheta_h$. Its magnitude is exactly B_h. Multiplication by the second impact sign has unit magnitude, proving both identities in (19.3).

Each factor in (19.2) has determinant one and nonnegative entries. Replacing all K_i by the identity can only decrease the product entrywise; the resulting upper-right entry is the sum of the free lengths. This proves (19.4). At the first collision $J = a \geq a_\circ$. At each subsequent collision, propagation from its predecessor obeys $J_{i+1} = J_i + \ell_i J_i'^+$ with $J_i'^+ \geq 2\kappa_- J_i$. Hence $J_{i+1} \geq (1 + 2\kappa_-\tau_-)J_i$. The final free flight also increases J, which proves (19.5). All assertions have been proved on regular physical charts; a one-sided limiting matrix with an infinite grazing kick is not called a finite derivative of a singular physical return. □

19.2. **Global multiplicity, including disconnected word domains.** A local nonzero Jacobian does not bound the number of its roots. The following global argument supplies precisely that missing fact. A *tangentially augmented itinerary* consists of the first marked center, the ordered h actual intervening disk centers, the second marked center, and the two signed impacts. The marked centers may coincide with one another or with centers visited at other times. They are not the incident centers of their own marked flights.

Theorem 19.3 (One trajectory per tangentially augmented itinerary)**.** *Fix $h \geq 1$, the augmented itinerary, R, and u_1, u_2 in their original clearance ranges. There is at most one regular physical broken path between the two feet realizing these data, including its oriented line at each endpoint. This is a global statement: no connectedness of its admissible word domain is assumed.*

Proof. For two candidate configurations $Q = (q_0, q_1, \ldots, q_h, q_{h+1})$ and $\widehat{Q}$, the endpoint positions lie on the two auxiliary circles of radii $R + u_1$ and $R + u_2$, and the other positions lie on the actual radius-R disks. These auxiliary circles only locate the feet; no new collision is counted. The Cartesian broken length $\mathcal{S}(Q) = \sum_{i=0}^{h} |q_{i+1} - q_i|$ is convex and continuously differentiable at each candidate because every segment has positive length. Gradient monotonicity gives

$$(\nabla\mathcal{S}(Q) - \nabla\mathcal{S}(\widehat{Q})) \cdot (Q - \widehat{Q}) \geq 0. \tag{19.6}$$

At an actual interior collision $\nabla_i \mathcal{S} = -2c_i n_i$, so its contribution on the left is $-R(c_i + \widehat{c}_i)|n_i - \widehat{n}_i|^2$. At a marked endpoint the velocity is tangent to its auxiliary circle, with its oriented tangent fixed by the prescribed impact sign. Thus the endpoint gradient is a fixed sign times $\mathcal{J} n_i$. Its difference is orthogonal to the contact difference, and its contribution to (19.6) is zero. Summing shows that every actual interior contact agrees, since $c_i, \widehat{c}_i > 0$.

For completeness, agreement of the contacts alone is not used to skip the endpoint argument. Write the two convex support gaps for $\mathcal{S}$ and add them. Their sum is the left side of (19.6), now zero. Each gap is a sum of nonnegative support gaps of the Euclidean norm for individual flight vectors. Equality in the norm support inequality, with nonzero reference vector, forces the two corresponding vectors to be positive multiples of one another. Therefore all corresponding flight directions agree. At either endpoint its direction and prescribed impact sign determine its auxiliary-circle normal uniquely, hence the foot itself. This proves global uniqueness. Repeated disk centers cause no identification of contact variables at different times and do not affect the argument. □

19.3. **An exact physical two-clearance change of variables.** Set

$$\mathcal{Z}_\circ = \{z \in \Lambda \setminus \{0\} : |z| \le \tau_+ + 2R_+ + \delta_*\}, \qquad N_\circ = |\mathcal{Z}_\circ|. \tag{19.7}$$

This is the same finite end-candidate geometry as the one-mark construction. After translating the first marked center to zero, the first actual collision center has at most $N_\circ$ choices; the second marked center relative to the last actual collision center has at most $N_\circ$ choices. The intervening physical word has length $h-1$, *not* h. It is empty when $h = 1$. Let $\mathcal{A}_{h,R}(u_1, u_2)$ be the set of these augmented itineraries with a regular physical realization, each counted once. Denote the associated original collision state and the two marked centers in its lifted initial frame by $(x_a, z_{1,a}, z_{2,a})$.

Theorem 19.4 (One-copy coarea across arbitrarily many true reflections)**.** *For any $h \ge 1$ and every nonnegative Borel weight f of the initial physical state and the two marks,*

$$\begin{aligned} &\int \sum_{z_1 \in \mathcal{Z}_{\delta_1}(x)} \sum_{z_2 \in \mathcal{Z}_{\delta_2}(T_R^h x)} f(x, z_1, z_2)\, d\nu(x) \\ &\quad = \frac{1}{4\pi R} \int_0^{\delta_1} \int_0^{\delta_2} \sum_{a \in \mathcal{A}_{h,R}(u_1,u_2)} \frac{f(x_a, z_{1,a}, z_{2,a})}{B_h(x_a)}\, du_2\, du_1. \end{aligned} \tag{19.8}$$

In the left side, the center on flight h is translated into the same original lifted frame as the first. The two marks are distinct as flight–center pairs even when their lattice centers agree. All original return, displacement, collision, physical-roof and section tests may be included in f without altering the identity.

Proof. Apply the exact first-mark formula of Theorem B.2; its prefactor is $(4\pi R)^{-1}$ and its parameters are $(\vartheta, u_1, \sigma_1)$. On each regular physical branch the second center has a unique impact sign. The second substitution has Jacobian B_h by Theorem 19.2. The two signs at each end are included among the augmented itineraries; no extra factor of two, radius, incidence, collision count, or word count is permitted. Theorem 19.3 ensures that overlapping charts of a single augmented itinerary cannot produce different roots.

To justify the global substitution, exhaust the regular parameter space by countably many relatively compact inverse-function charts, and disjointize their overlaps before summation. On each chart use the ordinary change-of-variables theorem first for bounded nonnegative weights. The chart exhaustion and then monotone approximation give (19.8) for arbitrary $f \ge 0$. Finite-horizon geometry gives finitely many physical words at each fixed h, although this number is never used as a bound. The usual singular collision set is null for collision area and hence for

first-mark area on every chart. Its image under a regular local change of variables is null as well. Values on those bulk-null sets therefore do not alter this identity. This argument makes no claim that their one-sided paired endpoint traces vanish. Membership in $\mathcal{A}_{h,R}$ retains all physical flights and feet; enlarging that set is allowed only for subsequent positive upper bounds. □

20. Uniform cross-flight summation and the complete mark count

The exponential lower bound on B_h does not by itself permit multiplication by the number of physical words. We instead compare B_h^{-1} with the *same* stable cocycle used in Input 11.1. This comparison includes both endpoint coordinate factors and is proved below. Only the unconditioned pressure inequality $\sup_{R,l} P_l(R) \le Q_*$ is used in the ensuing sum; no exact-label, roof-conditioned or new twisted pressure estimate is assumed.

20.1. **A stable optical comparison between the two physical feet.** On the full-area, two-sided regular set M_R', let x_i be the outgoing state at the ith intervening collision. Keep the Euclidean arclength convention in (s,φ), with $s = R\alpha$ and $p = \sin\varphi$. Set

$$L_\circ = \sqrt{1 + (\kappa_+ + \tau_-^{-1})^2}, \qquad C_J = \frac{L_\circ}{\kappa_- a_\circ^2}. \tag{20.1}$$

These constants do not depend on h, R or the exact labels.

Lemma 20.1 (The cross-flight Green entry pays its stable weight)**.** *At every regular marked trajectory in M_R',*

$$\frac{1}{B_h} \le C_J c_h\, j_{h-1}(x_1) \le C_J j_{h-1}(x_1), \qquad h \ge 1. \tag{20.2}$$

For $h = 1$ this uses the empty stable product $j_0 = 1$.

Proof. We spell out the stable-line convention so that a hidden grazing coordinate factor cannot enter the comparison. At an outgoing collision write the stable transverse slope as $-\beta_i$. Propagation to the next collision, followed by reflection, gives

$$\beta_i = \frac{k_{i+1} + \beta_{i+1}}{1 + \tau(x_i)(k_{i+1} + \beta_{i+1})}, \qquad k_i = \frac{2}{Rc_i}, \qquad 0 < \beta_i < \frac{1}{\tau(x_i)} \le \frac{1}{\tau_-}. \tag{20.3}$$

One can obtain this invariant line without a further summability assumption: start the backward recurrence with terminal value zero. Every recurrence map takes $[0,\infty)$ into $[0,\tau_-^{-1}]$ and its derivative is at most $(1 + 2\kappa_-\tau_-)^{-2} < 1$. Differences caused by a further terminal continuation therefore tend to zero geometrically, giving the unique invariant negative transverse slope. It is the stable line in the usual invariant collision cone. This agrees with the stable-line convention in Section 1.1 of [7]; no pressure conclusion is being imported from that observation.

If $V_i = d\varphi/ds$ on this line, the identities $J_i = c_i\, ds$ in the outgoing direction frame and $d\vartheta = (\kappa + V_i)ds$ give

$$V_i = -\kappa - c_i\beta_i, \qquad L_i = \sqrt{1 + V_i^2} \in [1, L_\circ], \qquad d\ell_i = |J_i| L_i / c_i.$$

Absolute values remove the harmless alternating transverse-frame sign. The physical stable Jacobian between collisions 1 and h therefore satisfies the *exact* coordinate identity

$$\frac{|J_h|}{|J_1|} = j_{h-1}(x_1)\frac{c_h}{c_1}\frac{L_1}{L_h}. \tag{20.4}$$

For the first foot to the first collision, backward propagation through that collision gives

$$\frac{|J_1|}{|J_{\text{first}}|} = \frac{1}{1 + a(k_1 + \beta_1)} \le \frac{c_1}{2\kappa_- a_\circ}.$$

For the final collision to the second foot the ratio is $1-b\beta_h \in (0,1)$. The second foot precedes the next actual collision by at least $a_\circ$; hence its stable slope magnitude β_{second} satisfies

$$0 < \beta_{\mathrm{second}} = \frac{\beta_h}{1-b\beta_h} \le \frac{1}{\tau(x_h)-b} \le \frac{1}{a_\circ}. \tag{20.5}$$

Combining these facts with (20.4), the stable transverse contraction between the feet obeys

$$\gamma := \frac{|J_{\mathrm{second}}|}{|J_{\mathrm{first}}|} \le \frac{L_\circ c_h}{2\kappa_- a_\circ} j_{h-1}(x_1). \tag{20.6}$$

All incidences at the first endpoint cancel in this formula; there is no assumption that either endpoint is uniformly nongrazing.

Since $\det \mathcal{M}_h = 1$, applying its inverse to a vector of slope $-\beta_{\mathrm{second}}$ gives $\gamma^{-1} = D_h + B_h\beta_{\mathrm{second}}$. If the final free-flight factor is removed, all remaining matrix entries are nonnegative; thus $B_h = B_0 + bD_h \ge a_\circ D_h$. Equations (20.5) and (20.6) now imply

$$\frac{1}{B_h} = \gamma\left(\frac{D_h}{B_h} + \beta_{\mathrm{second}}\right) \le \frac{2\gamma}{a_\circ} \le \frac{L_\circ c_h}{\kappa_- a_\circ^2} j_{h-1}(x_1).$$

This proves (20.2). The proof uses only the positive Jacobi propagation and the stable-line convention; the pressure bound is not used until the next theorem. □

Theorem 20.2 (A complete cross-flight clearance-density bound)**.** *Assume the retained bound* $\sup_{R,l} P_l(R) \le Q_*$ *from Input 11.1, with its original physical words and stable norm. Then for every* $h \ge 1$ *and almost every* $(u_1,u_2) \in (0,\delta_*)^2$*,*

$$\sum_{a \in \mathcal{A}_{h,R}(u_1,u_2)} \frac{1}{B_h(x_a)} \le 4N_\circ^2 C_J Q_*. \tag{20.7}$$

Consequently the original invariant collision measure satisfies

$$\int N_{\delta_1}(x) N_{\delta_2}(T_R^h x)\, d\nu(x) \le C_{\mathrm{cross}}\delta_1\delta_2, \qquad C_{\mathrm{cross}} = \frac{N_\circ^2 C_J Q_*}{\pi R_-}, \tag{20.8}$$

uniformly in the full separation $h \ge 1$*, the original radius interval, and* $0 < \delta_i \le \delta_*$*. Every original exact-record or roof selector bounded by one may be retained on the left. The essential bound in* (20.7) *is on the* two clearance variables*, not on the physical roof.*

Proof. First remove the bulk-null complement of M_R' on the countably many regular charts in Theorem 19.4. Because their two-clearance Jacobian is nonzero, the image of this complement is null in du_1du_2. For each fixed R and h, almost every pair of clearances therefore has all its admitted roots in M_R'. There are only countably many separations. A radius-dependent null set is sufficient; no common exceptional set for all radii is asserted.

Fix such a pair of clearances. Specify the first actual collision center relative to the first marked center, and the second marked center relative to the last actual collision center. This has at most $N_\circ^2$ choices by (19.7). There are four endpoint sign pairs. For each of these choices and each realizable physical word w of length $h-1$, Theorem 19.3 allows at most one root. Lemma 20.1 bounds its weight by $C_J \sup_{[w]_R \cap M_R'} j_{h-1}$. The full positive sum is therefore bounded by

$$4N_\circ^2 C_J \sum_{w \in \mathcal{W}_{h-1}(R)} \sup_{[w]_R \cap M_R'} j_{h-1} \le 4N_\circ^2 C_J Q_*.$$

This is a weighted sum, not D^{h-1} times a worst trajectory. It is valid even when word domains are disconnected and when the selected roots have different endpoint momenta. For $h=1$, the sum over the empty word is exactly $P_0 = 1$.

Use (19.8) with $f = 1$ and integrate the last bound over the original clearance rectangle. The exact prefactor gives $4/(4\pi R)$, proving (20.8). A Borel selector in the unchanged physical event is handled by positive domination only after the exact coarea identity; it never becomes an independent probability factor. The constants are independent of h because the assumed pressure bound is, not because a finite family of tested words was extrapolated. □

20.2. **All chronological duplicate marks, including repeated centers.** For a finite original record of m flights set

$$\mathscr{N}_{m,\boldsymbol{\delta}}(x) = \sum_{j=0}^{m-1} N_{\delta_j}(T_R^j x), \qquad S_{m,\boldsymbol{\delta}} = \sum_{j=0}^{m-1} \delta_j, \quad 0 < \delta_j \le \delta_*. \tag{20.9}$$

A center seen on two different flights gives two marks. Thus this count is different from both the number of distinct lattice centers and the indicator of at least one witness. Let

$$C_F = \max\{2C_{\mathrm{pair}}, C_{\mathrm{cross}}\}, \qquad C_{\mathrm{pair}} = \frac{N_{\mathrm{pair}}}{\pi R_- c_{\mathrm{pair}}} \tag{20.10}$$

with the inherited same-flight constant of Corollary 18.4.

Theorem 20.3 (Uniform second factorial control for the complete record)**.** *Under the pressure input of Theorem 20.2, the original collision measure satisfies*

$$\int \mathscr{N}_{m,\boldsymbol{\delta}}\, d\nu = S_{m,\boldsymbol{\delta}}/R, \tag{20.11}$$

$$\int \mathscr{N}_{m,\boldsymbol{\delta}}(\mathscr{N}_{m,\boldsymbol{\delta}} - 1)\, d\nu \le C_F S_{m,\boldsymbol{\delta}}^2. \tag{20.12}$$

The first equality needs no pressure assumption. In particular, writing $S = S_{m,\boldsymbol{\delta}}$ and $\mathscr{N} = \mathscr{N}_{m,\boldsymbol{\delta}}$,

$$\begin{aligned} 0 \le S/R - \nu(\mathscr{N} \ge 1) &\le \tfrac{1}{2} C_F S^2, \\ \nu(\mathscr{N} \ge 2) &\le \tfrac{1}{2} C_F S^2, \\ \nu(\mathscr{N} = 1) &\ge S/R - C_F S^2. \end{aligned} \tag{20.13}$$

These estimates include all pairs of collision times and all physical words, with constants independent of m. They are not a Poisson approximation and assert no independence of the marks.

Proof. Invariance of ν and the exact one-flight flux give (20.11). Put $N_j = N_{\delta_j}(T_R^j x)$. The exact integer identity

$$\mathscr{N}(\mathscr{N} - 1) = \sum_j N_j(N_j - 1) + 2\sum_{j<k} N_j N_k$$

separates same-flight ordered pairs from pairs at distinct times. Theorem 18.3 bounds the first sum by $2C_{\mathrm{pair}} \sum_j \delta_j^2$. Invariance and Theorem 20.2, applied at separation $k - j$, bound the second by $2C_{\mathrm{cross}} \sum_{j<k} \delta_j \delta_k$. The definition of C_F proves (20.12). For a nonnegative integer N, use $0 \le N - \mathbf{1}_{\{N \ge 1\}} \le N(N-1)/2$, $\mathbf{1}_{\{N \ge 2\}} \le N(N-1)/2$, and $N\mathbf{1}_{\{N \ge 2\}} \le N(N-1)$. Integration proves (20.13); no limiting interchange is needed. □

Corollary 20.4 (The unchanged exact event and complete depth schedule)**.** *Keep the original $\Pi = \Pi_{n,k,m,R}$, L, ν_R^* and all original labels. For every Borel $0 \le \psi \le 1$,*

$$\begin{aligned} 0 &\le \int \mathbf{1}_\Pi \psi(L)[\mathscr{N}_{m,\boldsymbol{\delta}} - \mathbf{1}_{\{\mathscr{N}_{m,\boldsymbol{\delta}} \ge 1\}}]\, d\nu_R^* \\ &\le \frac{C_F}{2c_*} S_{m,\boldsymbol{\delta}}^2. \end{aligned} \tag{20.14}$$

For the original schedule $\delta_j = 2\varepsilon r^{\min(j,m-1-j)}$, $r = q_0^{1/4}$, $0 < 2\varepsilon \le \delta_*$, *this becomes*

$$\frac{C_F}{2c_*} S_{m,\boldsymbol{\delta}}^2 \le \frac{8C_F\varepsilon^2}{c_*(1-r)^2}, \tag{20.15}$$

uniformly over all m, R, n, k *and bounded Borel roof tests. Equivalently, the difference between the mark-count-weighted and one-witness-weighted exact-record roof measures is a positive measure with this total-variation bound. This now pays* cross-flight as well as within-flight overcount*; the former was not covered by Corollary 18.5.*

Proof. The integrand is nonnegative and is bounded by $\mathscr{N}(\mathscr{N}-1)/2$. Retain the original exact event until using $\nu_R^* = c_*^{-1}\mathbf{1}_{Y_R^*}\nu$ and positive domination. Equation (20.12) gives (20.14). There are at most two flights at each nonnegative depth, including the middle flight when m is odd. Thus $S \le 2(2\varepsilon)\sum_{d\ge 0} r^d = 4\varepsilon/(1-r)$, which proves (20.15). Taking the supremum over Borel $0 \le \psi \le 1$ is exactly the total variation of the positive difference measure. The equality of this measure with the stated pushforward follows directly from its definition, not from a change of conditional law. □

20.3. **What this theorem does, and does not, close.** The new derivative, global tangential-end uniqueness and coarea statements are geometric theorems for the actual billiard. The uniform complete mark-count theorem is a deduction from the *already retained* unconditioned physical pressure input, whose independent reconstruction remains outstanding. It is not presented as a new proof of that input. The complete double-clearance density and the all-time duplicate payment are stronger than the inherited same-flight assertions, but their measures are still different from the target-roof essential height.

More precisely, for the full original exact event define the positive roof measures

$$\mu_{m,\boldsymbol{\delta}}^{\mathrm{count}} = L_{\#}(\mathbf{1}_{\Pi}\mathscr{N}_{m,\boldsymbol{\delta}}\nu_R^*), \qquad \mu_{m,\boldsymbol{\delta}}^{\mathrm{one}} = L_{\#}(\mathbf{1}_{\Pi}\mathbf{1}_{\{\mathscr{N}_{m,\boldsymbol{\delta}}\ge 1\}}\nu_R^*).$$

The exact identity $\mu^{\mathrm{count}} = \mu^{\mathrm{one}} + \mu^{\mathrm{excess}}$ has $\mu^{\mathrm{excess}} \ge 0$ and the bound (20.15). The first-order one-mark source is *not* subtracted from the problem. Even in the unconditioned law, (20.11) and (20.13) retain its positive first-order term S/R. The exact-label source need not have that same leading coefficient; no conditioning argument here claims otherwise.

To close the original theorem, a proof must still control the clearance-selected first-order source and any excess which enters the *actual paired source* in the $m^2L^\infty(dt)$ topology, including critical, degenerate, one-sided boundary and section-cut contributions, or dominate that complete source by the original nongrazing capacity. This bulk identity alone does not identify a paired endpoint trace with a bulk restriction. The low-incidence shallow source must also be paid. The inherited precise common closure condition remains (15.8); it has not been renamed or declared true.

At every fixed outer B the clearance widths are fixed first. Neither the $O(\varepsilon^2)$ mass above nor the exponential single-trajectory lower bound for B_h is multiplied by m^2 and asserted to vanish. The complete physical sum, the radius and exact-label suprema, the roof essential supremum, the collision limsup and the final outer B limit retain their original meanings and order. Thus the raw pointwise local limit and four-coordinate inversion target is unchanged and not yet proved.

21. The angular clearance source as a weighted unstable family

The proofs in Sections 19–20 are inherited, including their stated pressure premise. We now give a different summation route. It starts from the actual angular source in the one-mark flux, rather than taking a supremum over every physical cylinder. The geometric results of this section use no pressure or growth assumption. The propagation theorem in the next section uses the explicitly specified homogeneous curve-growth input, not the stronger physical-word pressure input of Input 11.1.

All radii, sections, time conventions and exact records are unchanged. Put $b_* = R_+ + \delta_* = 97/200$, $a_* = a_\circ = \sqrt{1827}/200$, $\kappa_- = 1/R_+$ and $\kappa_+ = 1/R_-$. We use actual boundary arclength $s = R\alpha$ and outgoing angle φ, with $c = \cos\varphi > 0$ and $p = \sin\varphi$. Arclength on a collision curve always means Euclidean arclength in (s, φ).

21.1. **A genuine first collision, not a reflection at the mark.** Fix $0 < u \le \delta_*$ and $\sigma \in \{-1, 1\}$, and put $b = \sigma(R + u)$. For $v_\vartheta = (\cos\vartheta, \sin\vartheta)$, the marked foot is $q_f = -bv_\vartheta^\perp$. Starting at this point with velocity v_ϑ, let $q_1 = q_f + av_\vartheta$ be the first actual collision, and let $Y_{R,\sigma,u}(\vartheta)$ be the outgoing state there. Thus, in the one-mark notation,

$$Y_{R,\sigma,u}(\vartheta) = T_R X_R(\vartheta, \sigma, u). \tag{21.1}$$

The line is exterior to the marked disk. Moreover, for every other lattice center z, $|z - q_f| \ge 1 - b_* > R_+$. The foot is therefore in the actual free region for every ϑ. The inherited finite-horizon bound supplies the two incident collisions. We exclude the finitely many first-collision tangencies when differentiating. The marked disk itself is never reflected from at the foot.

Theorem 21.1 (Exact first-image density and uniform geometric regularity)**.** *On every regular interval of fixed first collision center, let s be an unwrapped boundary coordinate at that center. Then*

$$\frac{ds}{d\vartheta} = -\frac{a}{c}, \qquad V := \frac{d\varphi}{ds} = \frac{1}{R} + \frac{c}{a}, \qquad \rho := \left|\frac{d\vartheta}{d\ell}\right| = \frac{c}{a\sqrt{1+V^2}}. \tag{21.2}$$

In particular, with

$$U_0 = \kappa_+ + a_*^{-1}, \qquad K_0 = U_0/a_* + (1 + b_*/a_*)/a_*^2, \tag{21.3}$$

every such image is an unstable graph satisfying

$$a \ge a_*, \qquad \kappa_- \le V \le U_0 < \kappa_+ + \tau_-^{-1}, \qquad |dV/ds| \le K_0, \qquad 0 < \rho \le c/a_*. \tag{21.4}$$

These bounds hold at every regular first collision, with no lower cutoff on c and no lower cutoff on u within its positive range.

Proof. The foot standoff is Lemma 19.1. Let n be the outward normal at q_1 and $t = n^\perp$. The incoming and outgoing velocities are $v^- = -cn + pt$ and $v^+ = cn + pt$. At fixed u, the variation of q_f is parallel to v^-. Differentiating $q_1 = q_f + av^-$ and projecting onto $(v^-)^\perp$ gives $-c\,ds/d\vartheta = a$, proving the first formula. The specular angle identity gives $d\vartheta^+ = 2R^{-1}ds - d\vartheta$; subtracting the normal-angle differential $R^{-1}ds$ yields $d\varphi = R^{-1}ds - d\vartheta$. Division by ds proves the formula for V. Since $d\ell = \sqrt{1+V^2}|ds|$, the density formula follows. Its vanishing factor c is essential.

For clarity, the second-derivative estimate does not appeal to a sampled trajectory or to bounded curvature by name. Projecting the same contact identity onto n gives $da/d\vartheta = -b - ap/c$. Hence

$$\frac{da}{ds} = p + \frac{bc}{a}, \qquad \frac{dc}{ds} = -pV, \qquad \frac{dV}{ds} = -\frac{pV}{a} - \frac{c}{a^2}\left(p + \frac{bc}{a}\right). \tag{21.5}$$

Use $|p|, c \le 1$, $|b| \le b_*$ and $a \ge a_*$ to obtain (21.4). The curvature of the graph in (s, φ) is $|dV/ds|/(1+V^2)^{3/2}$ and is therefore also bounded by K_0. Finally $a_* > \tau_-$ gives the strict upper-cone inequality. No finite derivative at $c = 0$ is asserted. □

21.2. **Uniform finite geometry before the homogeneous cut.** Use the finite candidate set already fixed in (19.7), and write $N_* = N_\circ$ in this section. For reference, its radius is $D_* = \tau_+ + 2R_+ + \delta_*$. It has the explicit packing upper bound

$$N_* + 1 \le 4(D_* + 1/2)^2. \tag{21.6}$$

Indeed, the disjoint open radius-$1/2$ disks centered at all the lattice points in this closed radius-D_* disk lie in the radius-$(D_* + 1/2)$ disk. This bound does not assign a sampled value to τ_+. Here

τ_+ means the inherited uniform supremum of actual free-flight lengths over the original radius family.

Lemma 21.2 (A uniform branch count)**.** *For each fixed R, u, σ, the angular circle can be cut at at most $8N_*$ points, in addition to an angular coordinate seam, so that on every resulting interval the first collision center is fixed, the contact coordinate has no seam, and the image in Theorem 21.1 is injective. The number of intervals is at most $M_* = 8N_* + 1$; each has image length at most $2\pi R_+\sqrt{1+U_0^2}$.*

Proof. Every first collision center belongs to the stated candidate set: its distance from the marked center is at most $\tau_+ + 2R_+ + \delta_*$. For a candidate z, loss or creation of an intersection of the line with its disk requires

$$z \cdot v_\vartheta^\perp + b = R \quad \text{or} \quad z \cdot v_\vartheta^\perp + b = -R.$$

Each equation has at most two roots on the angular circle, since $z \neq 0$. There are thus at most $4N_*$ tangency cuts. Away from these cuts, positive first-hit times vary continuously. Their ordering cannot change without two distinct disks having a common contact point, which is impossible because $2R < 1$. No intersection can pass through the foot, which is exterior to every disk as proved above. Thus the first target is fixed on each interval.

At each candidate disk choose one contact-coordinate seam, at $q = z + R(1,0)$. A line in the source passes through this point only if $q \cdot v_\vartheta^\perp = -b$. Since $|q| \geq 1 - R_+ > b_*$, there are at most two such roots for each disk. We allow $4N_*$ seam cuts in the displayed bound, which is a harmless overestimate. Coincident roots only reduce the count. On a seam-free interval $ds/d\vartheta = -a/c < 0$, so s is strictly monotone and cannot wind around its contact circle. Its range has length at most $2\pi R$. This proves injectivity and the length bound. The argument uses no long collision-word count. Overlapping images from different intervals remain separately labeled sources; they are not silently identified in the quotient collision space. □

21.3. **Paying the infinitely many grazing pieces.** Fix $k_0 \geq 2$. Define the homogeneous strips

$$H_k^+ = \{\pi/2 - k^{-2} < \varphi < \pi/2 - (k+1)^{-2}\}, \quad H_k^- = -H_k^+, \qquad k \geq k_0,$$

and let H_0 be the remaining central strip, apart from its boundary. Let $\ell_* > 0$ be a fixed allowed curve length. Subdivide each homogeneous image into pieces of length at most ℓ_*. Set

$$L_* = 1 + \left\lceil \frac{2\pi R_+\sqrt{1+U_0^2}}{\ell_*} \right\rceil, \qquad A_* = \frac{2M_*L_*}{a_*}\left(1 + \frac{2}{k_0 - 1}\right). \tag{21.7}$$

The factor two in A_* includes both first-impact signs.

Theorem 21.3 (A uniformly summable angular source)**.** *For every $R \in [R_-, R_+]$ and $0 < u \leq \delta_*$, the pushforward of $\sum_{\sigma=\pm1} d\vartheta$ under the actual maps $Y_{R,\sigma,u}$ has a countable, source-labeled representation*

$$\mu_{R,u} = \sum_{C \in \mathcal{F}_{R,u}} \rho_C \, d\ell_C, \qquad \sum_{C \in \mathcal{F}_{R,u}} \|\rho_C\|_\infty \leq A_*. \tag{21.8}$$

Each C is a homogeneous unstable graph of length at most ℓ_*, with the slope and curvature bounds in (21.4)*. The identity holds against every nonnegative Borel test, and has total mass 4π. An arbitrary Borel selector in the original angular source can be retained on the right as a pullback selector; its derivatives or connected components are not counted.*

Proof. On each interval of Lemma 21.2, $V > 0$ and s is strictly monotone, so φ is strictly monotone. Each homogeneity strip is consequently visited at most once by that interval. The length subdivision requires at most L_* pieces per strip. The density is exactly (21.2), not a

constant-density substitute. On H_0 it is at most $1/a_*$. On either $H_k^{\pm}$, it is at most k^{-2}/a_*, because $\cos\varphi \le k^{-2}$. Thus, summing over both signs,

$$\sum_C \|\rho_C\|_\infty \le \frac{2M_*L_*}{a_*}\left(1+2\sum_{k=k_0}^{\infty} k^{-2}\right) \le A_*.$$

The last inequality follows by integrating x^{-2} from k_0-1 to infinity. First perform the exact substitution on finitely many regular intervals and truncated homogeneous strips. The omitted angular cut points are finite, and strip boundaries have at most one preimage per regular interval. Monotone exhaustion proves the measure identity and its total mass. Borel selectors are multiplied before substitution and are not differentiated. In particular an infinite number of small selector components does not incur an infinite multiplicity charge. □

Theorem 21.3 is the new first-source result. Replacing ρ_C by the crude bound $1/a_*$ *before* summing the homogeneity strips would give a divergent series. The factor c is the geometric payment for that series. This is a source estimate in a specific curve norm, not a statement that $\mu_{R,u}$ has a two-dimensional collision density or a bounded original roof density.

22. A full cross-flight root sum from curve growth

We spell out the remaining dynamic input, including its curve class. This prevents replacement of a global pressure premise by an unstated regularity premise. The exact geometry of the preceding section is unconditional within the retained physical finite-horizon model.

Input statement 22.1 (Homogeneous curve growth, in its required class)**.** Choose the common cone upper bound $U = \kappa_+ + \tau_-^{-1}$ and a regular homogeneous unstable curve class containing all the graphs of Theorem 21.3. In particular its curvature allowance must accommodate K_0; being merely C^2 does not establish this membership. For some common $k_0 \ge 2$, sufficiently small $\ell_* > 0$, and $C_g \ge 1$, every curve C of that class and length at most ℓ_* has a source-labeled partition of its regular forward image into homogeneous short curves $\mathcal{G}_l^+(C)$ satisfying

$$\sum_{V\in\mathcal{G}_l^+(C)} \sup_V J_V(T_R^{-l}) \le C_g, \qquad R\in[R_-,R_+], \quad l\ge 0. \tag{22.1}$$

The branches cover arclength-almost every original point, the inverse branch in each summand is the actual physical one, and all resulting graphs have slopes in $[\kappa_-, U]$. The Jacobian is for Euclidean arclength in (s,φ).

This is the forward, time-reversed version of the curve input in Section D, with the source-class membership now made explicit. The relevant published curve-sum statement is [6, Lemma 3.2(b)]; it is based on the homogeneous curve, expansion and distortion properties (H2)–(H4), not on a sum of physical cylinder suprema. The stated class and its family-uniform constants remain explicit external inputs here. No independent reconstruction of the full geometric-pressure theorem, nor of the older singular pairing and spectral interfaces, is claimed.

Shrinking ℓ_* if necessary, require

$$\ell_* \le \frac{\pi}{2D_0}, \qquad D_0 = \kappa_+ + U + (1-R_+)^{-1}. \tag{22.2}$$

The admissible growth partition uses this common choice; any change of its subdivision scale must be included in its constant C_g.

Lemma 22.2 (Counted clearance on a short unstable graph)**.** *For a Borel set $I \subset (0, \delta_*)$, let $N_I(x)$ count all unvisited disks on the next actual flight of x whose clearance belongs to I. For every graph V in Input 22.1 of length at most ℓ_*,*

$$\int_V N_I \, d\ell \le C_t |I|, \qquad C_t = \frac{2N_\circ \sqrt{1+U^2}}{\kappa_- a_*}. \tag{22.3}$$

This is a bound on the mark count*, including both impact signs, not just on its nonzero indicator.*

Proof. Use a lift with the incident disk centered at zero. Every possibly marked center belongs to $\mathcal{Z}_\circ$. For a fixed such z, let $\vartheta(s)$ be the outgoing direction, and put $b_z = (z-q)\cdot v_\vartheta^\perp$, $t_z = (z-q)\cdot v_\vartheta$. Direct differentiation gives

$$b_z' = -\cos\varphi - t_z \vartheta', \qquad \vartheta' = 1/R + \varphi' \ge \kappa_-.$$

The angle $\eta = \arg(z-q) - \vartheta$ is strictly decreasing: $\eta' \le (1-R_+)^{-1} - \kappa_- < 0$. Its total variation is at most $D_0 \ell_* \le \pi/2$. Thus $t_z > 0$, equivalently $\cos\eta > 0$, holds on at most one interval of the graph. On this interval b_z is strictly decreasing. On the physical mark subset the interior-foot estimate gives $t_z \ge a_*$; therefore $|b_z'| \ge \kappa_- a_*$. For each sign, change variables in this one-to-one map on the marked subset. The integral of its indicator is at most $\sqrt{1+U^2}|I|/(\kappa_- a_*)$. Dropping a first-hit or unvisited mask occurs only in this positive bound. Sum the two signs and the $N_\circ$ candidates. Approximation by open sets, or the one-dimensional area formula for a monotone smooth map, gives the claim for every Borel I. No number of components of I or of an exact-event selector occurs. □

Theorem 22.3 (A fixed-first-clearance kernel bound)**.** *Assume Input 22.1. For every $R \in [R_-, R_+]$, every fixed $0 < u_1 \le \delta_*$, every integer $h \ge 1$, and every Borel $I \subset (0, \delta_*)$,*

$$\sum_{\sigma=\pm 1} \int_0^{2\pi} N_I\big(T_R^h X_R(\vartheta, \sigma, u_1)\big) \, d\vartheta \le H_* |I|, \qquad H_* = A_* C_g C_t. \tag{22.4}$$

The integrand is taken on its actual regular domain. Its exceptional set has zero angular measure under the curve input. Arbitrary Borel weights between zero and one, including the original section, exact record and physical roof selectors, may be retained inside the sum without increasing the bound. In particular the second-clearance measure has a density bounded by H_*, uniformly in the first clearance and in the complete collision separation.

Proof. The first actual collision has already been taken in $Y = T_R X$. Apply Theorem 21.3 to its source. Exactly $l = h - 1$ further iterates remain. For each labeled curve C, use the physical image partition in (22.1) and one-dimensional change of variables:

$$\begin{aligned} \int_C \rho_C(y) N_I(T_R^l y) \, d\ell_C(y) &= \sum_{V \in \mathcal{G}_l^+(C)} \int_V (\rho_C \circ T_R^{-l}) N_I \, J_V T_R^{-l} \, d\ell_V \\ &\le \|\rho_C\|_\infty C_g C_t |I|. \end{aligned}$$

Use Lemma 22.2 separately on each V, and then sum its weighted inverse Jacobian, not its number of components. Summing C uses (21.8) and proves the bound. For $h = 1$ the middle map is the identity and $l = 0$; no collision is missing or added. First truncate the nonnegative sums and then use monotone convergence. Carry a bounded Borel selector through each exact substitution before dropping it in the inequality. The measure domination for all I implies absolute continuity and the displayed density bound by the Radon–Nikodym theorem. It concerns clearance, not the roof variable $L_{m,R}$. □

Theorem 22.4 (One-copy marked root pressure without physical-word pressure)**.** *Under precisely the preceding curve-growth input, let $\mathcal{A}_{h,R}(u_1,u_2)$ be the set of actual tangentially augmented itineraries in Theorem 19.4. Then*

$$\sup_{R\in[R_-,R_+],\ h\ge 1}\ \operatorname*{ess\,sup}_{(u_1,u_2)\in(0,\delta_*)^2}\ \sum_{a\in\mathcal{A}_{h,R}(u_1,u_2)} B_h(x_a)^{-1}\le H_*. \tag{22.5}$$

The complete positive physical sum precedes the essential supremum. No premise on $\mathcal{P}_l(R)$, no product of independently chosen flights and no growth-rate bound on the number of words is used. Consequently

$$\int N_{\delta_1}(N_{\delta_2}\circ T_R^h)\,d\nu\le C_{\mathrm{ag}}\delta_1\delta_2,\qquad C_{\mathrm{ag}}=H_*/(4\pi R_-). \tag{22.6}$$

Proof. Use the exact one-mark flux, followed by the actual cross-flight coarea identity of Theorem 19.4. Global one-copy multiplicity is supplied by Theorem 19.3, not by local invertibility. For an arbitrary Borel set $I_1\subset(0,\delta_*)$ and I_2 in the same interval, the resulting joint clearance measure is both

$$\int_{I_1}\sum_{\sigma}\int_0^{2\pi} N_{I_2}(T_R^hX_R(\vartheta,\sigma,u_1))\,d\vartheta\,du_1\quad\text{and}\quad\int_{I_1\times I_2}\sum_{a\in\mathcal{A}_{h,R}} B_h(x_a)^{-1}\,du_1du_2.$$

The common collision prefactor $1/(4\pi R)$ is omitted on *both* sides of this equality. Theorem 22.3 bounds this measure by $H_*du_1du_2$, first on rectangles and then on all Borel sets. Uniqueness of its density proves (22.5). Reintroducing $1/(4\pi R)$ and integrating over the two clearance intervals proves (22.6). For each fixed radius the countable separations admit a common clearance-null exceptional set. A null set common to an uncountable radius family is neither needed nor asserted. □

22.1. **The inherited complete-record conclusion under the curve input.** The conclusion below was already obtained in Theorem 20.3 with the physical pressure input. The new contribution is its proof from the directly lifted angular source and curve growth, and the stronger fixed-first-clearance kernel bound, not a second claim of novelty for the integer identity.

Corollary 22.5 (Complete chronological multiplicity with the revised dependency)**.** *Put $C_A=\max\{2C_{\mathrm{pair}},C_{\mathrm{ag}}\}$, where C_{pair} is the inherited same-flight constant. For any finite deterministic sequence $0\le\delta_j\le\delta_*$, put $\mathscr{N}=\sum_{j=0}^{m-1}N_{\delta_j}(T_R^jx)$ and $S=\sum_j\delta_j$. Under Input 22.1,*

$$\int\mathscr{N}\,d\nu=S/R,\qquad\int\mathscr{N}(\mathscr{N}-1)\,d\nu\le C_AS^2. \tag{22.7}$$

The first equality requires no curve-growth assumption. For the original exact event Π and any Borel $0\le\psi\le1$,

$$0\le\int\mathbf{1}_\Pi\psi(L_{m,R})[\mathscr{N}-\mathbf{1}_{\{\mathscr{N}\ge1\}}]d\nu_R^*\le\frac{C_AS^2}{2c_*}. \tag{22.8}$$

In particular, for $\delta_j=2\varepsilon r^{\min(j,m-1-j)}$, $r=q_0^{1/4}$ and $2\varepsilon\le\delta_*$, this is at most $8C_A\varepsilon^2/[c_*(1-r)^2]$ for every m.

Proof. Invariance of ν and the exact one-mark mean give the first identity. Expand the factorial exactly as

$$\mathscr{N}(\mathscr{N}-1)=\sum_j N_j(N_j-1)+2\sum_{j<k}N_jN_k.$$

The same-flight theorem bounds the first sum by $2C_{\mathrm{pair}}\sum_j\delta_j^2$ after integration. Invariance and (22.6) bound the second by $2C_{\mathrm{ag}}\sum_{j<k}\delta_j\delta_k$. Their sum is at most C_AS^2. No independence was used. The deterministic inequality $0\le n-\mathbf{1}_{\{n\ge1\}}\le n(n-1)/2$ for integers $n\ge0$ and $\Pi\subset Y_R^*$

give (22.8); only here is the source dominated by ν/c_*. There are at most two indices at each depth, so $S \le 4\varepsilon/(1-r)$, proving the last bound. Repeated sightings of a disk on distinct flights are included in the cross terms. □

22.2. **The original roof obstruction has not been replaced.** The new norm in (21.8) describes a singular one-dimensional angular source in collision space. The density in (22.4) is in *second clearance*. It is not a density in the original roof, and the complete root bound in (22.5) has no hidden m^{-2} factor. The unconditioned first-order mass S/R remains positive. An exact-label selection can change its leading coefficient; it is not replaced by a probability conditioned on a mark.

To reach the raw target, the original ordered shortage limit (15.8) must still be proved. In particular the first-order marked source must be estimated after the complete original roof pushforward and physical word sum, including fold, critical, higher-degeneracy, one-sided and pairing-boundary terms, with the required m^2 normalization. The single-mark fixed-clearance roof bridge in (E.3) remains only a sufficient unproved condition; it is not a premise added to our main problem. The local fold calculations show why it is safer to integrate clearance before taking a roof supremum. Low-incidence shallow source and nongrazing capacity likewise remain in the original shortage. At fixed outer B, multiplying (22.8) by m^2 does not prove decay. No collision-dependent roof window, altered section, exceptional-set subtraction or changed order of limits is introduced here.

23. A grazing-uniform curvature graph transform

The preceding numbered proof bodies are retained verbatim. We now discharge one of their explicitly exposed premises: admissibility of the actual angular source for homogeneous curve growth. This is not a deduction of roofwise pressure from a curvature bound. We prove curvature invariance here, one-step weighted complexity and distortion in the next section, and only then apply a precisely identified curve-sum lemma. No event or collision is changed by these constructions.

Write $t_* = \tau_- = 3/50$, $k_- = 100/47$, $k_+ = 20/9$, and $U = k_+ + t_*^{-1} = 170/9$. At a fixed radius write $\kappa = 1/R$. Our curves are graphs $\varphi = \varphi(s)$ in actual boundary arclength, with $V = d\varphi/ds \in [\kappa, U]$. Their signed geometric curvature is

$$K = \frac{dV/ds}{(1+V^2)^{3/2}}.$$

Neither K nor dV/ds is an optical wavefront curvature. The notation in this section is local to one actual regular collision.

Lemma 23.1 (Exact circular graph transform)**.** *Let $c = \cos\varphi$, $p = \sin\varphi$, $d = \cos\varphi_1$, $p_1 = \sin\varphi_1$ at the two actual collisions, and let τ be their physical flight time. Put*

$$w = \kappa + V, \qquad A = c + \tau w, \qquad y = dw/A. \tag{23.1}$$

On a regular branch, $c, d > 0$, and the image slope and signed derivatives are

$$s_{1,s} = -A/d, \qquad V_1 = \kappa + y, \quad \tau_s = -p - p_1 A/d, \tag{23.2}$$

$$V_{1,s_1} = -\frac{p_1 w}{A}(\kappa + 2y) - \frac{p d^2 w(\kappa + 2V)}{A^3} - \frac{c d^2}{A^3} V_s. \tag{23.3}$$

In particular, $\kappa \le V_1 \le U$.

Proof. Let n, t and n_1, t_1 be the outward normal and its counterclockwise rotation at the two contacts. The outgoing velocity is $cn + pt$, and at the next disk its incoming representation is $-dn_1 + p_1 t_1$. The outgoing direction derivative is $\vartheta_s = \kappa + V = w$. Differentiate $q_1 = q + \tau v$

and project perpendicular to v. The initial projection is c, the terminal one is $-ds_{1,s}$, so $-ds_{1,s} = c+\tau w$. Projection along v gives $\tau_s = p_1 s_{1,s} - p$. Specular reflection gives $d\varphi_1 = \kappa\, ds_1 - d\vartheta$, whence $V_1 = \kappa - w/s_{1,s}$. This proves (23.2).

For the second derivative use

$$c_s = -pV, \quad d_s = p_1 V_1 A/d, \quad A_s = -p(\kappa + 2V) - p_1 A w/d + \tau V_s.$$

Differentiate $V_1 = \kappa + dw/A$ and divide by $s_{1,s} = -A/d$. The coefficient of V_s is $-d^2/A^2 + d^2\tau w/A^3 = -cd^2/A^3$; the remaining terms give (23.3). Finally $0 < y \le 1/\tau \le 1/t_*$, so the image stays in the stated cone. Differentiation is only on regular branches; no value of DT at a tangency is asserted. □

Theorem 23.2 (Uniform contraction of geometric curvature)**.** *For every radius in the original interval and every regular actual flight,*

$$|K_1| \le q_*|K| + C_*, \qquad q_* := \frac{4\big(U + (2U)^{-1}\big)^3}{27k_-(k_- + U)^2\{1 + t_*(k_- + U)\}} < \frac{1}{2}, \quad C_* := \frac{58}{27 t_* k_-^2}. \tag{23.4}$$

These constants are independent of both positive incidences and of the flight count. In exact rational form,

$$q_* = \frac{946236091513673179721}{2005319667933288000000}, \qquad C_* = \frac{64061}{8100}.$$

Consequently any

$$B_{\mathrm{cv}} \ge \max\{K_0, 2C_*\} \tag{23.5}$$

defines a forward invariant class of homogeneous C^2 unstable graphs of geometric curvature at most B_{cv}. It contains the actual sources of Theorem 21.3. Along every regular branch of length l,

$$|K_l| \le q_*^l |K_0^{\mathrm{curve}}| + C_* \frac{1 - q_*^l}{1 - q_*}. \tag{23.6}$$

This assertion includes arbitrarily close approaches to grazing but not a smooth extension through a singular collision.

Proof. After division of (23.3) by $(1 + V_1^2)^{3/2}$, the absolute coefficient of K is

$$\begin{aligned} Q &= \frac{cd^2(1 + V^2)^{3/2}}{\{A^2 + (\kappa A + dw)^2\}^{3/2}} \\ &= \frac{c}{A}\frac{(1 + V^2)^{3/2}}{w^2}\frac{y^2}{\{1 + (\kappa + y)^2\}^{3/2}}. \end{aligned} \tag{23.7}$$

Here $c/A \le [1 + t_* w]^{-1}$ and

$$\frac{y^2}{\{1 + (\kappa + y)^2\}^{3/2}} \le \frac{y^2}{(\kappa + y)^3} \le \frac{4}{27\kappa}.$$

The last inequality follows by differentiating $y^2/(\kappa + y)^3$; its maximum for $y \ge 0$ is at $y = 2\kappa$. For fixed κ, the function

$$h(V) = \frac{(1 + V^2)^{3/2}}{(\kappa + V)^2\{1 + t_*(\kappa + V)\}}$$

is increasing for $V \ge \kappa > 1$. Indeed $3V/(1+V^2) > 3/(\kappa+V)$ because $\kappa V > 1$, and the remaining lower bound for its logarithmic derivative is $1/(\kappa + V) - t_*/[1 + t_*(\kappa + V)] > 0$. For fixed V it decreases with κ. Thus $Q \le 4h(U)|_{\kappa = k_-}/(27k_-)$. The elementary bound $\sqrt{1 + U^2} \le U + (2U)^{-1}$ gives $Q \le q_*$. The displayed rational evaluation verifies $q_* < 1/2$ without rounding.

For the first forcing term in (23.3), use $w/A \le 1/t_*$ and $\kappa + 2y \le 2(\kappa + y)$ to bound its normalized magnitude by $2/(t_* k_-^2)$. For the second, write its normalized magnitude as

$$\frac{\kappa + 2V}{Aw} \frac{y^2}{\{1 + (\kappa + y)^2\}^{3/2}} \le \frac{2}{t_* w} \frac{4}{27\kappa} \le \frac{4}{27 t_* k_-^2}.$$

Their sum is C_*. This proves (23.4). Induction gives (23.6), and $q_* B_{\mathrm{cv}} + C_* \le B_{\mathrm{cv}}$ follows from (23.5). A homogeneous cut or length subdivision restricts a smooth curve and does not increase its curvature. The exact source curvature was bounded by K_0 in Theorem 21.1. All denominators used here are positive on the regular domain; the uniform estimates themselves require no incidence cutoff. □

Remark 23.3 (What the norm conversion does and does not do). The graph second derivative on this invariant class is at most $B_2 = B_{\mathrm{cv}}(1 + U^2)^{3/2}$. This is a consequence of geometric curvature invariance, not a claim that the same bound on φ'' is itself an invariant rectangular constraint. The class in (H2) of [6] is formulated with geometric curvature; equivalence of these two boundedness notions is used only for derivative estimates. The optical Jacobi curvature and the long-record roof Hessian are different quantities and are not controlled by (23.4).

24. Physical one-step complexity and homogeneous distortion

We verify the remaining curve-growth hypotheses for precisely the class constructed above. The argument concerns actual one-flight branches and all their homogeneous pieces. It is not an estimate obtained by counting long collision words. Constants may depend on the fixed original geometric family and the chosen finite curvature allowance B_{cv}, but not on R, incidences, clearance marks, collision separation or exact-event selectors.

24.1. **Uniform finite geometry and a one-sided first-hit lemma.** The retained τ_+ can be defined intrinsically as the supremum of lengths of straight free segments in the original family. For clarity, its finiteness also follows directly from the fixed lattice. A complete line of irrational direction is dense on $\mathbb{R}^2/\Lambda$ and meets the open radius-R_- disk. A line of rational direction is parallel to a primitive lattice vector z, with neighboring lattice rows at distance $(\sqrt{3}/2)/|z| \le \sqrt{3}/2$. Its distance from one such row is at most $\sqrt{3}/4 < R_-$. It therefore meets open disks on that row. The same argument applies to half-lines. The direction dichotomy follows, for example, by describing the closure of a line subgroup of a two-dimensional torus: a proper connected subgroup has an integral annihilator and is a rational circle; otherwise the closure is the full torus. The positive semiorbit has the same closure by recurrence of translation on this compact group. These elementary observations use the Euclidean triangular lattice, not a linear deformation of the physical disks.

If free segments of unbounded length existed, translate their midpoints into a compact fundamental cell and take a convergent subsequence of midpoints and directions. Every finite subsegment of the limiting complete line would avoid every open radius-R_- disk, a contradiction. Thus $\tau_+ < \infty$, uniformly also for $R \ge R_-$. This compactness proof does not assign a sampled numerical value to τ_+. Put

$$D_f = \tau_+ + 2R_+, \qquad N_f = \#\{z \in \Lambda : 0 < |z| \le D_f\}, \qquad J_f = 2N_f + 1. \tag{24.1}$$

The packing bound $N_f + 1 \le 4(D_f + 1/2)^2$ is fully geometric.

Choose a preliminary curve length small enough that the relative angle $\eta_z = \arg(z - q) - \vartheta$ varies by at most $\pi/2$ for every candidate. This is possible uniformly because $|z - q| \ge 1 - R_+$ and $\vartheta_s = \kappa + V \le k_+ + U$. On every such graph,

$$\eta_z' \le (1 - R_+)^{-1} - 2k_- < 0, \qquad b_z' = -c - t_z(\kappa + V), \quad t_z = (z - q) \cdot v, \quad b_z = (z - q) \cdot v^{\perp}.$$

The positive-foot set is an interval, and b_z is strictly decreasing there. Every disk-intersection set is consequently an interval, with at most two tangency endpoints. The initial point is exterior to every other disk, so $t_z > 0$ and $|b_z| < R$ are exactly the conditions for a positive intersection. Distinct first-entry times cannot exchange their order without a common point of two disjoint disks. Cutting at these at most $2N_f$ tangencies gives at most J_f genuine first-hit branches. Coordinate seams are glued on the collision circle, not counted as new physical branches. In a seam-free local lift the image is a graph and φ_1 is strictly monotone by Lemma 23.1. Each first-hit branch thus meets each target homogeneity strip at most once.

Lemma 24.1 (At most one non-grazing limiting branch)**.** *Fix $k_0 \geq 2$, with the strips of Theorem 21.3. There is a common $\ell_0(k_0) > 0$ such that every allowed curve of length at most $\ell_0(k_0)$ has at most one connected physical image piece in H_0. This assertion counts maximal physical homogeneous pieces before the length subdivision used in the growth partition.*

Proof. We give the local argument also at a limiting initial grazing state. Fix a limiting radius and outgoing state in the compact closure of the collision cylinder. Consider its oriented straight ray and the first disk which it intersects strictly. Such a disk exists by the preceding finite geometry, even if the ray is tangent to one or more earlier disks. The strict first-entry time and center persist continuously nearby. The finitely many earlier possible contacts in the limit are tangencies. Any nearby first-hit branch on one of these earlier disks has terminal incidence tending to zero, hence cannot have its image in H_0 in a sufficiently small neighborhood. A later disk cannot be the next collision when the persistent strict disk is met.

It remains to check connectivity for the persistent strict disk, rather than infer it just from uniqueness of a limiting center. At an earlier tangency $b_z = \pm R$ the positive foot satisfies $t_z \geq \sqrt{(1-R_+)^2 - R_+^2} > 0$. Thus b_z' is uniformly negative there on all allowed graphs. In a sufficiently small neighborhood this candidate can cross only one of its two tangency levels: the levels are separated by $2R \geq 2R_-$ and $|b_z'|$ has a uniform finite upper bound on the finite candidate set. Its intersection condition is therefore one-sided along any short source graph. Avoiding all earlier disks is an intersection of one-sided intervals, hence an interval. The strict-disk branch on that interval has monotone φ_1, so restricting it to H_0 is still an interval. Incipient tangencies after the strict contact have no effect on which collision is first.

These neighborhoods are open in radius and the compact closed collision cylinder. A finite subcover, and a smaller common curve length, give $\ell_0(k_0)$; the bounded cone slopes keep each sufficiently short graph inside a member of this cover. Equivalently a sequence of shrinking counterexample curves would converge to one of the limiting states just analyzed. Only this limiting line is allowed to pass through tangency points for purposes of classification. Every branch being estimated is an actual regular billiard flight with its actual first collision. No tangency is erased from a physical record. □

24.2. **The complete one-step weighted sum.** For a tangent vector to a positive or negative cone graph define

$$|v|_* = \frac{\kappa + |V|}{\sqrt{1+V^2}}|v|.$$

On these cones $|v| \leq |v|_* \leq \Gamma|v|$, where $\Gamma = \sqrt{1+k_+^2}$. This follows respectively from $\kappa > 1$ and Cauchy–Schwarz. Extend this cone norm to an equivalent norm off the cones if needed; no subsequent estimate uses such an extension. For a positive cone vector the exact expansion is

$$\frac{|DT_R v|_*}{|v|_*} = 1 + \frac{2\kappa A}{dw} \geq 1 + \frac{2k_- t_*}{d} \geq \Lambda_* := 1 + 2k_- t_* = \frac{59}{47}. \tag{24.2}$$

Time reversal gives the corresponding backward expansion on negative cones.

Theorem 24.2 (Uniform one-step expansion and weakened exponent)**.** *Choose k_0 so large that*

$$T_0 := \frac{J_f}{k_- t_*(k_0-1)} < \frac{1-\Lambda_*^{-1}}{2}. \tag{24.3}$$

For $\ell_* > 0$ sufficiently small as in Lemma 24.1, let $\mathcal{H}(C)$ be the maximal regular homogeneous pieces of $T_R C$. Then

$$\sum_{V\in\mathcal{H}(C)} \sup_V J_V^*(T_R^{-1}) \le \theta_* := \Lambda_*^{-1} + T_0 < 1. \tag{24.4}$$

For every $\zeta > 1/2$ there is a finite uniform C_ζ such that

$$\sum_{V\in\mathcal{H}(C)} (\sup_V J_V(T_R^{-1}))^\zeta \le C_\zeta. \tag{24.5}$$

Here J^* and J denote curve Jacobians in the adapted and Euclidean metrics respectively; these are supremum bounds, not just average bounds.

Proof. The possible H_0 piece contributes at most Λ_*^{-1}. On $H_k^\pm$, $d \le k^{-2}$, so the inverse of (24.2) is at most $(2k_- t_* k^2)^{-1}$. There are at most J_f pieces in each such strip. Sum both signs and use $\sum_{k\ge k_0} k^{-2} \le 1/(k_0-1)$ to prove (24.4). This sum includes every first-hit branch. The norm comparison gives $J \le \Gamma J^*$ at the two endpoints, not a new factor for each subsequent collision. Thus one may take

$$C_\zeta = (\Gamma/\Lambda_*)^\zeta + 2J_f \left(\frac{\Gamma}{2k_- t_*}\right)^\zeta \sum_{k\ge k_0} k^{-2\zeta} < \infty.$$

These are precisely the needed expansion/complexity properties, with weakened-exponent threshold $\zeta_0 = 1/2 < 1$. No claim of the smaller infinite-horizon threshold appearing elsewhere in the literature is used. □

24.3. **A direct distortion estimate in the target coordinate.**

Lemma 24.3 (One-step logarithmic distortion)**.** *Let $J_C T_R$ be the Euclidean curve Jacobian. On a smooth image graph,*

$$\begin{aligned}\partial_{s_1} \log J_C T_R = \frac{p_1 w}{A} + \frac{dp(\kappa+2V)}{A^2} - \frac{d\tau V_s}{A^2} + \frac{p_1 V_1}{d} \\ + \frac{V_1 V_{1,s_1}}{1+V_1^2} + \frac{V V_s d}{A(1+V^2)}.\end{aligned} \tag{24.6}$$

There is a finite uniform D_B such that its magnitude is at most $U/d + D_B$. For example, with $a_0 = 2k_- t_*$ and $B_2 = B_{\mathrm{cv}}(1+U^2)^{3/2}$, one may use

$$D_B = t_*^{-1} + 2/(t_* a_0) + (B_2/a_0)(1 + (2k_-)^{-1}) + B_2. \tag{24.7}$$

Proof. The Jacobian is $J_C T_R = (A/d)\sqrt{1+V_1^2}/\sqrt{1+V^2}$. Differentiate its logarithm in s and divide by $-A/d$. Use the exact A_s, d_s of Lemma 23.1; their apparent $1/d$ singularity leaves exactly $p_1 V_1/d$ and the finite terms in (24.6). This proves the identity, including its signs. For the bound use $A \ge t_* w \ge a_0$, $w/A \le 1/t_*$, $\kappa + 2V \le 2w$, $\tau/A \le 1/(2k_-)$ and $|V_s|, |V_{1,s_1}| \le B_2$. The six terms are respectively bounded by t_*^{-1}, $2/(t_* a_0)$, $B_2/(2k_- a_0)$, U/d, B_2 and B_2/a_0. There is no inverse initial incidence in this bound. Since $ds_1/d\ell_1 \le 1$ in absolute value, it also bounds the logarithmic derivative per unit image arclength. □

Theorem 24.4 (Uniform homogeneous distortion at every collision count)**.** *There is $D < \infty$ such that, if x, x' belong to a single regular curve segment whose images through time l are homogeneous, then*

$$|\log J_C T_R^l(x) - \log J_C T_R^l(x')| \le D\, d_{T_R^l C}(T_R^l x, T_R^l x')^{1/3}. \tag{24.8}$$

Distances are along the same physical curve branch. The constant is uniform in l and R. The corresponding Jacobian-ratio distortion bound and the stable inverse version hold on short homogeneous curves.

Proof. For two points of an image graph in $H_k^{\pm}$, let r be their arclength separation. Since $V_1 \ge k_-$,

$$r \le \frac{\sqrt{1+U^2}}{k_-}(k^{-2} - (k+1)^{-2}) \le L_h k^{-3}, \qquad L_h = 3\sqrt{1+U^2}/k_-.$$

Also $d \ge (2/\pi)(k+1)^{-2}$ throughout that segment. The logarithmic derivative bound is therefore at most Ck^2, with a constant independent of k. Hence its integral is bounded by $Ck^2 r \le C L_h^{2/3} r^{1/3}$. On H_0, $d \ge \sin(k_0^{-2})$ and the same conclusion follows from the Lipschitz bound, after fixing the maximal short-curve length. This proves one-step logarithmic distortion in the image distance, with a common constant D_1.

Let $r_l = d_{T_R^l C}(T_R^l x, T_R^l x')$. The adapted expansion and its endpoint metric comparison imply $r_j \le \Gamma \Lambda_*^{-(l-j)} r_l$. Expand the logarithm of the l-step Jacobian as the sum of its actual one-step logarithms and apply the preceding estimate at image times $j+1$. The geometric series gives

$$D = D_1 \Gamma^{1/3} \sum_{r=0}^{\infty} \Lambda_*^{-r/3} = \frac{D_1 \Gamma^{1/3}}{1 - \Lambda_*^{-1/3}}.$$

For bounded r_l, the inequality $|e^z - 1| \le e^{|z|}|z|$ gives the ratio form. Time reversal $\mathcal{I}(s,\varphi) = (s, -\varphi)$ is an arclength isometry and conjugates T_R to T_R^{-1}, yielding the stable version. The smooth collision measure is $c\, ds\, d\varphi/(4\pi R)$ and is invariant, so its measure Jacobian is identically one. The measure-Jacobian portion of the distortion hypothesis therefore has zero distortion. This use of time reversal changes neither the original section nor its occupation convention. □

25. Admissible physical curve growth and the clearance consequence

We now separate the mathematical application of a published combinatorial lemma from the inherited global spectral and roof-pressure inputs. The application below is a theorem of this revision, not a claim that a successful build has certified the billiard dynamics.

Theorem 25.1 (The required angular-source growth input is satisfied)**.** *For the original triangular circular Lorentz family, choose the per-radius unstable graph class*

$$\mathcal{W}_R^u(B_{\mathrm{cv}}) = \{C :\ C \text{ is a homogeneous } C^2 \text{ graph},\ \kappa_R \le d\varphi/ds \le U,\ |K_C| \le B_{\mathrm{cv}}\},$$

where B_{cv} satisfies (23.5)*. There exist common $k_0, \ell_* > 0$ and $C_g < \infty$ such that, for every R, every $C \in \mathcal{W}_R^u(B_{\mathrm{cv}})$ with $|C| \le \ell_*$ and every integer $l \ge 0$,*

$$\sum_{V \in \mathcal{G}_l^+(C)} \sup_V J_V(T_R^{-l}) \le C_g. \tag{25.1}$$

The partition is obtained by the actual regular homogeneous image branches and a fixed short-length subdivision. It covers arclength-almost all of C before pushforward. It contains no added reflection at a clearance mark. All hypotheses of Input 22.1, for this chosen class, are therefore satisfied.

Proof. Fix k_0 by (24.3); choose ℓ_* no larger than the length from Lemma 24.1, the short-curve length needed for its preliminary relative-angle argument, and the length in (22.2). All choices are made once for the radius family, before the collision count or any outer limit.

Theorem 23.2 verifies invariance of the homogeneous C^2 curve class in the literal geometric-curvature sense of (H2) in [6]. It contains the original first-image source graphs; this membership is now proved, not inferred from their being C^2. Equation (24.2) gives the common adapted hyperbolicity and a uniform endpoint metric comparison. Theorem 24.2 verifies (H3.A) and (H3.B), with the explicit permissible threshold $\zeta_0 = 1/2$. Its estimate uses suprema of contraction, so it is at least as strong as the one-step hypothesis used there. Theorem 24.4 gives the arclength and invariant-measure parts of (H4) on homogeneous branches. The logarithmic and ratio forms are converted on the one fixed short length. None of these verifications uses the physical-word pressure input.

For completeness, singular-null coverage is checked on these curves. At each forward step there are only finitely many physical tangency cuts on a short graph, by the positive-foot monotonicity above. Its regular image has strictly monotone φ_1, so homogeneous boundaries have at most one preimage per first-hit branch per strip. The additional length cuts are countable. Starting with a countable collection, iteration thus excludes a countable collection of points for each finite l. Each regular restriction is injective. No positive arclength part of a graph lies in a tangency or homogeneous-boundary preimage. This proves the asserted null coverage without treating a bulk-null set as a zero endpoint trace in any pairing construction.

We use precisely [6, Lemma 3.2(b), p. 14]: for the stable class and the homogeneous inverse partition $\mathcal{G}_l(W)$, with long pieces subdivided at the fixed short scale, the sum of sup arclength contraction Jacobians is at most a constant depending on the common curve, expansion and distortion constants, uniformly in W, T and l. The published proof specifies (H2)–(H4) as its requirements; it is not a theorem about all cylinder suprema or roof Fourier transforms. Apply this statement to $\mathcal{W}^s_R(B_{\mathrm{cv}}) = \mathcal{I}\mathcal{W}^u_R(B_{\mathrm{cv}})$ and reverse it by $\mathcal{I}T_R\mathcal{I} = T_R^{-1}$. Since $\mathcal{I}$ is an arclength isometry, its Jacobian conversion is exact and gives (25.1). At $l = 0$ the sum is one. One may apply the lemma separately to each R: its constant depends only on the common estimates just proved. A common Banach-space identification across radii, or a perturbation theorem for variable boundary lengths, is not required for this curve-sum conclusion. Further fixed shortening of the scale is included in C_g. □

Corollary 25.2 (Complete physical two-clearance bound with the input discharged)**.** *Retain the genuine one-copy cross-flight coarea and augmented-itinerary uniqueness theorems of Section 19, with their stated physical scopes. In the original Lorentz family the bounds* (22.4)*,* (22.5)*,* (22.6)*,* (22.7) *and* (22.8) *now follow without an unverified homogeneous curve-growth premise and without physical-word pressure. In particular, for every fixed* $0 < u_1 \le \delta_*$ *and every* $h \ge 1$*,*

$$\sum_{\sigma=\pm1} \int_0^{2\pi} N_I(T_R^h X_R(\vartheta, \sigma, u_1))\, d\vartheta \le A_* C_g C_t\, |I|. \tag{25.2}$$

The first-image source is taken after exactly one actual collision, and there are $h - 1$ *further collisions before applying the second mark. All exact-record and original-roof Borel selectors between zero and one may be kept inside this positive inequality. The full one-copy root sum is bounded in clearance essential supremum uniformly in* R *and* h*.*

Proof. Theorem 25.1 supplies exactly the class, norm, partition, null-set and uniformity requirements of Input 22.1. Insert it in the preserved proofs of Theorems 22.3 and 22.4 and Corollary 22.5. The angular source has mass 4π; the collision flux factor is $1/(4\pi R)$, and conditioning on the original section contributes exactly $1/c_*$. No further angular sign factor is inserted into $A_* C_g C_t$. The factorial conclusions and their constants are inherited consequences; the new assertion is the

discharge of their curve-growth dependency. A null exceptional set can be chosen for all integer h at a fixed radius. No null set common to uncountably many radii is claimed. □

The distinction from the original roofwise theorem. Theorem 25.1 is a complete curve-sum conclusion and Corollary 25.2 sums all regular physical two-mark branches, including arbitrarily large separation and arbitrarily small positive incidences. They remove a genuine hypothesis from that part of the proof chain. They do not turn a clearance density into a roof density. The count mean S/R is still first order, and the original-clock pushforward can have folds, critical and one-sided boundary contributions. In particular no factor m^{-2} occurs in (25.2).

The original target remains, for every fixed central parameter M,

$$\lim_{B\to\infty}\limsup_{m\to\infty}\sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} \left(\mathsf{U}^{\mathrm{low}}_{B,m}(t) - K_B P^{\mathrm{ng}}_{r_B,m}(t)\right)_+ = 0. \tag{25.3}$$

Here B is the inherited outer parameter. The curvature allowance B_{cv} is fixed permanently by (23.5). All original outer widths are fixed before the collision limsup. Their prescribed $B\to\infty$ limit is not used to tune the curve partition. No result of this revision establishes (25.3), the full raw pointwise LLT, or its final four-coordinate inversion. The remaining minimal obligations along this route are the normalized first-order physical-roof source and complete chart/boundary sum, the incidence-only low-load source or valid comparison capacity, and the stated original spectral, arithmetic, pairing and normalization interfaces. The new one-flight complexity and curve-sum proof does not certify any stronger whole-cylinder or roofwise pressure statement still used elsewhere.

26. Physical momentum fibres and a complete endpoint kernel

The preceding source, curvature and growth theorems are retained, with no change to their proof bodies. We use them for a different source: a fixed initial physical tangential momentum. This is a coordinate disintegration, not a new conditioning event. In particular we neither replace ν_R^* nor remove any collision or section test. Throughout this section $R\in[R_-,R_+]$, $m\ge 1$, and $a\in(-1,1)$. The source fibre is

$$F_a=\{(\alpha_0,p_0):0\le\alpha_0<2\pi,\ p_0=a\},\qquad d\lambda_a=d\alpha_0.$$

Its total mass is 2π, not one. We keep $t_*=3/50$, $k_-=100/47$, $k_+=20/9$, and $U=170/9$ from Section 23. A fixed enlargement of the admissible geometric-curvature allowance will be explicitly paid below.

Lemma 26.1 (First true collision of a momentum fibre)**.** *Put $c=\sqrt{1-a^2}$, $d=\cos\varphi_1$, $p_1=\sin\varphi_1$, $\kappa=1/R$, $A=c+\tau\kappa$, and $y=d\kappa/A$. On every actual regular first-hit branch of F_a,*

$$\frac{ds_1}{ds_0}=-\frac{A}{d},\qquad V_1:=\frac{d\varphi_1}{ds_1}=\kappa+y,\qquad \kappa\le V_1\le U, \tag{26.1}$$

$$\frac{dV_1}{ds_1}=-\frac{p_1\kappa}{A}(\kappa+2y)-\frac{ad^2\kappa^2}{A^3}, \tag{26.2}$$

$$\rho_1:=\frac{d\lambda_a}{d\ell_1}=\frac{d}{RA\sqrt{1+V_1^2}}\le\frac{d}{t_*}. \tag{26.3}$$

The geometric curvature of this image is bounded by

$$K_{\mathrm{mom}}:=\frac{k_++2/t_*}{t_*}+\frac{1}{t_*^3k_-}, \tag{26.4}$$

uniformly in both positive endpoint incidences. This bound is deliberately coarse; it has no collision-count or microscopic-window dependence.

Proof. The differentiation proving Lemma 23.1 is valid also for an initial graph of slope zero: its identities precede any use of the unstable cone. Substitute $V = 0$ and $V_s = 0$, not $V = \kappa$, in those identities. This gives (26.1)–(26.2). Alternatively the two projections of $q_1 = q_0 + \tau v$ give $-ds_{1,s_0} = c + \tau\kappa$ and $\tau_{s_0} = -a - p_1 A/d$; differentiating the reflection formula gives the same result. Since $A \geq t_*\kappa$, one has $y \leq 1/t_*$ and $\kappa/A \leq 1/t_*$. Taking absolute values in (26.2), and then dividing by $(1 + V_1^2)^{3/2} \geq 1$, proves (26.4). Finally $d\ell_1 = \sqrt{1 + V_1^2}|ds_1|$, $ds_0 = R\, d\alpha_0$, and $RA = Rc + \tau \geq t_*$, proving (26.3). All derivatives are on the actual first-hit component. Nothing is continued through a physical grazing collision in this statement. □

Fix once and for all

$$\widehat{B}_{\mathrm{cv}} \geq \max\{B_{\mathrm{cv}}, K_{\mathrm{mom}}, 2C_*\}. \tag{26.5}$$

Theorem 23.2 and the physical complexity and distortion proofs apply to this fixed allowance. Let $\widehat{k}_0, \widehat{\ell}_*, \widehat{C}_g$ be the resulting constants of Theorem 25.1. The growth constant is allowed to change; we do not reuse a constant for a smaller, unverified source class. We may shorten $\widehat{\ell}_*$ once. None of these choices depends on a, m, the exact record, or the outer parameter B.

Lemma 26.2 (A complete, grazing-summable first-image family)**.** *For every a the first physical image of F_a, with the source labels retained, decomposes into countably many homogeneous short curves C in $\mathcal{W}_R^u(\widehat{B}_{\mathrm{cv}})$, carrying their exact pushed weights ρ_C. It covers λ_a-almost every initial point and satisfies*

$$\sum_C \|\rho_C\|_\infty \leq A_{\mathrm{mom}} < \infty \tag{26.6}$$

with a constant uniform in R and a. One explicit choice in terms of the fixed geometric constants is

$$\ell_{\mathrm{pre}} = \frac{\pi}{4\{(1 - R_+)^{-1} + k_+\}}, \qquad J_{\mathrm{pre}} = 2 + \left\lceil \frac{2\pi R_+}{\ell_{\mathrm{pre}}} \right\rceil, \tag{26.7}$$

$$L_0 = \frac{\pi\sqrt{1 + U^2}}{k_-}, \qquad J_{\mathrm{len}} = 2 + \left\lceil \frac{2L_0}{\widehat{\ell}_*} \right\rceil, \tag{26.8}$$

$$A_{\mathrm{mom}} = \frac{J_{\mathrm{pre}} J_f J_{\mathrm{len}}}{t_*} \left(1 + \frac{2}{\widehat{k}_0 - 1}\right). \tag{26.9}$$

Here J_f and the intrinsic finite τ_+ have exactly the meanings of (24.1)*; no sampled value is assigned to them.*

Proof. Divide the initial boundary circle into at most J_{pre} arcs of s_0-length at most ℓ_{pre}. For every candidate center z the relative angle $\eta_z = \arg(z - q_0) - \vartheta$ varies by at most $\pi/4$ on such an arc, and

$$\eta_z' \leq (1 - R_+)^{-1} - k_- < 0, \qquad b_z' = -c - t_z\kappa.$$

The positive-foot set is an interval; on it the impact parameter is strictly decreasing. Thus each disk contributes at most two tangency cuts. Between these cuts the first-entry times of two disjoint disks cannot exchange order, since equality would give the same physical point on both boundaries. The same finite candidate argument as in Section 24 gives at most J_f first-hit branches on each preliminary arc. The infinite lattice is reduced here by the proved finite-horizon bound, not by truncating a computation.

On each branch $V_1 \geq k_- > 0$, so φ_1 is strictly monotone as a function of a local lift of s_1. Each target homogeneity strip is met at most once. Since $\varphi_1 \in (-\pi/2, \pi/2)$, every such connected image has length at most L_0. Subdivide it into lengths at most $\widehat{\ell}_*$, using no more than J_{len}

pieces, with the fixed short-length convention of the growth partition. Circular coordinate seams are glued, or split once within this conservative count, not counted as different trajectories.

In the central strip (26.3) gives $\rho_C \le t_*^{-1}$; in each of the two strips of index $k \ge \widehat{k}_0$ it gives $\rho_C \le (t_* k^2)^{-1}$. Sum all physical branches, subdivisions and both strip signs, using $\sum_{k\ge\widehat{k}_0} k^{-2} \le (\widehat{k}_0 - 1)^{-1}$. This proves (26.9). The density bound, rather than a finite number of grazing strips, is essential. Physical tangencies are finite on each preliminary arc, and strip and length boundaries are countable. Their preimages have zero $d\alpha_0$ measure. The first hit is unique everywhere else. Coincident images, if any after a coordinate identification, retain their different source labels and are summed. □

Lemma 26.3 (Exact one-collision lattice kernel and physical roof)**.** *For $a, v \in (-1, 1)$ and a candidate lattice displacement $z \ne 0$, put $r_z = |z|$, $\gamma_z = \arg z$ and*

$$Q_z(a,v) = \sqrt{r_z^2 - R^2(v-a)^2}, \tag{26.10}$$

$$\alpha_z(a,v) = \gamma_z - \arcsin\Big(\frac{R(v-a)}{r_z}\Big) - \arcsin a \pmod{2\pi},$$

$$T_z(a,v) = Q_z(a,v) - R\sqrt{1-a^2} - R\sqrt{1-v^2}. \tag{26.11}$$

Let $\mathcal{A}_z(a,v)$ be the exact indicator that the regular physical next collision from $(\alpha_z(a,v), a)$ is with center z. For a bounded Borel source selector h, the one-collision endpoint kernel is, almost everywhere in v,

$$D^h_{1,R}(a,v) = \sum_{\substack{z\in\Lambda \\ 0<|z|\le D_f}} h(\alpha_z(a,v), a)\mathcal{A}_z(a,v)\frac{R}{Q_z(a,v)}. \tag{26.12}$$

The unweighted expression is also the exact sum of regular inverse root derivatives at each a, v. On an admitted term, T_z is the original *flight time and z its actual lattice displacement. In particular*

$$D_{1,R}(a,v) \le C_1 := \frac{47N_f}{2\sqrt{291}}. \tag{26.13}$$

This is not a Markov transition rule for the momentum process.

Proof. Let $\vartheta = \alpha_0 + \arcsin a$ be the outgoing direction. The signed impact on the next disk is $b_z = (z - q_0)\cdot v_\vartheta^\perp = z\cdot v_\vartheta^\perp + Ra = Rp_1$. Thus $z\cdot v_\vartheta^\perp = R(v-a)$. The root with $z\cdot v_\vartheta > 0$ is exactly the displayed α_z, and $z\cdot v_\vartheta = Q_z$ there. The other angular root cannot be a physical forward next collision: on a physical flight this projection is $\tau + R\sqrt{1-a^2} + R\sqrt{1-v^2} > 0$. There is no vanishing angular discriminant, since $r_z \ge 1$ and $2R \le 94/100 < 1$.

At fixed a, differentiation gives $\partial_{\alpha_0} p_1 = -z\cdot v_\vartheta/R = -Q_z/R$. The projection identity just used proves (26.11). One-dimensional coarea now gives (26.12), with the actual next-hit mask retained. The mask excludes a disk hidden by an earlier collision, not the earlier collision from the physical record. Every actual first center lies in the proved candidate set, so the expression is complete. Finally $Q_z \ge \sqrt{1-4R_+^2} = \sqrt{291}/50$ and $R/Q_z \le 47/(2\sqrt{291})$. Sum at most N_f terms. The formula keeps the reconstructed initial angle and its exact physical visibility mask. Discarding that information and composing these kernels as if the momenta alone were a Markov chain is not part of this proof or of any subsequent argument. □

Theorem 26.4 (All-time endpoint-momentum kernel on the physical source)**.** *There exists a finite constant*

$$C_{\mathrm{mom}} = \max\left\{C_1, \frac{A_{\mathrm{mom}}\widehat{C}_g J_f}{k_- t_*}\right\} \tag{26.14}$$

such that, for every R, $m \geq 1$, $a \in (-1,1)$, Borel set $I \subset (-1,1)$, and Borel selector $0 \leq h \leq 1$ on the original physical source,

$$\int_{F_a^{\mathrm{reg}}(m)} h(\alpha_0, a)\mathbf{1}_{\{p_m \in I\}}\, d\alpha_0 \leq C_{\mathrm{mom}}|I|. \tag{26.15}$$

The selector may retain the complete original $\Pi_{m,n,k}$, both section tests, every collision, any original-roof Borel test and any bounded physical guard. It is not required to factor over time. For every fixed $a, v \in (-1,1)$ the complete regular root sum obeys

$$D_{m,R}(a,v) := \sum_{\substack{\alpha_0:\ p_m(\alpha_0,a)=v \\ (\alpha_0,a)\in F_a^{\mathrm{reg}}(m)}} \frac{1}{|\partial_{\alpha_0} p_m(\alpha_0,a)|} \leq C_{\mathrm{mom}}. \tag{26.16}$$

Roots are unique physical initial points, not multiple copies from homogeneity or coordinate seams. In particular the original section source has the joint endpoint-momentum domination

$$(p_0, p_m)_{\#}(h\, d\nu_R^*) \leq \frac{C_{\mathrm{mom}}}{4\pi c_*}\, da\, dv. \tag{26.17}$$

The statement concerns ν_R^* as originally defined; no conditional probability at $p_0 = a$ is introduced as a replacement model.

Proof. First expose the cancellation at the *last actual collision*. For a short incoming unstable graph of slope V and arclength ℓ, write $w = \kappa + V$, $A = c + \tau w$ as in Lemma 23.1. On each physical next-hit branch,

$$\left|\frac{dp_1}{d\ell}\right| = \frac{dV_1}{\sqrt{1+V^2}}\left|\frac{ds_1}{ds}\right| = \frac{\kappa A + dw}{\sqrt{1+V^2}} \geq k_- t_*. \tag{26.18}$$

Indeed $A \geq t_* w$ and $w/\sqrt{1+V^2} \geq 1$ because $\kappa > 1$ and $V \geq \kappa$. The small terminal incidence d cancels exactly before an upper bound is taken. There are at most J_f physical next-hit branches on this short graph, each with strictly monotone p_1. Do *not* cut this last coarea into infinitely many terminal homogeneity strips. Direct one-dimensional substitution therefore gives

$$\int_V \mathbf{1}_{\{p_1 \in I\}}\, d\ell \leq \frac{J_f}{k_- t_*}|I|. \tag{26.19}$$

For $m \geq 2$, push each first-image density of Lemma 26.2 through $m-2$ additional actual collisions, using the complete homogeneous short-curve partition. The density on an incoming final graph is bounded by $\|\rho_C\|_\infty$ times its exact inverse curve Jacobian. Their complete sum of suprema is at most $\widehat{C}_g\|\rho_C\|_\infty$, by Theorem 25.1. Now apply (26.19) to the one remaining physical collision. Sum the labelled graphs and first-image sources with Tonelli's theorem. This proves the bound with $A_{\mathrm{mom}}\widehat{C}_g J_f/(k_- t_*)$. For $m = 1$, directly on each of the at most $J_{\mathrm{pre}} J_f$ initial first-hit pieces,

$$\left|\frac{dp_1}{d\alpha_0}\right| = RAV_1 = A + d \geq t_*\kappa \geq k_- t_*.$$

Lemma 26.3 gives the sharper complete bound C_1 used in (26.14). There is neither an extra initial nor an extra terminal collision in either case. A selector between zero and one is retained until the positive upper bound. The first-image and growth singular-null arguments apply on each labelled source, and the final collision removes only finitely many tangencies on each incoming short curve. Their pullbacks are null for $d\alpha_0$ by the exact absolutely continuous weights above.

On each regular original m-itinerary the map $\alpha_0 \mapsto p_m(\alpha_0, a)$ has nonzero derivative: after the first hit it is an unstable graph and all subsequent transformations are regular. One-dimensional coarea gives (26.16) for almost every v. To obtain the stated bound at every regular root value, fix a finite collection of distinct roots at v_0 and take disjoint regular source neighborhoods. The

inverse branches and their positive weights are continuous for v near v_0, where their sum is bounded almost everywhere by the already proved density bound. Pass to v_0 and then take the supremum over finite collections. This proves the bound for the complete nonnegative root series, including roots on artificial subdivision boundaries. No value of a physical derivative is assigned at a singular trajectory by this argument. Finally integrate the fibre inequality in a and use $d\nu_R^* = \mathbf{1}_{Y_R^*}\, d\alpha_0\, da/(4\pi c_*)$. The factor c_* occurs exactly once, and there is no R factor because the initial fibre coordinate is α_0, not s_0. □

Remark 26.5 (What is, and is not, smoothed). The bounds sum all relevant regular collision words before any supremum in R, m, a, v with $a, v \in (-1, 1)$, with no endpoint-incidence loss. They have no whole-cylinder pressure premise. They give no mixing rate, product law, lattice local limit, or density in the original clock L_m. The grazing-uniform bound follows from the last-collision cancellation (26.18), not from dropping an inverse incidence factor in a terminal-coordinate coarea estimate. Singular one-sided critical traces require a separate argument below; they do not follow by declaring a bulk-null set to have zero trace.

27. A pressure-free sum of physical critical sectors

We now use the endpoint kernel on its own physical source, rather than multiply a local critical estimate by a count of collision words. The conclusion is a complete, collision-uniform leading-coefficient sum. It removes a genuine pressure dependency of Proposition 17.5; it is not yet a normalized roof-density theorem. All endpoint coordinates below are the original angles $q = (\alpha_0, \alpha_m)$.

Lemma 27.1 (Endpoint Hessian bounds and the momentum inverse weight)**.** *Let x be a critical point of a regular reduced action F with every visited incidence positive. Write*

$$G = D_q^2 F(x) = \begin{pmatrix} a_0 & e \\ e & d_0 \end{pmatrix}, \qquad \Delta = \det G > 0, \qquad u_x = \frac{1}{|\partial_{\alpha_0} p_m(x)|}.$$

Then $p_0 = p_m = 0$, and

$$u_x = \frac{R|e|}{\Delta} = \frac{R^2\omega(x)}{\Delta}, \qquad \mathfrak{J}(x) = \frac{\sqrt{\Delta}}{2R^2 c_*}\, u_x, \tag{27.1}$$

$$\frac{\sqrt{\Delta}}{R^2} \le \kappa + t_*^{-1} \le U = 170/9, \qquad \frac{\lambda_{\max}(G)}{R^2} \le \kappa + 2t_*^{-1} \le U_2 := 320/9. \tag{27.2}$$

For $m \ge 2$ the second bound improves to $\lambda_{\max}(G)/R^2 \le \kappa + t_*^{-1}$. The same auxiliary-action identities hold at a one-sided unvisited boundary when all visited *incidences stay positive. No physical DT^m at a grazing collision is thereby defined.*

Proof. The endpoint first variations in the retained optical proof are $F_{\alpha_0} = -Rp_0$, $F_{\alpha_m} = Rp_m$. The generating-function matrix in Proposition 17.2 therefore has lower-left entry $-\Delta/(Re)$. Its inverse absolute value gives the first identity, since $\omega = |e|/R$; the retained critical coarea coefficient $\mathfrak{J} = \omega/(2c_*\sqrt{\Delta})$ gives the second.

Here are the upper bounds, including their signs and units. For one physical flight of length τ, differentiate its Euclidean length with respect to the first contact angle. The diagonal second derivative is $Rc_0 + R^2 c_0^2/\tau$: the circular acceleration is $-Rn_0$ and its scalar product with the initial length gradient $-v$ is Rc_0. At the terminal contact the incoming velocity has normal component $-c_m$, giving the same positive circular term Rc_m. At a critical orbit $c_0 = c_m = 1$. Consequently the two endpoint diagonals of the unreduced full length action are at most $R + R^2/t_*$. Eliminating the positive interior optical Hessian is a Schur complement, which subtracts a positive semidefinite matrix from the endpoint block. Thus both reduced diagonals are at most this

value, and $\sqrt{\Delta} \le \sqrt{a_0 d_0} \le R + R^2/t_*$. If $m \ge 2$, the unreduced endpoint block is diagonal, so its maximum eigenvalue has the same upper bound. If $m = 1$ there is no interior to eliminate: the mixed second derivative of the one-flight length has absolute value at most R^2/τ, and the two-row bound is $R + 2R^2/t_*$. Division by R^2 proves (27.2). The positivity and elimination are exactly the physical optical Hessian construction retained in Section 4; no full-trajectory Hessian is replaced by an independent matrix model. At a skipped-unvisited tangency its strictly reflected auxiliary word is still a smooth action with positive visited incidences. The same calculation is valid for that auxiliary action, restricted to its actual physical side as in Section 17. □

Theorem 27.2 (Complete regular critical coefficient sum)**.** *For every R, every $m \ge 1$ and every exact record (n, k), the one-copy sum over all regular critical orbits of the original m-flight roof in that record satisfies*

$$\sum_x \mathfrak{J}(x) \le \frac{UC_{\mathrm{mom}}}{2c_*}. \tag{27.3}$$

This also bounds the sum with any coefficients between zero and one, including restrictions to original roof intervals. Neither the whole-cylinder pressure input nor an independent periodic-orbit counting theorem is used. Critical orbits are the actual normal-endpoint reversing trajectories identified in the retained proof, not periodic points of a replacement symbolic model.

Proof. At every such x, $p_0 = p_m = 0$. Distinct physical initial points are distinct regular roots in (26.16) at $(a, v) = (0, 0)$. The pointwise root bound, rather than an arbitrarily chosen version of an almost-everywhere density, gives $\sum_x u_x \le C_{\mathrm{mom}}$. Use (27.1)–(27.2) and sum nonnegative terms. Exact section and record restrictions can only decrease the sum. The proof works for the complete countable collection, by finite exhaustion, before taking the supremum over radius or collision count. □

27.1. **One-sided sectors are paid on the physical side.** Bulk singular-nullness alone proves nothing about an endpoint trace. We give the additional argument for the one-sided critical strata used in the retained local coarea theorem. Consider a finite or countable one-copy family of critical auxiliary-action germs x. Every germ has all visited incidences positive and normal endpoints. Its physical-side set E_x retains all unvisited-obstacle inequalities and the original section and record cuts. In a neighborhood of x, assume these cuts are a finite Boolean combination of C^1 inequalities whose active boundary differentials are nonzero. The limiting tangent set is then a finite union of angular sectors. Multiple simultaneous boundaries are permitted; no positive trace is discarded.

This class includes the original finite-time section/record cuts on a regular auxiliary word. A section boundary is a coordinate boundary pulled back by a locally invertible finite collision map, so its boundary differential is nonzero. The exact occupation count is a finite Boolean combination of these tests. An unvisited tangency is a clearance boundary: the line-coordinate determinant in Lemma B.2 makes its differential nonzero, and an interior-foot tangency stays away from the incident contacts. Only finitely many possible centers occur for a finite word by finite horizon. Strict displacement labels are locally constant on such a word. General Borel pairing selectors, or strata involving an actual visited grazing collision, are *not* asserted to have this tangent sector property; their exact local selectors and outstanding budgets remain in the manuscript.

Put $y = G_x^{1/2}(q - q_x)$ and let $\beta_y(x)$ be the angular fraction of the physical tangent set in these normalized coordinates. It is the leading coefficient fraction in the local theorem of Section 17. Let $z = (p_0, p_m)$ and denote the corresponding angular fraction in the z-plane by $\beta_z(x)$. Auxiliary branches are identified by their strict visited word; their source restrictions are disjointized on

overlaps. No multiple copy is introduced at a coordinate seam. The family below may also contain fully regular critical points, whose interior sector fraction is one.

Lemma 27.3 (The correct angular change of variables)**.** *Let $D = \operatorname{diag}(-1, 1)$. At a critical point,*

$$z = R^{-1} D G_x^{1/2} y + o(|y|), \qquad L_x = R^{-1} D G_x^{1/2}. \tag{27.4}$$

Writing $e_\theta = (\cos\theta, \sin\theta)$, the absolute angular Jacobian is

$$\left| \frac{d \arg(L_x e_\theta)}{d\theta} \right| = \frac{\sqrt{\det G_x}}{e_\theta^{\mathsf{T}} G_x e_\theta}. \tag{27.5}$$

For every such tangent sector set,

$$\mathfrak{J}(x)\beta_y(x) \le \frac{\lambda_{\max}(G_x)}{2R^2 c_*} u_x \beta_z(x). \tag{27.6}$$

The Euclidean angle in the unnormalized endpoint coordinates is not substituted for either fraction.

Proof. Use $g = \nabla_q F = G_x(q - q_x) + o(|q - q_x|)$ and $z = R^{-1} D g$. For an invertible planar matrix L, differentiation of its polar angle gives $|\det L|/|Le_\theta|^2$. Here $|\det L_x| = \sqrt{\det G_x}/R^2$ and $|L_x e_\theta|^2 = e_\theta^{\mathsf{T}} G_x e_\theta / R^2$. Integrating (27.5) over the sector yields $\beta_z \ge \sqrt{\Delta}\, \beta_y / \lambda_{\max}(G_x)$. Combine this with (27.1). This argument also permits finite unions, half-planes, corners and complements; boundary rays have zero angular measure. □

Theorem 27.4 (Complete one-sided leading critical budget)**.** *For the physical-side family just specified, with unique strict visited words and one-copy source restrictions, one has*

$$\sum_x u_x \beta_z(x) \le C_{\mathrm{mom}}, \tag{27.7}$$

and consequently, uniformly in R, $m \ge 1$, exact labels and any restriction of the critical values to an original roof interval,

$$\sum_x \mathfrak{J}(x)\beta_y(x) \le \frac{U_2 C_{\mathrm{mom}}}{2c_*}. \tag{27.8}$$

Here u_x at an unvisited boundary is the auxiliary inverse weight $R|e|/\Delta$, not a claimed derivative of the singular physical map. This conclusion removes the whole-cylinder pressure premise for this complete leading critical-sector sum. It does not extend a physical trajectory through a grazing reflection or assert that every Borel pairing seam possesses a sector limit.

Proof. First take a finite collection $\mathcal{F}$ of distinct physical auxiliary germs. A strictly reflected finite word with positive visited incidences has an invertible endpoint-momentum map: its endpoint derivative has nonzero determinant, as follows from $e \ne 0$ and $\Delta > 0$. The inverse-function theorem gives a common small z-disk for the finitely many local inverses. The absolute Jacobian from (p_0, p_m) to (p_0, α_0) on germ x is a positive continuous function $u_x(z)$ with $u_x(0) = u_x$.

Distinct regular initial points have disjoint sufficiently small source neighborhoods. The same is true for distinct strict-word unvisited-boundary germs: at a limiting state, continuing straight past its unvisited tangencies fixes the next strictly met disk uniquely; iterate this observation through the fixed m strict collisions. Any alternative word using one of the tangencies would have a visited zero incidence and is outside this class. Thus two such germs at the same initial state are the same germ; their selectors are united, not counted twice. This is a uniqueness argument for the auxiliary extension and does not alter a physical collision record. Artificial partition seams are identified as well. Finite disjointization of actual source restrictions handles any remaining coordinate overlap.

On the actual side of each germ, its auxiliary orbit is the actual regular physical orbit. Therefore its positive contribution to the joint endpoint root density is bounded, almost everywhere in z, by the complete kernel already proved:

$$\sum_{x\in\mathcal{F}} u_x(z)\mathbf{1}_{E_x}(z) \le C_{\mathrm{mom}}. \tag{27.9}$$

No equality on the singular state itself is needed. By polar Fubini, the inequality holds at almost every angle for almost every radius $r > 0$. Choose a sequence of such radii tending to zero. On each fixed finite family, the C^1 inequalities with nonzero active linear parts imply convergence of $\mathbf{1}_{E_x}(re_\theta)$ to its momentum tangent-sector indicator for almost every θ. Continuity of $u_x(z)$ and bounded convergence on the angular circle give, after dividing the integral of (27.9) by 2π, $\sum_{x\in\mathcal{F}} u_x\beta_z(x) \le C_{\mathrm{mom}}$. Take the supremum over finite subfamilies to prove (27.7). Countability causes no exchange of a signed series or of the collision and outer limits. Finally apply Lemma 27.3 and (27.2). Keeping original record, critical-roof or bounded continuous guard weights only reduces the nonnegative sum. Neither independence nor a physical-word multiplicity factor is used. □

27.2. **Exact gain and the remaining normalized obstruction.** Theorems 27.2 and 27.4 prove a complete leading-coefficient bound at every collision count, including genuine one-sided physical critical sectors. Their source is the full original momentum coordinate disintegration, and their constants are uniform on the original radius family. This replaces the pressure-dependent coefficient estimate, not its historical proof body.

There is no illicit interchange of a local-radius limit with $m \to \infty$. The sector proof first treats finitely many germs at a *fixed* R, m, takes the local radius to zero, and exhausts their positive coefficients. The resulting bound is uniform in R, m because its right-hand side is. It does not give a common neighborhood size or a summable bound for the local $O(\sqrt{s})$ errors. In particular the critical values of different germs can differ, so the bound at the common momentum origin cannot be used as a bound at different momentum points selected by a common physical roof level.

The full original problem still requires the unchanged limit (25.3), with B-dependent widths fixed before the collision limsup. Along the present route the minimal additional inputs are: a complete original-roof chart/error estimate with the required m^{-2} normalization, including the first-order clearance source, noncritical folds and actual visited singular strata; payment of the original incidence-only low-load source or a valid full nongrazing-capacity comparison; and the stated spectral, arithmetic, pairing and conditional-denominator inputs with their actual uniform quantifiers. General pairing selectors retain their exact local angular functions unless their tangent sectors are proved to exist. Neither the nonzero single-mark mass nor a positive boundary trace is set to zero. The complete raw pointwise local limit and final four-coordinate inversion are not asserted by the present theorems.

28. Exact radial coarea and Borel critical-origin bands

The preceding critical-sector theorem takes a local limit at fixed collision count and requires a tangent sector for each physical cut. Here we replace the local Taylor calculation, for an explicitly defined physical class, by an exact nonlinear calculation. This gives a uniform source estimate at every band width, including arbitrary Borel cuts. It neither identifies different physical roof levels nor bounds the paired endpoint measure by the collision measure without a comparison proof.

Throughout this section the dynamics, the probability ν_R^*, the section, and the exact event $\Pi_{n,k,m,R}$ are those of the original model. We write $\Phi_m(x) = (p_0(x), p_m(x)) = \zeta$ and $D =$

$\mathrm{diag}(-1,1)$. The new polar radius $r=|\zeta|$ is an auxiliary coordinate, not the physical clock and not one of the microscopic widths r_B. Fix $0<\rho<1$ and put

$$K_\rho=\frac{100}{47}\sqrt{1-\rho^2},\qquad U_m=\begin{cases}320/9, & m=1,\\ 170/9, & m\ge 2.\end{cases} \tag{28.1}$$

All estimates below are uniform in $R\in[9/20,47/100]$ and in m. There is no cutoff on a positive interior incidence.

Lemma 28.1 (An endpoint-momentum metric away from the critical point)**.** *On every regular physical m-flight branch with $|\Phi_m|<\rho$, the reduced action and its Hessian satisfy*

$$\widehat{G}:=R^{-2}D_q^2F,\qquad K_\rho I\preceq\widehat{G}\preceq U_mI. \tag{28.2}$$

The map from endpoint angles $q=(\alpha_0,\alpha_m)$ to ζ is a local diffeomorphism. For every local inverse $q=q(\zeta)$,

$$D_\zeta q=RG^{-1}D,\qquad \nabla_\zeta(F\circ q)=D\widehat{G}^{-1}D\zeta. \tag{28.3}$$

These are identities for the actual finite itinerary on its regular physical side; they do not define a physical collision map at tangency.

Proof. The full second variation and its minimizing Schur complement were proved in Lemma 4.2. They give $G\succeq R\,\mathrm{diag}(c_0,c_m)$. Since $|p_0|,|p_m|<\rho$, division by R^2 gives the lower bound in (28.2). For $m\ge 2$ the unreduced endpoint block is diagonal. Its two entries, divided by R^2, are $c_0/R+c_0^2/\tau_0$ and $c_m/R+c_m^2/\tau_{m-1}$. Each is at most $20/9+50/3=170/9$. Minimizing over the interior variations subtracts a positive semidefinite matrix. For $m=1$ there is no interior block. The absolute mixed entry, divided by R^2, is c_0c_m/τ_0; the row-sum bound is at most $20/9+100/3=320/9$. Thus the upper estimate, previously used only at normal endpoints in Lemma 27.1, holds everywhere on the stated regular physical set.

The exact first variation is $\nabla_qF=RD\zeta$, and hence $D_q\zeta=R^{-1}DG$. Its determinant is nonzero by positivity of G. Inverting and applying the chain rule gives

$$(D_\zeta q)^{\mathsf T}\nabla_qF=RDG^{-1}(RD\zeta)=D\widehat{G}^{-1}D\zeta.$$

In particular, no derivative of a Hessian-dependent angular change of variables has been dropped. Positivity is used only at regular points with the true specular stationarity equations satisfied. □

Theorem 28.2 (Exact radial roof disintegration and synchronized bound)**.** *Let $e_\theta=(\cos\theta,\sin\theta)$ and, on any regular inverse branch $x=x(\zeta)$, set*

$$h_x(\zeta)=e_\theta^{\mathsf T}D\widehat{G}_x^{-1}De_\theta,\quad u_x(\zeta)=|\partial_{\alpha_0}p_m(x)|^{-1},\quad \zeta=re_\theta\neq 0. \tag{28.4}$$

For fixed θ, while the same actual branch is regular,

$$\frac{dL(x(re_\theta))}{dr}=rh_x(re_\theta),\qquad U_m^{-1}\le h_x\le K_\rho^{-1}. \tag{28.5}$$

The collision-source density in (L,θ) on this branch is exactly

$$b_x(re_\theta)=\frac{u_x(re_\theta)}{4\pi c_*h_x(re_\theta)}. \tag{28.6}$$

A Borel selector $0\le w\le 1$, including the original initial/terminal section tests, exact labels, roof intervals, guards and pairing cuts, multiplies this density without a smoothness assumption. For the complete one-copy regular physical root family,

$$\sum_{x:\,\Phi_m(x)=re_\theta}w(x)b_x(re_\theta)\le\frac{U_mC_{\mathrm{mom}}}{4\pi c_*} \tag{28.7}$$

for $0 < r < \rho$ at every regular root value (and for the corresponding almost-everywhere versions of measurable disintegrations). The summands in (28.7) *share the same endpoint momenta; their physical roofs need not be equal.*

Proof. Equation (28.3) gives (28.5). The eigenvalue bounds in (28.2) give its two-sided constants. The determinant of the map $(\alpha_0, p_0) \mapsto (p_0, p_m)$ is $-\partial_{\alpha_0} p_m$. This derivative is nonzero: equivalently, if $G_{0m} = e \neq 0$, the generating matrix has lower-left entry $-\det G/(Re)$. The nonvanishing of e is the exact positive optical denominator identity, not an assumed genericity condition. Consequently the original probability contributes $u_x\, d\zeta/(4\pi c_*)$, with the initial-section indicator retained. Polar coordinates give $d\zeta = r\, dr\, d\theta$. At fixed angle, $dt = r h_x\, dr$, where $t = L(x)$. The factors r cancel, proving (28.6), with no missing R or incidence factor.

For precision with arbitrary cuts, cover the regular source by countably many relatively compact inverse charts, fix one angular seam, and disjointize their source overlaps. Apply change of variables first to bounded nonnegative tests on finitely many charts and then use monotone convergence. On each angular slice, subdivide the open regular inverse domain into intervals; on every such interval (28.5) is strictly positive. Borel cuts stay inside the integrands rather than being turned into new open branches. This proves the exact formula also when a cut has no angular density at its boundary. The angular seam is only a coordinate seam and is counted once. This argument does not discard a singular endpoint trace.

Finally $1/h_x \leq U_m$, and Theorem 26.4 bounds the complete root sum $\sum_x u_x(\zeta)$ by C_{mom}. Apply it to finite subfamilies and exhaust the nonnegative terms. Multiplying by selectors between zero and one cannot increase this sum. The root bound at regular root values was proved in that theorem; no chosen version of an almost-everywhere density is evaluated at a new point. □

28.1. **Critical-origin arms with their physical cuts still present.** The next class is defined by an actual continuation property, not by deleting an inconvenient source. A *critical-origin radial arm* is a regular physical inverse along a fixed momentum ray, defined for $0 < r < r_+(\theta) \leq \rho$, with no missing collision on that interval. The arm stays on the same strict visited word. Its endpoint at $r = 0$ may be a regular critical orbit, an unvisited/section boundary, or a limit with a visited tangency; no physical derivative is asserted at that endpoint. All physical points with $r > 0$ remain genuine regular orbits. Only one copy of each physical initial point may be used.

We work with measurable one-copy families of these arms and arbitrary Borel selectors on them. In particular this includes every original pairing sheet whose incoming limit has $|\nabla_q F| = 0$ and whose regular interval reaches that limit. The original sheet construction and its measurable, one-copy parametrization remain the explicitly retained input of Section 3; the present theorem does not purport to reconstruct that historical construction. Its geometric estimate applies to every family satisfying the definition, not only to finitely many coordinate balls or to smooth selectors. Other components of the original event, including arms which terminate at positive momentum radius before reaching zero, are not assigned zero source and remain outside this particular estimate.

Lemma 28.3 (Exact band geometry, without a Taylor radius)**.** *On a critical-origin arm the finite limit $a(\theta) = \lim_{r \downarrow 0} L(x(re_\theta))$ exists and*

$$\frac{r^2}{2U_m} \leq L(x(re_\theta)) - a(\theta) \leq \frac{r^2}{2K_\rho}. \tag{28.8}$$

No local asymptotic error occurs. The limiting endpoint angles also exist in local angular lifts. The assertions remain valid when an interior incidence tends to zero as $r \downarrow 0$.

Proof. For $0 < \eta < r$, integrate (28.5) to obtain

$$\frac{r^2-\eta^2}{2U_m} \le L(r) - L(\eta) \le \frac{r^2-\eta^2}{2K_\rho}.$$

Thus $L(\eta)$ is Cauchy as $\eta \downarrow 0$ and has a finite limit. Taking this limit proves (28.8). Also $|dq/dr| \le 1/(RK_\rho)$ by (28.3); hence both endpoint angles are Cauchy in consistent lifts on a sufficiently short terminal interval. These estimates use no interior-incidence lower bound. All calculations were made at $\eta > 0$ on actual regular orbits. Taking a limit of their physical lengths and endpoint angles does not continue a collision map through a grazing collision. □

Proposition 28.4 (No drift of the limiting roof on a physical fan)**.** *Suppose a continued physical inverse has domain $\{re_\theta : 0 < r < r_+(\theta),\ \theta \in I\}$, where I is a connected angular interval and every compact $J \subset I$ admits a positive radius r_J with $(0, r_J) \times J$ in that domain. Then $a(\theta)$ in Lemma 28.3 is constant on I. In particular the bands on such a fan are measured from one actual limiting physical roof, not a freely chosen angular clock correction. Arbitrary Borel cuts may still be imposed on the actual source after this identity has been obtained.*

Proof. Put $e_\theta^\perp = (-\sin\theta, \cos\theta)$. The exact angular chain rule, using (28.3), is

$$\partial_\theta L(re_\theta) = r^2 (e_\theta^\perp)^{\mathsf{T}} D\widehat{G}^{-1} De_\theta.$$

Its absolute value is at most r^2/K_ρ. Given two angles in I, choose a compact connecting interval J and use the common positive radius in the hypothesis. Integration at any $0 < r < r_J$ gives $|L(re_\theta) - L(re_\psi)| \le r^2|\theta - \psi|/K_\rho$. Letting $r \downarrow 0$ proves equality of the two limiting roofs. Only a compact subinterval is used; no collision-uniform continuation radius is assumed. No angular boundary limit of a Borel selector is needed because the identity is for the underlying physical action. □

Theorem 28.5 (Complete Borel critical-band and singularity budgets)**.** *For any measurable one-copy family of critical-origin arms let E_s be the set of its actual points satisfying $0 < L - a(\theta) \le s$, where $s > 0$, and put*

$$A_s = \min\{\rho, \sqrt{2U_m s}\}.$$

For any Borel $0 \le w \le 1$, retaining the original event and original probability, one has the simultaneous bounds

$$\int_{E_s} \mathbf{1}_{\Pi_{n,k,m,R}} w \, d\nu_R^* \le \frac{C_{\mathrm{mom}}}{4c_*} A_s^2, \tag{28.9}$$

$$\int_{E_s} \mathbf{1}_{\Pi_{n,k,m,R}} w \, |\Phi_m|^{-\gamma} \, d\nu_R^* \le \frac{C_{\mathrm{mom}}}{2c_*(2-\gamma)} A_s^{2-\gamma}, \qquad 0 \le \gamma < 2, \tag{28.10}$$

$$\int_{E_s} \mathbf{1}_{\Pi_{n,k,m,R}} w \, |\nabla_q F|^{-1} \, d\nu_R^* \le \frac{C_{\mathrm{mom}}}{2c_* R} A_s. \tag{28.11}$$

These estimates hold for the complete family before any supremum over R, m, n, k, and for every s, not merely in a local-radius limit at a fixed collision count. They also hold for a disjoint union of exact records whose total selector is at most one. No tangent-sector limit, finite C^1 description of a pairing cut, common chart radius, or uniform bound on third derivatives of the roof is required.

Proof. By (28.8), every selected point has $|\Phi_m| \le A_s$. The positive joint-kernel domination from Theorem 26.4, on the *actual* physical source, gives

$$\int_{E_s} \mathbf{1}_{\Pi} w \, |\Phi_m|^{-\gamma} \, d\nu_R^* \le \frac{C_{\mathrm{mom}}}{4\pi c_*} \int_{|\zeta| \le A_s} |\zeta|^{-\gamma} \, d\zeta.$$

Apply this first with bounded truncations on finitely many disjoint charts. Monotone convergence supplies the entire physical source and the unbounded nonnegative weight. The polar integral is $2\pi A_s^{2-\gamma}/(2-\gamma)$, proving (28.10). Its case $\gamma = 0$ is (28.9). Finally $|\nabla_q F| = R|\Phi_m|$ gives (28.11) at $\gamma = 1$.

The constants never depended on how many arms were selected or on their individual continuation lengths. A shorter actual arm contributes only its actual points; it was not extended through an unvisited disk. The same argument covers limiting visited tangencies because all integrated points have positive visited incidences. It controls their interior source approach, not an arbitrarily prescribed trace measure supported on the singular boundary. General Borel cuts remain measurable selectors throughout; their pointwise or angular boundary limits are not assumed to exist. The assertion for unions follows from one-copy positive domination, not from adding separately normalized probabilities. □

28.2. **What has and has not been transferred to the original roof.** Theorems 28.2 and 28.5 remove the need for Taylor remainders and tangent sectors for the stated synchronized coarea and physical-band budgets. In particular (28.9) is a collision-uniform $O(s)$ statement and (28.11) is a collision-uniform $O(\sqrt{s})$ statement on the full defined arm family. These are continuum estimates, not conclusions of a finite search or a fitted exponent.

They do not close (25.3). In the actual roof density at a common t, formula (28.6) must be evaluated at the branch-dependent radius $r_x(t,\theta)$. The synchronized sum (28.7) evaluates all branches at the *same* r and is not a bound for that different sum. Likewise $L - a(\theta)$ is used only to specify a physical subset; it is not substituted for L in the exact record, Fourier reference, conditioning or LLT.

The band budget bounds the original collision-source measure. The paired measure $c^{\varepsilon}\,du\,d\theta$ uses the minimum of its two endpoint traces and is not automatically a submeasure of the interior density $f^{\varepsilon}\,du\,d\theta$. Its transfer still requires the retained positive-variation/comparison argument. The inverse-gradient estimate does not include the unbounded insertion I_m, and no such insertion is silently absorbed into its constant. Critical-origin arms do not exhaust the positive-radius singular cuts, noncritical folds or low-load source. These sources, the first-order clearance contribution, nongrazing capacity, and the original spectral, arithmetic and conditional-denominator interfaces retain their exact proof obligations. No factor m^2 is attached to the new unscaled bounds at a fixed outer width and declared harmless. The original ordered limit and four-coordinate raw local inversion target remain unchanged and are not asserted to be proved here.

29. Grazing-uniform endpoint localization and the third variation

The estimates in this section are on every actual regular finite itinerary, not on an auxiliary stochastic process. They hold for all $m \geq 1$ and $R \in [9/20, 47/100]$, without a lower cutoff on any positive incidence. The derivative is with respect to the two endpoint angles of the actual stationary action. It is not the derivative of the forward collision map with fixed initial momentum. This distinction is essential: no uniform bound on that expanding forward derivative is asserted.

Keep $\tau_- = 3/50$, $R_+ = 47/100$, and define

$$\sigma = \frac{47}{59}, \qquad s_i = \sigma^i + \sigma^{m-i}, \qquad J_* = \max\left\{1, \frac{R_+}{2\tau_-}(\sigma + \sigma^{-1})\right\} = \frac{2845}{354}. \tag{29.1}$$

On a local physical branch let $\alpha(q) = (\alpha_0(q), \dots, \alpha_m(q))$, $q = (\alpha_0, \alpha_m)$, be the stationary lift. For an endpoint direction $\eta \in \mathbb{R}^2$ put $\xi_i = D_q\alpha_i[\eta]$. Every assertion about continuation below is restricted to a path in this branch up to its first actual boundary.

Theorem 29.1 (Two-sided localization without an inverse-incidence loss)**.** *For every such trajectory and every endpoint direction,*

$$|c_i\xi_i| \le \sigma^i c_0|\eta_0| + \sigma^{m-i}c_m|\eta_m| \le s_i\|\eta\|_\infty, \tag{29.2}$$

$$|\xi_i| \le J_* s_i\|\eta\|_\infty, \qquad 0 \le i \le m. \tag{29.3}$$

In particular, endpoint displacement decays exponentially towards the middle of the true collision chain even when an intermediate incidence is arbitrarily small. No comparison trajectory, new section, or collision suppression is used.

Proof. For $m = 1$ the endpoint assertions follow immediately from $s_0, s_m \ge 1$. Suppose $m \ge 2$. Differentiate the interior stationary equations and use the full physical second variation (4.2). With $w_i = \tau_i^{-1}$ and $y_i = (-1)^i c_i\xi_i$ this gives

$$(\kappa_i + w_{i-1} + w_i)y_i - w_{i-1}y_{i-1} - w_i y_{i+1} = 0, \qquad \kappa_i = \frac{2}{Rc_i}, \quad 1 \le i < m. \tag{29.4}$$

The sign change is important: both off-diagonal coefficients in this Dirichlet equation are negative. For a real sequence with nonnegative boundary values and nonnegative left side in (29.4), a negative interior minimum would make that left side strictly negative, because $\kappa_i > 0$. This proves the needed maximum principle directly.

Write the left side as $\mathcal{D}y$. Since $w_i \le \tau_-^{-1}$ and $\kappa_i \ge 2/R_+$,

$$\frac{\mathcal{D}(\sigma^i)_i}{\sigma^i} = \kappa_i + w_{i-1}(1-\sigma^{-1}) + w_i(1-\sigma) \ge \frac{2}{R_+} - \frac{\sigma^{-1}-1}{\tau_-} = 0.$$

The reversed sequence σ^{m-i} is a supersolution by the same calculation with the adjacent flights interchanged. Thus $b_i = \sigma^i c_0|\eta_0| + \sigma^{m-i}c_m|\eta_m|$ is a nonnegative supersolution and $b_0 \ge |y_0|$, $b_m \ge |y_m|$. Apply the maximum principle separately to $b - y$ and $b + y$ to get (29.2).

Dividing that bound by c_i would lose uniformity and is not the next step. Instead use the equation itself:

$$\kappa_i|y_i| \le w_{i-1}|y_{i-1}| + w_i|y_{i+1}|.$$

Since $\kappa_i c_i = 2/R$, it follows that

$$|\xi_i| \le \frac{R_+}{2\tau_-}(s_{i-1} + s_{i+1})\|\eta\|_\infty = \frac{R_+}{2\tau_-}(\sigma + \sigma^{-1})s_i\|\eta\|_\infty.$$

The endpoint inequalities follow from $\xi_0 = \eta_0$, $\xi_m = \eta_m$. This proves (29.3) for the entire chain. The calculation used positive incidences only at the actual regular point, with no uniform positive lower bound on them. □

29.1. **Physical flights, incidence, and every unvisited disk.** Let $v_i = (x_{i+1} - x_i)/\tau_i$ be the actual unit flight direction. Put

$$C_c = J_*\left[1 + \frac{R_+}{\tau_-}(1 + \sigma^{-1})\right], \qquad C_\Delta = R_+ J_*(1+\sigma). \tag{29.5}$$

Proposition 29.2 (Differentiation of the physical observables)**.** *On each actual regular branch,*

$$|D_q\tau_i[\eta]| \le R_+ J_*(s_i + s_{i+1})\|\eta\|_\infty, \qquad 0 \le i < m, \tag{29.6}$$

$$\|D_q v_i[\eta]\|_2 \le \frac{R_+ J_*}{\tau_-}(s_i + s_{i+1})\|\eta\|_\infty, \qquad 0 \le i < m, \tag{29.7}$$

$$|D_q c_i[\eta]| \le C_c s_i\|\eta\|_\infty, \qquad 0 \le i \le m. \tag{29.8}$$

The true least clearance Δ_i, with precisely the two incident disks excluded as in (1.2)*, is locally Lipschitz and has the following directional bound along every continuously differentiable physical branch path, almost everywhere on that path:*

$$|D_q\Delta_i[\eta]| \le R_+J_*\max(s_i,s_{i+1})\|\eta\|_\infty \le 2C_\Delta\sigma^{d_i'}\|\eta\|_\infty. \tag{29.9}$$

The last assertion includes changes of the nearest unvisited disk and of its nearest point on the flight. There is no factor equal to the number of candidate disks.

Proof. $D_qx_i[\eta] = Rn'(\alpha_i)\xi_i$. Differentiate length and unit direction and use respectively $\|D|X|\| = 1$ and $D(X/|X|) = |X|^{-1}(I - vv^{\mathsf T})$ to prove (29.6)–(29.7). For $i < m$, $c_i = v_i\cdot n_i$, so $|Dc_i| \le \|Dv_i\|_2 + |\xi_i|$. For the terminal collision use $c_m = -v_{m-1}\cdot n_m$ instead. The inequalities $s_{i+1} \le \sigma^{-1}s_i$ and $s_{i-1} \le \sigma^{-1}s_i$ give (29.8). This argument differentiates the actual geometric cosine, not an expression with an artificially introduced $1/c_i$.

The Hausdorff distance of segments $[x_i,x_{i+1}]$ and $[\widetilde{x}_i,\widetilde{x}_{i+1}]$ is at most $\max(\|x_i-\widetilde{x}_i\|_2, \|x_{i+1}-\widetilde{x}_{i+1}\|_2)$: compare points with the same affine parameter. Distance to any fixed center changes by at most that number. Taking an infimum over exactly the same set of unvisited centers preserves this inequality. Subtracting R does not change it. The fixed itinerary fixes the two excluded centers; it does not discard any other disk. This proves the local Lipschitz assertion and its derivative bound along a branch path by integration of the endpoint speeds. A Lipschitz function on a compact path interval is absolutely continuous; nearest-disk switches do not create atomic jumps. Finally

$$s_i + s_{i+1} = (1+\sigma)(\sigma^i + \sigma^{m-1-i}) \le 2(1+\sigma)\sigma^{d_i'}$$

proves the last inequality in (29.9). □

29.2. **A third-variation bound uniform in collision count.** Define constants depending only on the original geometric interval,

$$\begin{aligned} C_3 &= R_+ + \frac{3R_+^2}{\tau_-} + \frac{3R_+^3}{\tau_-^2},\\ C_F &= \frac{8C_3J_*^3(1+\sigma)^3}{1-\sigma^3}. \end{aligned} \tag{29.10}$$

Theorem 29.3 (Uniform physical third variation)**.** *For every actual regular endpoint action, all $m \ge 1$, all original radii, and $\eta,\zeta,\upsilon \in \mathbb{R}^2$,*

$$|D_q^3F[\eta,\zeta,\upsilon]| \le C_F\|\eta\|_\infty\|\zeta\|_\infty\|\upsilon\|_\infty. \tag{29.11}$$

Consequently $\|D_qG[\eta]\|_{\mathrm{op}} \le 2C_F\|\eta\|_\infty$. For any piecewise continuously differentiable path $q(a)$ that remains in a single actual regular branch,

$$\|G(q(b)) - G(q(a))\|_{\mathrm{op}} \le 2C_F\int_a^b \|q'(v)\|_\infty\,dv.$$

In particular this bound applies on every line segment contained in that branch. It does not assert the existence of a collision-uniform ball of physical continuation across unvisited tangencies or section cuts.

Proof. For the Euclidean norm $N(X) = |X|$, at $X \ne 0$, $\|DN\| \le 1$, $\|D^2N\| \le |X|^{-1}$ and $\|D^3N\| \le 3|X|^{-2}$. The last inequality follows by writing its trilinear form as minus $|X|^{-2}$ times the sum of $(v\cdot a)b^{\mathsf T}(I-vv^{\mathsf T})c$ and the two cyclic terms. For one flight $X = Rn(\alpha_{i+1}) - Rn(\alpha_i) + O_{i+1} - O_i$, each derivative of n up to order three has norm one. If $V_i(\xi) = |\xi_i| + |\xi_{i+1}|$, the third-order chain rule therefore gives

$$|D^3\tau_i[\xi,\widetilde{\xi},\widehat{\xi}]| \le C_3V_i(\xi)V_i(\widetilde{\xi})V_i(\widehat{\xi}). \tag{29.12}$$

The coefficient three in the middle term of C_3 counts the three second–first derivative terms; no such term is omitted.

Let $\mathcal{J}\eta = D_q\alpha[\eta]$ be the stationary lift. Differentiated stationarity says that $H\mathcal{J}\eta$ has zero interior coordinates, where $H = D^2S$. From $D^2F[\eta,\zeta] = H[\mathcal{J}\eta,\mathcal{J}\zeta]$, differentiation in υ gives the exact identity

$$D^3F[\eta,\zeta,\upsilon] = D^3S[\mathcal{J}\eta,\mathcal{J}\zeta,\mathcal{J}\upsilon]. \tag{29.13}$$

Indeed, derivatives of $\mathcal{J}$ have zero endpoint coordinates, and their contractions with $H\mathcal{J}\eta$ or $H\mathcal{J}\zeta$ vanish. This cancellation occurs after differentiating the full stationary action; it is not a claim that derivatives of the interior variables vanish.

By Theorem 29.1, each $V_i(\mathcal{J}\eta)$ is at most $J_*(s_i + s_{i+1})\|\eta\|_\infty$. Moreover,

$$\sum_{i=0}^{m-1}(s_i+s_{i+1})^3 \le 4(1+\sigma)^3\sum_{i=0}^{m-1}(\sigma^{3i}+\sigma^{3(m-1-i)}) \le \frac{8(1+\sigma)^3}{1-\sigma^3}.$$

Sum (29.12) and use (29.13) to obtain (29.11). Each entry of $D_qG[\eta]$ is bounded by $C_F\|\eta\|_\infty$; its operator norm is at most twice that number. Integrating along the stated physical path completes the proof. □

Remark 29.4 (Domain, time, and source). Theorem 29.3 concerns the true reduced action at every regular point. It requires neither a pressure theorem nor the endpoint-momentum kernel. Uniform boundedness of its third derivative is not uniform control of all higher derivatives, a chart-count bound, or a roof-density theorem. All labels may restrict the points at which it is used, but are not differentiated. For a path of genuine trajectories with a fixed itinerary reaching a physical boundary the estimates hold up to that boundary. They do not define a collision-map derivative at a visited tangency or permit continuing an exact event through a new hit.

30. A native proof of the original regular-variation estimate

This section supplies a complete proof of (3.3), which was stated as inherited in Input 3.1. Its two averaged conclusions (3.4)–(3.5) are not proved by this calculation and remain separate inputs. The proof uses the original guard, including every unvisited obstacle, and the original insertion I_m. Nothing is conditioned on a smaller event.

Lemma 30.1 (Logarithmic sensitivity of the positive optical denominator)**.** *Regarding $\mathfrak{b}_m$ in* (4.6) *as a polynomial in the positive variables τ_i and κ_j, it has nonnegative coefficients and degree at most one in each variable. Consequently*

$$0 \le \frac{\tau_i\partial_{\tau_i}\mathfrak{b}_m}{\mathfrak{b}_m} \le 1, \qquad 0 \le \frac{\kappa_j\partial_{\kappa_j}\mathfrak{b}_m}{\mathfrak{b}_m} \le 1. \tag{30.1}$$

Proof. Each variable occurs in exactly one optical factor, that factor is linear in it, and every entry of every factor has nonnegative coefficients. Matrix multiplication preserves coefficient positivity and does not repeat a variable. For any variable x, write $\mathfrak{b}_m = a + xb$ with $a, b \ge 0$ and $\mathfrak{b}_m > 0$. Then $x\partial_x\mathfrak{b}_m/\mathfrak{b}_m = xb/(a+xb) \in [0,1]$. For $m = 1$ this applies to $\mathfrak{b}_1 = \tau_0$ and there are no curvature variables. □

Set $c_\circ = \sqrt{19}/10$ and define

$$C_\omega = \frac{2R_+J_*(1+\sigma)}{\tau_-(1-\sigma)} + \frac{4C_c}{c_\circ} + 2C_c. \tag{30.2}$$

Proposition 30.2 (The precise inherited incidence insertion suffices)**.** *At every original exact-event regular point and for every endpoint direction,*

$$|D_q \log \omega[\eta]| \le C_\omega I_m \|\eta\|_\infty, \qquad I_m = 1 + \sum_{i=1}^{m-1} \frac{q_0^{d_i}}{c_i}. \tag{30.3}$$

Proof. The radius is fixed under D_q, so (4.7) and Lemma 30.1 imply

$$|D \log \omega| \le \frac{|Dc_0|}{c_0} + \frac{|Dc_m|}{c_m} + \sum_{i=0}^{m-1} \frac{|D\tau_i|}{\tau_i} + \sum_{i=1}^{m-1} \frac{|Dc_i|}{c_i}.$$

Here $D \log \kappa_i = -D \log c_i$; the sign is irrelevant only after taking absolute values. The original section cap gives $c_0, c_m \ge c_\circ$ and $s_0, s_m \le 2$. Also

$$\sum_{i=0}^{m-1} (s_i + s_{i+1}) \le \frac{2(1+\sigma)}{1-\sigma}, \qquad s_i \le 2\sigma^{d_i} \le 2q_0^{d_i}.$$

Apply Proposition 29.2. The first two terms in (30.2) bound the endpoint and flight contributions, and $2C_c(I_m - 1)$ bounds the interior contribution. Since $I_m \ge 1$, (30.3) follows with the displayed constant. No additional inverse incidence or collision-dependent multiplier occurs. □

30.1. **The actual guard, with nonsmooth clearance switches.** Put $M_\Theta = \sup_{x\in\mathbb{R}} |\Theta'(x)| < \infty$ for the fixed smooth cutoff already present in (1.2), and set

$$\beta = \sigma q_0^{-1/4} < 1, \qquad C_W = \frac{4M_\Theta(C_c + C_\Delta)}{1-\beta}. \tag{30.4}$$

Theorem 30.3 (Collision-uniform variation of the original physical guard)**.** *For each $\varepsilon > 0$, W^ε is locally Lipschitz on an actual regular branch. Along every continuously differentiable physical branch path it is absolutely continuous on compact parameter intervals and, almost everywhere,*

$$|D_q W^\varepsilon[\eta]| \le C_W \varepsilon^{-1} \|\eta\|_\infty. \tag{30.5}$$

The bound is independent of the collision count, exact labels, and radius. No section-distance factor is inserted in W^ε.

Proof. All guard factors lie in $[0,1]$. Apply the product rule along a compact path and use Proposition 29.2; for clearance factors use the almost-everywhere chain rule for Lipschitz functions on an interval. The collision and flight depth schedules are exactly those in the original guard, not new schedules. Each depth occurs at most twice, so

$$\sum_{i=0}^{m} \frac{|Dc_i|}{q_0^{d_i/4}} \le \frac{4C_c}{1-\beta}\|\eta\|_\infty, \qquad \sum_{i=0}^{m-1} \frac{|D\Delta_i|}{q_0^{d'_i/4}} \le \frac{4C_\Delta}{1-\beta}\|\eta\|_\infty.$$

Multiplication by M_Θ/ε proves (30.5). The inequality $\beta < 1$ follows, for example, from $\sigma < q_0 < 1$. The clearance is locally Lipschitz even at a nearest-disk switch, so there is no omitted jump contribution at such a switch. Actual section boundaries and collision-word boundaries are not crossed by this argument; their one-sided physical traces retain their earlier status. □

30.2. **Roofwise variation and physical normalization.** Keep $\lambda_\circ = 9\sqrt{19}/200$ and $\Lambda_\circ = 47/6$. Define

$$C_{\mathrm{reg}} = \frac{\Lambda_\circ}{\lambda_\circ}\left(C_\omega + \frac{2C_F}{\lambda_\circ} + C_W\right). \tag{30.6}$$

Theorem 30.4 (Native regular variation on every actual roof-angle segment)**.** *On every regular segment of the original physical pairing space, with $u = F$, $e = \nabla F/|\nabla F|$ fixed along the segment, $z = |\nabla F| > 0$, and $A = \omega/(4\pi c_* \chi)$, the original density $f^\varepsilon = AW^\varepsilon$ is locally absolutely continuous and*

$$|\partial_u f^\varepsilon| \le C_{\mathrm{reg}}(1+\varepsilon^{-1})\frac{AI_m}{z} \quad \text{almost everywhere.} \tag{30.7}$$

This is precisely (3.3)*, with a finite explicit constant depending only on the original geometry and cutoff. Its proof uses no pressure, curve-growth, mixing, or local-limit theorem.*

Proof. The two-dimensional determinant defining χ equals $e^{\perp\mathsf{T}} G e^\perp$, as in the retained Gauss-coordinate calculation. Thus $\chi \ge \lambda_\circ$. Differentiating on a fixed-angle segment, not in an arbitrary direction with changing e, gives

$$\left\| \frac{dq}{du} \right\|_\infty = \left\| \frac{G^{-1}e}{ze^{\mathsf{T}}G^{-1}e} \right\|_\infty \le \frac{\Lambda_\circ}{\lambda_\circ z}, \qquad |\partial_u \chi| \le 2C_F \left\| \frac{dq}{du} \right\|_\infty .$$

The latter bound is Theorem 29.3; there is no angular derivative term because e is fixed. By Proposition 30.2,

$$|\partial_u \log A| \le \left(C_\omega I_m + \frac{2C_F}{\lambda_\circ} \right) \frac{\Lambda_\circ}{\lambda_\circ z}.$$

The factor $4\pi c_*$ is the unchanged physical probability normalization and is constant under this differentiation. Using $0 \le W^\varepsilon \le 1$ and Theorem 30.3 now yields

$$|\partial_u (AW^\varepsilon)| \le \frac{A\Lambda_\circ}{\lambda_\circ z} \left(C_\omega I_m + \frac{2C_F}{\lambda_\circ} + \frac{C_W}{\varepsilon} \right).$$

Since $I_m \ge 1$, this implies (30.7) with (30.6). Compact subsegments have positive incidences and $z > 0$, so all differentiations and absolute-continuity claims are legitimate there. The uniform constants do not depend on those compact-subsegment minima. □

Corollary 30.5 (Explicit integration up to the actual endpoint)**.** *For two roof values in one original regular fixed-angle segment,*

$$|f^\varepsilon(u_2) - f^\varepsilon(u_1)| \le C_H(1+\varepsilon^{-1})\sigma^{(m-2)_+}\sqrt{|u_2-u_1|}, \quad C_H = \frac{500 C_{\mathrm{reg}}\sqrt{2\Lambda_\circ}}{4\pi c_* \lambda_\circ^2}. \tag{30.8}$$

This supplies a native regular-interior premise for the retained endpoint-trace results, rather than a new claim of zero trace on a bulk-null unvisited or section boundary.

Proof. Use the already proved physical trajectory bound $\omega I_m \le 500\sigma^{(m-2)_+}$ in Corollary 4.4, $\chi \ge \lambda_\circ$, and integrate (30.7). Lemma 5.1 gives $\int_{u_1}^{u_2} du/z \le \lambda_\circ^{-1}\sqrt{2\Lambda_\circ |u_2-u_1|}$. This proves (30.8) on compact subsegments and then at one-sided limits. Identification of those limits with the actual physical pairing and treatment of simultaneous singular endpoints remain the retained trace construction; no derivative of a grazing collision map is introduced here. □

Proposition 30.6 (Complete positive variation domination, before any supremum)**.** *On the retained countable one-copy sheet space, let $h \ge 0$ be any Borel selector (including the original exact record and actual roof). Then, allowing both sides to be infinite,*

$$\int h(u,\theta)|\partial_u f^\varepsilon(u,\theta)|\, du\, d\theta \le C_{\mathrm{reg}}(1+\varepsilon^{-1}) \int h(u,\theta)\frac{AI_m}{z}(u,\theta)\, du\, d\theta. \tag{30.9}$$

For selectors h pulled back from physical states, the right side is $C_{\mathrm{reg}}(1+\varepsilon^{-1})\int_\Pi h\, I_m z^{-1}\, d\nu_R^*$ under the original coarea convention.

Proof. Multiply (30.7) by h on compact regular subsegments, integrate, then exhaust each open segment and the countable sheet space. All quantities are nonnegative, so monotone convergence proves (30.9) without a word-count factor or a signed-series exchange. The final identification is precisely the retained physical one-copy coarea input. It is not a new construction of pairing multiplicity and does not identify the paired minimum of endpoint traces with the interior density. □

30.3. **What is closed, and the unchanged remaining mathematical limit.** The regular-variation clause of Input 3.1 now has a full native geometric proof. The previous trace, clearance, radial, and critical coefficient theorems remain in the manuscript and are not reclassified as new results. In particular the two averaged clauses of that input are not consequences of bounded D^3F. Neither (30.9) nor the per-trajectory exponential factor in (30.8) can be multiplied by a physical-word count to obtain a normalized roof-density estimate.

The original unfinished assertion is still (15.8), namely the same positive low-load shortage after subtraction of the actual nongrazing capacity, with the complete physical sum and roof essential supremum preceding the collision limsup, and the outer limit in B last. A proof must still supply the complete normalized same-physical-roof source or a valid capacity comparison, with the actual pairing endpoint measure and the first-order clearance and incidence-only sources retained. The spectral/arithmetic and conditional-denominator conclusions retain their exact hypotheses. This section removes a specific local optical proof dependency; it does not replace this missing global result by a new residual or a relaxed window. The full raw pointwise local limit theorem and final four-coordinate inversion are not asserted as completed.

31. A grazing-uniform resolvent for the physical reflection equations

We retain the actual circular-scatterer trajectories, radius interval, unit-speed clock, and all exact selectors of (1.1). The issue here is the derivative of an endpoint boundary-value problem, not a derivative bound for the expanding forward collision map. All derivatives are taken at regular physical points of a fixed actual itinerary. No unvisited disk is removed from its admissible domain.

Put $a_* = R_+/\tau_- = 47/6$, $\lambda = 50/47$, and $I = \{1, \dots, m-1\}$. A sum over I is empty when $m = 1$. Write $X_i = x_{i+1} - x_i$, $\vartheta_i = \arg X_i$, using local real lifts of these angles. The specular law is equivalent, on the physical branch, to constancy of

$$E_i(\alpha) = 2\alpha_i - \vartheta_{i-1}(\alpha) - \vartheta_i(\alpha), \qquad i \in I. \tag{31.1}$$

Its constant is an odd multiple of π. The value of that multiple does not enter a derivative; it is fixed on each local lift. The reflection equation is not obtained by assuming that division by a small incidence is harmless.

Lemma 31.1 (The full physical Jacobian)**.** *At an actual regular trajectory, set $w_i = \tau_i^{-1}$ and $c_i = \sqrt{1 - p_i^2}$. The interior Jacobian $B = D_{\alpha_I} E$ is tridiagonal, with*

$$B_{ii} = 2 + Rc_i(w_{i-1} + w_i), \tag{31.2}$$

$$B_{i,i-1} = Rw_{i-1}c_{i-1}, \tag{31.3}$$

$$B_{i,i+1} = Rw_i c_{i+1}, \tag{31.4}$$

$$(D_{\alpha_0} E)_1 = Rw_0 c_0, \tag{31.5}$$

$$(D_{\alpha_m} E)_{m-1} = Rw_{m-1}c_m. \tag{31.6}$$

Only entries with indices in I belong to B; the last two entries are boundary forcing. When $m = 2$ both boundary terms are in the same row. If $H = D^2S$ is the full stationary-action Hessian, then

$$H_{II} = R\,\mathrm{diag}(c_i)_{i\in I}\,B. \tag{31.7}$$

Proof. For an oriented physical flight $v_i = X_i/\tau_i$, the differential of its direction is $D\vartheta_i = v_i^{\perp}\cdot DX_i/\tau_i$. The departure and arrival normal components are respectively c_i and $-c_{i+1}$. Since $n'(\alpha) = n(\alpha)^{\perp}$,

$$\partial_{\alpha_i}\vartheta_i = -Rc_iw_i, \qquad \partial_{\alpha_{i+1}}\vartheta_i = -Rc_{i+1}w_i.$$

Applying these two identities to both flights in (31.1) proves every entry, including both boundary terms. Comparing with (4.2) proves (31.7). Thus a factor c_i in the action Hessian is a row factor in the gradient formulation; it is absent from the reflection-angle Jacobian. This identity does not assert invertibility of H_{II} when a limiting c_i is zero. □

Theorem 31.2 (Column coercivity and a positive resolvent)**.** *Let $\mathcal{S} = \mathrm{diag}((-1)^i)_{i\in I}$ and $A = \mathcal{S}B\mathcal{S}$. For every $m \geq 2$, every original radius, and every actual regular itinerary, A^{-1} is entrywise nonnegative and*

$$\|B^{-1}f\|_1 \leq \tfrac{1}{2}\|f\|_1. \tag{31.8}$$

More generally, for positive weights v_i such that $\lambda^{-1} \leq v_{i+1}/v_i \leq \lambda$, one has

$$\sum_{i\in I} v_i|(B^{-1}f)_i| \leq \sum_{i\in I} v_i|f_i|. \tag{31.9}$$

In particular,

$$\sum_{i\in I} \lambda^{|i-j|}(A^{-1})_{ij} \leq 1 \quad (j \in I). \tag{31.10}$$

The assertions extend algebraically to simultaneous values $c_i = 0$ by the same formulas. No lower incidence cutoff is required.

Proof. The off-diagonal entries of A are nonpositive. Let $D_{jj} = 2 + Rc_j(w_{j-1} + w_j)$ and write $A = (\mathrm{Id} - N)D$, where N has zero diagonal and nonnegative off-diagonal entries. Its column sum is at most

$$\frac{Rc_j(w_{j-1} + w_j)}{2 + Rc_j(w_{j-1} + w_j)} \leq \frac{a_*}{1 + a_*} = \frac{47}{53} < 1.$$

Consequently $A^{-1} = D^{-1}\sum_{k\geq 0} N^k$ exists and is nonnegative. This is an algebraic matrix series, not a probabilistic replacement of the Lorentz gas. Missing interior neighbors at the two ends only decrease the column sum.

Each column of A has sum at least 2: $\mathbf{1}^{\mathsf{T}}A \geq 2\mathbf{1}^{\mathsf{T}}$. Multiplication on the right by the nonnegative inverse gives $\mathbf{1}^{\mathsf{T}}A^{-1} \leq \mathbf{1}^{\mathsf{T}}/2$. Since $|B^{-1}f| \leq A^{-1}|f|$, this proves (31.8). The argument uses *column*, not row, dominance. Row dominance can fail when one incidence is much smaller than its neighbors.

For the weighted assertion, a column calculation gives

$$(v^{\mathsf{T}}A)_j \geq v_j\{2 - (\lambda - 1)Rc_j(w_{j-1} + w_j)\} \geq v_j\{2 - 2a_*(\lambda - 1)\} = v_j.$$

Here $2a_*(\lambda - 1) = 1$ exactly. Right multiplication by A^{-1} proves (31.9). Taking $v_i = \lambda^{|i-j|}$ proves (31.10). All denominators in this argument are at least 2. It therefore remains valid when any collection of incidences is zero. No inverse c_i has been introduced in taking that limit. □

31.1. **Complete higher endpoint derivatives.** For a vector $u = (u_0, \dots, u_m)$ set $V_i(u) = |u_i| + |u_{i+1}|$ and $\|u\|_{1,v} = \sum_{i=0}^{m} v_i |u_i|$. We use $v_i = \lambda^{d_i}$, $d_i = \min(i, m-i)$, including both endpoints. For $r \geq 1$ define the explicit constants

$$T_r = \sum_{k=1}^{r} \left\{ {r \atop k} \right\} (k-1)! a_*^k, \qquad T_1 = a_*, \quad T_2 = a_* + a_*^2, \quad T_3 = a_* + 3a_*^2 + 2a_*^3, \tag{31.11}$$

where $\left\{ {r \atop k} \right\}$ counts partitions of $\{1, \dots, r\}$ into k nonempty blocks.

Lemma 31.3 (Local angle tensors, with every chain-rule term)**.** *For each flight and each $r \geq 1$,*

$$|D^r \vartheta_i[u^1, \dots, u^r]| \leq T_r \prod_{h=1}^{r} V_i(u^h).$$

For $r \geq 2$, the full vector $E = (E_i)_{i \in I}$ consequently satisfies

$$\|D^r E[u^1, \dots, u^r]\|_1 \leq 4 T_r \prod_{h=1}^{r} \|u^h\|_1, \tag{31.12}$$

$$\|D^r E[u^1, \dots, u^r]\|_{1,v} \leq 2(1+\lambda) T_r \prod_{h=1}^{r} \|u^h\|_{1,v}. \tag{31.13}$$

Proof. Identify the plane with $\mathbb{C}$ and use a local logarithm of $X_i \neq 0$. Its kth derivative is $(-1)^{k-1}(k-1)! X_i^{-k}$; taking the imaginary part gives the direction derivative. Every derivative of $X_i = O_{i+1} - O_i + R(e^{i\alpha_{i+1}} - e^{i\alpha_i})$ of order $b \geq 1$ has absolute value at most $R_+ \prod_{h=1}^{b} V_i(u^h)$. The composition rule indexed by set partitions gives exactly the sum in (31.11), using $|X_i| \geq \tau_-$. This proves the first assertion without omitting mixed derivatives.

Each flight appears in at most two interior reflection equations. Also $V_i(u) \leq \|u\|_1$ and $\sum_i V_i(u) \leq 2\|u\|_1$. Summing one factor and bounding the others proves the first norm inequality. For the second, the sum of the weights of the adjacent reflection equations is at most $v_i + v_{i+1}$, and

$$(v_i + v_{i+1}) V_i(u) \leq (1+\lambda)(v_i |u_i| + v_{i+1} |u_{i+1}|).$$

Summing over flights counts each vertex at most twice. The other factors are bounded by their weighted norms, since $v_i \geq 1$. This proves the second inequality, including the boundary flights. □

Let $\mathfrak{P}_r$ denote the set of all set partitions of $\{1, \dots, r\}$. Define two finite positive recursions

$$C_1 = 2 + a_* = 59/6, \qquad C_r = 2 \sum_{\substack{\pi \in \mathfrak{P}_r \\ |\pi| \geq 2}} T_{|\pi|} \prod_{Q \in \pi} C_{|Q|}, \tag{31.14}$$

$$M_1 = 2 + 2\lambda a_* = 56/3, \qquad M_r = 2(1+\lambda) \sum_{\substack{\pi \in \mathfrak{P}_r \\ |\pi| \geq 2}} T_{|\pi|} \prod_{Q \in \pi} M_{|Q|}, \tag{31.15}$$

$$r \geq 2. \tag{31.16}$$

The symbols C_r and M_r denote the jet constants in this section and the next, not a redefinition of earlier flight-tensor constants. Every block in these sums has size less than r, so the recursions are not circular. In particular $C_2 = 2T_2 C_1^2$ and $C_3 = 2T_3 C_1^3 + 6T_2 C_1 C_2$.

Theorem 31.4 (All finite endpoint jets on actual physical branches)**.** *For every integer $r \geq 1$, every $m \geq 1$, and every actual regular stationary lift $\alpha(q)$, $q = (\alpha_0, \alpha_m)$,*

$$\|D_q^r \alpha[\eta^1, \dots, \eta^r]\|_1 \leq C_r \prod_{h=1}^{r} \|\eta^h\|_\infty, \tag{31.17}$$

$$\sum_{i=0}^{m} \lambda^{d_i} |D_q^r \alpha_i[\eta^1, \dots, \eta^r]| \leq M_r \prod_{h=1}^{r} \|\eta^h\|_\infty. \tag{31.18}$$

The constants depend on derivative order and the original geometric interval only. They are independent of collision count, itinerary, small positive incidence, return/displacement labels, or roof window.

Proof. For $r = 1$, the interior equation is $BJ_I = -D_q E\,\eta$. Its right side has norm at most $a_*(|\eta_0| + |\eta_m|)$ and weighted norm at most $\lambda a_*(|\eta_0| + |\eta_m|)$. Theorem 31.2, followed by adding the two actual endpoint derivatives, gives C_1 and M_1. This also covers $m = 2$, when the two boundary terms share a row. For $m = 1$ only the two endpoint derivatives occur.

For $r \geq 2$ the endpoint derivatives of order r vanish. Differentiate $E(\alpha(q)) =$ constant in the r prescribed directions. Isolating the partition with one block gives

$$B(D_q^r \alpha)_I = -\sum_{\substack{\pi \in \mathfrak{P}_r \\ |\pi| \geq 2}} D^{|\pi|} E[D_q^{|Q|} \alpha[\eta^h : h \in Q] : Q \in \pi].$$

This identity contains every mixed chain-rule term exactly once. Induction, Lemma 31.3, and respectively the $1/2$ and 1 inverse bounds yield precisely (31.16). Local smoothness follows already from the ordinary implicit function theorem and invertibility of B; the induction establishes uniform bounds for its derivatives. No assertion of a common physical continuation ball is used. □

Corollary 31.5 (Fixed-itinerary grazing limits, without a fictitious event)**.** *Fix m and a sequence of actual regular trajectories with the same centers and convergent collision angles and radius. Suppose the limit is in the original radius interval. Any number of interior incidences may tend to zero. The local reflection equations at the limiting configuration have a unique smooth formal solution in the endpoint angles; every finite endpoint jet of the original physical branch has that formal limit. The bounds in* (31.18) *persist. This does not make the formal continuation an admissible billiard trajectory beyond its actual first boundary.*

Proof. Distinct successive disk centers and $R \leq R_+$ keep every limiting flight length at least τ_-. Local direction logarithms are smooth there. The limiting Jacobian is invertible by Theorem 31.2, including simultaneous zero incidences. The implicit function theorem at that configuration supplies the formal reflection solution, with the radius also treated as a smooth parameter. Local uniqueness identifies every sufficiently close member of the given physical sequence with it. Its finite jets converge, and their uniform bounds pass to the limit. An extra disk can obstruct the formal solution, and the prescribed section or return event can end at the boundary. Neither obstruction is discarded; no collision-map derivative at tangency is asserted. □

32. Fourth physical variation and the exact high-order cancellation

The third-variation result of Section 29 is retained. It does not by itself bound a fourth derivative: differentiating the stationary lift introduces its second derivative. The reflection resolvent now controls that derivative without a small-incidence loss. The following proof includes those terms rather than continuing the third-order Schur cancellation beyond its valid scope.

Set

$$D_3 = R_+ + 3R_+^2/\tau_- + 3R_+^3/\tau_-^2, \tag{32.1}$$

$$D_4 = R_+ + 7R_+^2/\tau_- + 18R_+^3/\tau_-^2 + 15R_+^4/\tau_-^3, \tag{32.2}$$

$$\mathcal{F}_4 = 2D_4C_1^4 + 6D_3C_2C_1^2, \qquad C_1 = 59/6, \quad C_2 = 2(a_* + a_*^2)C_1^2. \tag{32.3}$$

All constants concern the unchanged interval $R \in [9/20, 47/100]$.

Lemma 32.1 (Fourth tensor of one physical flight)**.** *For $r = 3, 4$ and arbitrary collision-angle variations,*

$$|D^r\tau_i[u^1, \dots, u^r]| \le D_r \prod_{h=1}^{r} V_i(u^h), \qquad |D^r S[u^1, \dots, u^r]| \le 2D_r \prod_{h=1}^{r} \|u^h\|_1.$$

Proof. For $N(X) = |X|$, put $v = X/|X|$ and $P = \mathrm{Id} - vv^{\mathsf{T}}$. Then $DN = v^{\mathsf{T}}$, $D^2N = P/|X|$, and

$$D^3N[a, b, c] = -|X|^{-2}\{(v \cdot a)b^{\mathsf{T}}Pc + (v \cdot b)a^{\mathsf{T}}Pc + (v \cdot c)a^{\mathsf{T}}Pb\}.$$

Thus $\|D^3N\| \le 3|X|^{-2}$. Differentiating this display gives $\|D^4N\| \le 15|X|^{-3}$: differentiating the radial prefactor, the three copies of v, and the three copies of P contributes at most 6, 3, and 6, respectively, since $Dv[d] = Pd/|X|$ and $DP[d] = -\{(Pd)v^{\mathsf{T}} + v(Pd)^{\mathsf{T}}\}/|X|$.

Each derivative of a circular endpoint position has size at most R_+ times the product of its angular variations. The fourth composition rule has respectively $1, 7, 6, 1$ set partitions into $1, 2, 3, 4$ blocks. Multiplying by the norm-derivative bounds $1, |X|^{-1}, 3|X|^{-2}, 15|X|^{-3}$ proves the displayed D_4. The third-order rule gives D_3, as in the retained proof. Finally $\sum_i \prod_h V_i(u^h) \le 2\prod_h \|u^h\|_1$, by summing one factor and taking the suprema of the others. This proves both full-action inequalities independently of m. □

Theorem 32.2 (Collision-uniform fourth derivative of the reduced action)**.** *On every actual regular endpoint branch and for all four endpoint directions,*

$$|D_q^4F[\eta^1, \eta^2, \eta^3, \eta^4]| \le \mathcal{F}_4 \prod_{h=1}^{4} \|\eta^h\|_\infty. \tag{32.4}$$

There is no interior-incidence cutoff. In particular,

$$\|D_q^2G[\eta, \zeta]\|_{\mathrm{op}} \le 2\mathcal{F}_4\|\eta\|_\infty\|\zeta\|_\infty.$$

At fixed-itinerary multiple-grazing limits, these tensors have the one-sided limits specified in Corollary 31.5.

Proof. Write $J_a = D_q\alpha[\eta^a]$ and $K_{ab} = D_q^2\alpha[\eta^a, \eta^b]$. The latter has zero endpoint coordinates. Differentiate the exact third-order identity (29.13) in direction η^4. One obtains

$$\begin{aligned} D_q^4F[\eta^1, \eta^2, \eta^3, \eta^4] &= D^4S[J_1, J_2, J_3, J_4] \\ &\quad + D^3S[K_{14}, J_2, J_3] + D^3S[J_1, K_{24}, J_3] + D^3S[J_1, J_2, K_{34}]. \end{aligned} \tag{32.5}$$

The three terms containing K do not vanish. Their omission would be an incorrect fourth-order use of stationary elimination. Equivalently, let $g_{ab} = (D^3S[\mathbf{e}_i, J_a, J_b])_{i\in I}$. Twice differentiated stationarity gives $H_{II}(K_{ab})_I = -g_{ab}$. Hence the last three terms are

$$-g_{14}^{\mathsf{T}}H_{II}^{-1}g_{23} - g_{24}^{\mathsf{T}}H_{II}^{-1}g_{13} - g_{34}^{\mathsf{T}}H_{II}^{-1}g_{12}.$$

This form verifies the symmetry but is not used to bound H_{II}^{-1} near grazing. The reflection equation gives instead $\|J_a\|_1 \le C_1\|\eta^a\|_\infty$ and $\|K_{ab}\|_1 \le C_2\|\eta^a\|_\infty\|\eta^b\|_\infty$. Lemma 32.1 applied to all four terms of (32.5) gives exactly (32.3). For $m = 1$, $K = 0$ and the same bound holds. Each entry of $D_q^2G[\eta, \zeta]$ is bounded by $\mathcal{F}_4\|\eta\|_\infty\|\zeta\|_\infty$; a two-by-two matrix has operator norm at

most twice its largest entry. The fixed-itinerary limit assertion follows from the smooth formal reflection continuation already proved, restricted to the original physical branch. □

Corollary 32.3 (Uniform Taylor errors on the actual physical domain)**.** *Suppose the entire straight segment $q + sh$, $0 \le s \le 1$, lies in a single actual regular branch. Then*

$$\left|F(q+h) - F(q) - DF(q)[h] - \tfrac{1}{2}D^2F(q)[h,h] - \tfrac{1}{6}D^3F(q)[h,h,h]\right| \le \tfrac{1}{24}\mathcal{F}_4\|h\|_\infty^4, \tag{32.6}$$

$$\|G(q+h) - G(q) - DG(q)[h]\|_{\mathrm{op}} \le \mathcal{F}_4\|h\|_\infty^2. \tag{32.7}$$

The same bounds hold by a one-sided limit at a fixed-itinerary grazing endpoint. They require neither a pressure estimate nor a prescribed positive lower bound on intermediate incidences.

Proof. Apply the one-dimensional integral Taylor remainder to the stated segment. The remainder integral for F is $\frac{1}{6}\int_0^1 (1-s)^3 D^4F(q+sh)[h,h,h,h]\,ds$. For G, integrate its second derivative with weight $1-s$. The bounds follow from Theorem 32.2; its one-sided limit justifies the endpoint statement. The hypothesis that the segment stays physical is not inferred from the derivative bound. □

32.1. **What this resolves in the original source problem.** The higher reflection jets and (32.5) remove an actual local analytic obstruction: repeated stationary differentiation need not incur powers of the smallest visited incidence or an uncontrolled factor equal to the collision count. They also justify the displayed higher-order Taylor errors on every existing physical chart. These are uniform statements about all actual regular itineraries, not assertions from a finite sample or a replacement Markov model.

They do not count physical charts or continue through a newly hit disk. The least unvisited clearance can still cut a chart, the original section/return selector can still jump, and a positive one-sided paired trace is not automatically a bulk-collision submeasure. In particular no norm of the algebraic reflection resolvent is a pressure estimate for the twisted transfer operator.

The remaining target is still (15.8), with $B \to \infty$ outermost and the full one-copy physical sum made before the original radius, exact-label and roof suprema and collision limsup. Neither (31.18) nor (32.4) supplies the missing m^{-2} common-physical-roof summation or nongrazing capacity. The two averaged clauses of Input 3.1, the stronger pressure/pairing interfaces and the original exact arithmetic and conditioned-denominator hypotheses retain their distinct status. No theorem or target is deleted to hide these remaining obligations.

33. Uniform analytic lifts and genuine physical continuation tubes

All radii, disks, collision labels, physical clocks and the original section are unchanged. The constants in this section are deliberately conservative; their purpose is a positive radius independent of the number of actual collisions. Unlike a bound for each fixed derivative, the first theorem provides one common domain of analytic continuation. A separate geometric argument determines when this continuation is a true billiard trajectory. The latter step is not inferred from analyticity.

Let $a = 47/6$, $\lambda = 50/47$, $v_i = \lambda^{\min(i,m-i)}$, and set

$$\begin{aligned} K &= 2(1+\lambda)(4a+16a^2), & r_* &= \min\{[8(1+a)]^{-1}, (8K)^{-1}\}, \\ \eta_* &= \frac{r_*}{8(1+\lambda a)}, & P_* &= 2(1+\lambda a), \end{aligned} \tag{33.1}$$

$$\begin{aligned} C_{\mathrm{inc}} &= P_*[1+2a(1+\lambda)], & C_{\mathrm{clr}} &= R_+P_*, \\ \delta_* &= \min\{\eta_*, (2C_{\mathrm{inc}})^{-1}, (2C_{\mathrm{clr}})^{-1}\}. \end{aligned} \tag{33.2}$$

The symbols r_*, η_*, δ_* in this section are auxiliary endpoint-domain constants, not the original microscopic width r_B. For complex vectors put $\|z\|_v = \sum_{i=0}^{m} v_i|z_i|$.

Lemma 33.1 (Uniform complex neighborhood of the reflection equations)**.** *At every actual regular reference trajectory α^0, the reflection vector E of* (31.1) *has a holomorphic continuation to $\|\alpha - \alpha^0\|_v < r_*$. On that ball,*

$$\|D^2E(\alpha)[z,w]\|_v \le K\|z\|_v\|w\|_v.$$

The output norm uses only interior indices. Both complexified flight coordinates stay nonzero, uniformly in collision count and incidence.

Proof. Write the two complexifications of a flight as

$$X_i^+ = (O_{i+1} - O_i)_x + i(O_{i+1} - O_i)_y + R(e^{i\alpha_{i+1}} - e^{i\alpha_i}),$$

and let X_i^- be the same expression with i replaced by $-i$. For complex angles X_i^- is not defined by complex conjugation. At the real reference point $|X_i^+| = |X_i^-| = \tau_i \ge \tau_-$. For $\|\alpha - \alpha^0\|_v \le r_*$,

$$|X_i^\pm(\alpha) - X_i^\pm(\alpha^0)| \le R_+e^{r_*}(|\alpha_i - \alpha_i^0| + |\alpha_{i+1} - \alpha_{i+1}^0|) \le 2R_+r_* < \tau_-/4.$$

Here $r_* < 1/8$, $e^{r_*} < 2$, and $2ar_* < 1/4$. Choose the logarithms on these nonvanishing disks and put $\vartheta_i = (\log X_i^+ - \log X_i^-)/(2i)$, with the additive real lift fixed at the reference point. This agrees with the real flight angle. Each first or second derivative of $X_i^\pm$ has bound $2R_+\prod_h(|z_i^h| + |z_{i+1}^h|)$, and $|X_i^\pm| \ge \tau_-/2$. The two logarithmic derivative terms are consequently bounded by $4a$ and $16a^2$. The weighted flight summation in Lemma 31.3 then gives $2(1+\lambda)(4a + 16a^2)$, without a factor m. □

Theorem 33.2 (A collision-uniform analytic endpoint domain)**.** *Fix any actual regular m-flight trajectory, $R \in [R_-, R_+]$, and its designated disk centers. For every complex endpoint increment $h = (h_0, h_m)$ with $s = |h_0| + |h_m| < \eta_*$ there is a unique solution of the same locally lifted reflection equations in the interior ball $\|\alpha_I - \alpha_I^0\|_v \le r_*/2$. It depends holomorphically on h, is real for real h, and satisfies*

$$\sum_{i=0}^{m} v_i|\alpha_i(h) - \alpha_i^0| \le P_*s. \tag{33.3}$$

Uniqueness is asserted in this ball, not among arbitrary distant solutions of the word equations. A simultaneous visited-grazing limit of reference configurations has the same formal statement. No physical passage through grazing is asserted.

Proof. For $m = 1$ the assertion is just the two endpoints. Otherwise let $B_0 = D_IE(\alpha^0)$ and let Q_0h be its boundary forcing. Theorem 31.2 gives $\|B_0^{-1}\|_{v\to v} \le 1$ also over complex vectors, and $\|Q_0h\|_v \le \lambda as$. Write

$$E(\alpha^0 + (h,u)) - E(\alpha^0) = B_0u + Q_0h + N(h,u).$$

On the stated ball $s + \|u\|_v < r_*$, the preceding lemma and integration along the complex line segment give $\|N(h,u)\|_v \le K(s + \|u\|_v)^2/2$ and $\|D_uN\| \le K(s + \|u\|_v)$. The map $u \mapsto -B_0^{-1}(Q_0h + N(h,u))$ sends $\|u\|_v \le r_*/2$ into itself: its forcing is at most $r_*/8$ and its nonlinear part is at most $r_*/16$. Its Lipschitz constant is at most $Kr_* \le 1/8$. The successive iterates from zero thus converge geometrically, uniformly on every smaller endpoint domain, to the unique fixed point. The iterates are holomorphic, so their locally uniform limit is holomorphic. Real coefficients and uniqueness give reality for real h. For the solution, $s + \|u\|_v \le r_*$ yields

$$\|u\|_v \le \lambda as + \tfrac{1}{16}(s + \|u\|_v), \qquad s + \|u\|_v \le \tfrac{16}{15}(1 + \lambda a)s \le P_*s.$$

This proves (33.3). At a fixed-word grazing limit the real limiting flight lengths are still at least τ_- and the limiting reflection inverse retains its weighted bound. The identical contraction

proof applies there. It concerns reflection equations only until physical admissibility is verified separately below. □

Theorem 33.3 (Physical tubes for the original depth-dependent guard)**.** *Put $r = q_0^{1/4}$, with the unchanged $q_0 = 47/53$. Suppose an actual regular reference orbit satisfies, for $0 < \varepsilon \le 1/4$,*

$$c_i^0 \ge \varepsilon r^{d_i} \quad (0 \le i \le m), \qquad \Delta_i^0 \ge \varepsilon r^{d_i'} \quad (0 \le i < m). \tag{33.4}$$

For every real h with $|h_0| + |h_m| < \delta_* \varepsilon$, the analytic lift is an actual regular trajectory with exactly the same ordered collisions and displacement. Every designated incidence and every least third-disk clearance obeys

$$c_i(h) \ge \tfrac{1}{2}\varepsilon r^{d_i}, \qquad \Delta_i(h) \ge \tfrac{1}{2}\varepsilon r^{d_i'}. \tag{33.5}$$

The physical clock is the sum of the true lengths. The original section, visit count and terminal test are evaluated at the resulting physical orbit; they are not declared constant on the tube. Thus any Borel exact-record selector can be retained pointwise without adding a section-distance condition to the guard.

Proof. The norm bound in (33.3) gives $|\alpha_i(h) - \alpha_i^0| \le P_* s \lambda^{-d_i}$. For real angles the corresponding endpoint displacement is at most $R_+ P_* s \lambda^{-d_i}$. The Hausdorff distance between the two closed flight segments is at most $C_{\mathrm{clr}} s \lambda^{-d_i'}$. Distance to every third disk is one-Lipschitz for this distance; taking the infimum over all centers except exactly the two incident centers preserves it. This controls every unvisited obstacle at once, including switches of the minimizing obstacle or closest segment point.

For a nonzero real flight, comparison of normalized vectors gives $|X/|X| - X^0/|X^0|| \le 2|X - X^0|/|X^0|$. Since adjacent depth weights have ratio at most λ, each signed departure normal component, and each negative arrival normal component, changes by at most $C_{\mathrm{inc}} s \lambda^{-d_i}$. The same bound holds for the signed tangential component. Initially both physical normal components at an interior reflection equal c_i^0. The exact rational inequality

$$\lambda^4 q_0 = \frac{50^4}{47^3\, 53} > 1$$

implies $\lambda^{-d} \le r^d$. The choice of δ_* therefore proves both half-margins, first for signed normal components and segment clearances of the formal real lift.

Their strict positivity makes that lift physical. A positive outgoing normal component keeps a segment outside its initial disk; a negative incoming normal component keeps it outside its final disk until the stated endpoint. All other disks have strictly positive clearance. Thus each segment is the true next free flight, not a flight with a hidden earlier collision. The fixed reflection equations, together with these normal signs, give specular reflection at every interior endpoint. Concatenation proves the actual ordered word and unchanged displacement, with no deleted collision. Its unit-speed duration is its geometric length. Section membership can jump as its boundary is crossed, so the original selector is retained rather than differentiated or frozen. This makes no assertion about the source outside the tube. □

The guard margins in (33.4) already occur in the original product (1.2); they are not a new conditioning event for the target probability. The theorem establishes an explicit domain property of that existing geometric class. Points failing the margins, and all one-sided singular traces, remain present in the full source problem. No claim that their normalized mass tends to zero is made by constructing these tubes.

34. Optical distortion and summation at one common physical roof

The synchronized momentum sum in (28.7) is not itself a sum at one physical roof: a fixed roof generally selects a different momentum radius on every critical branch. We now supply an explicit comparison mechanism on the uniform physical tubes just constructed. It uses logarithmic optical distortion before summation, not an exchange of the roof supremum and the sum. The only global root input is the retained physical kernel Theorem 26.4, with its native growth dependencies left unchanged.

Set $\sigma = 47/59$, $r = q_0^{1/4}$ and $\beta = \sigma/r < 1$. The constants J_*, C_c, C_F retain the exact meanings given in Section 29. Define

$$g = R_-/2 = 9/40, \qquad U = 320/9,$$
$$D_* = \frac{8C_c}{1-\beta} + \frac{4aJ_*}{1-\sigma} + \frac{4C_F}{g}, \qquad C_{\mathrm{dist}} = \exp(D_*\delta_*/2). \tag{34.1}$$

No constant in this display depends on collision count, labels or an unproved lower bound for an interior incidence.

Lemma 34.1 (Relative distortion of the actual inverse momentum weight)**.** *On any convex physical endpoint domain with $c_i \ge (\varepsilon/2)r^{d_i}$ and $c_0, c_m \ge 1/2$, put $G = D_q^2 F$, $e = G_{0m}$ and*

$$u(q) = \frac{R|e(q)|}{\det G(q)} = |\partial_{\alpha_0} p_m|^{-1}.$$

For $0 < \varepsilon \le 1/4$,

$$|D_q \log u[\eta]| \le D_* \varepsilon^{-1} \|\eta\|_\infty. \tag{34.2}$$

Consequently $u(q) \le e^{D_*\|q-q_x\|_\infty/\varepsilon} u(q_x)$ whenever the connecting segment is in that domain.

Proof. The true optical formula is $|e| = R^2 c_0 c_m/\mathfrak{b}_m$. Positivity and separate multiaffinity of $\mathfrak{b}_m$, proved in Lemma 30.1, give

$$|D \log |e|| \le \sum_{i=0}^{m} \frac{|Dc_i|}{c_i} + \sum_{i=0}^{m-1} \frac{|D\tau_i|}{\tau_i}.$$

Each incidence occurs once in this bound, including the two endpoints; there is no new product of inverse incidences. With $s_i = \sigma^i + \sigma^{m-i} \le 2\sigma^{d_i}$ and at most two indices at every depth,

$$\sum_{i=0}^{m} s_i r^{-d_i} \le \frac{4}{1-\beta}, \qquad \sum_{i=0}^{m-1} (s_i + s_{i+1}) \le \frac{4}{1-\sigma}.$$

Proposition 29.2 therefore bounds the incidence sum by $8C_c/[\varepsilon(1-\beta)]$ and the flight sum by $4aJ_*/(1-\sigma)$, times $\|\eta\|_\infty$. The physical Schur complement has $G \succeq gI$. By Theorem 29.3, $\|DG[\eta]\|_{\mathrm{op}} \le 2C_F\|\eta\|_\infty$, so

$$|D \log \det G[\eta]| = |\operatorname{tr}(G^{-1} DG[\eta])| \le 4C_F g^{-1} \|\eta\|_\infty.$$

Subtract these logarithmic derivatives in $\log u$ and use $\varepsilon \le 1$ to obtain (34.2). Integration on the actual physical segment proves the last assertion. The identity for u is the unchanged generating-map Jacobian of Theorem 28.2; no radius or section factor is absorbed in it. □

For $0 < \varepsilon \le 1/4$ let $\mathscr{Z}_{\varepsilon,m,R}$ be all distinct regular physical critical roots $p_0 = p_m = 0$ satisfying (33.4). Lattice translation copies and angular seams are identified, so each root is a unique physical initial point. These are actual reversing critical orbits by the retained theorem, not

critical points of an unrelated surrogate action. Their roots are isolated on the initial momentum fibre since $\partial_{\alpha_0} p_m \neq 0$; hence the family is countable. Put

$$s_\varepsilon = \tfrac{1}{2}\delta_* \varepsilon, \qquad \rho_\varepsilon = \min\{1/4,\, g s_\varepsilon/(2R_+)\}, \qquad K_{1/2} = \frac{100}{47}\sqrt{1-1/4}. \tag{34.3}$$

Both positive radii are fixed before taking any collision limsup.

Lemma 34.2 (A common physical momentum disk at every guarded critical root)**.** *For every $x \in \mathscr{Z}_{\varepsilon,m,R}$ there is an actual regular inverse $q_x(\zeta)$ for $|\zeta| < \rho_\varepsilon$, where $\zeta = (p_0, p_m)$ and $q_x(0)$ is the original endpoint pair. It is contained in the physical endpoint ball $|q - q_x(0)|_2 < s_\varepsilon$ and obeys*

$$|q_x(\zeta) - q_x(0)|_2 \le R_+|\zeta|/g, \qquad u_x(\zeta) \le C_{\mathrm{dist}} u_x(0).$$

This is a common domain for all collision words in the stated class, not a separate radius chosen after m is fixed.

Proof. The Euclidean ball of radius s_ε lies strictly inside the ℓ^1 endpoint tube of Theorem 33.3. At a critical center both endpoint tangential components vanish. The signed tangential estimate in that proof and $C_{\mathrm{inc}}\delta_* \le 1/2$ imply $|p_0|, |p_m| \le \varepsilon/2$ on the entire ball; in particular $c_0, c_m \ge 1/2$ and $G \succeq gI$ there. The domain and every segment used next are genuine physical domains, established before using the Hessian.

Given $|\zeta| < \rho_\varepsilon$, minimize $F(q) - R(D\zeta)\cdot(q - q_x(0))$ on the closed endpoint ball, where $D = \mathrm{diag}(-1, 1)$. On its boundary the radial first variation is at least $g s_\varepsilon^2 - R_+|\zeta| s_\varepsilon > 0$. A boundary minimizer is impossible, because moving radially inward would strictly decrease the function. Strong convexity gives one interior minimizer, satisfying $\nabla F = RD\zeta$, and hence the correct endpoint momenta. The ordinary inverse theorem gives its smooth dependence, while strong convexity gives $g|q - q_x(0)|_2 \le R_+|\zeta|$. The optical mixed entry is nonzero, so this is also a regular local parametrization of the original collision source. Lemma 34.1, integrated on the physical radial segment in endpoint space, gives the stated relative bound. In it $\|q - q_x(0)\|_\infty \le s_\varepsilon$ and $s_\varepsilon/\varepsilon = \delta_*/2$. □

Theorem 34.3 (One-copy critical-tube sum at a common physical roof)**.** *For fixed R, m, ε take the union of the actual momentum charts in Lemma 34.2. Give this union its original collision measure, counting each physical initial point once. Retain any Borel selector $0 \le w \le 1$, in particular the full $\Pi_{n,k,m,R}$, both section tests, the original guard and any original physical-roof interval. Let $Q^w_{\varepsilon,m,R}(t)$ be the density of its pushforward by the original total flight time. Then*

$$\|Q^w_{\varepsilon,m,R}\|_{L^\infty(dt)} \le \frac{U C_{\mathrm{dist}} C_{\mathrm{mom}}}{2c_*}. \tag{34.4}$$

This bound is uniform in $\varepsilon \in (0, 1/4]$, all $m \ge 1$, all original radii and exact labels. It makes the complete positive chart sum before taking the physical-roof essential supremum. It is unscaled: it does not assert an m^{-2} gain.

Proof. Enumerate the distinct critical roots. Each chart is a Borel image of its open momentum disk; the endpoint-to-source map is locally invertible since $e \neq 0$. It is injective on the convex endpoint ball: at fixed α_0, $\partial_{\alpha_m} p_0 = -e/R$ has a fixed nonzero sign along the connected vertical slice. Thus a physical source point is represented at most once in a given chart. If charts overlap, assign the physical point to the first chart in the enumeration. These Borel assignments partition the unchanged union; they remove only coordinate multiplicity, never an actual collision or a point of the union. Multiply w by the assignment indicator in each chart, denoting the result by w_x.

Let $a_x = L(x)$ and write $\zeta = re_\theta$. The exact radial coarea in Theorem 28.2 gives, for $r > 0$,

$$\partial_r L_x(re_\theta) = rh_x(re_\theta), \qquad U^{-1} \le h_x \le K_{1/2}^{-1}, \qquad b_x(re_\theta) = \frac{u_x(re_\theta)}{4\pi c_* h_x(re_\theta)}.$$

These formulas hold on the full common disk by the preceding lemma. For fixed t, θ, strict radial monotonicity gives at most one $r_x(t,\theta) \in (0, \rho_\varepsilon)$ per center. These radii are generally different for different centers; we do not identify them. The contribution is zero when no such radius exists. Otherwise

$$w_x b_x(r_x(t,\theta)e_\theta) \le \frac{UC_{\mathrm{dist}}}{4\pi c_*}\, u_x(0).$$

The retained complete regular-root theorem at the single common momentum pair $(0,0)$ gives $\sum_{x\in\mathscr{Z}_{\varepsilon,m,R}} u_x(0) \le C_{\mathrm{mom}}$. Here the transfer from different momentum points to $(0,0)$ is paid by the preceding distortion estimate, not inferred from the synchronized kernel alone.

Apply the coarea identity first to bounded nonnegative roof tests and finitely many charts, retaining the Borel selectors in the integrands. The upper bound just obtained is summable independently of t and θ. Monotone convergence gives the full nonnegative source sum. Integrating in $\theta \in [0, 2\pi)$ gives precisely (34.4); the polar origin has zero collision area in each regular chart and creates no atom. This area statement does not set a separate one-sided pairing trace to zero. At no step is a sum of individual roof suprema bounded by a synchronized value without a comparison factor. □

Corollary 34.4 (Uniform band coverage and a localized critical-roof envelope)**.** *Every chart above reaches every physical roof*

$$0 < t - a_x < \rho_\varepsilon^2/(2U)$$

on every momentum ray. Moreover the density in the theorem satisfies for almost every t the sharper bound

$$Q^w_{\varepsilon,m,R}(t) \le \frac{UC_{\mathrm{dist}}}{2c_*} \sum_{\substack{x\in\mathscr{Z}_{\varepsilon,m,R}\\ t-\rho_\varepsilon^2/(2K_{1/2})\le a_x\le t}} u_x(0). \tag{34.5}$$

The endpoints a_x are the actual physical critical roofs, not clock corrections. No section visit count is transferred from a tube point to its center unless section stability is proved separately.

Proof. Integrating $r/U \le \partial_r L_x \le r/K_{1/2}$ from zero gives $r^2/(2U) \le L_x(re_\theta) - a_x \le r^2/(2K_{1/2})$. Continuity and strict monotonicity prove coverage; the upper inequality restricts the contributing centers at any fixed physical roof. Insert that restriction into the nonnegative sum in the theorem. The original Borel exact-record selector is still evaluated on each actual orbit; the envelope simply dominates it by one. □

34.1. **The remaining normalized source is not eliminated.** The new analytic domain, physical guard tube and common-roof summation remove two specific gaps of the earlier critical argument: no collision-dependent continuation radius is chosen, and different branch momentum radii are now compared by a proved multiplicative optical bound. They neither assert that all critical centers satisfy a common guard margin nor estimate every original singular, noncritical or paired-boundary source. The latter measures need not be restrictions of the bulk collision measure used here.

Even on the covered union, (34.4) is an $O(1)$ roof-height bound, not an $O(m^{-2})$ bound. It cannot be multiplied by m^2 and declared negligible. The envelope (34.5) isolates genuine physical roof locations but provides no unproved critical-roof nonconcentration rate or transfer of return labels across a section cut. The missing original limit remains exactly (15.8), not a renamed substitute. To close it requires a normalized complete physical-source estimate or sufficient

original nongrazing capacity, including the uncovered incidence-only low-load source, unvisited and visited boundary terms, and the retained spectral, arithmetic and conditioned-denominator interfaces. Its original widths stay fixed before $m \to \infty$ and $B \to \infty$ remains outermost. No limit has been interchanged in the new finite-width, all-collision theorem.

35. A buffered horizon and bending on every physical record

The physical horizon established in Section 2 contains a strict covering margin which has not yet been used. This margin gives a deterministic lower bound for accumulated reflection, not just an upper bound for a single flight. All centers in this section belong to the original triangular lattice; all disks still have radius $R \in [R_-, R_+] = [9/20, 47/100]$. Incidence means $c_i = \sqrt{1-p_i^2}$ at an actual collision. In particular this section imposes no new cutoff on the exact event (1.1).

Lemma 35.1 (The unused normal margin in the lattice covering)**.** *Set*

$$H = \frac{28}{5}, \qquad \omega_0 = \frac{179}{400}, \qquad b_0 = R_- - \omega_0 = \frac{1}{400}, \qquad \gamma_0 = \frac{b_0}{2H} = \frac{1}{4480}. \tag{35.1}$$

For every $q \in \mathbb{R}^2$ and every unit vector v, there is a center $z \in \Lambda$ whose perpendicular projection onto $\{q + tv : 0 \le t \le H\}$ is at a parameter $s \in (0, H)$ and satisfies $|z - (q + sv)| < \omega_0$. The assertion is simultaneous in the starting point and direction, including limiting directions.

Proof. Apply Lemma 2.1 at the midpoint $q + Hv/2$. The four direction intervals and primitive row vectors in the proof of Theorem 2.2 cover every direction by lattice symmetry. Their squared normal widths are bounded, respectively, by

$$\frac{961}{4804}, \quad \frac{13697401}{73866304}, \quad \frac{529}{3136}, \quad \frac{75}{448}.$$

Each is at most the first number, and

$$\frac{961}{4804} < \frac{32041}{160000} = \left(\frac{179}{400}\right)^2.$$

These are comparisons of positive rational numbers. The absolute longitudinal displacement from the midpoint is at most $\frac{1}{2}\sqrt{3847/124} < 14/5 = H/2$, as proved there. Thus its foot lies strictly inside the stated segment. No simulated orbit or unstated compactness constant is involved. The number ω_0 is a covering distance, not a changed scatterer radius. □

Theorem 35.2 (Every physical time block carries positive bending)**.** *Let $\gamma : [0, H] \to \mathbb{R}^2$ be any unit-speed trajectory segment of the original Lorentz gas, with the actual specular reflections at times $0 < t_i < H$. Then*

$$\sum_{0<t_i<H} c_i \ge \gamma_0. \tag{35.2}$$

One may start or end inside a flight. For a collision at the initial time, the initial direction is the outgoing one. The same statement holds for a fixed finite-word grazing limit which is still a nonpenetrating specular polygon; a tangential contact has $c_i = 0$. The bound is deterministic and uniform in R, the word, the starting point and the initial direction.

Proof. There are finitely many collisions in a bounded physical time interval: consecutive flights have length at least $\tau_- = 3/50$, with only the first and last partial flights possibly shorter. Write v_0 for the initial outgoing velocity and $\Delta v_i = v(t_i+) - v(t_i-)$. Specular reflection at a circular disk with outward unit normal n_i gives

$$\Delta v_i = -2(v(t_i-) \cdot n_i)n_i, \qquad |\Delta v_i| = 2c_i.$$

Integrating the piecewise constant velocity, for $0 \le t \le H$ we have the exact identity and bound

$$\gamma(t) - \gamma(0) - tv_0 = \sum_{0<t_i<t} (t - t_i)\Delta v_i, \qquad |\gamma(t) - \gamma(0) - tv_0| \le 2H \sum_{0<t_i<H} c_i. \tag{35.3}$$

Choose z and s from Lemma 35.1. If the sum in (35.2) were less than γ_0, then

$$|\gamma(s) - z| < \omega_0 + 2H\gamma_0 = R_- \le R.$$

This places a true physical point inside an open scatterer, a contradiction. The argument uses neither disjoint reflection events nor statistical independence. It also applies directly to the stated limiting polygons, for which the same velocity identity holds and which remain outside every open disk. It does not define a billiard continuation through an arbitrary singular endpoint. □

Corollary 35.3 (A uniform block bound over all collision words)**.** *For any actual regular record with $m \ge 1$ flights, put $N_0 = 94$. Then*

$$\sum_{i=1}^{m-1} c_i \ge \gamma_0 \left\lfloor \frac{m}{N_0} \right\rfloor, \tag{35.4}$$

$$\# \left\{ 1 \le i < m : c_i \ge \frac{1}{416640} \right\} \ge \left\lfloor \frac{m}{N_0} \right\rfloor. \tag{35.5}$$

Every full block of 94 *consecutive flights supplies its own collision in the second inequality; these supplied collisions are distinct. All other collisions remain in the record and in every observable.*

Proof. The duration of a 94-flight block is at least $94\tau_- = 141/25 > 28/5 = H$. Starting at its first collision with the outgoing velocity, Theorem 35.2 supplies a sum at least γ_0 among its 93 interior collision indices. The sets of such interior indices for disjoint successive blocks are disjoint. Summation proves (35.4). Within each block at least one of the 93 nonnegative summands is at least $\gamma_0/93 = 1/416640$, proving (35.5). The possibly incomplete last block is not assigned any fictitious contribution. For $m < 94$ both statements retain their stated meaning, with a zero right side. □

Remark 35.4 (What positive bending does and does not measure)*.* The result concerns visited incidences along every original physical record. It is not abundance of the original comparison pool: $\mathsf{U}^{\mathrm{low}}_{B,m}$ and $P^{\mathrm{ng}}_{r_B,m}$ carry additional roof, clearance, pairing and exact-label requirements. A positive fraction of moderately transverse collisions does not pay a single unvisited tangency or force any endpoint trace to vanish. We use the bound below only in an exact variation of actual critical physical lengths; we do not multiply an unscaled mass bound by m^2 and call it small.

36. Normal-boundary continuation and transversality of the physical clock

A regular critical record is the actual m-flight record of Section 17 with $p_0 = p_m = 0$. It is one half of a true reversing periodic trajectory. We vary the common physical disk radius within $[R_-, R_+]$, keeping the ordered lattice centers $O_0, \dots, O_m$, all m collisions and the unit-speed clock. A formal reflection continuation will always be distinguished from a physical continuation. The latter is required to avoid every third disk and preserve the actual first-hit order.

36.1. **The full normal-boundary reflection equations.** Choose local real lifts of the contact angles α_i and of the flight directions ϑ_i. Their normal-reflection equations are, with the constants fixed at the reference record,

$$E_0 = \alpha_0 - \vartheta_0, \quad E_i = 2\alpha_i - \vartheta_{i-1} - \vartheta_i \ (1 \le i < m), \quad E_m = \alpha_m - \vartheta_{m-1}. \tag{36.1}$$

At a physical critical record E_0 is zero modulo 2π and E_m is π modulo 2π. The interior constants are the appropriate fixed odd multiples of π. These lifts remove any branch-cut ambiguity in the following derivatives.

Theorem 36.1 (A full reflection inverse without a grazing divisor)**.** *Let $\mathcal{B} = D_\alpha E$ at a regular critical record with $m \geq 1$. Write $w_i = \tau_i^{-1}$ and set $w_{-1} = w_m = 0$. Its only nonzero entries are*

$$\begin{aligned} \mathcal{B}_{ii} &= b_i + Rc_i(w_{i-1} + w_i), & b_0 = b_m = 1, \\ \mathcal{B}_{i,i+1} &= Rw_i c_{i+1}, & b_i = 2 \quad (0 < i < m), \\ \mathcal{B}_{i+1,i} &= Rw_i c_i, & c_0 = c_m = 1. \end{aligned} \tag{36.2}$$

For $\lambda = 50/47$ and $C_{\mathrm{B}} = 194/3$,

$$\|\mathcal{B}^{-1}\|_{1\to 1} \leq 1, \qquad \sum_{i=0}^{m} \lambda^{|i-j|} |(\mathcal{B}^{-1})_{ij}| \leq 2, \qquad \|\mathcal{B}^{-1}\|_{\infty\to\infty} \leq C_{\mathrm{B}}. \tag{36.3}$$

The constants are independent of R, m and the word. The algebraic statement includes simultaneous zero values of interior c_i when the positive flight lengths are retained. This extension of the matrix statement is not a claim that all such arrays are physical.

Proof. For $X_i = O_{i+1} - O_i + R(n_{i+1} - n_i) = \tau_i v_i$, $d\vartheta_i = v_i^{\perp} \cdot dX_i/\tau_i$. At the departure contact $v_i \cdot n_i = c_i$ and at arrival $v_i \cdot n_{i+1} = -c_{i+1}$. Differentiating (36.1) therefore gives (36.2), including its two boundary rows. In particular the boundary diagonal is $1 + Rc_i w$, not the interior expression with a spurious second flight.

Conjugate by $S = \mathrm{diag}((-1)^i)$ to obtain $A = S\mathcal{B}S = D - C$, where D is the positive diagonal and $C \geq 0$ has zero diagonal. The excess of the diagonal over the sum of the off-diagonal absolute values in column i is exactly b_i. Consequently $\|CD^{-1}\|_1 < 1$ for this finite matrix, $A^{-1} = D^{-1} \sum_{r\geq 0} (CD^{-1})^r \geq 0$, and $\mathbf{1}^T A \geq \mathbf{1}^T$ implies $\mathbf{1}^T A^{-1} \leq \mathbf{1}^T$. This proves the first bound. No row-dominance assertion is being used.

Fix j and conjugate further by $W = \mathrm{diag}(\lambda^{|i-j|})$. Across each edge the weight ratio is at most λ. Since $a_* = R_+/\tau_- = 47/6$ and $a_*(\lambda - 1) = 1/2$, the column excess of WAW^{-1} is at least $1 - 1/2 = 1/2$ at either endpoint and at least $2 - 1 = 1$ at an interior index. The same positive-inverse argument gives its inverse ℓ^1 norm at most 2. As $W_{jj} = 1$, this is the middle assertion in (36.3). In particular $|(\mathcal{B}^{-1})_{ij}| \leq 2\lambda^{-|i-j|}$. Sum this bound over a row, retaining at most two entries at each positive distance:

$$\sum_j |(\mathcal{B}^{-1})_{ij}| \leq 2\left(1 + 2\sum_{d\geq 1} \lambda^{-d}\right) = 2\left(1 + \frac{2}{\lambda - 1}\right) = \frac{194}{3}.$$

All steps remain valid when some interior c_i are zero. The endpoint case $m = 1$ is the same two-by-two calculation, without an interior row. □

Theorem 36.2 (A common formal radius domain and physical radius response)**.** *Every actual regular critical record has a holomorphic continuation of* (36.1) *on a radius disk of size independent of its collision count and its smallest interior incidence. More precisely, put*

$$\begin{aligned} L_R &= 4/\tau_-, & K_R &= 48/\tau_- + 512/\tau_-^2, \\ s_R &= \min\{1/8, \tau_-/24\}, & r_R &= \min\{s_R, (8C_{\mathrm{B}}K_R)^{-1}\}, \\ \eta_R &= \frac{r_R}{8(1 + C_{\mathrm{B}}L_R)}. \end{aligned} \tag{36.4}$$

For $|\rho| < \eta_R$ the fixed local equations have a unique solution $\alpha(R + \rho)$ in $\|\alpha - \alpha(R)\|_\infty \leq r_R/2$, and

$$\|\alpha(R + \rho) - \alpha(R)\|_\infty \leq 2(1 + C_{\mathrm{B}}L_R)|\rho|. \tag{36.5}$$

This is a formal normal-reflection domain. On every connected interval on which its record is actually physical, regular, and $R \in [R_-, R_+]$, one has the sharper derivative bound

$$\max_i |\alpha_i'(R)| \le D_R, \qquad D_R = C_{\mathrm{B}} L_R = \frac{38800}{9}. \tag{36.6}$$

Physical continuation through an unvisited tangency or a grazing collision is not asserted. A section crossing is not excluded and its actual selector is reevaluated at the new radius.

Proof. First hold the angles fixed at the reference record. The formula $\partial_R \vartheta_i = v_i^{\perp} \cdot (n_{i+1} - n_i)/\tau_i$ gives $|\partial_R \vartheta_i| \le 2/\tau_-$. Hence $\|E_R\|_\infty \le L_R$. At every actual regular critical point, implicit differentiation and Theorem 36.1 give $\alpha' = -\mathcal{B}^{-1} E_R$ and (36.6).

Here are explicit bounds for a domain, rather than just pointwise implicit differentiation. Let z_i denote the complex lattice center O_i. For complex independent increments (ρ, u) use the two holomorphic flight coordinates

$$X_i^{\pm} = (z_{i+1}^{\pm} - z_i^{\pm}) + (R + \rho)\big(e^{\pm i(\alpha_{i+1} + u_{i+1})} - e^{\pm i(\alpha_i + u_i)}\big), \qquad z_i^{+} = z_i, \quad z_i^{-} = \overline{z}_i.$$

The minus coordinate is defined by this expression, not by conjugating complex variables. At the real reference point both moduli are $\tau_i \ge \tau_-$. In the norm $\|(\rho, u)\| = |\rho| + \|u\|_\infty$, for increments of norm at most s_R, each flight changes by at most $6 s_R \le \tau_-/4$. Indeed $|e^{iu} - 1| \le e^{s_R} |u|$, $e^{s_R} < 2$, and $R < 1/2$. Thus logarithms continuous from the reference flights are defined on convex nonzero disks. On this domain $|R + \rho| < 1$, and direct differentiation gives the conservative bounds

$$\|D X_i^{\pm}\| \le 8, \qquad \|D^2 X_i^{\pm}\| \le 12, \qquad |X_i^{\pm}| \ge \tau_-/2.$$

The first two inequalities follow by distributing the radius derivative and at most two adjacent angular derivatives; their operator norms use the displayed joint norm and do not sum over collision indices. For $\vartheta_i = (\log X_i^{+} - \log X_i^{-})/(2i)$ up to its fixed reference constant, the second derivative of the logarithm gives

$$\|D^2 \vartheta_i\| \le 24/\tau_- + 256/\tau_-^2, \qquad \|D^2 E\| \le K_R.$$

These are estimates for each component in the supremum norm, uniformly in the dimension.

Write $E(R + \rho, \alpha + u) - E(R, \alpha) = \mathcal{B} u + E_R \rho + N(\rho, u)$. Taylor's integral formula gives $\|N\|_\infty \le (K_R/2)(|\rho| + \|u\|_\infty)^2$ and $\|D_u N\| \le K_R(|\rho| + \|u\|_\infty)$. For $|\rho| \le \eta_R$ the map

$$u \longmapsto -\mathcal{B}^{-1}(E_R \rho + N(\rho, u))$$

preserves $\|u\|_\infty \le r_R/2$: the linear forcing is at most $r_R/8$ and the nonlinear term at most $r_R/16$. Its Lipschitz constant is at most $1/8$. Iteration from zero gives the unique fixed point there, holomorphic by locally uniform convergence of the holomorphic iterates. Comparing the fixed point with the first iterate gives

$$\|u\|_\infty \le \frac{C_{\mathrm{B}} L_R |\rho| + (C_{\mathrm{B}} K_R/2)|\rho|^2}{1 - 1/8} \le 2(1 + C_{\mathrm{B}} L_R)|\rho|.$$

This proves the asserted formal domain, also for the zero-incidence matrix limits with nonzero reference flights and the same reflected angle identities. For a regular physical reference, continuity and its finitely many positive flight/incidence/third-disk margins give some open physical subinterval; its size need not be uniform. The argument has not promoted the formal disk to a common physical radius disk. Nor has it frozen a discontinuous exact-record selector. □

36.2. **A collision-uniform derivative of the true critical roof.** Let J be a connected interval of regular physical critical continuation as in Theorem 36.2. Its actual roof is

$$a(R) = \sum_{i=0}^{m-1} |O_{i+1} - O_i + R(n_{i+1}(R) - n_i(R))|.$$

Every summand is a real physical flight, not a symbolic roof.

Theorem 36.3 (Exact radius variation and uniform critical-clock slope)**.** *On J one has*

$$a'(R) = -c_0 - c_m - 2\sum_{i=1}^{m-1} c_i = -2 - 2\sum_{i=1}^{m-1} c_i. \tag{36.7}$$

In particular, with $\kappa_0 = 1/210560$,

$$2 + 2\gamma_0 \left\lfloor \frac{m}{94} \right\rfloor \le -a'(R) \le 2m, \qquad \kappa_0 m \le -a'(R) \le 2m. \tag{36.8}$$

For any $R_1 < R_2$ in this same physical interval,

$$\kappa_0 m(R_2 - R_1) \le a(R_1) - a(R_2) \le 2m(R_2 - R_1). \tag{36.9}$$

There is also the uniform normalized second-derivative estimate

$$|a''(R)| \le 2(m-1)C_{cR} \le 2mC_{cR}, \qquad C_{cR} = D_R + \frac{2 + 2R_+D_R}{\tau_-}. \tag{36.10}$$

Consequently a/m has both a slope separated from zero and a Lipschitz derivative with constants independent of the collision word. The same one-sided first-variation identities and bounds hold at a fixed finite-word nonpenetrating grazing limit of such records. They do not identify different words across a singularity.

Proof. The full length action $S(R, \alpha)$ has zero angular gradient at a critical record: the interior equations are specular stationarity, and the two endpoint derivatives are $-Rp_0$ and Rp_m. Differentiating $S(R, \alpha(R))$ therefore cancels every angular chain-rule term and leaves

$$a'(R) = \partial_R S(R, \alpha) = \sum_{i=0}^{m-1} v_i \cdot (n_{i+1} - n_i).$$

Each summand is $-c_{i+1} - c_i$. The interior incidences occur exactly twice; the two normal endpoints occur once and equal one. This proves (36.7), including the signs and both boundary factors. Corollary 35.3 supplies the first lower bound in (36.8). Since $\lfloor m/94 \rfloor \ge m/94 - 1$ and $0 < \gamma_0 < 1$,

$$2 + 2\gamma_0 \lfloor m/94 \rfloor \ge (2\gamma_0/94)m, \qquad 2\gamma_0/94 = 1/210560.$$

The upper bound follows from $0 \le c_i \le 1$. Integration over the actual connected physical interval proves (36.9). In particular no exchange of a radius supremum and a time or collision limit is used.

For the second derivative, the full response bound gives $|n_i'| \le D_R$ and

$$|X_i'| \le 2 + 2R_+D_R, \qquad |v_i'| = |(I - v_i v_i^T)X_i'|/\tau_i \le (2 + 2R_+D_R)/\tau_-.$$

Differentiate the incoming formula $c_i = -v_{i-1} \cdot n_i$. It follows that $|c_i'| \le C_{cR}$, without dividing by c_i. The endpoints remain normal throughout J, so their derivatives are zero. Differentiating (36.7) yields (36.10). At a stated finite-word limit, the positive flight lengths, the uniform reflection inverse and the formal local continuation supply the one-sided derivatives. The finite sums of incidences and the nonpenetrating polygon bound pass to that limit. No assertion about an arbitrary branch beyond that endpoint follows. □

Corollary 36.4 (Branchwise band width and exact radius–roof coarea)**.** *On every such connected physical branch, for a physical roof interval I_t,*

$$|\{R \in J : a(R) \in I_t\}| \le |I_t|/(\kappa_0 m). \tag{36.11}$$

More generally let $\mathscr{K}_m$ be the full set of regular physical critical m-flight records with the initial disk represented in a fixed lattice fundamental domain and $R \in [R_-, R_+]$. Write x for a record at a given R. Give it any nonnegative Borel weight $f(R,x)$, which may include the original exact return/displacement/collision selector and an original optical weight. For every nonnegative Borel roof test ϕ,

$$\int_{R_-}^{R_+} \sum_{x:(R,x)\in\mathscr{K}_m} f(R,x)\phi(a_x(R))\,dR = \int_{\mathbb{R}} \phi(t) \sum_{\substack{(R,x)\in\mathscr{K}_m \\ a_x(R)=t}} \frac{f(R,x)}{|a_x'(R)|}\,dt. \tag{36.12}$$

The equality is an identity of positive measures, possibly infinite; no global integrability is silently assumed. In every term

$$\frac{1}{2m} \le \frac{1}{|a_x'(R)|} \le \frac{1}{\kappa_0 m}.$$

Physical records are counted once; coordinate chart overlaps are not extra records. The original probability normalization c_* is unchanged. For example, inserting a weight with factor $(4\pi c_*)^{-1}$ leaves that factor exactly once on both sides.

Proof. The derivative in (36.8) has one sign and is separated from zero, so each connected branch is a C^1 strictly monotone graph over R and a C^1 graph over its physical roof. The ordinary one-dimensional substitution formula proves (36.11) and (36.12) on one chart. To justify the full physical sum, there are countably many finite lattice words and each finite-dimensional angle space is second countable. The invertible full reflection matrix and the open physical regularity conditions give a countable cover by charts on open radius intervals. Assign a physical point to its first chart in a fixed enumeration. The assigned subsets are Borel, and their indicators can be included in f on the uncut charts. Substitution for finitely many nonnegative chart terms, then monotone convergence, gives the displayed equality. This allocation deletes representations, never collisions or physical source points. The exact selector is evaluated at (R,x), not transported from another radius or from a tube center. □

Remark 36.5 (Fixed-radius shortage is not a radius average)*.* The lower clock slope is a new all-word geometric input on the original radius family. Formula (36.12) is an auxiliary exact change of variables, not a replacement of the original probability problem by a radius-averaged limit theorem. A bound for crossings summed at different radii does not follow from the retained momentum kernel at one radius; conversely, a small radius integral cannot control $\sup_R$. Neither (36.11) nor the presence of moderately transverse collisions proves the original m^{-2} roof nonconcentration, the full nongrazing comparison capacity, or the shortage limit (15.8). Singular/paired endpoint traces, nonguarded critical centers and the complete noncritical positive source remain part of that limit. All original microscopic widths remain fixed before the collision limsup and B remains outermost. No theorem or proof body of the inherited raw-LLT and inverse program has been removed.

37. The complete record: endpoint work and radius–roof disintegration

The preceding clock calculation extends beyond reversing records, but its endpoint work must then be retained. This section treats every regular physical m-flight record, with all original exact labels, not just the critical tubes of Section 34. It produces a transverse coordinate for

a parameter-extended source. It does not change the fixed-radius probability space or the raw target.

Theorem 37.1 (Uniform continuation at prescribed endpoint momenta)**.** *Fix any actual regular record and its two endpoint momenta* $(a, v) = (p_0, p_m) \in (-1, 1)^2$*. Keep them fixed while varying the common radius. Replace the endpoint constants in* (36.1) *by*

$$E_0 = -\arcsin a \pmod{2\pi}, \qquad E_m = \pi + \arcsin v \pmod{2\pi}.$$

The full derivative matrix is still (36.2)*, now with* $c_0 = \sqrt{1-a^2}$ *and* $c_m = \sqrt{1-v^2}$ *in place of one. All three bounds* (36.3) *remain valid, as do the common formal radius disk* (36.4) *and the physical radius response* (36.6)*. None of their constants depends on* (a, v)*, the word, or the smallest incidence. Physicality is asserted only on connected regular physical continuation intervals, not on the whole formal radius disk.*

Proof. The departure direction is $\alpha_0 + \arcsin a$. The incoming terminal direction is $\alpha_m + \pi - \arcsin v$; tangential momentum is unchanged by the last specular reflection. These give exactly the two displayed constants modulo 2π. Their derivatives with respect to R vanish because a, v are fixed. The flight differentiation in Theorem 36.1 then gives the same matrix with its true endpoint cosines. Its column excess remains exactly b_i, including the endpoint columns: each endpoint cosine appears both in its diagonal correction and in the sole off-diagonal entry of that column. The unweighted and weighted column arguments therefore apply without alteration, and the row bound follows by the same geometric sum. No derivative of $\arcsin a$ or $\arcsin v$ is taken. The complex flight estimates in Theorem 36.2 use only the positive flight lengths, the disk radius bound and this inverse, not normality. They supply the same formal continuation and the same bound on α'. The interpretation of its physical subinterval is unchanged. □

Theorem 37.2 (The physical roof has only bounded endpoint work)**.** *Let* $L(R)$ *be the true physical roof on any connected physical continuation interval from Theorem 37.1. Then*

$$L'(R) = -c_0 - c_m - 2\sum_{i=1}^{m-1} c_i + R\{-a\alpha_0'(R) + v\alpha_m'(R)\}. \tag{37.1}$$

Define the collision-independent constants

$$C_{\mathrm{end}} = 2R_+D_R, \qquad C_{\mathrm{loss}} = C_{\mathrm{end}} + 2\gamma_0, \qquad m_0 = \lceil 2C_{\mathrm{loss}}/\kappa_0 \rceil .$$

For every $m \geq 1$ *and every such record,*

$$\kappa_0 m - C_{\mathrm{loss}} \leq -L'(R) \leq 2m + C_{\mathrm{end}}. \tag{37.2}$$

In particular for all $m \geq m_0$*, the actual roof is strictly monotone on each connected physical component, and* $-L'(R) \geq \kappa_0 m/2$*. These statements are uniform over the entire original radius interval, over the endpoint momenta and over all physical collision words. They impose no new incidence or clearance guard on the original source. The conservative constant* m_0 *is a theoretical threshold, not an experimentally reached collision count.*

Proof. On the stationary interior of the full action, its endpoint gradient is $R(-a, v)$. Differentiating the action along the prescribed-momentum continuation consequently leaves the two endpoint chain-rule terms, in addition to the explicit radius derivative. The latter telescopes as in (36.7), proving (37.1). Its endpoint work has absolute value at most $R(|a| + |v|)D_R \leq C_{\mathrm{end}}$. The visited incidence sum is bounded below by $\gamma_0\lfloor m/94 \rfloor$ from Corollary 35.3. Dropping only the two nonnegative endpoint cosines in a lower bound, and using $\lfloor m/94 \rfloor \geq m/94 - 1$, gives

$$-L'(R) \geq 2\gamma_0\lfloor m/94 \rfloor - C_{\mathrm{end}} \geq \kappa_0 m - C_{\mathrm{loss}}.$$

There are exactly $2m$ incidence contributions with multiplicity, each at most one, which gives the upper bound. The definition of m_0 proves the last assertion. This is a pointwise geometric estimate for every physical record, not an ergodic average. In particular no independence, pressure input or collision-word count occurs in its proof. □

Theorem 37.3 (An exact disintegration of the parameter-extended physical source)**.** *Fix $m \ge m_0$ and a nonnegative Borel function $h(R,x)$ on the regular physical collision space, zero off the original initial section when section probability is used. It may include the exact original selector $\mathbf{1}_{\Pi_{n,k,m,R}}$, with the true section visits evaluated at that radius. For a regular root with endpoint momenta (a,v) let*

$$u(R,x) = |\partial_{\alpha_0} p_m|^{-1} \quad \text{with } R, p_0 = a \text{ fixed.}$$

Use the physical momentum branches and one-copy convention of Section 26. For every nonnegative Borel test ϕ,

$$\begin{aligned} &\int_{R_-}^{R_+} \int h(R,x)\phi(L_{m,R}(x))\, d\nu_R^*(x)\, dR \\ &= \frac{1}{4\pi c_*} \int_{-1}^{1}\int_{-1}^{1}\int_{\mathbb{R}} \phi(t) \sum_{\substack{(R,x):\ p_0=a,\ p_m=v \\ L_{m,R}(x)=t}} \frac{h(R,x)u(R,x)}{|L'_{a,v,x}(R)|}\, dt\, dv\, da. \end{aligned} \tag{37.3}$$

The derivative is the full derivative with the endpoint work in (37.1)*, not a derivative at fixed initial angle. The sums include only genuine regular physical records and count each once. No finiteness of either side is assumed. The new Jacobian factor satisfies*

$$\frac{1}{2m + C_{\mathrm{end}}} \le \frac{1}{|L'_{a,v,x}(R)|} \le \frac{2}{\kappa_0 m}. \tag{37.4}$$

The equality is auxiliary geometric information in the added radius coordinate. It makes no assertion that the radius integral can replace the required supremum over R.

Proof. On a regular physical branch the initial collision measure is $d\alpha_0\, da/(4\pi)$ and the map to (a,v) has positive absolute Jacobian $|\partial_{\alpha_0} p_m|$; this is the actual root Jacobian used in the physical momentum change of variables in Section 26. After the original section restriction its density is $u/(4\pi c_*)$. This is a local change of variables; the all-time upper kernel bound in that section is not used here. In particular it is not a composition of one-step momentum kernels.

Theorem 37.1 supplies local physical branches parametrized by (R,a,v). For $m \ge m_0$ the coordinate change $(R,a,v) \mapsto (L(R,a,v),a,v)$ is locally invertible, with absolute determinant $|L'_{a,v,x}(R)|$, by Theorem 37.2. The ordinary substitution theorem therefore gives (37.3) on one chart, with the reciprocal Jacobian in the stated direction. Countably many physical charts cover the regular source: use countably many lattice words and second countability in (R,α_0,a). A fixed first-chart allocation makes their representations disjoint. The allocation indicators and any section or exact-label cuts stay inside h; none is differentiated. Applying substitution to finite positive chart sums and then monotone convergence gives the equality for the complete regular source, allowing infinite values. The identity concerns the regular source and makes no claim about the separate paired-boundary traces. Finally (37.2) proves (37.4). □

Remark 37.4 (The precise unpaid step is unchanged)*.* This exact formula retains all regular physical records, all original collision labels and their true physical roofs. It is not a new probability model and does not discard low-incidence records. Nevertheless its crossing sum is taken at different radii. The unscaled root-kernel bound at a common (R,a,v) does not control this sum. Nor does an estimate integrated in R control the original fixed-radius roof essential height. A uniform bound for those concentrations, the complete positive source and paired

traces, or the original comparison capacity is still required to prove (15.8). This remains a substantive mathematical obligation rather than an automatic consequence of the factor $1/m$ in a different-coordinate Jacobian. All original probability quantifiers and ordered microscopic limits are retained.

38. A global variational certificate for every physical word

This section supplies a comparison valid on the whole original physical source, without requiring a common clearance margin or a physical critical center. It extends, rather than reclaims as new, the Cartesian support argument of Lemma 10.1 and the fixed-endpoint minimizing property in Lemma 10.3. The new ingredients are a complete disk-word duality formula, an exact endpoint defect, and quantitative certificates with nonnegative terms.

Fix $m \geq 1$, $R \in [R_-, R_+]$, and an ordered lifted center word $w = (O_0, \dots, O_m)$, with $O_0 = 0$ and $O_{i+1} \neq O_i$. A word of an actual trajectory always satisfies this last condition. Set

$$K_w(R) = \prod_{i=0}^{m} \overline{B}(O_i, R), \qquad S(Y) = \sum_{i=0}^{m-1} |Y_{i+1} - Y_i|, \qquad \ell_w(R) = \min_{Y \in K_w(R)} S(Y).$$

The disks here are the original geometric disks, including their interiors only for this auxiliary minimization. An arbitrary minimizer is *not* called a Lorentz trajectory: it may have an interior vertex, a tangency, or a segment intersecting an unlisted disk. Neither ℓ_w nor the minimization defines a replacement dynamics, a roof, a probability law, or a new conditioning event. The actual clock throughout remains $L = S(Q)$ on the original first-hit trajectory Q. Successive disks are separated by at least $1 - 2R_+ = 3/50$, so S is continuously differentiable on a neighborhood of $K_w(R)$.

For vectors $V = (v_0, \dots, v_{m-1})$, $|v_i| \leq 1$, put

$$\begin{aligned} &d_0 = -v_0, \quad d_i = v_{i-1} - v_i \ (0 < i < m), \quad d_m = v_{m-1}, \\ &C_w(V) = \sum_{i=0}^{m-1} v_i \cdot (O_{i+1} - O_i), \qquad T(V) = \sum_{i=0}^{m} |d_i|, \qquad D_{w,R}(V) = C_w(V) - RT(V). \end{aligned} \tag{38.1}$$

All lengths in these expressions have the original physical units.

Theorem 38.1 (Exact disk-word duality and a nonnegative gap)**.** *For every word and radius just specified,*

$$\ell_w(R) = \max_{|v_i| \leq 1} D_{w,R}(V). \tag{38.2}$$

Both extrema are attained. For every $Y \in K_w(R)$ and every feasible V,

$$\begin{aligned} S(Y) - D_{w,R}(V) = &\sum_{i=0}^{m-1} \big(|Y_{i+1} - Y_i| - v_i \cdot (Y_{i+1} - Y_i)\big) \\ &+ \sum_{i=0}^{m} \big(d_i \cdot (Y_i - O_i) + R|d_i|\big). \end{aligned} \tag{38.3}$$

Every summand is nonnegative. At any minimizing Y, the unit edge vectors $v_i = (Y_{i+1} - Y_i)/|Y_{i+1} - Y_i|$ attain the dual maximum. In particular an optimal dual vector can always be chosen with $|v_0| = |v_{m-1}| = 1$ and $2 \leq T(V) \leq 2m$.

Proof. Expand $\sum_i v_i \cdot (Y_{i+1} - Y_i) = \sum_i d_i \cdot Y_i$ and $C_w(V) = \sum_i d_i \cdot O_i$. Subtraction proves (38.3). The first sum is nonnegative by $|v_i| \leq 1$; the second is nonnegative by $|Y_i - O_i| \leq R$. This proves weak duality without a minimax interchange.

Compactness gives a primal minimizer Y. Convexity and differentiability of S show that its first-order inequality on the convex product $K_w(R)$ is necessary and sufficient for minimality.

With the displayed unit edge vectors its ith gradient is exactly d_i. Varying one vertex at a time therefore shows that Y_i minimizes $d_i \cdot Z$ on $\overline{B}(O_i, R)$. If $d_i \neq 0$, this minimizer is $O_i - Rd_i/|d_i|$; if $d_i = 0$, the corresponding gap is identically zero. Thus every term on the right of (38.3) vanishes at this Y, V. This proves equality, dual attainment and the asserted choice of dual vectors. At the two endpoints $|d_0| = |d_m| = 1$; each of the $m-1$ interior differences has norm at most two, giving $2 \leq T(V) \leq 2m$. The case $m = 1$ uses just the two endpoint terms. Nonconsecutive repetitions of a disk cause no difficulty: the vertices are separate variables in the stated deterministic comparison. □

Theorem 38.2 (The exact endpoint deficit of an arbitrary true record)**.** *Let $Q_i = O_i + Rn_i$ be* any *regular physical m-flight record, with its actual unit velocities v_i, endpoint momenta p_0, p_m, and incidences $c_i > 0$. All original section and exact-label tests may be retained. Then*

$$L - D_{w,R}(V) = E_R(Q) := R(2 - c_0 - c_m), \tag{38.4}$$

whence, uniformly in m, R, w and all intermediate incidences and clearances,

$$0 \leq L - \ell_w(R) \leq E_R(Q) = R\left(\frac{p_0^2}{1 + c_0} + \frac{p_m^2}{1 + c_m}\right) \leq 2R. \tag{38.5}$$

Consequently the range of the true roof on a physical word is at most $2R$, improving the still valid $4R$ enclosure of Lemma 10.3. No collision is deleted in this bound.

Proof. For actual velocities the flight terms of (38.3) vanish. The true reflection law gives $d_i = -2c_i n_i$ for $0 < i < m$, so every interior contact term also vanishes. At departure, $d_0 = -v_0$ and $v_0 \cdot n_0 = c_0$; at arrival, $d_m = v_{m-1}$ and $v_{m-1} \cdot n_m = -c_m$. The two remaining contact terms are exactly $R(1 - c_0)$ and $R(1 - c_m)$. This proves (38.4). Weak duality and feasibility of Q sandwich ℓ_w between $D_{w,R}(V)$ and L. The identity $1 - c = p^2/(1 + c)$ proves the second expression. Every true roof of the word lies in the same interval $[\ell_w, \ell_w + 2R]$. This proof uses only real flight vectors and the exact specular law; it introduces no independence, pressure hypothesis or enlarged observational window. The interval is a deterministic enclosure, not the event on which a local limit theorem is asserted. □

Corollary 38.3 (A globally certified regular critical record)**.** *If $p_0 = p_m = 0$, then $L = \ell_w(R)$ and Q is the unique minimizer of S on $K_w(R)$. For any competitor $Y \in K_w(R)$,*

$$S(Y) - L \geq \frac{1}{2R}\sum_{i=0}^{m} \lambda_i |Y_i - Q_i|^2 + \frac{1}{2}\sum_{i=0}^{m-1} |Y_{i+1} - Y_i||\widehat{v}_i - v_i|^2, \quad \lambda_0 = \lambda_m = 1, \quad \lambda_i = 2c_i, \tag{38.6}$$

where $\widehat{v}_i$ is the unit vector of the competitor edge. This global minimizing certificate is new here; the uniqueness among regular physical critical records was already proved in Section 10 and is not counted again as a new result.

Proof. Equation (38.5) gives equality and all $d_i = -\lambda_i n_i$, with $\lambda_i > 0$. For $z = Y_i - O_i$ in the radius-R disk,

$$R - n_i \cdot z = \frac{|z - Rn_i|^2 + R^2 - |z|^2}{2R} \geq \frac{|Y_i - Q_i|^2}{2R}.$$

For the flight term, unit length of v_i gives $|e| - v_i \cdot e = |e|\,|e/|e| - v_i|^2/2$. Insert these expressions into (38.3). Equality of lengths forces every $Y_i = Q_i$, proving uniqueness even against nonphysical competitors. This does not assert uniqueness of an arbitrary relaxed minimizer at a word with inactive contacts. □

39. Global radius order and a positive physical shape energy

The radius–roof identities of Sections 36 and 37 are inherited. Here a global variational value allows critical comparisons between disconnected physical continuation intervals, and its second variation has a nonnegative, geometrically explicit form. Nothing is continued as a true trajectory through a forbidden collision or an unvisited disk.

Theorem 39.1 (Global radius convexity of the disk-word value)**.** *For each fixed word of Section 38, $R \mapsto \ell_w(R)$ is convex and strictly decreasing on $[R_-, R_+]$. For $R_1 < R_2$,*

$$2(R_2 - R_1) \le \ell_w(R_1) - \ell_w(R_2) \le 2m(R_2 - R_1). \tag{39.1}$$

This statement concerns the geometric value even where its minimizer is not physical. If regular physical critical records of the same word exist at both R_1, R_2, with true roofs a_1, a_2, then the stronger bound

$$\kappa_0 m(R_2 - R_1) \le a_1 - a_2 \le 2m(R_2 - R_1), \qquad \kappa_0 = 1/210560, \tag{39.2}$$

holds without requiring a physical continuation on (R_1, R_2). Thus a fixed word has at most one regular physical critical record with a prescribed physical roof over the whole *radius interval.*

Proof. The dual formula (38.2) is a maximum of affine functions $C_w(V) - RT(V)$ over the same compact vector set for all R. It proves convexity. At each R_j, choose the optimal unit-edge vector V_j furnished by the primal minimizer. Its affine function supports ℓ_w from below and $2 \le T(V_j) \le 2m$. Therefore

$$T(V_2)(R_2 - R_1) \le \ell_w(R_1) - \ell_w(R_2) \le T(V_1)(R_2 - R_1).$$

This proves (39.1), without differentiability of ℓ_w or a common minimizing configuration.

At a regular physical critical point, Corollary 38.3 identifies its true roof with ℓ_w. Its actual velocity vector is an optimal dual vector and $T(V) = 2 + 2\sum_{i=1}^{m-1} c_i \ge \kappa_0 m$ by the already proved complete-bending bound (36.8). Use that vector at R_2 in the lower supporting inequality and the vector at R_1 in the upper inequality. This proves (39.2) across any nonphysical holes. At a fixed radius the critical point is unique by the inherited physical uniqueness theorem; distinct radii have strictly ordered roofs by the new global comparison. No physical orbit in a hole has been asserted or counted. □

Theorem 39.2 (Transfer of complete physical bending to every minimizing dual)**.** *Let Q be any regular physical record of w at R, without normal endpoint or clearance assumptions. Let Y be any minimizer of S on $K_w(R)$, let V and $\widehat{V}$ be their respective unit flight vectors, and write $E = R(2 - c_0 - c_m)$. Then*

$$\frac{1}{2}\sum_{i=0}^{m-1} |Y_{i+1} - Y_i|\,|\widehat{v}_i - v_i|^2 \le E, \qquad T(\widehat{V}) \ge 2 + 2\sum_{i=1}^{m-1} c_i - 2\sqrt{2mE/\tau_-}. \tag{39.3}$$

In particular, with $C_{\mathrm{bend}} = 4\sqrt{R_+/\tau_-}$, every such minimizing dual satisfies

$$T(\widehat{V}) \ge b_m := \kappa_0 m - C_{\mathrm{bend}}\sqrt{m}, \qquad \kappa_0 = 1/210560. \tag{39.4}$$

The comparison holds even when Y is nonphysical and contains inactive contacts. No inverse incidence occurs.

If $R_1 < R_2$ both admit arbitrary regular physical records of this same word, with actual roofs L_1, L_2, then

$$b_m(R_2 - R_1) - 2R_+ \le L_1 - L_2 \le 2m(R_2 - R_1) + 2R_+. \tag{39.5}$$

For $b_m > 0$, all radii admitting an actual record of the word with its true roof in any specified interval I of length h therefore have diameter at most $(h + 2R_+)/b_m$, even when that set is

disconnected. For $m \ge \lceil 64R_+/(\tau_-\kappa_0^2)\rceil$, $b_m \ge \kappa_0 m/2$. This is a deterministic support bound across the original radius family, not a radius-averaged substitute for the fixed-radius raw theorem.

Proof. Use the actual velocity vector V of Q in the nonnegative gap (38.3) at the minimizing Y. Its total gap is $\ell_w - D_{w,R}(V) \le L - D_{w,R}(V) = E$. Its flight terms are exactly those in the first part of (39.3). Every comparison edge has length at least τ_-, so $\sum_i |\widehat{v}_i - v_i|^2 \le 2E/\tau_-$. The two endpoint contributions to T are one for both unit-vector families. The reverse triangle inequality for each interior difference and Cauchy–Schwarz give

$$|T(\widehat{V}) - T(V)| \le 2\sum_i |\widehat{v}_i - v_i| \le 2\sqrt{2mE/\tau_-}.$$

For the actual physical velocities $T(V) = 2 + 2\sum c_i$. The complete-block incidence estimate in Section 35 gives $2 + 2\sum c_i \ge \kappa_0 m$ for every such record: this uses only the same full-block counting as (36.8), not its critical endpoint hypothesis. Since $E \le 2R_+$, the stated C_{bend} follows. This is an energy comparison with the actual trajectory, not a claim that a minimizing polygon obeys the bending theorem physically.

Apply the supporting affine vector of a minimizer at R_2 to $\ell_w(R_1)$. The preceding lower bound on its T gives $\ell_w(R_1) - \ell_w(R_2) \ge b_m(R_2 - R_1)$. The global upper bound remains $2m(R_2 - R_1)$. Writing $L_j = \ell_w(R_j) + e_j$ with $0 \le e_j \le 2R_+$ proves (39.5). If both actual roofs lie in I, $L_1 - L_2 \le h$, which yields the diameter bound for every pair of admitted radii and thus for its entire, possibly disconnected, set. The final integer threshold is the exact rearrangement of $C_{\mathrm{bend}}\sqrt{m} \le \kappa_0 m/2$. It is not claimed to be sharp or to have been reached by the finite simulations. □

Theorem 39.3 (Exact nonnegative curvature of the true critical clock)**.** *On a connected interval of regular physical critical continuation of one fixed word, write $Q_i(R) = O_i + Rn_i(R)$, $X_i = Q_{i+1} - Q_i = \tau_i v_i$, and let a prime denote its genuine radius derivative. Then*

$$\begin{aligned} a''(R) &= \sum_{i=0}^{m-1} \frac{|(I - v_i v_i^{\mathsf{T}})(Q'_{i+1} - Q'_i)|^2}{\tau_i} + R\sum_{i=0}^{m} \lambda_i |n'_i|^2 \\ &= \sum_{i=0}^{m-1} \tau_i |v'_i|^2 + R\left(|n'_0|^2 + |n'_m|^2 + 2\sum_{i=1}^{m-1} c_i |n'_i|^2\right) \ge 0. \end{aligned} \tag{39.6}$$

Every incidence and flight in this formula is physical. In particular

$$\frac{d}{dR}\sum_{i=1}^{m-1} c_i = -\frac{1}{2}a''(R) \le 0. \tag{39.7}$$

At a regular physical point equality $a''(R) = 0$ occurs precisely for a normal two-disk bouncing record (possibly stopped after any positive number of flights). Otherwise $a''(R) > 0$. No positive lower curvature constant uniform over all words is asserted.

Proof. The full Cartesian gradient of S at a critical record is $\nabla_i S = -\lambda_i n_i$, with the coefficients of Corollary 38.3. Twice differentiating the sum of actual lengths gives

$$a'' = \sum_i \frac{|(I - v_i v_i^{\mathsf{T}})(Q'_{i+1} - Q'_i)|^2}{\tau_i} + \sum_i \nabla_i S \cdot Q''_i.$$

Now $Q''_i = 2n'_i + Rn''_i$, $n_i \cdot n'_i = 0$, and $n_i \cdot n''_i = -|n'_i|^2$. Substitution yields exactly (39.6), including its positive sign and both endpoint factors. Since $v'_i = (I - v_i v_i^{\mathsf{T}})(Q'_{i+1} - Q'_i)/\tau_i$, the second display is identical to the first. The common formal reflection construction supplies the needed smooth derivatives on each physical interval, but no pressure or probability argument is used. Differentiating the retained identity $a' = -2 - 2\sum_{i=1}^{m-1} c_i$ proves (39.7).

If $a'' = 0$, every coefficient $R\lambda_i$ is positive, so all $n_i' = 0$, and all flight projections in (39.6) vanish. Consequently $n_{i+1} - n_i$ is parallel to v_i. At the initial normal contact $v_0 = n_0$. The next normal must therefore be parallel to v_0, and its negative incoming incidence forces $n_1 = -v_0$. The first reflection is normal and reverses the flight. Repeating the argument gives normal incidence and reversal at every contact. Since these are true first-hit flights, reversal returns to the very disk just departed, not an invented farther center. Hence the word is a repeated two-disk bounce. Conversely that record has fixed contact normals and flight directions while R varies, and $a(R) = m(|O_1 - O_0| - 2R)$; its curvature is zero. This also covers $m = 1$. □

Corollary 39.4 (A word-indexed critical radius–roof coarea)**.** *In* (36.12)*, normalize $O_0 = 0$ as before and write $\mathscr{W}_m$ for the countable set of all length-m lattice words. For each t, w, either no regular physical critical record of w has roof t, or there is exactly one pair $(R_w(t), x_w(t))$. For every nonnegative Borel weight f and test ϕ,*

$$\int_{R_-}^{R_+} \sum_{x \in \mathscr{K}_m(R)} f(R,x)\phi(L(x))\, dR = \int_{\mathbb{R}} \phi(t) \sum_{w \in \mathscr{W}_m} \mathbf{1}_{\{(R_w, x_w) \text{ exists}\}} \frac{f(R_w(t), x_w(t))}{|a_w'(R_w(t))|}\, dt. \tag{39.8}$$

The sum counts a word at most once at each roof, even when its set of physical radii is disconnected. It allows infinite values and retains arbitrary original section/return selectors. It is not a fixed-radius LLT and does not bound the complete sum of the displayed weights.

Proof. Use the countable regular local charts and first-chart Borel allocation of Corollary 36.4. Each chart is locally invertible in radius and roof. The new global order proves that charts of the same word cannot yield distinct physical points at the same roof. Their inverse graphs therefore agree on every overlap representing a point, and combine into the indicated Borel partial graph. Nonnegative finite chart substitution and monotone convergence give the identity, after regrouping by word. This removes coordinate and disconnected-component multiplicity, not any physical record. The reciprocal slope lies between $1/(2m)$ and $1/(\kappa_0 m)$ at every admitted term, as previously proved. No radius average is exchanged with a radius supremum. □

40. Physical clearance certificates and complete-source localization

The endpoint defect in (38.5) is available for every original regular trajectory, including those whose geometric word minimum is not a true critical orbit. We quantify this distinction rather than discarding that source. The final proposition then inserts the proved enclosure into the existing exact source formula, at the original fixed radius and roof.

Theorem 40.1 (A quantitative certificate relative to every minimizing contact)**.** *Let Q be an arbitrary true regular record and let $E = E_R(Q)$. For every minimizer Y defining $\ell_w(R)$ set*

$$\delta_0 = \delta_m = 2\sqrt{2RE}, \qquad \delta_i = \sqrt{RE/c_i} \quad (0 < i < m).$$

Then $|Y_i - Q_i| \le \delta_i$ for every i. Consequently, for every unvisited center z on every original flight, without a lattice truncation,

$$\operatorname{dist}(z, [Y_i, Y_{i+1}]) - R \ge \operatorname{dist}(z, [Q_i, Q_{i+1}]) - R - \max(\delta_i, \delta_{i+1}). \tag{40.1}$$

There is a sufficient proved *physicality certificate: if*

$$\Delta_i > \max(\delta_i, \delta_{i+1}) \quad (0 \le i < m), \tag{40.2}$$

$$c_i > (\delta_{i-1} + 2\delta_i + \delta_{i+1})/\tau_- \quad (0 < i < m), \tag{40.3}$$

then every minimizer Y is the same regular physical normal-endpoint record of the original word. No section label is thereby transferred from Q to Y. If these tests fail, Q stays in the original source; its minimum is not declared physical.

Proof. Insert the actual velocity vector of Q and a minimizing Y into (38.3). Weak duality and (38.4) give $0 \le S(Y) - D_{w,R}(V) \le E$. Thus each nonnegative contact term is at most E. At an interior contact it equals $2c_i\{R - n_i \cdot (Y_i - O_i)\} \ge (c_i/R)|Y_i - Q_i|^2$, which proves the interior bound.

At departure the supporting disk point for $-v_0$ is $O_0 + Rv_0$; the endpoint contact slack bounds its squared distance from Y_0 by $2RE$. Also $|Q_0 - (O_0 + Rv_0)|^2 = 2R^2(1 - c_0) \le 2RE$. The triangle inequality gives $|Y_0 - Q_0| \le 2\sqrt{2RE}$. At arrival use $O_m - Rv_{m-1}$ and the same argument. For a common affine interpolation parameter the distance between two corresponding segment points is at most $\max(\delta_i, \delta_{i+1})$. Both directed Hausdorff distances satisfy this bound. Distance to any fixed z is 1-Lipschitz for Hausdorff distance; this proves (40.1) for all lattice centers simultaneously.

It remains to justify physicality, not merely small displacement. The edges of Y have length at least $\tau_- = 3/50$, because successive original disks are disjoint. If $\widehat{v}_i$ is their unit direction, the normalization inequality $|u/|u| - v/|v|| \le 2|u - v|/\min(|u|, |v|)$ gives

$$|\widehat{v}_i - v_i| \le 2(\delta_i + \delta_{i+1})/\tau_-.$$

Hence the ith interior gradient of S at Y has norm at least $2c_i - 2(\delta_{i-1} + 2\delta_i + \delta_{i+1})/\tau_- > 0$. By the minimizing condition in Theorem 38.1, Y_i lies on its circle and that gradient is a negative multiple of the outward normal. Equality of the two adjacent unit speeds then makes the incoming normal component strictly negative and the outgoing one strictly positive, with equal tangential components: this is exactly the regular exterior specular law. At the endpoints the gradients have norm one and give normal departure and arrival. The strict third-disk inequalities (40.2) exclude every unlisted collision; the signs just proved exclude another intersection with either incident disk inside its flight. Thus Y is a true regular first-hit record. Corollary 38.3 gives uniqueness. Nothing in the proof keeps section visits constant along a competitor or along a homotopy; the original labels of Q have not changed. □

Proposition 40.2 (A deterministic positive junction cost)**.** *Partition a word into consecutive nonempty blocks at any chosen indices. For each block choose a minimizing unit-edge dual vector as in Theorem 38.1. Denote the last vector before a junction by v^- and the first after it by v^+. Then*

$$\ell_w(R) \ge \sum_{\text{blocks } b} \ell_b(R) + R \sum_{\text{junctions}} (2 - |v^- - v^+|). \tag{40.4}$$

Every junction term is nonnegative. If both block minima at a junction are physical critical records with terminal/departure normals n^-, n^+, the corresponding term is $R(2 - |n^- + n^+|)$. The comparison does not assert physical realizability of any block minimum and does not factor any trajectory probability.

Proof. Concatenate the chosen dual flight vectors. They form a feasible dual vector for the whole word. The center term C_w is the sum of the block center terms. At a junction the two separate endpoint penalties are $R|v^-| + R|v^+| = 2R$, whereas the penalty of the concatenated dual vector is $R|v^- - v^+|$. All other penalties agree. Substitution into (38.2) proves (40.4), including each junction once. In the physical critical case $v^- = -n^-$ and $v^+ = n^+$, giving the asserted normal expression. This argument is an exact vector comparison, not a renewal or Markov decomposition. □

Proposition 40.3 (Word localization of the unchanged complete roof source)**.** *Fix the original m, R, n, k and any nonnegative Borel trajectory weight h. Let $x_w(t, \vartheta)$ be exactly the partial physical inverse in* (10.6)*, with all original event cuts. That formula admits the additional exact mask*

$$\mathbf{1}_{\{\ell_w(R) \le t \le \ell_w(R) + E_R(x_w(t,\vartheta))\}} \tag{40.5}$$

in every existing summand, without changing either side. In particular only words with $\ell_w(R) \in [t-2R,t]$ *can contribute. This assertion requires no physical critical center, clearance guard, pressure hypothesis or incidence lower bound for the regular source.*

For clarity about what a pressure consequence does and does not add, define the deterministic localized physical-word pressure

$$P_{m,R}(J) = \sum_{\substack{w\in\mathcal{W}_m(R)\\ \ell_w(R)\in J}} \sup_{[w]_R\cap M'_R} j_m.$$

Under exactly *Input 11.1 and the inherited roof–angle formula, for* $0 \le h \le 1$,

$$\frac{dL_{\#}(h\mathbf{1}_{\Pi_{n,k,m,R}}\nu_R^*)}{dt}(t) \le \frac{C_0}{2c_*\lambda_\circ} P_{m,R}([t-2R,t]) \quad \textit{for almost every } t. \tag{40.6}$$

This is an unscaled comparison, not a new estimate on that localized pressure and not a proof of the original normalized shortage.

Proof. Theorem 38.2 applies to every existing physical inverse point, so its mask is one. Apply the nonnegative source exhaustion of Proposition 10.4; the mask is Borel, and no test or boundary is differentiated. This proves the exact identity with the mask, for extended weights as well. For the last bound use $\chi \ge \lambda_\circ$ and $\omega \le C_0 c_m j_m \le C_0 j_m$ at the actual point, and dominate h and the actual event only on the upper side by one. Global physical roof–angle uniqueness gives at most one point per word. Each word retained by the mask lies in the indicated geometric interval, so its stable weight is bounded by the same-word supremum at the *same* R. Integration over the angle of length 2π gives the factor in (40.6). The section factor c_*^{-1} occurs once, exactly as in the inherited formula. The supremum of this unlocalized pressure is the retained Q_*; no decay in m for the localized expression has been inferred. □

40.1. **The original positive shortage is still the closing condition.** The new results remove the need for a physical critical anchor in the geometric word enclosure; certify an actual-versus-unphysical minimizing configuration using nonnegative gaps; give a positive junction defect; and compare critical roofs across disconnected physical radius domains. They do not identify the auxiliary ℓ_w with a physical clock when the certificate fails. In particular (40.5) does not replace the observed microscopic window by $[t-2R,t]$. The observed quantity is still the density at the original t with the original exact labels; the interval is solely an upper-support enclosure in a separate deterministic word statistic.

No new theorem here supplies a normalized roof-height estimate for the remaining positive source or enough nongrazing comparison capacity. The raw target is still (1.5), with the very same minimal ordered closure condition (15.8). It includes the low-load incidence-only source, first-order unvisited clearance, noncritical singular charts, one-sided paired/section traces and all original arithmetic/denominator requirements. The finite source sum is taken before $\sup_R \operatorname{ess\,sup}_t$ and before the collision limsup; all microscopic widths remain fixed until that limsup, and $B \to \infty$ is outermost. Neither an auxiliary convex value, a positive curvature identity, a finite diagnostic, nor the radius-integrated coarea pays that still unproved limit automatically.

41. Uniform localization of the entire variational sublevel

41.1. **Actual records and the inherited comparison functional.** All lengths are in units of the triangular lattice spacing. Let

$$\Lambda = \{(a+b/2, \sqrt{3}b/2) : a,b \in \mathbb{Z}\}, \qquad R \in [9/20, 47/100], \quad R_+ = 47/100.$$

Fix any actual regular record of $m \geq 1$ free flights and $m+1$ collision points $Q_0, \dots, Q_m$. Its successive lattice centers are $O_0, \dots, O_m$, with $O_i \neq O_{i+1}$. Write

$$Q_i = O_i + Rn_i, \quad |n_i| = 1, \quad Q_{i+1} - Q_i = \tau_i v_i, \quad |v_i| = 1.$$

Regular means genuine exterior, first-hit, nontangent specular collisions at all these points and no extra collision on any open flight. Nonconsecutive repeated centers are permitted. At interior collisions let

$$c_i = -v_{i-1} \cdot n_i = v_i \cdot n_i > 0, \qquad v_i - v_{i-1} = 2c_i n_i \quad (1 \leq i < m). \tag{41.1}$$

The endpoint cosines are $c_0 = v_0 \cdot n_0 > 0$ and $c_m = -v_{m-1} \cdot n_m > 0$. Define

$$L = \sum_{i=0}^{m-1} \tau_i, \qquad E = R(2 - c_0 - c_m), \qquad 0 \leq E \leq 2R \leq 47/50 < 1. \tag{41.2}$$

The original endpoint momenta obey $p_0^2 + c_0^2 = p_m^2 + c_m^2 = 1$, so $E = R[p_0^2/(1+c_0) + p_m^2/(1+c_m)]$.

For this fixed ordered word, the comparison domain and length are

$$\mathcal{K} = \prod_{i=0}^{m} \overline{B}(O_i, R), \qquad A(Y) = \sum_{i=0}^{m-1} |Y_{i+1} - Y_i|.$$

Repeated centers give separate factors. This is the native disk-product comparison, not a substitute physical dynamics. Define its full sublevel

$$\mathcal{S}(Q) = \{Y \in \mathcal{K} : A(Y) \leq L\}.$$

It is nonempty and compact. Every global minimum of A lies in it, whether or not the minimum is physical. For $Y \in \mathcal{K}$ put

$$\delta_i = Y_i - Q_i, \qquad Y_{i+1} - Y_i = \ell_i \widehat{v}_i, \qquad e_i = \widehat{v}_i - v_i.$$

Consecutive disks are separated by $\tau_- = 1 - 2R_+ = 3/50$, so every $\ell_i \geq \tau_- > 0$. The retained geometric facts, reconstructed in Appendix A, are

$$\tau_i \leq H := 28/5, \qquad \sum_{i=a+1}^{a+93} c_i \geq \gamma := 1/4480 \quad \text{whenever } 0 \leq a,\ a+94 \leq m. \tag{41.3}$$

Thus every complete 94-flight block has an interior index with $c_i \geq c_{\mathrm{a}} := 1/416640$. No such condition is imposed on all individual incidences.

41.2. **The complete positive gap, not a coordinatewise singular estimate.** The next calculation extends the inherited endpoint-gap bookkeeping from minima to the entire sublevel. The general disk duality and the endpoint defect themselves were already established in module 297.

Lemma 41.1 (Full sublevel energy and contact budget)**.** *For every $Y \in \mathcal{S}(Q)$,*

$$\sum_{i=0}^{m-1} \ell_i |e_i|^2 \leq 2E, \tag{41.4}$$

$$\sum_{i=1}^{m-1} c_i |\delta_i|^2 \leq RE, \tag{41.5}$$

$$|\delta_0|, |\delta_m| \leq 2\sqrt{2RE}. \tag{41.6}$$

These statements hold for interior or inactive comparison vertices as well as boundary vertices.

Proof. Set $d_0 = -v_0$, $d_i = v_{i-1} - v_i = -2c_i n_i$ for $1 \le i < m$, and $d_m = v_{m-1}$. Telescoping, including both endpoints exactly once, gives

$$\sum_{i=0}^{m-1} v_i \cdot (Y_{i+1} - Y_i) = \sum_{i=0}^{m} d_i \cdot Y_i.$$

Consequently, with $D = \sum_i (d_i \cdot O_i - R|d_i|)$,

$$A(Y) - D = \frac{1}{2} \sum_{i=0}^{m-1} \ell_i |e_i|^2 + \sum_{i=0}^{m} \big[d_i \cdot (Y_i - O_i) + R|d_i| \big]. \tag{41.7}$$

Every term is nonnegative: the first uses two unit vectors, and the second is the difference from the minimum of a linear functional on its actual disk. At Q every flight and interior contact term is zero; the endpoint terms are $R(1 - c_0)$ and $R(1 - c_m)$. Thus $L - D = E$. Since $A(Y) \le L$, the whole right-hand side of (41.7) is at most E, proving (41.4).

For any $z \in \overline{B}(0, R)$ and unit n, the exact identity

$$2R(R - n \cdot z) = |z - Rn|^2 + R^2 - |z|^2 \tag{41.8}$$

implies $|z - Rn|^2 \le 2R(R - n \cdot z)$. An interior contact term in (41.7) equals $2c_i(R - n_i \cdot (Y_i - O_i))$ and is therefore at least $c_i |\delta_i|^2 / R$. Summing proves (41.5), without dividing by any individual c_i.

At the first endpoint let $Z_0 = O_0 + Rv_0$. The endpoint term gives $|Y_0 - Z_0|^2 \le 2RE$ by (41.8), while $|Q_0 - Z_0|^2 = 2R^2(1 - c_0) \le 2RE$. The triangle inequality proves the first bound in (41.6). At the final endpoint use $Z_m = O_m - Rv_{m-1}$; the signs give the same bound. □

Lemma 41.2 (Transverse telescoping with the turning work retained)**.** *Put $w_i = v_i^{\perp}$ and $r_i = \ell_i e_i$. For $0 \le j < i \le m$,*

$$w_{i-1} \cdot \delta_i = w_j \cdot \delta_j + \sum_{l=j}^{i-1} w_l \cdot r_l + \sum_{l=j+1}^{i-1} (w_l - w_{l-1}) \cdot \delta_l. \tag{41.9}$$

If $i - j \le 94$ and $Y \in \mathcal{S}(Q)$, the absolute values of the last two sums are respectively bounded by $38\sqrt{E}$ and $14\sqrt{E}$.

Proof. The exact displacement equation is

$$\delta_{l+1} - \delta_l = (\ell_l - \tau_l) v_l + r_l.$$

Its projection on w_l removes only its longitudinal term. Summing these projected equations and collecting coefficients of each δ_l yields (41.9). The last sum is present because the transverse frame turns. It cannot in general be omitted.

All comparison vertices obey $|\delta_l| \le 2R$, so

$$\ell_l \le \tau_l + |\delta_l| + |\delta_{l+1}| \le H + 4R_+ = 187/25 =: T.$$

Weighted Cauchy–Schwarz and (41.4) give

$$\sum_{l=j}^{i-1} |r_l| \le \left(\sum_{l=j}^{i-1} \ell_l \right)^{1/2} \left(\sum_{l=j}^{i-1} \ell_l |e_l|^2 \right)^{1/2} \le \sqrt{2 \cdot 94 \cdot TE} < 38\sqrt{E}.$$

The rotation is an isometry, so (41.1) gives $|w_l - w_{l-1}| = 2c_l$. Therefore (41.5) gives

$$\sum_{l=j+1}^{i-1} |(w_l - w_{l-1}) \cdot \delta_l| \le 2 \left(\sum_{l=j+1}^{i-1} c_l \right)^{1/2} \left(\sum_{l=j+1}^{i-1} c_l |\delta_l|^2 \right)^{1/2} \le 2\sqrt{94 R_+ E} < 14\sqrt{E}.$$

Only actual reflection cosines enter the turning sum. In particular no dynamics is ascribed to the comparison polygon. □

Theorem 41.3 (Collision-uniform sublevel localization without an incidence cutoff)**.** *For every actual regular record in the original radius family, every $m \geq 1$, and every $Y \in \mathcal{S}(Q)$,*

$$\max_{0\leq i\leq m} |Y_i - Q_i| \leq 22\, E^{1/4}. \tag{41.10}$$

The constant is independent of the word, collision count, section/return/displacement labels and every interior incidence. At $E = 0$ the sublevel is the singleton $\{Q\}$. No optimality of the exponent or nontriviality for large E is asserted.

Proof. If $E = 0$, (41.5) and positivity of the actual interior cosines force all interior displacements to vanish, and (41.6) handles the endpoints. Assume $0 < E < 1$.

Call an interior index an anchor if $c_i \geq c_{\mathrm{a}}$. At such an index,

$$|\delta_i| \leq \sqrt{R_+/c_{\mathrm{a}}}\sqrt{E} < 443\sqrt{E}.$$

Together with $|\delta_i| \leq 2R_+$ this gives $|\delta_i|^2 < 2R_+443\sqrt{E} < 22^2\sqrt{E}$. At either endpoint (41.6) gives $|\delta_i|^2 \leq 8R_+E < 4\sqrt{E} < 22^2\sqrt{E}$.

Consider a nonanchor interior index i. Let $j < i$ be the last anchor before it, or $j = 0$ if none exists. Necessarily $i - j < 94$: otherwise the 94-flight block starting at j has by (41.3) an anchor strictly between j and i, a contradiction. The initial displacement in (41.9) is at most $443\sqrt{E}$, also when $j = 0$ by (41.6). Lemma 41.2 therefore yields

$$|w_{i-1} \cdot \delta_i| \leq (443 + 38 + 14)\sqrt{E} = 495\sqrt{E}.$$

Because $n_i \cdot v_{i-1} = -c_i$ and $|n_i \cdot w_{i-1}| \leq 1$,

$$|n_i \cdot \delta_i| \leq c_i|\delta_i| + |w_{i-1} \cdot \delta_i|.$$

Crucially, the first term needs no inverse incidence: by (41.5), $c_i|\delta_i| \leq \sqrt{c_i}\sqrt{c_i|\delta_i|^2} \leq \sqrt{R_+E} < \sqrt{E}$. Thus $|n_i \cdot \delta_i| \leq 496\sqrt{E}$. Since Y_i lies in the disk and Q_i on its boundary, (41.8) becomes

$$|\delta_i|^2 \leq -2Rn_i \cdot \delta_i \leq 2R_+496\sqrt{E} = \frac{11656}{25}\sqrt{E} < 484\sqrt{E}.$$

Taking square roots proves (41.10). The argument treats short records and the first incomplete block without inserting an artificial collision. It uses at most one bounded physical block for each index, not a sum over the full record or over its words. □

Remark 41.4 (What is new relative to the complete predecessor). The earlier pointwise contact certificate had a factor $c_i^{-1/2}$. Here the contact budget is used *before* division: its weighted sum pays the turning work in (41.9), and disk curvature converts the resulting normal control into an unweighted position bound. The dense-anchor input is inherited real geometry, not a fresh probability hypothesis. The result concerns every point of a continuous compact sublevel, not a list of sampled minima. It neither constructs an orbit through a singularity nor controls forward-time derivatives at grazing.

42. All-center clearance and a true physical reference

For a record Q put $S_i(Q) = [Q_i, Q_{i+1}]$. Define the true third-disk clearance of a flight by

$$\Gamma_i(Q) = \inf_{z\in\Lambda\setminus\{O_i,O_{i+1}\}} \operatorname{dist}(z, S_i(Q)) - R, \qquad \Gamma(Q) = \min_{0\leq i<m} \Gamma_i(Q). \tag{42.1}$$

The infimum includes disks visited elsewhere in the word. It excludes only the two incident centers of this particular flight. Distances are Euclidean distances to closed segments, so a minimizing foot may change or reach an endpoint. There is no chosen differentiable nearest-center label.

Theorem 42.1 (One modulus for every third lattice disk)**.** *For every $Y \in \mathcal{S}(Q)$ and $d(E) = 22E^{1/4}$,*

$$d_H(S_i(Y), S_i(Q)) \le d(E), \tag{42.2}$$

$$|\operatorname{dist}(z, S_i(Y)) - \operatorname{dist}(z, S_i(Q))| \le d(E) \quad \text{for every } z \in \mathbb{R}^2, \tag{42.3}$$

$$|\Gamma_i(Y) - \Gamma_i(Q)| \le d(E). \tag{42.4}$$

All these estimates hold simultaneously over all flights and all lattice centers. They require no lower bound on any interior incidence.

Proof. Match points with the same affine segment parameter $s \in [0,1]$. Their difference is $(1-s)\delta_i + s\delta_{i+1}$, whose norm is at most $\max\{|\delta_i|, |\delta_{i+1}|\} \le d(E)$ by Theorem 41.3. This bounds both directed Hausdorff distances. For two nonempty compact sets at Hausdorff distance at most d, the triangle inequality gives $\operatorname{dist}(z, A) \le \operatorname{dist}(z, B) + d$ and the reverse inequality, proving (42.3). Take infima over the *same* center set in both directions to obtain (42.4). The infima are finite. No counting or differentiability of the minimizing center is used. □

Theorem 42.2 (A grazing-independent clearance certificate for a physical minimum)**.** *Let Q be any original regular record, and put*

$$\varepsilon(E) = \sqrt{2E/\tau_-}, \qquad d(E) = 22E^{1/4}.$$

Suppose

$$c_i(Q) > \varepsilon(E) \ (1 \le i < m), \qquad \Gamma(Q) > d(E). \tag{42.5}$$

The first condition is vacuous for $m = 1$. Then A has exactly one global minimum on $\mathcal{K}$. This minimum is an actual regular exterior first-hit trajectory of the same ordered collision word, with normal initial and final endpoints. All these assertions concern the unchanged radius R and actual circular obstacles. They assert no equality of its section/return labels or physical roof with those of Q.

Proof. Compactness supplies at least one minimizing Y. Since $A(Y) \le A(Q)$, Theorem 41.3 and Lemma 41.1 apply. Each edge error obeys

$$|\widehat{v}_i - v_i| \le \sqrt{2E/\ell_i} \le \varepsilon(E).$$

All edges are nonzero because consecutive disks are separated. The gradient of A with respect to an interior vertex is $\widehat{d}_i = \widehat{v}_{i-1} - \widehat{v}_i$. By the physical reflection law of Q and (42.5),

$$|\widehat{d}_i| \ge 2c_i - |e_{i-1}| - |e_i| \ge 2c_i - 2\varepsilon(E) > 0.$$

First-order optimality on the ith closed disk means Y_i minimizes its linear functional $\widehat{d}_i \cdot z$. Hence $Y_i = O_i + R\widehat{n}_i$ with $\widehat{d}_i = -|\widehat{d}_i|\widehat{n}_i$. For two unit vectors a, b with $a - b = -\lambda n$, $\lambda > 0$, taking the scalar product with $a + b$ shows $a \cdot n = -b \cdot n$; taking it with n gives $a \cdot n = -\lambda/2$ and $b \cdot n = \lambda/2$. Their tangential components are equal. This proves a true nontangent specular reflection at each interior vertex, with the correct exterior signs. At the endpoints the gradients are $-\widehat{v}_0$ and $\widehat{v}_{m-1}$, of norm one. Their disk minimizers give $\widehat{n}_0 = \widehat{v}_0$ and $\widehat{n}_m = -\widehat{v}_{m-1}$, precisely normal endpoints.

Theorem 42.1 and the strict second inequality in (42.5) give $\Gamma_i(Y) > 0$ for every flight. Thus no third disk meets any flight, including a disk appearing at a different time in the word. The two incident disks are avoided in the open flight as well: positive departing normal component implies $|Y_i + s\widehat{v}_i - O_i|^2 = R^2 + 2Rs(\widehat{n}_i \cdot \widehat{v}_i) + s^2 > R^2$ for $s > 0$, and the analogous backward formula applies at arrival. This proves first-hit physicality without discarding any collision.

For uniqueness, use this physical Y as the reference in (41.7). Its normal endpoints have defect zero. Any other minimum Z has $A(Z) = A(Y)$, so the nonnegative gap is zero. All

contact coefficients are strictly positive (one at each endpoint and $2\widehat{c}_i$ internally), and (41.8) forces $Z_i = Y_i$ at every vertex. Thus the minimum is unique. □

Remark 42.3 (Sufficiency, not source deletion or capacity). The test may fail while a physical minimum still exists. Failed records remain in the original source. The incidence requirement in (42.5) checks nonzero comparison reflections; it is not an inverse-incidence loss in the clearance modulus. An actual critical reference is one trajectory, not a positive measure comparison pool. Exact section visits and return labels must be evaluated anew on it, not transported from Q by proximity. The original observed roof remains $L(Q)$, never $A(Y)$.

Proposition 42.4 (Positive inclusion on the unchanged exact-record source)**.** *Fix R, m and any of the original exact return/displacement/collision events Π, with its original section probability measure ν_R^* and nonnegative measurable weight h. Let $\mathcal{B}$ be the subset of regular records whose ordered disk word does not have a unique regular physical normal-endpoint global minimum. Set*

$$\mathcal{W} = \{Q : \exists i \in \{1, \dots, m-1\},\ c_i(Q) \le \varepsilon(E(Q)) \quad \text{or} \quad \Gamma(Q) \le d(E(Q))\}. \tag{42.6}$$

On the regular source, $\mathcal{B} \subseteq \mathcal{W}$. Consequently the following positive measure inequality holds on the original physical roof axis*:*

$$L_*(\mathbf{1}_{\mathcal{B}}\mathbf{1}_{\Pi} h \, d\nu_R^*) \ \le \ L_*(\mathbf{1}_{\mathcal{W}}\mathbf{1}_{\Pi} h \, d\nu_R^*). \tag{42.7}$$

This includes all physical words with one-copy multiplicity and extends to nonnegative marked sums by monotone convergence. It changes neither the conditioned event nor any roof observation window. It asserts no small upper bound on the right-hand side.

Proof. The inclusion is the contrapositive of Theorem 42.2. At fixed R, m the physical ordered-word collection is countable; the property defining $\mathcal{B}$ is a property of a word, not of a chosen minimizer. Thus $\mathcal{B}$ is a union of its measurable regular word cylinders. The functions in (42.6) are measurable on every regular chart: each center-to-segment distance is continuous, the center set is countable, and finite minima and the displayed elementary functions preserve measurability. No measurable selection of a minimum is needed.

Multiply the pointwise inclusion by $\mathbf{1}_{\Pi} h \mathbf{1}_{\{L \in I\}}$ for any Borel roof set I, then integrate. This gives (42.7), initially for bounded h and then by increasing truncation. Apply the same argument on disjoint original marked copies and use monotone convergence for complete nonnegative sums. The section normalization is the original one already in ν_R^*; no additional Jacobian or copy factor appears. □

42.1. **Exact remaining analytic obligation.** If the two pushforwards are absolutely continuous, (42.7) orders their densities almost everywhere at the same R, m, n, k, t. Without that hypothesis it remains a measure inequality, not an assertion that a singular component vanishes. Paired boundary traces are not bulk measures and are not set to zero by this argument.

The positive witness source in (42.7) is *not* proved to be small in $m^2 L^\infty(dt)$. Nor are all physical-reference records shown to meet the original nongrazing comparison-pool constraints. The original first-order single/cross-flight clearance, low-load incidence source, degenerate/critical charts, section cuts and singular pairing terms therefore remain in (0.2). The present theorem supplies a proved geometric mechanism for obtaining a physical reference without the previous inverse-incidence clearance loss. A normalized source/pressure or original-capacity argument must still exploit that mechanism, with the original full sums and order of limits, to close the raw theorem.

43. Summable physical coercivity and critical singular closure

The maximum estimate in Theorem 41.3 is inherited. Summing its fourth power would introduce a factor $m+1$. We instead sum the positive gap before estimating the transverse work. This gives one budget for the complete record, including arbitrarily small incidences. Throughout, $R \in [9/20, 47/100]$, $m \geq 1$, and the notation $Q, O, n, v, \tau, Y, \ell, \delta, e, L, D, E$ is exactly that of Section 41. Set

$$r = 47/100, \quad N = 94, \quad a = 1/416640, \quad T = 187/25, \quad \tau_- = 3/50. \tag{43.1}$$

No probability, roof window or return event is changed by the following comparison.

Definition 43.1 (Closure of a fixed physical word)**.** A closed physical reference is a coordinatewise limit of actual regular records with the same finite ordered centers $(O_0, \dots, O_m)$, with radii converging inside the original closed radius interval. The record length and word are fixed while taking this closure. At a limiting contact there may be zero incidence or an unlisted disk tangency. We regard the limit as a polygonal reference, not as a definition of the billiard map at that singularity, and do not assign it return labels by continuity.

Lemma 43.2 (Positive gap on the closed physical reference)**.** *Actual regular records and closed physical references satisfy the reflection and block bounds*

$$v_i - v_{i-1} = 2c_i n_i, \quad 0 \leq c_i \leq 1, \qquad \sum_{i=b+1}^{b+93} c_i \geq 1/4480 \quad (b+94 \leq m). \tag{43.2}$$

For every $Y \in \mathcal{K}$, not only a minimizing polygon, put

$$G = A(Y) - D, \qquad h_i = d_i \cdot (Y_i - O_i) + R|d_i|, \qquad E_0 = R(1-c_0), \quad E_m = R(1-c_m).$$

Then $G \geq 0$, $E = E_0 + E_m = L - D$, and

$$G = \tfrac{1}{2} \sum_{i=0}^{m-1} \ell_i |e_i|^2 + \sum_{i=0}^{m} h_i, \qquad h_i \geq 0, \tag{43.3}$$

$$\sum_{i=1}^{m-1} c_i |\delta_i|^2 \leq R \sum_{i=1}^{m-1} h_i, \qquad \sum_i \ell_i |e_i|^2 \leq 2G, \tag{43.4}$$

$$|\delta_0|^2 \leq 4R(h_0 + E_0), \qquad |\delta_m|^2 \leq 4R(h_m + E_m). \tag{43.5}$$

Also $|\delta_i| \leq 2r$ and $\tau_- \leq \ell_i \leq T$.

Proof. For regular records, the exact gap is the telescoping identity (41.7); it holds on the entire disk product. The disk identity (41.8) gives each interior contact inequality, including when $c_i = 0$. For the first endpoint, use $Z_0 = O_0 + Rv_0$. Then $|Y_0 - Z_0|^2 \leq 2Rh_0$ and $|Q_0 - Z_0|^2 = 2RE_0$. The inequality $|u+v|^2 \leq 2|u|^2 + 2|v|^2$ gives (43.5); use $Z_m = O_m - Rv_{m-1}$ at the final endpoint. Since consecutive disks are separated, every edge is nonzero, even on the whole comparison product. The upper bound is $\ell_i \leq \tau_i + |\delta_i| + |\delta_{i+1}| \leq 28/5 + 4r = T$. For a closed reference, positions, unit edge directions, normals, incidences and lengths converge: their denominators are bounded below by R_- or τ_-. The regular reflection equations, horizon bound and each fixed finite block inequality therefore pass to the limit. The same algebra then applies directly at the limit. This passage does not continue a flow across a grazing collision. □

Theorem 43.3 (Complete-record quartic coercivity)**.** *For every actual regular record or closed physical reference, every $m \geq 1$ and every $Y \in \mathcal{K}$,*

$$\sum_{i=0}^{m} |Y_i - Q_i|^4 \leq C_G G + C_E E, \tag{43.6}$$

where the following constants are independent of the word, its length, the original radius, and all incidences:

$$C_G = \frac{4r^3}{a} + 16r^3 + 8r^2\left(r + \frac{3Nr}{a} + 6N^2T + 12N^2r\right) = \frac{12318659063517}{125000}, \tag{43.7}$$

$$C_E = 16r^3 + 96Nr^3 = \frac{11731999}{12500}. \tag{43.8}$$

In particular, with $C = 10^8$, the entire inherited sublevel obeys

$$\sup_{Y\in\mathcal{S}(Q)} \sum_{i=0}^{m} |Y_i - Q_i|^4 \le CE. \tag{43.9}$$

The supremum is over the continuous compact sublevel, not a sample of minima. This is a summability estimate, rather than the m-multiple of a one-contact bound.

Proof. Call an interior index an anchor when $c_i \ge a$. Let $\mathcal{A}$ be the anchor set and $\mathcal{I}$ the other interior indices. For each $i \in \mathcal{I}$, let $j(i)$ be the last anchor strictly before i, or 0 when there is none. Equation (43.2) implies $i - j(i) < N$: a complete N-flight block starting at $j(i)$ would otherwise have an interior anchor before i. Thus each anchor or initial endpoint is used as a start at most N times. Each flight or interior contact lies in at most N of the intervals $[j(i), i]$. These are deterministic overlap bounds on one record; there is no multiplication by the number of its words.

Put $w_l = v_l^{\perp}$ and $b_i = w_{i-1} \cdot \delta_i$. The exact transverse telescope (41.9), which also holds in the closure, and the inequality $(x + y + z)^2 \le 3(x^2 + y^2 + z^2)$ give

$$|b_i|^2 \le 3|\delta_{j(i)}|^2 + 3NT \sum_{l=j(i)}^{i-1} \ell_l |e_l|^2 + 12N \sum_{l=j(i)+1}^{i-1} c_l |\delta_l|^2.$$

Indeed the squared flight term is at most $(\sum \ell_l)(\sum \ell_l |e_l|^2)$; the squared turning term is at most $4(\sum c_l)(\sum c_l |\delta_l|^2)$, since $|w_l - w_{l-1}| = 2c_l$. All terms here are physical reflection quantities of the reference, not of Y.

The starting-point budget is shared, not paid anew for every i. Disjointness of the anchor and initial contact terms and $4r \le r/a$ imply

$$\sum_{j\in\mathcal{A}} |\delta_j|^2 + |\delta_0|^2 \le \frac{r}{a} \sum_{j\in\mathcal{A}} h_j + 4rh_0 + 4rE_0 \le \frac{r}{a} G + 4rE.$$

Summing the preceding transverse bound and using the overlap at most N and (43.4) now proves

$$\sum_{i\in\mathcal{I}} |b_i|^2 \le \left(\frac{3Nr}{a} + 6N^2T + 12N^2r\right) G + 12NrE. \tag{43.10}$$

This is where a complete-record budget replaces the unbounded number of local estimates.

For $i \in \mathcal{I}$, disk curvature and $n_i \cdot v_{i-1} = -c_i$ give

$$|\delta_i|^2 \le -2Rn_i \cdot \delta_i \le 2r(c_i|\delta_i| + |b_i|), \qquad |\delta_i|^4 \le 8r^2(c_i^2|\delta_i|^2 + |b_i|^2).$$

Use $c_i^2 \le c_i$, sum, and apply (43.4) and (43.10). The nonanchor contribution is at most

$$8r^2\left(r + \frac{3Nr}{a} + 6N^2T + 12N^2r\right) G + 96Nr^3E.$$

For anchors, $|\delta_i|^4 \le 4r^2|\delta_i|^2$ and the contact budget give a total at most $4r^3G/a$. The two endpoints together contribute at most $16r^3(G + E)$ by (43.5). Adding these three disjoint classes

proves (43.6). It also covers $m = 1$, empty anchor sets and both incomplete end blocks. The exact rational evaluations (43.7)–(43.8) satisfy

$$C_G + C_E = \frac{12318776383507}{125000} < 10^8, \qquad 10^8 - C_G - C_E = \frac{181223616493}{125000} > 0.$$

On $\mathcal{S}(Q)$, $0 \le G = A(Y) - D \le L - D = E$, proving the claimed supremum estimate. No exchange of a collision limit with that supremum has been made. □

Corollary 43.4 (Coercivity at a critical singular limit)**.** *If the closed physical reference has normal endpoints, so $E = 0$, then for every $Y \in \mathcal{K}$*

$$A(Y) \ge L + C_G^{-1} \sum_{i=0}^{m} |Y_i - Q_i|^4. \tag{43.11}$$

Thus Q is the unique disk-product minimum, even if some internal incidences vanish. An approximate minimum of excess at most η has $\sum_i |Y_i - Q_i|^4 \le C_G \eta$.

Proof. Here $D = L$, so $G = A(Y) - L$. Apply (43.6). Nonnegativity and equality force every coordinate to agree with Q; the approximate assertion follows from $G \le \eta$. In particular this proof does not divide by a vanishing incidence, unlike the regular critical contact argument alone. □

Corollary 43.5 (Unique fixed-word compactification)**.** *Fix m and its ordered centers. Any sequence of actual regular normal-endpoint records of that word with $R_j \to R_*$ in the original interval converges to the same closed physical reference. Its roof equals $\ell_w(R_*)$. This asserts uniqueness of the limiting polygon, not physical continuation, section-label continuity or a coarea density at the singular limit.*

Proof. The finite product of bounded disks is compact. Every convergent subsequence is a closed physical reference with normal endpoints. By Corollary 43.4, each such limit is the unique minimizer of the same functional at R_*. All subsequential limits therefore agree, which proves convergence of the entire sequence and the roof identity by continuity of Euclidean length. The conclusion does not require the set of admissible radii to be connected. No orbit is inserted at a radius where none is physical. □

Remark 43.6 (What the singular statement does not remove)*.* The limiting word may have an unlisted tangency and a singular contact can carry paired boundary data not seen by bulk collision measure. Uniqueness of its minimizing polygon does not bound the inverse roof Jacobian, continue the return map, or show that its normalized paired source vanishes. All such trace terms in the inherited source remain present. The constants above are intentionally nonoptimal; their independence of the collision count and incidence is the conclusion.

44. Summable clearance and exact-mark transport on the physical clock

We retain the physical model and the actual event (1.1), including $d\nu_R = d\alpha\, dp/(4\pi)$ and $d\nu_R^* = c_*^{-1} \mathbf{1}_{Y_R^*} d\nu_R$, where $c_* = 91/(10000\pi)$. Statements involving physical observations below are only made when both compared configurations are actual regular records of the same ordered word. The geometric statement comes first and needs no such physicality of the comparison polygon.

Theorem 44.1 (One budget for every center and every flight)**.** *For any reference and any $Y \in \mathcal{S}(Q)$ allowed by Theorem 43.3, define*

$$\kappa_i = \sup_{z \in \Lambda} \left| \operatorname{dist}(z, [Y_i, Y_{i+1}]) - \operatorname{dist}(z, [Q_i, Q_{i+1}]) \right|.$$

Then

$$\sum_{i=0}^{m-1} \kappa_i^4 \le 2CE, \qquad \#\{i : \kappa_i \ge \rho\} \le 2CE\rho^{-4} \quad (\rho > 0). \tag{44.1}$$

The same estimates apply to the absolute differences of the unvisited-center clearances Γ_i in Section 42. All centers, including a center visited on a different flight, and all possible feet on the closed segments are included. Negative clearance of a nonphysical Y is permitted and is not suppressed.

Proof. Using the same affine parameter on the two segments gives their Hausdorff distance at most $\max\{|\delta_i|, |\delta_{i+1}|\}$. Distance to any fixed point is 1-Lipschitz for Hausdorff distance, so the bound holds simultaneously for every lattice center, without any enumeration. Thus $\kappa_i^4 \le |\delta_i|^4 + |\delta_{i+1}|^4$. The full sum is at most twice the contact sum in (43.9); each endpoint is in fact counted only once. Taking infima over the identical set $\Lambda \setminus \{O_i, O_{i+1}\}$ preserves this uniform difference bound for Γ_i. Finally $\rho^4 \#\{i : \kappa_i \ge \rho\} \le \sum_i \kappa_i^4$. □

44.1. **Physical reference class and the original section.** Fix R and m. Let $\mathcal{C}_{R,m}$ be the union of actual regular word cells whose disk-product minimum is an actual regular normal-endpoint record Y^w. Such a minimum is unique by the retained critical result, or Corollary 43.4. The sufficient strict physicality certificate of Theorem 42.2 yields a subclass; no claim is made that all words lie in $\mathcal{C}_{R,m}$. Word cells and their original exact-event subsets are Borel, with the usual countable singularity partition. For this fixed R, m, the assignment of the constant record Y^w on each word cell is therefore Borel; it involves no unproved measurable choice of auxiliary minima.

For $Q \in \mathcal{C}_{R,m}$ compare its $m+1$ collision states $x_i = (\alpha_i, p_i)$ with $\widehat{x}_i$ of Y^w. Angles are taken in $\mathbb{R}/(2\pi\mathbb{Z})$, and

$$d_\infty(x, \widehat{x}) = \max\{\mathrm{dist}_{\mathbb{S}^1}(\alpha, \widehat{\alpha}), |p - \widehat{p}|\}.$$

At the last collision, the tangential momentum can be computed from the incoming flight: specular reflection leaves that component unchanged. Consequently the terminal state, although its next flight is not in the record, is fully determined for the comparison.

Theorem 44.2 (Full phase budget and section discrepancy)**.** *Put $R_- = 9/20$ and*

$$K = 128/\tau_- + 24C/R_-^4. \tag{44.2}$$

For every actual $Q \in \mathcal{C}_{R,m}$, writing $E = E_R(Q)$,

$$\sum_{i=0}^{m} d_\infty(x_i, \widehat{x}_i)^4 \le KE. \tag{44.3}$$

Let $\eta_R = \mathbf{1}_{Y_R^*}$ denote the original section indicator, not a smoothed or enlarged section, and define

$$N_{\mathrm{cut}}(Q) = \sum_{i=0}^{m} |\eta_R(x_i) - \eta_R(\widehat{x}_i)|,$$

$$B_\rho(Q) = \sum_{i=0}^{m} \mathbf{1}_{\{\mathrm{dist}_\infty(x_i, \partial Y_R^*) \le \rho\}}.$$

Then for every $\rho > 0$

$$N_{\mathrm{cut}}(Q) \le B_\rho(Q) + K\rho^{-4} E_R(Q). \tag{44.4}$$

Both endpoints are included once, whereas the original occupation count $A_m = \sum_{i=0}^{m-1} \eta_R(x_i)$ still excludes the final state. In particular $|A_m(Q) - A_m(Y^w)| \le N_{\mathrm{cut}}(Q)$.

Proof. Since Y^w is physical and lies in $\mathcal{S}(Q)$, all its vertices are on the original disks. The unit normals differ by $|\widehat{n}_i - n_i| = |\delta_i|/R$. For angles at circular distance $\theta \in [0,\pi]$, $2\sin(\theta/2) \geq 2\theta/\pi$. Thus

$$\mathrm{dist}_{\mathbb{S}^1}(\alpha_i, \widehat{\alpha}_i) \leq \frac{\pi}{2R}|\delta_i| \leq \frac{2}{R_-}|\delta_i|.$$

Tangential projection of the outgoing unit vector gives $|p_i - \widehat{p}_i| \leq |e_i| + |\delta_i|/R_-$ for $i < m$. The terminal incoming projection gives the same estimate with e_{m-1}. Using $(u+v)^4 \leq 8(u^4+v^4)$ and bounding a maximum by a sum, we get

$$d_\infty(x_i, \widehat{x}_i)^4 \leq 8|e_{\iota(i)}|^4 + 24R_-^{-4}|\delta_i|^4, \quad \iota(i) = \min(i, m-1).$$

The last flight index is used twice and the others once. Since the difference of two unit vectors has norm at most 2,

$$\sum_i |e_i|^4 \leq 4\sum_i |e_i|^2 \leq 8G/\tau_- \leq 8E/\tau_-.$$

Summing the phase estimate and using (43.9) proves (44.3).

The original section is a finite union of open rectangles in these circle–momentum coordinates. If its indicator differs at two points, a shortest circle–momentum path between them meets its boundary. Hence a discrepancy at index i implies $\mathrm{dist}_\infty(x_i, \partial Y_R^*) \leq d_\infty(x_i, \widehat{x}_i)$. If the left side is at most ρ, it is counted by B_ρ; otherwise its indicator is at most $\rho^{-4} d_\infty(x_i, \widehat{x}_i)^4$. Sum over all i and use (44.3). Points on the boundary are explicitly counted, not assigned zero cost by a bulk-null argument. The occupation assertion follows by dropping the last nonnegative discrepancy. □

Proposition 44.3 (Exact lattice labels and all four twist variables)**.** *On $\mathcal{C}_{R,m}$ the true count m and lattice displacement k of Q and Y^w agree. Their true roofs satisfy $0 \leq L(Q) - L(Y^w) \leq E_R(Q)$. For every $\xi \in \mathbb{R}^2$, $s, u \in \mathbb{R}$,*

$$\left| e^{i[\xi\cdot k + sL(Q) + uA_m(Q)]} - e^{i[\xi\cdot k + sL(Y^w) + uA_m(Y^w)]} \right| \leq |s| E_R(Q) + 2|\sin(u/2)| N_{\mathrm{cut}}(Q). \tag{44.5}$$

The factors $\eta_R(x_0)\eta_R(x_m)$, if included, incur in addition at most the two endpoint discrepancies. No label stability is asserted when that discrepancy has not been paid.

Proof. The physical ordered centers are identical; thus the displacement and number of flights are identical, without dropping a reflection. The actual critical minimum has length $\ell_w(R)$, so the inherited endpoint-gap theorem gives the roof inequality. The triangle inequality and $|e^{ia} - e^{ib}| \leq |a-b|$ bound the roof part. For an integer q, telescoping q unit phase increments gives $|e^{iuq} - 1| \leq |q|\,|e^{iu} - 1| = 2|q||\sin(u/2)|$, including negative q. Take $q = A_m(Q) - A_m(Y^w)$ and use Theorem 44.2. A product of two indicators has difference at most the sum of the differences of its two factors. This also shows exactly where the initial and terminal section tests enter. □

Theorem 44.4 (Positive discrepancy payment on the unchanged roof)**.** *For every R, m, n, k and nonnegative Borel weight h, let Π be the original exact event (1.1) and define, on the class $\mathcal{C}_{R,m}$ only,*

$$\mathcal{M}_h[f] = (L_R)_*(\mathbf{1}_{\mathcal{C}_{R,m}} \mathbf{1}_\Pi h f \, d\nu_R^*).$$

The pushforward is always by the actual original roof $L_R(Q)$, not the comparison roof. As an inequality of positive measures,

$$\mathcal{M}_h[N_{\mathrm{cut}}] \leq \mathcal{M}_h[B_\rho] + K\rho^{-4}\mathcal{M}_h[E_R] \quad (\rho > 0). \tag{44.6}$$

It remains valid after a countable sum over all physical word cells and nonnegative marked copies, with their actual multiplicities. If all three measures have densities, the same inequality holds between the densities almost everywhere, at the same R, m, n, k, t.

Proof. Multiply (44.4) by the nonnegative actual weight $\mathbf{1}_{\mathcal{C}_{R,m}}\mathbf{1}_{\Pi}h$ and integrate its product with any nonnegative Borel test of $L_R(Q)$. This proves the measure inequality, first for bounded weights and then by monotone convergence. Countable marked sums are handled by Tonelli before any supremum or limiting operation. The conditional collision probability already contains exactly the factor c_*^{-1}; no second section normalization is introduced. There is no coordinate change in this proof, and hence no unrecorded Jacobian or physical time multiplier. The density statement is the order property of Radon–Nikodym derivatives when absolute continuity has actually been established. □

Remark 44.5 (The exact remaining normalization). The new term B_ρ is a specified observation-boundary occupation, and E_R is the actual endpoint defect. Their occurrence is an explicit error accounting for real labels, not a claim that either has small normalized density. The class complementary to $\mathcal{C}_{R,m}$ is not discarded. Equations (44.6) and (44.5) alone give no m^{-2} estimate, no high-s uniform decay, and no comparison-pool capacity. In particular choosing ρ as a function of m requires estimates uniform at those widths; it is not justified by a fixed-width result. The singular paired endpoint traces are not bulk pushforwards and are not declared paid by this theorem. The complete source must still satisfy the original (15.8), with all exact labels, roof intervals, ordered suprema and outermost B limit unchanged. This paper does not assert completion of the raw pointwise theorem or its four-coordinate inversion consequence.

45. Boundary localization of a global disk-word minimum

The summable sublevel theorem gives a collision-independent budget, but does not by itself locate its mass. Here global minimality pays the variational work in the interior by work in two prescribed transition layers. A nonlinear recurrence then gives a uniform algebraic tail. No regular physical continuation from the reference to the comparison is assumed.

Let $m \geq 1$ and let $Q_0, \dots, Q_m$ be an original regular physical record, or a fixed-word closed physical reference, at an original radius $R \in [R_-, r] = [9/20, 47/100]$. Its ordered centers are O_i. Let Y be any global minimum of the unchanged disk-word action

$$A(Y) = \sum_{i=0}^{m-1} |Y_{i+1} - Y_i|, \qquad Y_i \in \overline{B}(O_i, R).$$

Consecutive centers are distinct. All original and comparison flights therefore have length at least $\tau_- = 3/50$. Use the retained physical constants

$$N = 94, \qquad a = 1/416640, \qquad T = 187/25, \qquad C = 10^8.$$

Every complete N-flight reference block has an interior contact with incidence at least a. The retained physical horizon is $\tau_i \leq 28/5$; hence every feasible comparison edge satisfies $\ell_i \leq \tau_i + |\delta_i| + |\delta_{i+1}| \leq 28/5 + 4r = T$. These statements include fixed-word closed limits. Write

$$\begin{gathered} \delta_i = Y_i - Q_i, \quad \tau_i = |Q_{i+1} - Q_i|, \quad \ell_i = |Y_{i+1} - Y_i|, \\ v_i = (Q_{i+1} - Q_i)/\tau_i, \quad \widehat{v}_i = (Y_{i+1} - Y_i)/\ell_i, \quad e_i = \widehat{v}_i - v_i, \\ F_i = \ell_i |e_i|^2, \qquad H_i = c_i |\delta_i|^2 \quad (0 < i < m), \qquad E = E_R(Q) = R(2 - c_0 - c_m). \end{gathered} \tag{45.1}$$

The reference dual certificate D_Q gives the complete positive gap $G = A(Y) - D_Q$. The inherited gap identity and disk support inequality give

$$0 \leq G \leq E \leq 2r, \qquad G = \tfrac{1}{2}\sum_i F_i + \sum_{i=0}^{m} h_i, \qquad h_i \geq H_i/R \ (0 < i < m).$$

The already proved sublevel theorem gives $\sum_{i=0}^{m} |\delta_i|^4 \leq CE$. We do not reclassify that theorem as a contribution of this section.

Lemma 45.1 (A variational inequality with only cutoff work)**.** *For any $0 \le \eta_i \le 1$ with $\eta_0 = \eta_m = 0$, put $a_i^\eta = (\eta_i^2 + \eta_{i+1}^2)/2$. Then*

$$\frac{1}{R}\sum_{i=1}^{m-1} \eta_i^2 H_i + \frac{1}{4}\sum_{i=0}^{m-1} a_i^\eta(\ell_i + \tau_i)|e_i|^2 \le \frac{1}{\tau_-}\sum_{i=0}^{m-1}(\eta_{i+1} - \eta_i)^2(|\delta_i|^2 + |\delta_{i+1}|^2). \tag{45.2}$$

In particular, no separate endpoint deficit is paid on the right.

Proof. Let $n_i = (Q_i - O_i)/R$. At an interior reference contact the action gradient is $g_i^Q = v_{i-1} - v_i = -2c_i n_i$. For every point in the same disk,

$$|\delta_i|^2 \le -2Rn_i \cdot \delta_i, \qquad g_i^Q \cdot \delta_i \ge H_i/R.$$

The one-coordinate minimum condition gives $g_i^Y \cdot (Q_i - Y_i) \ge 0$. Since $g_i^Q - g_i^Y = e_i - e_{i-1}$, it follows that $(e_i - e_{i-1}) \cdot \delta_i \ge H_i/R$. This argument applies to inactive and interior comparison vertices.

Set $b_i^\eta = \eta_{i+1}^2 - \eta_i^2$ and $\overline{\delta}_i = (\delta_i + \delta_{i+1})/2$. The exact identities

$$\eta_{i+1}^2\delta_{i+1} - \eta_i^2\delta_i = a_i^\eta(\delta_{i+1} - \delta_i) + b_i^\eta\overline{\delta}_i, \qquad e_i \cdot (\delta_{i+1} - \delta_i) = \tfrac{1}{2}(\ell_i + \tau_i)|e_i|^2$$

and finite integration by parts put the contact work and the full kinetic term on the left. Their sum is at most $-\sum_i b_i^\eta e_i \cdot \overline{\delta}_i$. When $a_i^\eta > 0$, Young's inequality bounds the absolute value of each summand by

$$\tfrac{1}{4}a_i^\eta(\ell_i + \tau_i)|e_i|^2 + \frac{(b_i^\eta)^2|\overline{\delta}_i|^2}{a_i^\eta(\ell_i + \tau_i)}.$$

For $a_i^\eta = 0$ both associated cutoff values and the work vanish. Use

$$(b_i^\eta)^2/a_i^\eta \le 4(\eta_{i+1} - \eta_i)^2, \quad |\overline{\delta}_i|^2 \le (|\delta_i|^2 + |\delta_{i+1}|^2)/2, \quad \ell_i + \tau_i \ge 2\tau_-.$$

Absorbing the first Young term proves the stated coefficient and normalization. Only the global minimum condition was used for Y; the inequality is not asserted for an arbitrary comparison. □

Lemma 45.2 (Interior coercivity without endpoint payment)**.** *For $0 \le u < v \le m$, define*

$$A_H = \frac{4r^2}{a} + 8r^2\left(1 + \frac{3N}{a} + 12N^2\right), \qquad A_F = 24r^2N^2T. \tag{45.3}$$

Then

$$\sum_{i=u+N}^{v-1} |\delta_i|^4 \le A_H \sum_{u<i<v} H_i + A_F \sum_{u\le i<v} F_i. \tag{45.4}$$

An empty sum is zero. Every anchor used to prove this estimate is strictly inside $[u, v]$. This lemma holds for every feasible comparison polygon, without a minimum assumption.

Proof. For an included contact write $b_i = v_{i-1}^\perp \cdot \delta_i$. At an anchor, $|\delta_i|^4 \le 4r^2H_i/a$. At any other included contact choose the last preceding anchor j; the complete N-flight block preceding i supplies one with $u < j < i$ and $i - j < N$. The exact transverse telescope is

$$b_i = b_j + \sum_{l=j}^{i-1} \ell_l v_l^\perp \cdot e_l + \sum_{l=j}^{i-1}(v_l - v_{l-1})^\perp \cdot \delta_l.$$

Here $|v_l - v_{l-1}| = 2c_l$ and $c_l^2 \le c_l$. Cauchy–Schwarz and the three-square inequality consequently give

$$b_i^2 \le \frac{3H_j}{a} + 3NT\sum_{l=j}^{i-1} F_l + 12N\sum_{l=j}^{i-1} H_l.$$

The same disk support inequality used above, resolved into the incoming direction and its perpendicular, implies

$$|\delta_i|^2/(2R) \le c_i|\delta_i| + |b_i|, \qquad |\delta_i|^4 \le 8r^2(H_i + b_i^2).$$

Every preceding anchor, flight, and contact occurs in at most N of the chosen intervals. Sum the anchor bounds separately, and sum the last two displays over the nonanchors. The resulting coefficients are exactly A_H, A_F. The indices range from $u+N$ to $v-1$, so no endpoint surrogate enters the telescope. □

Put $S_d = \sum_{i=d}^{m-d} |\delta_i|^4$ for integer $d \ge 0$, with $S_d = 0$ if $2d > m$.

Theorem 45.3 (A nonlinear depth recurrence for every global minimum)**.** *For integers $k \ge 0$ and $L \ge 1$,*

$$S_{k+L+N} \le C_{\mathrm{cut}} L^{-3/2}\sqrt{S_k}, \qquad C_{\mathrm{cut}} = \frac{4(rA_H + 4A_F)}{\tau_-}. \tag{45.5}$$

The same constant works for every original collision count, word, radius, and global minimum.

Proof. There is nothing to prove when the target is empty. Otherwise choose

$$\eta_i = \min\{1, (i-k)_+/L, (m-k-i)_+/L\}.$$

There are two complete transition layers of L flights. At most $4L$ vertices occur in them, each at most twice, and all have depth at least k. Therefore

$$\sum_{\eta_{i+1} \ne \eta_i} (|\delta_i|^2 + |\delta_{i+1}|^2) \le 4\sqrt{LS_k}.$$

The right side of (45.2) is thus at most $W = 4\tau_-^{-1}L^{-3/2}\sqrt{S_k}$. On $[u,v] = [k+L, m-k-L]$ the cutoff is one. Its contact sum is at most rW, and its flight sum is at most $4W$, since the retained kinetic term is at least $F_i/4$ there. The target positions lie in $[u+N, v-1]$. Lemma 45.2 proves the claimed recurrence with exactly the displayed C_{cut}. □

Theorem 45.4 (Uniform algebraic localization through inactive contacts)**.** *With*

$$K_{\mathrm{tail}} = \max\{2rC(4N)^3, (64C_{\mathrm{cut}})^2\}, \tag{45.6}$$

every global minimum satisfies

$$S_d \le \min\{CE_R(Q), K_{\mathrm{tail}}/(d+1)^3\} \qquad (d \ge 0). \tag{45.7}$$

The estimate includes an isolated central contact when $m = 2d$.

Proof. The first bound is the inherited full sublevel estimate. For $d < 4N$ it also proves the second bound by the first term in (45.6). For $d \ge 4N$ use strong induction, put $k = \lfloor d/2 \rfloor$ and $L = d - k - N$, and note that

$$L \ge 1, \qquad L(k+1) \ge (d+1)^2/16.$$

The recurrence and the induction hypothesis give

$$S_d \le \frac{C_{\mathrm{cut}}\sqrt{K_{\mathrm{tail}}}}{[L(k+1)]^{3/2}} \le \frac{64C_{\mathrm{cut}}\sqrt{K_{\mathrm{tail}}}}{(d+1)^3} \le \frac{K_{\mathrm{tail}}}{(d+1)^3}.$$

If the target is empty the same inequality is immediate. All constants were fixed before either the word or the depth was chosen. □

Corollary 45.5 (Deep contact and flight energy)**.** *For every integer $d \geq 1$,*

$$\sum_{\substack{0<i<m \\ d\leq i\leq m-d}} H_i \leq \frac{32r\sqrt{K_{\mathrm{tail}}}}{\tau_- d^3}, \tag{45.8}$$

$$\sum_{i=d}^{m-d-1} F_i \leq \frac{128\sqrt{K_{\mathrm{tail}}}}{\tau_- d^3}. \tag{45.9}$$

Proof. Use the same cutoff with $k = \lfloor d/2 \rfloor$ and $L = d - k$, without an anchor offset. Since $L(k+1) \geq d^2/4$, its work is at most $32\sqrt{K_{\mathrm{tail}}}/(\tau_- d^3)$. The contact and flight bounds follow respectively by multiplying this work by r and by 4. For a central singleton the contact is still in the unit-cutoff region, and the flight sum is empty. □

46. Transverse clearance and a uniform physical boundary layer

Longitudinal displacement of a contact need not be charged as a loss of clearance: a sufficiently narrow comparison strip confines that part of the comparison flight to its actual incident disks. This gives a quadratic obstruction budget and a summable depth schedule. We keep the original physical reference, disks, constants and notation of Section 45. Statements about arbitrary sublevel points will be distinguished from those requiring a global minimum.

For any feasible comparison polygon define

$$\beta_i = \max\{|v_i^\perp \cdot \delta_i|, |v_i^\perp \cdot \delta_{i+1}|\}, \qquad \Gamma_i(Y) = \inf_{z\in\Lambda\setminus\{O_i,O_{i+1}\}} \operatorname{dist}(z, [Y_i, Y_{i+1}]) - R. \tag{46.1}$$

Every third center is retained, including a center visited elsewhere in the same word. Put $\Gamma_i(Q)$ for the original closed flight.

Lemma 46.1 (A strip comparison with both incident disks retained)**.** *If $2\beta_i < \tau_-$, then*

$$\Gamma_i(Y) \geq \min\{\Gamma_i(Q), \tau_-\} - \beta_i. \tag{46.2}$$

The statement holds for arbitrary feasible comparison endpoints.

Proof. Use coordinates in which the original endpoints are $(0,0)$ and $(\tau_i, 0)$. Intersect each of its two incident closed disks with the strip $|y| \leq \beta_i$. The longitudinal projections of these compact convex sets are disjoint intervals: otherwise two points, one in each disk, would have the same longitudinal coordinate and distance at most $2\beta_i < \tau_-$, contradicting disk separation. The first interval precedes the second, since they contain respectively 0 and $\tau_i > 0$. Thus the comparison segment has strictly increasing longitudinal coordinate and stays in the strip.

The portion projected into $[0, \tau_i]$ is within β_i of the original closed segment. If a left extension exists, let Z be its intersection with the line of longitudinal coordinate zero. The segment from Y_i to Z is within β_i of the segment from Y_i to Q_i, using the same convex parameter. The latter segment lies in the first incident disk. The right extension is treated identically with the second disk and Q_{i+1}. Every third disk is at distance at least τ_- from each incident disk. Its distance from a point of the comparison segment is therefore at least $\Gamma_i(Q) - \beta_i$ on the central portion, and at least $\tau_- - \beta_i$ on either extension. Taking the infimum over the full segment and every third disk proves the result. The argument includes endpoints, changes of the closest center, and changes of the minimizing foot. It does not replace a closed flight by its infinite supporting line. □

Define the uniform coefficients

$$B_H = 3\left(\frac{1+3N}{a} + 12N^2\right) + 8, \qquad B_F = 9N^2T, \tag{46.3}$$

$$B_G = \frac{3r}{a} + \frac{9Nr}{a} + 18N^2T + 36N^2r + 12r, \qquad B_E = 36Nr + 4r. \tag{46.4}$$

In particular $B_G = 167591049$ and $B_E = 1592.36$.

Theorem 46.2 (Quadratic transverse budgets)**.** *For every feasible comparison polygon, with its original positive gap G, one has*

$$\sum_{i=0}^{m-1} \beta_i^2 \le B_G G + B_E E_R(Q). \tag{46.5}$$

For every $0 \le u < v \le m$ one also has

$$\sum_{i=u+N}^{v-2} \beta_i^2 \le B_H \sum_{u<i<v} H_i + B_F \sum_{u\le i<v} F_i. \tag{46.6}$$

For a global minimum, put

$$K_\perp = \frac{(256rB_H + 1024B_F)\sqrt{K_{\mathrm{tail}}}}{\tau_-}, \qquad C_\perp = \sqrt{K_\perp}. \tag{46.7}$$

Then for every integer $d \ge 2N$,

$$\sum_{i=d}^{m-d-1} \beta_i^2 \le K_\perp d^{-3}. \tag{46.8}$$

Proof. Write $b_i = v_{i-1}^\perp \cdot \delta_i$ for $1 \le i \le m$ and $b_0 = v_0^\perp \cdot \delta_0$. At an interior contact,

$$|v_i^\perp \cdot \delta_i|^2 \le 2b_i^2 + 8H_i.$$

The transverse telescope from Lemma 45.2, with anchors paid directly, gives on an interior interval

$$\sum_{i=u+N}^{v-1} b_i^2 \le \left(\frac{1+3N}{a} + 12N^2\right) \sum_{u<i<v} H_i + 3N^2T \sum_{u\le i<v} F_i.$$

Since $\beta_i^2 \le |v_i^\perp \cdot \delta_i|^2 + b_{i+1}^2$, summing gives the stated local coefficients, with all target indices contained in the preceding interval.

For completeness the two true endpoints can be paid without assuming their incidences bounded away from zero. Put $E_0 = R(1 - c_0)$ and $E_m = R(1 - c_m)$, so $E_0 + E_m = E$. At either endpoint the disk support gap and the decomposition into longitudinal and transverse components give

$$b_0^2 \le 4r(h_0 + E_0), \qquad b_m^2 \le 4r(h_m + E_m).$$

Indeed the square of the comparison transverse coordinate relative to its disk center is at most $2Rh_0$ or $2Rh_m$; the reference square is at most $2RE_0$ or $2RE_m$. The difference-square inequality supplies the factor two.

Use a preceding physical anchor for every nonanchor after the first anchor, and use endpoint 0 for the at most N initial indices. The exact telescope for these initial indices has no reflection term at 0. Its initial-value payment is at most $3Nb_0^2$. The flight and contact reuse bounds are still at most N, including these initial intervals. Because $(1+3N)/a \ge 12N$, the anchor payment

and the $12Nrh_0$ part of the initial payment can be paid together by the same complete positive gap. Using $\sum H_i \le rG$ and $\sum F_i \le 2G$ for the remaining terms therefore gives

$$\sum_{i=1}^{m-1} b_i^2 \le \left(\frac{(1+3N)r}{a} + 6N^2T + 12N^2r\right)G + 12NrE.$$

Finally

$$\sum_i \beta_i^2 \le 3\sum_{i=1}^{m-1} b_i^2 + 8\sum_{i=1}^{m-1} H_i + b_0^2 + b_m^2.$$

The endpoint pair is at most $4r(G+E)$. Substitution proves (46.5), including its exact coefficients.

For a global minimum and a nonempty deep target, apply the local bound on $[u,v] = [d-N, m-d+N]$. It contains all the target flights. The contact and flight sums are contained in the energy at depth $d-N \ge d/2$. Corollary 45.5 gives

$$\sum_{i=d}^{m-d-1} \beta_i^2 \le \frac{(32rB_H + 128B_F)\sqrt{K_{\mathrm{tail}}}}{\tau_-(d-N)^3} \le K_\perp d^{-3}.$$

An empty target needs no interval argument. □

Corollary 46.3 (A stronger geometric obstruction certificate)**.** *For any $0 < \rho \le \tau_-/2$, a third-disk violation $\Gamma_i(Y) \le 0$ at a flight with $\Gamma_i(Q) > \rho$ forces $\beta_i \ge \rho$. Consequently*

$$\#\{i : \Gamma_i(Y) \le 0,\ \Gamma_i(Q) > \rho\} \le \rho^{-2}(B_G G + B_E E). \tag{46.9}$$

On the original variational sublevel the right side is at most $\rho^{-2}(B_G + B_E)E$. This is a quadratic transverse payment, not a fourth-power Hausdorff payment.

Proof. If $2\beta_i \ge \tau_-$ the implication is immediate. Otherwise the strip lemma gives $\min(\Gamma_i(Q), \tau_-) \le \beta_i$, which proves it as well. Sum the nonnegative squares and use the global budget. The sublevel inequality is $G \le E$. □

Lemma 46.4 (An inactive deep minimum forces a true small incidence)**.** *Put*

$$C_{\mathrm{inactive}}^2 = 512\sqrt{K_{\mathrm{tail}}}/\tau_-^2. \tag{46.10}$$

If a global minimum is inactive at an interior contact of depth $d_i = \min(i, m-i) \ge 2$, then

$$c_i(Q) \le C_{\mathrm{inactive}} d_i^{-3/2}. \tag{46.11}$$

Proof. Inactivity means $\widehat{v}_{i-1} = \widehat{v}_i$. Thus $2c_i = |v_i - v_{i-1}| \le |e_i| + |e_{i-1}|$, and

$$c_i^2 \le \tfrac{1}{2}(|e_i|^2 + |e_{i-1}|^2) \le \frac{1}{2\tau_-}\sum_{l=d_i-1}^{m-d_i} F_l \le \frac{64\sqrt{K_{\mathrm{tail}}}}{\tau_-^2(d_i-1)^3} \le C_{\mathrm{inactive}}^2 d_i^{-3}.$$

The two adjacent flights are retained, including when the contact is the central singleton. No inverse comparison incidence occurs. □

Let $d_i' = \min(i, m-1-i)$ be flight depth, and put $\delta_{\mathrm{flux}} = 3/200$. Fix the explicit integer

$$D_0 = \left\lceil \max\left\{2N+2, (2C_\perp/\delta_{\mathrm{flux}})^{2/3}, (2C_{\mathrm{inactive}})^{2/3}\right\}\right\rceil. \tag{46.12}$$

For fixed R, m and $D \ge D_0$, let $\mathcal{B}_{D,R,m}$ be the original regular records whose ordered disk word admits a global minimum with an inactive contact at depth at least D, or a flight at depth at least D with nonpositive third-disk clearance. The definition uses all global minima. At fixed radius and word the existence in question is a property of the word itself. Consequently $\mathcal{B}_{D,R,m}$ is a countable union of original Borel physical word cells. No measurable choice of a minimum is needed.

Theorem 46.5 (A complete positive-source boundary-layer bound)**.** *Keep the original exact event* Π*, conditional measure* $\nu_R^* = c_*^{-1}\mathbf{1}_{Y_R^*}\nu$*, physical roof* L_R*, radius range and exact labels* n, k*. For any Borel* $0 \le h \le 1$*, define the positive original-roof measure*

$$\mu_{D,R,m,n,k,h} = (L_R)_*(\mathbf{1}_{\mathcal{B}_{D,R,m}}\mathbf{1}_\Pi h\, d\nu_R^*). \tag{46.13}$$

On every original Borel roof set its mass, and hence its total variation on the whole roof line, is at most

$$\frac{4C_\perp}{R_- c_*\sqrt{D-1}} + \frac{C_{\text{inactive}}^2}{c_*(D-1)^2}. \tag{46.14}$$

In particular

$$\lim_{D\to\infty} \sup_{R,m,n,k,\,0\le h\le 1} \|\mu_{D,R,m,n,k,h}\|_{\text{TV}} = 0. \tag{46.15}$$

Proof. The deterministic estimates hold simultaneously over all minima. A deep blocked comparison flight has $\beta_i \le C_\perp (d_i')^{-3/2} \le \delta_{\text{flux}}/2 < \tau_-/2$. The strip lemma therefore forces $\Gamma_i(Q) \le C_\perp (d_i')^{-3/2}$. The inactive-contact witness is $c_i(Q) \le C_{\text{inactive}} d_i^{-3/2} \le 1/2$. Thus the entire bad source is contained in the union of these explicit small-clearance and small-incidence events of the original physical record.

The retained exact one-copy flux gives $\nu\{\Gamma_i(Q) \le u\} \le u/R$ for $0 < u < \delta_{\text{flux}}$, by collision invariance. For the closed threshold, decrease slightly larger open widths to u; continuity of finite measures gives the same bound. The exact incidence marginal and invariance give

$$\nu\{c_i \le u\} = 1 - \sqrt{1-u^2} \le u^2 \qquad (0 \le u \le 1).$$

For any Borel roof set multiply the full finite union bound by $\mathbf{1}_\Pi h \mathbf{1}_{\{L_R \in A\}}$ and integrate with the original conditional measure. Drop factors between zero and one only in the positive upper bound. This pays exactly one c_*^{-1}, with no replacement of the event or roof.

There are at most two flights or contacts at any prescribed depth. The complete infinite tails satisfy

$$\sum_{d=D}^{\infty} d^{-3/2} \le 2(D-1)^{-1/2}, \qquad \sum_{d=D}^{\infty} d^{-3} \le [2(D-1)^2]^{-1}.$$

The resulting constants are those in (46.14). All actual words and all collision counts were included before taking this upper bound. Its constants are independent of every displayed supremum variable, proving the stated limit without interchanging the original microscopic limits. □

Corollary 46.6 (A true central comparison subrecord)**.** *If* $D \ge D_0$*,* $m > 2D$*, and the original regular record is outside* $\mathcal{B}_{D,R,m}$*, then every global minimum has the actual regular first-hit subrecord* $Y_D, \dots, Y_{m-D}$ *in the original Lorentz gas. The omitted end portions are not asserted physical.*

Proof. Every included contact is active, so the one-coordinate minimum condition places it on its original circle with the exterior specular law and strictly positive incidence. Each included flight therefore stays outside its two incident disks; strict positive clearance excludes every third disk. These facts prove genuine first-hit behavior on each successive central flight. The implications are simultaneous in the global minimum. □

The source theorem controls total mass for bounded positive weights. It does not control normalized roof-density peaks, arbitrary optical Jacobians, unbounded marked multiplicity, or singular pairing traces. Its deterministic clearance mechanism and exact source limit do not alter the original section, exponential guard or microscopic window.

47. The central physical clock and exact section observations

The geometric boundary-layer result gives actual central comparison records outside a specified positive original source. The remaining clock comparison must retain its two endpoint work terms. Dropping them would turn a variational identity into a false estimate for the physical roof.

Keep the notation and original family of Section 45. Fix integers $d \geq 1$, $2d < m$, and put $b = m - d$. In this section Y is a global disk-word minimum whose consecutive vertices $Y_d, \dots, Y_b$ form an actual regular physical subrecord, with the same consecutive centers as $Q_d, \dots, Q_b$. The complement of the source in Theorem 46.5, for an applicable $D \leq d$, supplies this physicality. The statements below need only the stated physical subrecord, not a physical continuation to Y or regularity of either discarded end portion. These are comparisons inside the original record; they do not replace the original full conditional event.

Write

$$\begin{gathered} I_d = \{d, \dots, b\}, \qquad J_d = \{d, \dots, b-1\}, \qquad m_d = m - 2d, \\ L_d(Q) = \sum_{i \in J_d} \tau_i, \quad L_d(Y) = \sum_{i \in J_d} \ell_i, \quad W_d = v_{b-1} \cdot \delta_b - v_d \cdot \delta_d, \\ \mathsf{T}_{\mathrm{clock},d} = L_d(Y) - L_d(Q) - W_d. \end{gathered} \tag{47.1}$$

Theorem 47.1 (The exact central clock with its boundary work)**.** *For every such original record and minimum,*

$$\mathsf{T}_{\mathrm{clock},d} = \frac{1}{R} \sum_{i=d+1}^{b-1} H_i + \frac{1}{2} \sum_{i=d}^{b-1} F_i \geq 0. \tag{47.2}$$

In particular, with the same K_{tail} as in Theorem 45.4,

$$0 \leq \mathsf{T}_{\mathrm{clock},d} \leq K_{\mathrm{clock}} d^{-3}, \qquad K_{\mathrm{clock}} = \left(\frac{32r}{R_-} + 64 \right) \frac{\sqrt{K_{\mathrm{tail}}}}{\tau_-}. \tag{47.3}$$

The length on either side is the unit-speed physical time of that central subrecord. No clock is averaged or rescaled.

Proof. For each included flight, the identity between two unit directions is

$$\ell_i - \tau_i = v_i \cdot (\delta_{i+1} - \delta_i) + \tfrac{1}{2} \ell_i |e_i|^2.$$

Sum it over $d \leq i < b$. Finite integration by parts gives boundary term W_d and interior coefficients $v_{i-1} - v_i = -2c_i n_i$. Both actual contact points Q_i, Y_i lie on the same circle, so

$$|Q_i - O_i + \delta_i|^2 = R^2 \quad \Longrightarrow \quad 2R n_i \cdot \delta_i + |\delta_i|^2 = 0.$$

Thus each interior coefficient contributes H_i/R with a positive sign. This proves the exact identity, including the case of one central flight, when its contact sum is empty. Apply the contact and flight estimates of Corollary 45.5, and enlarge R^{-1} to R_-^{-1}, to obtain the displayed constant. This argument uses circular equality only at the included physical contacts. The disk support inequality for a nonphysical interior vertex would not justify the same clock equality. □

Let $x_i = (\alpha_i, p_i)$ and $\widehat{x}_i$ be the original and comparison collision states on I_d. Use outgoing tangential velocity at $i < b$ and the incoming tangential projection at b. Specular reflection makes that terminal projection the actual collision coordinate without requiring a next flight. The distance d_∞ is the maximum of the shortest circular angle distance and the momentum distance. Define

$$\mathsf{T}_{\mathrm{phase},d} = \sum_{i=d}^{b} d_\infty(x_i, \widehat{x}_i)^4, \qquad K_{\mathrm{phase}} = \frac{8192\sqrt{K_{\mathrm{tail}}}}{\tau_-^2} + \frac{24 K_{\mathrm{tail}}}{R_-^4}. \tag{47.4}$$

Theorem 47.2 (A complete central phase and section payment)**.** *One has*

$$\mathsf{T}_{\mathrm{phase},d} \le K_{\mathrm{phase}} d^{-3}. \tag{47.5}$$

Let $\eta_R = \mathbf{1}_{Y_R^*}$ *be exactly the original section indicator. For* $\rho > 0$ *put*

$$B_{\rho,d}(Q) = \sum_{i=d}^{b} \mathbf{1}_{\{\mathrm{dist}_\infty(x_i,\partial Y_R^*)\le\rho\}}, \qquad N_{\mathrm{cut},d}(Q,Y) = \sum_{i=d}^{b} |\eta_R(x_i) - \eta_R(\widehat{x}_i)|.$$

Then, for the actual observations on both subrecords,

$$N_{\mathrm{cut},d} \le B_{\rho,d} + \rho^{-4}\mathsf{T}_{\mathrm{phase},d} \le B_{\rho,d} + \rho^{-4} K_{\mathrm{phase}} d^{-3}. \tag{47.6}$$

Occupation uses only J_d*; the last section test at* b *is separate. In particular the absolute occupation difference is bounded by the sum in* (47.6) *with its last summand removed.*

Proof. The circle estimate in Theorem 44.2 applies contact by contact:

$$d_\infty(x_i,\widehat{x}_i)^4 \le 8|e_{\iota(i)}|^4 + 24R_-^{-4}|\delta_i|^4, \qquad \iota(i) = \begin{cases} i, & i < b, \\ b-1, & i = b. \end{cases}$$

Indeed angle distance is at most $2|\delta_i|/R_-$ and tangential momentum distance at most $|e_{\iota(i)}| + |\delta_i|/R_-$. The last edge occurs twice, including when the subrecord has only one edge. All other included edges occur once. Since $|e_i| \le 2$ and $\ell_i \ge \tau_-$, the sum of their direction contributions is at most

$$8\sum_{i\in I_d} |e_{\iota(i)}|^4 \le \frac{64}{\tau_-}\sum_{i=d}^{b-1} F_i.$$

Use Corollary 45.5 for this sum and Theorem 45.4 for the position sum. Enlarging $(d+1)^{-3}$ to d^{-3} gives exactly K_{phase}.

A shortest product-coordinate path between states with different section indicators meets the original section boundary. Therefore a discrepancy is either in its closed ρ collar or has $d_\infty(x_i,\widehat{x}_i) > \rho$. Its indicator in the latter case is at most $\rho^{-4} d_\infty(x_i,\widehat{x}_i)^4$. Sum over every included contact. Boundary states have been counted explicitly; no null-set deletion of their weighted source is used. Dropping the last nonnegative summand gives the occupation claim. □

Proposition 47.3 (All central Fourier marks with the exact clock phase)**.** *The central collision count* m_d *and lattice displacement* $k_d = O_b - O_d$*, expressed in the original lattice basis, are identical on the two physical subrecords. Put* $A_d(Q) = \sum_{i=d}^{b-1} \eta_R(x_i)$ *and define* $A_d(Y)$ *in the same way. For every* $\xi \in \mathbb{R}^2$ *and* $s, \zeta \in \mathbb{R}$*,*

$$\begin{aligned} &\left| e^{i[\xi\cdot k_d + sL_d(Y) + \zeta A_d(Y)]} - e^{isW_d} e^{i[\xi\cdot k_d + sL_d(Q) + \zeta A_d(Q)]} \right| \\ &\qquad \le |s|\mathsf{T}_{\mathrm{clock},d} + 2|\sin(\zeta/2)|\, N_{\mathrm{cut},d}. \end{aligned} \tag{47.7}$$

If the two central endpoint section tests are inserted, their additional error is at most the sum of those two actual indicator discrepancies. All these comparisons retain every central collision.

Proof. The ordered centers give k_d and m_d exactly. Substitute (47.2) into the left side, retaining the phase e^{isW_d} of modulus one. The clock error is bounded by $|e^{is\mathsf{T}_{\mathrm{clock},d}} - 1| \le |s|\mathsf{T}_{\mathrm{clock},d}$. For an integer q, telescoping q unit increments gives $|e^{i\zeta q} - 1| \le 2|q||\sin(\zeta/2)|$; it also holds for negative q. Take $q = A_d(Y) - A_d(Q)$ and use the preceding theorem. The product of two indicators changes by at most the sum of their two changes, which proves the endpoint assertion. □

Remark 47.4. The full source event is still $\Pi_{n,k,m,R}$ and its roof still $L_{m,R}(Q)$. Neither the central occupation nor the phase-corrected central length is substituted into that event. The end portions, the section collar and the physicality-exception source remain present in the full problem until their own required estimates have been proved. In particular $\mathsf{T}_{\mathrm{clock},d}$ alone cannot bound $|L_d(Y) - L_d(Q)|$: the term W_d is indispensable.

48. Exponential confinement by physical anchor layers

The polynomial estimates above do not exhaust the variational mechanism. The physical turning bound supplies contacts at which the endpoint work of a whole comparison interval can be paid by energy in its removed boundary layer. This yields a strict contraction after a fixed number of actual collisions. The resulting exponential estimate concerns every global disk-product minimum, including nonphysical and inactive ones. It is not obtained by integrating a local endpoint derivative along an assumed physical continuation.

Keep the original reference, comparison, radius and constants of Sections 45 and 46. In particular Y is a global minimum of the unchanged disk-word problem, and the reference is actual regular or a fixed-word closed physical reference. To distinguish it from the depth of an index, denote the anchor threshold by $a_0 = 1/416640$. Thus $N = 94$, $r = 47/100$, $\tau_- = 3/50$, and

$$\kappa = \frac{r}{a_0\tau_-} = 3263680, \qquad q_{\mathrm{a}} = \frac{\kappa}{1+\kappa} = \frac{3263680}{3263681} \in (0,1). \tag{48.1}$$

The symbol q_{a} is a deterministic contraction constant; it does not replace the original guard constant $q_0 = 47/53$.

Lemma 48.1 (Common active geometry of global minima)**.** *For a fixed word and radius, all global disk-product minima have the same ordered unit edge directions and the same active index set. At every active index their vertices agree exactly. Both endpoints are active. Any difference between minima lies along the common straight direction in a chain of inactive contacts. This is a convexity fact about the comparison problem, not a new billiard map or a claim that its full minimum is unique.*

Proof. Let Y, Z be two minima. Their midpoint lies in the same product of disks and, by convexity of A, is another minimum. The sum of the nonnegative defects in the flightwise triangle inequalities is zero, so each such inequality is an equality. Both vectors of each edge are nonzero, by the fixed consecutive-disk separation. Equality in the Euclidean triangle inequality therefore makes them positive parallel vectors, and their unit directions agree.

The Cartesian gradients at every coordinate consequently agree for all minima. If this common gradient g_i is nonzero, the minimum condition on its disk gives the unique point $O_i - Rg_i/|g_i|$. If it is zero, its neighboring directions agree; this index is inactive for every minimum. At an endpoint the gradient is a unit vector up to sign, so it is always active. On a consecutive chain of inactive indices the edge direction is constant, and the differences between corresponding edges of Y and Z are multiples of that direction. Starting at the preceding common active vertex and summing these differences shows that all possible vertex differences on the chain are longitudinal. No continuation through an inactive vertex or through an unlisted disk is inferred. □

Lemma 48.2 (The exact work of an anchor-bounded interval)**.** *If $1 \le u < v \le m-1$ and both reference contacts are anchors, $c_u, c_v \ge a_0$, then*

$$\frac{1}{R}\sum_{i=u+1}^{v-1} H_i + \frac{1}{2}\sum_{i=u}^{v-1} F_i \le \frac{H_u + H_v}{a_0\tau_-}. \tag{48.2}$$

Both flights adjacent to the cut are retained on the left. In particular the inequality also covers $v = u+1$.

Proof. The variational comparison from Lemma 45.1 gives $(e_i - e_{i-1}) \cdot \delta_i \geq H_i/R$ for $u < i < v$. Summing on this hard interval, with its endpoint work retained, yields

$$\frac{1}{R} \sum_{i=u+1}^{v-1} H_i + \frac{1}{2} \sum_{i=u}^{v-1} (\ell_i + \tau_i)|e_i|^2 \leq e_{v-1} \cdot \delta_v - e_u \cdot \delta_u. \tag{48.3}$$

Indeed the boundary of the finite integration by parts is exactly the right side, and $e_i \cdot (\delta_{i+1} - \delta_i) = (\ell_i + \tau_i)|e_i|^2/2$. This identity is still valid when there are no interior contacts.

At either cut use the elementary inequality

$$|e|\,|\delta| \leq \frac{\tau_-}{4}|e|^2 + \frac{|\delta|^2}{\tau_-}.$$

An edge incident to only one cut loses at most $\tau_-|e_i|^2/4$ from its kinetic coefficient; its remaining coefficient is at least $\ell_i/2$. If $v = u + 1$, the same edge pays both cuts, but

$$\frac{\ell_i + \tau_i}{2} - \frac{\tau_-}{2} \geq \frac{\ell_i}{2}.$$

Thus every edge retains at least $F_i/2$, including the two boundary flights. At the actual reference anchors, $|\delta_u|^2 \leq H_u/a_0$ and $|\delta_v|^2 \leq H_v/a_0$. These facts prove (48.2). No assertion about the comparison incidences is used. □

For an integer $d \geq 0$ define the full relative energy at depth d by

$$\mathcal{E}_d(Q, Y) = \frac{1}{R} \sum_{\substack{0<i<m \\ d \leq i \leq m-d}} H_i + \frac{1}{2} \sum_{i=d}^{m-d-1} F_i. \tag{48.4}$$

Every sum over an empty interval is zero. In particular a central single contact is kept when $m = 2d$, although no flight remains. The actual endpoints are never assigned interior contact weights.

Theorem 48.3 (A contraction on every physical anchor layer)**.** *For every integer $d \geq 0$,*

$$\mathcal{E}_{d+N} \leq q_{\mathrm{a}} \mathcal{E}_d, \qquad \mathcal{E}_d \leq E_R(Q) q_{\mathrm{a}}^{\lfloor d/N \rfloor}. \tag{48.5}$$

The inequalities hold simultaneously for every global minimum and every original word and radius, including at a closed physical reference with zero internal incidences. They have no collision-count prefactor and no inverse individual incidence.

Proof. The complete positive gap, before any separate estimate of its summands, gives

$$\mathcal{E}_0 = \frac{1}{R} \sum_{i=1}^{m-1} H_i + \frac{1}{2} \sum_{i=0}^{m-1} F_i \leq \sum_{i=1}^{m-1} h_i + \frac{1}{2} \sum_{i=0}^{m-1} F_i \leq G \leq E_R(Q). \tag{48.6}$$

Thus both kinds of energy consume the same positive budget, not two copies of it. The last inequality uses global minimality only through $Y \in \mathcal{S}(Q)$.

If $2(d + N) > m$, the first inequality in (48.5) has zero left side. Otherwise the two complete physical N-flight blocks starting at d and $m - d - N$ supply reference anchors

$$u \in \{d + 1, \ldots, d + N - 1\}, \qquad v \in \{m - d - N + 1, \ldots, m - d - 1\}.$$

These are interior contacts, and $u < d+N \leq m-d-N < v$. Consequently every energy summand in $\mathcal{E}_{d+N}$ occurs on the left of (48.2). Conversely, the two anchor summands $H_u/R, H_v/R$ occur in $\mathcal{E}_d - \mathcal{E}_{d+N}$. All the other summands in that difference are nonnegative. Lemma 48.2 gives

$$\mathcal{E}_{d+N} \leq \frac{H_u + H_v}{a_0 \tau_-} = \frac{R}{a_0 \tau_-} \frac{H_u + H_v}{R} \leq \kappa(\mathcal{E}_d - \mathcal{E}_{d+N}).$$

Rearranging proves the strict contraction with exactly $q_{\mathrm{a}} = \kappa/(1+\kappa)$. The argument also covers $m = 2(d+N)$, where the inner energy contains one contact. No boundary flight is counted as a removed-layer payment: its Young payment has already been absorbed in (48.2).

Iterate the contraction at the depths $0, N, 2N, \dots$ and use (48.6). Since $\mathcal{E}_d$ is nonincreasing, $\mathcal{E}_d \le \mathcal{E}_{N\lfloor d/N\rfloor}$. This proves the second inequality. The anchors can be chosen anew at each layer; no compatible continuation or fixed selection of anchor points is required. □

Put

$$j(d) = \max\{0, \lfloor d/N\rfloor - 1\}, \qquad B_\perp = B_G + B_E. \tag{48.7}$$

Corollary 48.4 (Exponential position, direction and transverse budgets)**.** *For every integer $d \ge 0$ and every global minimum,*

$$\sum_{i=d}^{m-d} |\delta_i|^4 \le C E_R(Q) q_{\mathrm{a}}^{j(d)}, \tag{48.8}$$

$$\sum_{i=d}^{m-d-1} F_i \le 2E_R(Q) q_{\mathrm{a}}^{\lfloor d/N\rfloor}, \tag{48.9}$$

$$\sum_{i=d}^{m-d-1} \beta_i^2 \le B_\perp E_R(Q) q_{\mathrm{a}}^{j(d)}. \tag{48.10}$$

The fourth-power and transverse sums have their usual zero values on empty intervals. The position estimate includes both actual endpoints when $d = 0$; they are paid by the original deficit.

Proof. The flight estimate follows directly from the definition of $\mathcal{E}_d$. For $d < N$, the position and transverse estimates are respectively Theorem 43.3 and (46.5) on the sublevel.

For $d \ge N$ with a nonempty target interval, apply the interior coercivity lemma on $[u, v] = [d-N, m-d+N]$. Its target range contains all the positions in (48.8). The positive contact and flight components are jointly bounded by $\mathcal{E}_{d-N}$, so

$$S_d \le \max\{rA_H, 2A_F\}\mathcal{E}_{d-N} \le C\mathcal{E}_{d-N}.$$

For the constants in (45.3), the last comparison is the explicit rational inequality

$$\max\{rA_H, 2A_F\} < 10^8 = C.$$

Similarly the local transverse estimate (46.6) contains all the included flights and gives

$$\sum_{i=d}^{m-d-1} \beta_i^2 \le \max\{rB_H, 2B_F\}\mathcal{E}_{d-N} \le B_\perp \mathcal{E}_{d-N},$$

where $\max\{rB_H, 2B_F\} < B_G + B_E$ is another direct rational comparison of the displayed constants. Both inequalities use the shared energy before estimating its components. Since $\lfloor (d-N)/N\rfloor = j(d)$ for $d \ge N$, the layer theorem proves the two conclusions. The polynomial bounds already proved remain valid and may be used together with these exponential ones; no preceding proof body is removed. □

Corollary 48.5 (Explicit exponentially small physical failure witnesses)**.** *For an interior contact of depth $d_i \ge 1$, inactivity of a global minimum implies*

$$c_i(Q)^2 \le \frac{E_R(Q)}{\tau_-} q_{\mathrm{a}}^{\lfloor (d_i-1)/N\rfloor}. \tag{48.11}$$

At every flight,

$$\beta_i^2 \le B_\perp E_R(Q) q_{\mathrm{a}}^{j(d_i')}. \tag{48.12}$$

If the last upper bound is strictly below $\tau_-^2/4$, a third-disk violation of that comparison flight necessarily implies $\Gamma_i(Q) \le \sqrt{B_\perp E_R(Q)}\, q_{\mathrm{a}}^{j(d_i')/2}$. All these witnesses concern the true incidence and clearance of the original reference. No probability or roof-density assertion follows without a separate source argument.

Proof. Inactivity gives $2c_i \le |e_i| + |e_{i-1}|$, as before. Both flights belong to the energy at depth $d_i - 1$, so

$$c_i^2 \le \frac{|e_i|^2 + |e_{i-1}|^2}{2} \le \frac{1}{2\tau_-} \sum_{l=d_i-1}^{m-d_i} F_l \le \frac{E_R(Q)}{\tau_-} q_{\mathrm{a}}^{\lfloor (d_i-1)/N \rfloor}.$$

The transverse point bound is a single summand of (48.10). Its stated strict regime permits application of the already proved strip lemma, giving the clearance implication. This excludes no original collision, and treats every minimizer with the same constants. □

Remark 48.6 (Rate, original guard, and probability topology). The contraction q_{a} is close to one because the explicit physical anchor bound is conservative. It is nevertheless fixed before any collision limit, and its exponential decay is uniform on the entire original geometric family. It does not identify these rates with the much narrower original guard widths $\varepsilon q_0^{d/4}$. In particular a substitution of one depth schedule for the other would change the claimed microscopic problem and is not made. Exponential geometric cost, exact original-source mass, inherited-pressure conditional roof height, and the entire normalized shortage remain distinct assertions. None is inferred from a finite trajectory computation.

49. Comparison costs on the original roof and their normalization

A deterministic small comparison cost and the density of the exceptional physical source are different quantities. Here positive ordering pays the former on the unchanged physical roof. The unnormalized roof-height input is stated explicitly; none of the geometric theorems just proved depends on it.

Fix R, m, n, k and $d \ge 2N$, $2d < m$. On any Borel class $\mathcal{P}_d$ of original records with a physical central minimum as in Section 47, choose one such global minimum per physical disk word. At fixed radius there are only countably many finite words. A chosen minimum is constant on its word, so this choice gives Borel comparison coordinates on their physical domains. Alternatively, by Lemma 48.1, the active central vertices of all such minima coincide. No joint measurability in the radius is needed for the original order $\sup_R \operatorname{ess\,sup}_t$.

Let $0 \le h \le 1$ be any Borel physical source selector; it may depend on the whole record and on the fixed labels. With Π exactly as in (1.1), define the positive measures

$$\mathcal{M}_{h,d}[f] = (L_{m,R})_*(\mathbf{1}_{\mathcal{P}_d} \mathbf{1}_\Pi h f \, d\nu_R^*), \qquad \mu_0 = (L_{m,R})_*(\mathbf{1}_\Pi \, d\nu_R^*). \tag{49.1}$$

Every pushforward is by the original full physical roof, even when f is a central comparison cost. Let $\mathsf{T}_{\perp,d} = \sum_{i=d}^{m-d-1} \beta_i^2$.

Theorem 49.1 (Positive cost domination and its collision normalization)**.** *For every Borel roof set $I \subset \mathbb{R}$ and $f = \mathsf{T}_{\mathrm{clock},d}, \mathsf{T}_{\mathrm{phase},d}, \mathsf{T}_{\perp,d}$, respectively,*

$$\mathcal{M}_{h,d}[f](I) \le K_f d^{-3} \mu_0(I), \qquad (K_f) = (K_{\mathrm{clock}}, K_{\mathrm{phase}}, K_\perp). \tag{49.2}$$

This measure assertion is unconditional on the pressure input.

Under precisely Input 11.1, including its physical stable arclength and optical-coordinate conventions, these measures have densities on the whole original roof line, and

$$\left\| \frac{d\mathcal{M}_{h,d}[f]}{dt} \right\|_\infty \le C_\star K_f d^{-3}. \tag{49.3}$$

In particular, for every fixed $0 < \theta < 1/2$ *and* $d_m = \lceil \theta m \rceil$*, for all sufficiently large* m*,*

$$\sup_{R,n,k,h} m^2 \left\| \frac{d\mathcal{M}_{h,d_m}[f]}{dt} \right\|_\infty \le C_\star K_f \theta^{-3} m^{-1} \longrightarrow 0. \tag{49.4}$$

The same bound holds on each original microscopic roof interval and on the original central target. The norm on the whole line is the stronger assertion. The input constants are uniform in all the displayed variables.

Proof. Theorems 47.1, 47.2, and 46.2 give $0 \le f \le K_f d^{-3}$ pointwise on the stated source class. Multiply by $\mathbf{1}_{\mathcal{P}_d}\mathbf{1}_\Pi h\, \mathbf{1}_I(L_{m,R})$ and integrate against exactly ν_R^*. Removing only the upper-bound selectors $\mathbf{1}_{\mathcal{P}_d} h \le 1$ gives (49.2). Equivalently the same proof works against every nonnegative Borel test of the original roof. Tonelli permits the complete positive physical word sum before this comparison; there is no sum of a separate error constant for each word.

For $S = \varnothing$, Theorem 11.3, under its named input, gives $d\mu_0/dt \le C_\star$ almost everywhere, where $C_\star = C_0 Q_*/(2c_*\lambda_\circ)$. The order property of positive measures first gives absolute continuity of $\mathcal{M}_{h,d}[f]$ and then the bound on its Radon–Nikodym derivative. This uses the original inverse optical Jacobian and section factor already in $C_\star$; it inserts no new Jacobian and pays c_*^{-1} only once. Finally $d_m \ge \theta m$ and $m^2/d_m^3 \le \theta^{-3}/m$. All constants were fixed before the collision limsup. Restricting a density norm to a roof interval cannot enlarge it. □

Corollary 49.2 (The dominated persistent low-load interior source)**.** *For* $\gamma \ge 1$*, the same positive-cost and conditional normalized estimates apply to every Borel restriction of the inherited persistent low-load interior coefficient which is dominated by the original uncut roof measure as in Lemma 12.1. This includes its original low-load selectors, all bounded by one. It does not assert the corresponding domination for singular paired endpoint traces or for an arbitrary full endpoint coefficient.*

Proof. The inherited inequality is $0 \le c_{\mathrm{per}}^{\varepsilon,\gamma} \le f^\varepsilon(u,\vartheta) \le A(u,\vartheta)$ on each actual physical roof segment. Multiplication by its nonnegative low-load selector and by f preserves the inequality. Sum the actual source coefficients at their original roof and use the positive original-measure comparison of that lemma. The result is a measure dominated by the right side of (49.2); the preceding proof applies verbatim. This argument invokes that specific domination theorem and its existing physical multiplicities, not an assertion that a boundary trace has a bulk density. □

Corollary 49.3 (Central observation costs with a stated frequency range)**.** *On the same source, after retaining* e^{isW_d} *in* (47.7)*, the nonnegative error majorant is*

$$|s|\mathsf{T}_{\mathrm{clock},d} + 2|\sin(\zeta/2)|(B_{\rho,d} + \rho^{-4}\mathsf{T}_{\mathrm{phase},d}).$$

For fixed $\rho > 0$*, the two comparison-cost terms have collision-normalized original-roof height tending to zero when* $d = d_m$ *and* $|s| = o(m)$*, under the stated pressure input, uniformly in the two lattice twists and the return twist. The positive source weighted by* $B_{\rho,d}$ *remains an explicit separate term.*

Proof. Combine the exact twist estimate with the section payment and Theorem 49.1. The lattice phase has modulus one, and $2|\sin(\zeta/2)| \le 2$ for all ζ. The clock coefficient cost is bounded after multiplication by m^2 by a constant times $|s|/m$. At fixed ρ the other comparison term is $O(m^{-1})$. No estimate of $\mathcal{M}_{h,d}[B_{\rho,d}]$ was proved by this argument, so it remains on its own side of the inequality. □

Remark 49.4 (The exact scope of the normalization). The estimates above are for specified small geometric costs on a proved physical central comparison class. They do not pay the indicator of its complementary source, the original shallow end portions, the singular one-sided or paired

boundary traces, a physical multiplier not dominated by the declared source, or a shortage of roofwise comparison capacity. Nor does an estimate uniform on an increasing bounded frequency range by itself justify the full unbounded Fourier integral. In particular the measure contraction in Theorem 46.5 alone is not an m^2L^∞ source estimate. All these distinctions persist when the sharper exponential costs below replace the algebraic bounds.

50. An exponentially small source for exact central physical records

The anchor contraction gives a uniform positive-source theorem without a word-count prefactor. It controls all global minima simultaneously and fixes the original section observations on their physical central records. The source is defined entirely by the actual reference trajectory. No exceptional event is inserted into the original guard or into the original conditional return event.

Keep $R \in [R_-, r]$, $N = 94$, $\tau_- = 3/50$, q_{a} and $j(d)$ from (48.1) and (48.7). All probabilities in this section use $\nu = d\alpha\, dp/(4\pi)$ or its unchanged section conditioning $\nu_R^* = c_*^{-1}\nu|_{Y_R^*}$. The original record is regular for its m flights; the estimates do not define a singular flow. Bounded bulk sources, rather than singular paired traces, are under consideration.

Lemma 50.1 (The exact original section has a uniform collar bound)**.** *For $0 < \rho \le h_0 = 1/200$,*

$$\sup_R \nu\{\mathrm{dist}_\infty(x, \partial Y_R^*) \le \rho\} \le \frac{\rho}{2\pi}. \tag{50.1}$$

The bound refers to the five original rectangles in the collision circle–momentum coordinates.

Proof. For a rectangle of width w and height h, with $2\rho \le \min(w, h)$, its closed boundary collar has area at most

$$(w + 2\rho)(h + 2\rho) - (w - 2\rho)(h - 2\rho) = 4\rho(w + h).$$

On the circle this can be calculated on a lift; overlap at the coordinate seam can only reduce the projected area. Intersection with $-1 < p < 1$ also cannot increase it. The boundary of a finite union is contained in the union of the component boundaries. The four squares have $w = h = 1/100$, and U has $w = 3/25$ and $h = 3/10$. Therefore $\sum(w + h) = 4(2/100) + 3/25 + 3/10 = 1/2$. The total area is at most 2ρ, and division by 4π proves the claim. The radius changes the centers of two rectangles but not these widths. □

Put

$$\begin{gathered} K_\phi = \frac{128}{\tau_-} + \frac{24C}{R_-^4}, \\ A_\Gamma = (2rB_\perp)^{1/2}, \qquad A_c = (2r/\tau_-)^{1/2}, \qquad A_\phi = (2rK_\phi)^{1/4}. \end{gathered} \tag{50.2}$$

Lemma 50.2 (A phase estimate at every active comparison contact)**.** *Let i be an interior contact of depth $d_i = \min(i, m - i) \ge 1$, active in a global minimum. Give that contact its circle coordinate and the outgoing tangential component determined by its two adjacent comparison directions. Then*

$$d_\infty(x_i, \widehat{x}_i)^4 \le K_\phi E_R(Q) q_{\mathrm{a}}^{j(d_i)} \le A_\phi^4 q_{\mathrm{a}}^{j(d_i)}. \tag{50.3}$$

This is a geometric collision-state estimate before the comparison flights have been proved clear. Inactivity is not assigned an artificial collision state.

Proof. At an active minimum its nonzero Cartesian gradient is a negative multiple of the outward disk normal. Its two unit directions therefore obey the specular reflection law with strictly positive normal incidence. In particular their incoming and outgoing tangential components agree. The elementary circle and momentum argument used in Theorem 47.2, now using the incoming edge, gives

$$d_\infty(x_i, \widehat{x}_i)^4 \le 8|e_{i-1}|^4 + 24R_-^{-4}|\delta_i|^4.$$

That incoming edge has depth at least $d_i - 1$. Since $|e_{i-1}|^4 \le 4|e_{i-1}|^2 \le 4F_{i-1}/\tau_-$, the flight estimate (48.9) gives

$$|e_{i-1}|^4 \le \frac{8E_R(Q)}{\tau_-} q_{\mathrm{a}}^{\lfloor (d_i-1)/N \rfloor}.$$

Also (48.8) bounds $|\delta_i|^4$ by $CE_R(Q)q_{\mathrm{a}}^{j(d_i)}$. For every integer $d_i \ge 1$, $\lfloor (d_i-1)/N \rfloor \ge j(d_i)$. The resulting direction coefficient $64/\tau_-$ is at most the deliberately enlarged $128/\tau_-$ in K_ϕ. This proves the first inequality; use $E_R(Q) \le 2r$ for the second. The argument retains the incoming edge even when the target consists of one central contact. □

Let $\delta_{\mathrm{flux}} = 3/200$. Define the finite constants

$$\begin{aligned} J_{\mathrm{obs}} = 1 + \Bigg\lceil \frac{1}{-\log q_{\mathrm{a}}} \max\Big\{ 0,\ 2\log(2A_\Gamma/\delta_{\mathrm{flux}}), \\ 2\log(2A_c),\ 4\log(A_\phi/h_0) \Big\} \Bigg\rceil, \qquad D_{\mathrm{obs}} = NJ_{\mathrm{obs}}. \end{aligned} \tag{50.4}$$

For $D \ge D_{\mathrm{obs}}$, the following inequalities hold:

$$A_\Gamma q_{\mathrm{a}}^{j(D)/2} \le \delta_{\mathrm{flux}}/2 < \tau_-/2, \quad A_c q_{\mathrm{a}}^{j(D)/2} \le 1/2, \quad A_\phi q_{\mathrm{a}}^{j(D)/4} \le h_0. \tag{50.5}$$

These are conservative constants fixed before every collision limit. They do not modify the original guard widths $\varepsilon q_0^{d/4}$, where $q_0 = 47/53$.

Define $\mathcal{E}^{\mathrm{obs}}_{D,R,m}$ to be the following union of events on the original record:

$$\begin{aligned} \mathcal{E}^{\mathrm{obs}}_{D,R,m} = & \bigcup_{\substack{0 \le i < m \\ d_i' \ge D}} \{\Gamma_i(Q) \le A_\Gamma q_{\mathrm{a}}^{j(d_i')/2}\} \\ & \cup \bigcup_{\substack{0 < i < m \\ d_i \ge D}} \{c_i(Q) \le A_c q_{\mathrm{a}}^{j(d_i)/2}\} \\ & \cup \bigcup_{\substack{0 < i < m \\ d_i \ge D}} \{\mathrm{dist}_\infty(x_i, \partial Y_R^*) \le A_\phi q_{\mathrm{a}}^{j(d_i)/4}\}. \end{aligned} \tag{50.6}$$

Every quantity in this finite union is an actual original observable. In particular the clearance is the infimum over all third lattice centers, including centers visited elsewhere in the word. The source is Borel. For $0 \le h \le 1$ put

$$\mu^{\mathrm{obs}}_{D,R,m,n,k,h} = (L_{m,R})_*(\mathbf{1}_{\mathcal{E}^{\mathrm{obs}}_{D,R,m}} \mathbf{1}_{\Pi_{n,k,m,R}} h \, d\nu_R^*).$$

Theorem 50.3 (A uniform exact-record source and its full positive mass)**.** *Define*

$$\epsilon_D = \frac{2NA_\Gamma}{R_- c_*} \frac{q_{\mathrm{a}}^{j(D)/2}}{1 - q_{\mathrm{a}}^{1/2}} + \frac{2NA_c^2}{c_*} \frac{q_{\mathrm{a}}^{j(D)}}{1 - q_{\mathrm{a}}} + \frac{NA_\phi}{\pi c_*} \frac{q_{\mathrm{a}}^{j(D)/4}}{1 - q_{\mathrm{a}}^{1/4}}. \tag{50.7}$$

For every $D \ge D_{\mathrm{obs}}$, every m, R, n, k, every Borel $0 \le h \le 1$ and every Borel roof set I,

$$0 \le \mu^{\mathrm{obs}}_{D,R,m,n,k,h}(I) \le \epsilon_D, \qquad \sup_{m,R,n,k,h} \|\mu^{\mathrm{obs}}_{D,R,m,n,k,h}\|_{\mathrm{TV}} \le \epsilon_D \longrightarrow 0 \quad (D \longrightarrow \infty). \tag{50.8}$$

The complete physical word sum is already included in these inequalities. No word-count factor or independence assumption occurs.

If $Q \notin \mathcal{E}^{\mathrm{obs}}_{D,R,m}$ and $m > 2D$, then every *global minimum has an actual regular physical central record $Y_D, \dots, Y_{m-D}$ and*

$$\eta_R(\widehat{x}_i) = \eta_R(x_i) \qquad (D \le i \le m - D). \tag{50.9}$$

All those central contacts and states agree between global minima. Their central displacement, number of flights, occupation, and two endpoint section tests agree exactly with the original central record. The terminal indicator in (50.9) *is tested separately from its occupation count.*

Proof. The exact one-flight physical clearance flux gives, for $0 < u < \delta_{\mathrm{flux}}$,

$$\nu\{\Gamma_0(Q) \le u\} \le \int N_u \, d\nu = \frac{u}{R} \le \frac{u}{R_-}.$$

At equality boundaries, use the same estimate at $u + \varepsilon$ and pass down by continuity of a finite measure. Invariance of the actual collision map gives the same bound at each flight time. The original incidence distribution is exact:

$$\nu\{c_i \le u\} = 1 - \sqrt{1 - u^2} \le u^2 \qquad (0 \le u \le 1).$$

Lemma 50.1 and invariance likewise give the collar bound $\rho/(2\pi)$ at any contact time. The hypotheses of all three bounds hold at every included depth by (50.5). No assertion about independence of different times is used.

At most two contacts or flights have a specified depth d. For any $a > 0$ the elementary block sum gives

$$\sum_{d=D}^{\infty} q_{\mathrm{a}}^{aj(d)} \le N \, \frac{q_{\mathrm{a}}^{aj(D)}}{1 - q_{\mathrm{a}}^{a}} \qquad (D \ge D_{\mathrm{obs}}). \tag{50.10}$$

Indeed $j(d)$ is constant on blocks of N consecutive depths; the first, possibly partial, block is enlarged to a whole block. Apply the union bound to (50.6), the preceding three one-time estimates, and (50.10) with $a = 1/2, 1, 1/4$. Restriction to the original initial section costs exactly c_*^{-1}. Removing $\mathbf{1}_\Pi h \le 1$ only on the upper side then gives (50.7). The positive pushforward has total variation equal to its total mass; the roof test can only decrease it. Countable physical word cells form a partition of the regular source, so this is already their complete positive sum. The geometric series tends to zero uniformly in all the displayed variables, which proves (50.8).

Now fix an original record outside the source and any global minimum. At every included contact the original incidence is strictly larger than $A_c q_{\mathrm{a}}^{j(d_i)/2}$. Corollary 48.5, $E_R \le 2r$, and $\lfloor (d_i - 1)/N \rfloor \ge j(d_i)$ show that inactivity is impossible. At each included flight,

$$\beta_i \le A_\Gamma q_{\mathrm{a}}^{j(d_i')/2} < \min\{\Gamma_i(Q), \tau_-/2\}.$$

The strip lemma therefore gives strictly positive clearance from every third disk, with both incident disks retained.

At every active vertex the Cartesian minimum condition gives the specular law with positive incidence, as in the preceding lemma. An outward edge from a circle and an inward edge ending at the next circle stay outside their respective incident disks until the prescribed contacts, by convexity and strict normal incidence. Together with positive third-disk clearance, this proves that every included flight is the actual first-hit physical flight to the next specified disk. Thus the whole central record is regular and no collision has been hidden.

At a central contact, (50.3) bounds the state distance by $A_\phi q_{\mathrm{a}}^{j(d_i)/4}$, whereas the original state is strictly farther than that from the actual section boundary. A shortest circle–momentum path cannot cross the boundary, so the two indicators agree. This argument includes both central endpoint tests and every occupation state. Equality of centers gives the central lattice displacement and flight count exactly. Finally Lemma 48.1 gives common active vertices and unit directions for all minima. There was no measurable selection, parameter continuation, or minimum multiplicity payment in this proof. □

Corollary 50.4 (Exponential exact-label costs on the unchanged full roof)**.** *Let $D_{\mathrm{obs}} \le D \le d$, $2d < m$, and restrict the measures* (49.1) *to $Q \notin \mathcal{E}^{\mathrm{obs}}_{D,R,m}$. Then*

$$\begin{aligned} \mathsf{T}_{\mathrm{clock},d} &\le E_R(Q) q_{\mathrm{a}}^{\lfloor d/N \rfloor}, \\ \mathsf{T}_{\mathrm{phase},d} &\le K_\phi E_R(Q) q_{\mathrm{a}}^{j(d)}, \\ \mathsf{T}_{\perp,d} &\le B_\perp E_R(Q) q_{\mathrm{a}}^{j(d)}. \end{aligned} \tag{50.11}$$

Consequently, under precisely Input 11.1, their original full-roof densities are bounded respectively by

$$2rC_\star q_{\mathrm{a}}^{\lfloor d/N \rfloor}, \qquad 2rC_\star K_\phi q_{\mathrm{a}}^{j(d)}, \qquad 2rC_\star B_\perp q_{\mathrm{a}}^{j(d)}.$$

These assertions also apply to the dominated persistent low-load interior source of Corollary 49.2.

For fixed $0 < \theta < 1/2$ and $d_m = \lceil \theta m \rceil$, all three collision-normalized heights are $O(m^2 e^{-\lambda_0 m})$, uniformly in the original radius, exact labels and roof, where

$$\lambda_0 = \frac{\theta |\log q_{\mathrm{a}}|}{N} > 0. \tag{50.12}$$

On this same source the section error in (47.7) *is exactly zero. After retaining its endpoint phase e^{isW_d}, the remaining nonnegative Fourier error majorant is $|s|\mathsf{T}_{\mathrm{clock},d}$. For $0 < \lambda < \lambda_0$, its collision-normalized original-roof height tends to zero uniformly for $|s| \le e^{\lambda m}$ and for all lattice and return twists.*

Proof. The clock identity (47.2) is bounded by $\mathcal{E}_d$, so the first estimate follows from the anchor contraction. To sum the phase distances use the proof of Theorem 47.2, including its repeated terminal edge:

$$\mathsf{T}_{\mathrm{phase},d} \le \frac{64}{\tau_-} \sum_{i=d}^{m-d-1} F_i + 24 R_-^{-4} \sum_{i=d}^{m-d} |\delta_i|^4.$$

Insert (48.9) and (48.8), and use $\lfloor d/N \rfloor \ge j(d)$, to obtain exactly K_ϕ. The transverse estimate is (48.10). All three statements hold pointwise before integration. Positive ordering on the unchanged original full roof, as in Theorem 49.1, and $E_R(Q) \le 2r$ give the density assertions under the named input.

For large m, d_m is admissible, and $j(d_m) \ge \theta m/N - 2$. Thus $q_{\mathrm{a}}^{j(d_m)} \le q_{\mathrm{a}}^{-2} e^{-\lambda_0 m}$. The density bounds give the normalized rate. The exact section identity was proved for every central contact and both endpoint tests, so the occupation term in the twist comparison vanishes. Finally $m^2 e^{\lambda m} e^{-\lambda_0 m} \to 0$ when $\lambda < \lambda_0$, proving the stated frequency-range estimate. This bounds the positive error majorant on the original roof; it does not assert an interchange with an unbounded Fourier integral. □

Remark 50.5 (The retained order of limits and the remaining full source)*.* Theorem 50.3 proves $\lim_{D\to\infty} \sup_{m,R,n,k,h} \|\mu^{\mathrm{obs}}_{D,R,m,n,k,h}\|_{\mathrm{TV}} = 0$ on the entire original family. Corollary 50.4 proves a stronger roof topology for specified exponentially small costs on its complement, under a precisely named inherited input. Total variation of the exceptional source is not its roof essential height. In particular these two statements do not imply that its $m^2 L^\infty$ height vanishes.

The original $\varepsilon(B), h_B, r_B, \gamma_B, K_B$, microscopic roof windows and guard q_0 remain fixed before the collision limsup, and B remains outermost in (0.2). The positive shallow source, all required physical multipliers, critical and singular paired traces, roofwise pool capacity, full conditioning denominator and the complete spectral/arithmetic inversion errors still require their actual original-source bounds. The exponentially accurate central comparison does not discard the remaining end portions or change the full event. The final raw pointwise local limit theorem and four-coordinate inversion are therefore not claimed complete.

Appendix A. Reconstruction of the retained physical anchor input

This appendix reconstructs the specific consequences of native modules 287 and 294 needed in the new sublevel proof. Its statements are inherited results, not newly claimed contributions. The exact remote body of 287 is retained under `provenance/sublevel-continuation-freeze/provenance/287-original.tex`. The full module 294 is loaded natively in the present main; its predecessor copy is also preserved.

Lemma A.1 (Buffered all-direction line covering, inherited)**.** *Every straight segment of length $H = 28/5$ has a lattice center at perpendicular distance at most $\omega = 179/400$ from its line, with perpendicular foot strictly inside the segment.*

Proof. For a primitive triangular lattice vector of length l, the parallel lattice rows have spacing $h = \sqrt{3}/(2l)$. At a segment midpoint q choose a nearest row and then a nearest lattice point on it. The two coordinates of $z - q$ parallel and perpendicular to this primitive direction obey $|x| \le l/2$, $|y| \le h/2$. For a direction at angle δ to that vector, its transverse projection is at most $W = (h|\cos\delta| + l|\sin\delta|)/2$, and its longitudinal projection at most $\sqrt{l^2 + h^2}/2$.

By lattice symmetries reduce to an unoriented angle $0 \le \theta \le \pi/6$ and put $t = \tan\theta/\sqrt{3}$. The primitive lattice vectors and the bounds obtained from their row rectangles are

t interval	v	l^2	$\sup W^2$
$[0, 1/60]$	$(1, 0)$	1	961/4804
$[1/60, 1/12]$	$(11/2, \sqrt{3}/2)$	31	13697401/73866304
$[1/12, 1/5]$	$(5/2, \sqrt{3}/2)$	7	529/3136
$[1/5, 1/3]$	$(3/2, \sqrt{3}/2)$	3	75/448

Here is an explicit check that these are bounds for whole intervals. For the first row, $W^2 = 3(1+2t)^2/(16(1+3t^2))$, increasing since $2-3t > 0$. For each other row write $v = ((2a+1)/2, \sqrt{3}/2)$ with $a = 5, 2, 1$, $s = 2a + 1$, $N = a^2 + a + 1$. Then

$$W = \frac{\sqrt{3}}{4\sqrt{1 + 3t^2}} \left(\frac{s + 3t}{2N} + |st - 1| \right).$$

On the listed interval $t \le 1/s$, the positive numerator decreases with derivative $3/(2N) - s < 0$, and the denominator increases. Evaluate its left endpoint, and the right endpoint for the first row, to obtain the table. Every listed rational is strictly below $(179/400)^2$. The largest longitudinal bound is $\sqrt{31 + 3/124}/2 = \sqrt{3847/124}/2 < H/2$, since $96175 < 97216$. Therefore the selected foot lies strictly inside the original segment. This argument neither changes the disk radii nor discards a direction. □

Lemma A.2 (Physical-time turning and the 94-flight anchor, inherited)**.** *Every actual regular trajectory segment of physical duration H has total interior incidence cosine at least $\gamma = 1/4480$. Every actual free flight has length at most H, and every complete 94-flight block has an interior collision with $c_i \ge 1/416640$.*

Proof. A collision changes the unit velocity by a vector of length $2c_i$. Hence for a physical segment $q(s)$, $0 \le s \le H$, with initial outgoing velocity v, direct integration gives

$$|q(s) - q(0) - sv| \le 2H \sum_{0 < t_i < H} c_i.$$

Endpoint collisions do not affect this estimate: an initial collision is accounted for by the outgoing v, and the velocity change at H does not affect positions up to H. If the sum were less than $1/4480$, the deviation would be less than $1/400$. Apply Lemma A.1 to the initial straight reference segment and evaluate at its interior perpendicular foot. The actual position would then have

distance less than $179/400 + 1/400 = 9/20 \le R$ from a lattice center, contradicting exterior nonpenetration. This proves the incidence sum.

A straight free flight of length exceeding H would contain an H segment intersecting a disk interior by the same covering lemma, a contradiction. Consecutive distinct disks are separated by $\tau_- = 3/50$, so 94 flights take at least $94\tau_- = 141/25 > H$. Apply the physical-time bound to their first H units. The relevant collisions are among the 93 interior collisions of the block. Thus their total cosine is at least $1/4480$, and one is at least $1/(4480 \cdot 93) = 1/416640$. All collisions are actual; no probabilistic independence, symbolic model or number of words is used. □

Appendix B. An exact flux identity for unvisited near misses

The incidence load does not see a trajectory whose original guard is witnessed only by clearance. We introduce a counting observable on the *same* physical flights, not a new collision map. Throughout this section let

$$g_* = 1 - 2R_+ = 3/50, \qquad \delta_* = g_*/4 = 3/200, \qquad 0 < \delta \le \delta_*.$$

The symbol δ_* here is a one-flight geometric constant, not the outer microscopic width $\delta(B)$. All disks still have radius R.

For a regular outgoing collision x, write its flight as $q(x) + tv(x)$, $0 \le t \le \tau(x)$. Let $\mathcal{Z}_\delta(x)$ be the set of lattice centers other than the two incident centers for which

$$0 < \min_{0 \le t \le \tau(x)} |q(x) + tv(x) - z| - R < \delta, \qquad N_\delta(x) = |\mathcal{Z}_\delta(x)|.$$

This counts near misses, not collisions. In particular neither the collision index, the physical roof, nor the section-visit count is incremented by N_δ. A center is counted once on a flight. Tangencies may be assigned value zero on their collision-area null set; no assertion about a positive pairing-endpoint trace follows from this convention.

Lemma B.1 (An interior foot and uniform endpoint separation)**.** *Every center counted by N_δ has its perpendicular foot in the open physical flight. If t_z is the time of that foot, then*

$$t_z \ge g_* - \delta, \qquad \tau - t_z \ge g_* - \delta. \tag{B.1}$$

The candidate centers, after translating the initial incident center to zero, belong to the finite set

$$\{z \in \Lambda \setminus \{0\} : |z| \le \tau_+ + 2R_+ + \delta_*\}.$$

Write N_* for its cardinality. This constant is independent of the itinerary length and of all exact record labels.

Proof. At either incident point, the distance to a third center is at least $1 - R$, since distinct lattice centers have distance at least one. Its distance to the third disk is therefore at least $1 - 2R \ge g_*$. A minimum below $\delta \le g_*/4$ cannot occur at an endpoint. The nearest point on a line segment is then its perpendicular foot, whose distance from z is at most $R + \delta$. By the reverse triangle inequality its distance along the line from either endpoint is at least $(1 - R) - (R + \delta) \ge g_* - \delta$. Finally a foot is at distance at most $R + \tau_+$ from the initial center. Adding its distance from z gives the finite-set bound. This proof uses the actual segment, including both of its endpoints, rather than an infinite-line substitute for the clearance event. □

Put $v_\vartheta = (\cos\vartheta, \sin\vartheta)$ and $v_\vartheta^\perp = (-\sin\vartheta, \cos\vartheta)$. For a mark z define the signed line coordinate $b = (z - q) \cdot v_\vartheta^\perp$. Translate the marked center to zero. For $\sigma \in \{-1, 1\}$ and $0 < u < \delta_*$ the oriented line with $b = \sigma(R + u)$ has closest point $-\sigma(R + u)v_\vartheta^\perp$. Follow this line backwards to its last actual collision. Reduce that collision by a lattice translation to the initial fundamental disk, and denote the resulting outgoing state and marked-center displacement by

$$X_R(\vartheta, \sigma, u), \qquad Z_R(\vartheta, \sigma, u).$$

These maps use straight free flight until an *actual* disk is met. There is no reflection at the closest point. The finite-horizon assumption ensures actual collisions in both directions, apart from null singular parameters on which these maps can be defined arbitrarily. The interval from the preceding collision to the following one remains the original physical flight.

Theorem B.2 (One-copy clearance flux and exact multiplicity)**.** *For every nonnegative Borel function a of an outgoing state and a marked-center displacement,*

$$\int \sum_{z\in\mathcal{Z}_\delta(x)} a(x,z)\,d\nu(x) = \frac{1}{4\pi R}\sum_{\sigma=\pm1}\int_0^{2\pi}\int_0^{\delta} a(X_R(\vartheta,\sigma,u),Z_R(\vartheta,\sigma,u))\,du\,d\vartheta. \tag{B.2}$$

In particular,

$$\int N_\delta\,d\nu = \delta/R. \tag{B.3}$$

Both statements hold for every $R\in[R_-,R_+]$. The multiplicity on the left is the number of physically distinct unvisited disks, not an upper bound for the number of collision words.

Proof. We first verify the local measure, including its radius factor. Write $q = R(\cos\alpha,\sin\alpha)$ and $\vartheta = \alpha + \arcsin p$. At fixed ϑ,

$$\left.\frac{\partial b}{\partial\alpha}\right|_{\vartheta} = -R\cos(\vartheta-\alpha), \qquad \frac{\partial\vartheta}{\partial p} = (1-p^2)^{-1/2}.$$

Consequently

$$\left|\det D_{(\alpha,p)}(\vartheta,b)\right| = R, \qquad d\nu = \frac{d\vartheta\,db}{4\pi R}. \tag{B.4}$$

The two signs $b=\sigma(R+u)$ both have $|db|=du$.

There is also a global multiplicity assertion to prove. A marked physical flight maps to its unique oriented line relative to its marked center. Conversely, the closest point of every line just described is in the free region: its distance to any other lattice center is at least $1-R-u>R$. In fact the disk of radius $R+\delta_*$ around the marked center does not intersect any other actual disk, since $2R+\delta_*<1$. The central disk is missed because the line has impact magnitude greater than R. Following the line in both directions therefore gives a physical flight whose incident disks are different from the marked disk. Its foot is interior and its clearance from that disk is u. This is inverse to the marked-flight map, modulo the prescribed lattice translation. Multiple near misses on one flight give different marked copies; they are neither lost nor identified.

For measure-theoretic precision, partition the line parameters by the actual backward incident disk and the forward incident disk, and exclude their tangent boundaries. These partitions are Borel; on each regular piece the disk-intersection formulas and their inverses are smooth. Tangency boundaries satisfy the line equation $|b_z|=R$. For any fixed alternative center their equation in (ϑ,u) is a graph in u (with one of the two fixed signs), hence has planar Lebesgue measure zero. There are countably many centers. The corresponding singular collision sets are also null by (B.4) on regular charts. Exhaust these charts by relatively compact subcharts and disjointize them. Applying the local Jacobian formula to bounded functions on finite unions, then using monotone convergence for both chart unions and $a\wedge n$, proves (B.2) without a prior integrability assumption. Setting $a=1$ gives $2(2\pi)\delta/(4\pi R)=\delta/R$. □

Corollary B.3 (All physical depths with the original normalization)**.** *For arbitrary $0<\delta_j\le\delta_*$, $0\le j<m$,*

$$\int \sum_{j=0}^{m-1} N_{\delta_j}(T_R^j x)\,d\nu(x) = \frac{1}{R}\sum_{j=0}^{m-1}\delta_j.$$

For every original exact event Π *the same integral with* $\mathbf{1}_\Pi\, d\nu_R^*$ *is at most* $(Rc_*)^{-1}\sum_j \delta_j$*. In particular, if* $2\varepsilon \le \delta_*$ *and* $r = q_0^{1/4}$*, then*

$$\nu_R^*\Big(\Pi \cap \{\exists j : \Delta_j < 2\varepsilon r^{d'_j}\}\Big) \le \frac{4\varepsilon}{R_- c_*(1-r)}. \tag{B.5}$$

Proof. The collision measure ν is invariant under the original elastic collision map. This also follows from the oriented-line Jacobian (B.4), whose magnitude is the same at both incident circles, and specular reflection, which preserves boundary flux. Thus apply (B.3) at each j. The original initial restriction has density $\mathbf{1}_{Y_R^*}/c_* \le 1/c_*$ with respect to ν; further exact labels only decrease the positive integral. Outside the null exact-tangency set, existence of a clearance witness implies $N_{2\varepsilon r^{d'_j}} \ge 1$. There are at most two flight indices of each nonnegative depth, so $\sum_j 2\varepsilon r^{d'_j} \le 4\varepsilon/(1-r)$. No independence or return-conditioned invariant measure is introduced. □

The estimate (B.5) is an integrated mass estimate. Multiplication of it by m^2 at fixed $\varepsilon(B)$ is not a proof about a target-roof density.

Appendix C. Clearance strips transverse to physical test curves

A measure bound alone gives no short-curve control. The positive separation between an unvisited disk and the incident disks supplies a transverse derivative which remains nonzero even as an incident angle approaches grazing. We specify the time orientation explicitly: forward near misses are tested on increasing-slope curves, and *past-flight* near misses on decreasing-slope curves.

Use physical arclength $s = R\alpha$ increasing counterclockwise and $p = \sin\varphi$. Fix a finite $U > 0$ and put

$$\begin{aligned} \kappa_- &= 1/R_+, \qquad \kappa_+ = 1/R_-, \\ D_0 &= \kappa_+ + U + (1-R_+)^{-1}, \\ \ell_0 &= \pi/(2D_0), \qquad a_0 = \kappa_-(g_* - \delta_*), \\ C_{\mathrm{geo}} &= 2N_*\sqrt{1+U^2}/a_0. \end{aligned} \tag{C.1}$$

All lengths below are Euclidean arclength in (s,φ), not configuration-space arclength. A test is a connected graph over an s interval, of length at most ℓ_0. Let E_δ^+ be the forward physical near-miss event $\{N_\delta \ge 1\}$, and let E_δ^- be the analogous event on the preceding actual flight.

Theorem C.1 (A grazing-uniform, one-flight strip bound)**.** *For every* C^1 *test graph* W^+ *with* $0 \le d\varphi/ds \le U$*, and every* C^1 *test graph* W^- *with* $-U \le d\varphi/ds \le 0$*,*

$$|W^+ \cap E_\delta^+| \le C_{\mathrm{geo}}\delta, \qquad |W^- \cap E_\delta^-| \le C_{\mathrm{geo}}\delta. \tag{C.2}$$

The bound is uniform in R*, in the homogeneity strip of the test, and in its distance to the grazing boundary. Arbitrary Borel restrictions, including exact record restrictions, can be added to the sets on the left. Regular pieces are understood with their physical multiplicity one.*

Proof. On W^+ the direction angle is $\vartheta = s/R + \varphi(s)$. For a fixed candidate center $z \ne 0$ set

$$t_z = (z-q)\cdot v_\vartheta, \qquad b_z = (z-q)\cdot v_\vartheta^\perp.$$

Differentiation in the increasing s coordinate gives the signed identity

$$b_z' = -\cos\varphi - t_z\vartheta', \qquad \kappa_- \le \vartheta' \le \kappa_+ + U. \tag{C.3}$$

At a physical near miss, Lemma B.1 yields $t_z \ge g_* - \delta_*$. Hence $-b_z' \ge a_0$ there. It is the $t_z\vartheta'$ term, not a lower bound on $\cos\varphi$, which prevents degeneration.

We must not multiply this estimate by an uncontrolled number of components. Choose a continuous argument $\beta(s) = \arg(z - q(s))$ on the graph. Since $|z - q| \geq 1 - R_+$, one has $|\beta'| \leq (1 - R_+)^{-1}$. Moreover

$$(\beta - \vartheta)' \leq (1 - R_+)^{-1} - \kappa_- < 0,$$

because $R_+ < 1/2$. Its total variation is at most $D_0|W^+| \leq \pi/2$. Thus the set $t_z > 0$, equivalently $\cos(\beta - \vartheta) > 0$, is at most one interval: two distinct positive-cosine intervals are separated by a negative interval of angular length π. On the interval $t_z > 0$, formula (C.3) says that b_z strictly decreases. Each of the two impact bands $(R, R + \delta)$ and $(-R - \delta, -R)$ therefore has multiplicity at most one in this coordinate. Restricting to the portions which are actual near misses can break them into pieces, but cannot increase that multiplicity. By the one-dimensional change of variables on this monotone map,

$$\int_{\{s:\ z \text{ is a physical near miss}\}} ds \leq 2\delta/a_0.$$

Since $d\ell \leq \sqrt{1 + U^2}\, ds$, summing over the finite candidate set from Lemma B.1 proves the first bound. This sum counts only centers in one physical flight. It does not count iterated collision words or homogeneous continuity components.

For the second bound, the collision time-reversal involution is $\mathcal{I}(s, \varphi) = (s, -\varphi)$. The outgoing velocity at $\mathcal{I}x$ reverses the incoming velocity at x. Consequently its next straight flight is the reversal of the previous actual flight at x. This proves

$$E_\delta^- = \mathcal{I}E_\delta^+ = T_R E_\delta^+ \tag{C.4}$$

on the regular collision space. The involution is an isometry in (s, φ) and sends W^- to an increasing-slope graph. Apply the first bound. Borel restrictions decrease lengths. Taking monotone limits of truncated regular test pieces gives the same bound at grazing limits; an exactly tangent parameter contributes zero transverse length by the strict derivative above. This last statement does not delete any paired-source boundary trace in the older coarea construction. □

Corollary C.2 (Bounded multipliers on stable tests)**.** *If $|h| \leq H$ pointwise on regular collision states, then for every W^- as above and every bounded Borel test ψ,*

$$\left| \int_{W^-} \mathbf{1}_{E_\delta^-} h\psi \, d\ell \right| \leq H\|\psi\|_\infty \min\{|W^-|, C_{\text{geo}}\delta\}.$$

This is a bound for bounded functions with specified representatives. It is not an assertion that multiplication by the indicator is bounded on an entire space of anisotropic distributions.

Proof. Bound the integrand by $H\|\psi\|_\infty \mathbf{1}_{E_\delta^-}$ and use Theorem C.1 together with the trivial length bound. Representatives on singular sets are fixed to zero. □

Appendix D. Clearance gains after arbitrarily many actual collisions

We next give a consequence which is uniform in the full collision count, rather than only a one-flight estimate. Its additional input is the retained stable-curve growth bound, stated explicitly here. It is not the stronger target-roof local limit estimate.

Input statement D.1 (Retained homogeneous curve growth)**.** Use a family $\mathcal{W}_R^s$ of short homogeneous admissible stable graphs in the above physical coordinates, with slopes in $[-U, 0]$, length at most ℓ_0, and the inherited common curvature bound. For every $W \in \mathcal{W}_R^s$ and $l \geq 0$,

its regular inverse image has a countable partition $\mathcal{G}_l(W)$ into such graphs, up to arclength-null sets, with one-copy inverse branches and

$$\sum_{V\in\mathcal{G}_l(W)} \sup_V J_V T_R^l \le C_g. \tag{D.1}$$

Here $J_V T_R^l$ is the Euclidean arclength derivative along V; C_g is independent of R, l, W. At $l = 0$ the partition is $\{W\}$.

This is the curve-growth interface of historical module 241, not its whole-cell pressure conclusion. Length thresholds can be decreased: subdividing a pre-existing short component into lengths at most ℓ_0 multiplies its Jacobian upper sum by at most a fixed number, independent of l. The precise external interface is discussed in Appendix H. We retain this input separately from the unconditional one-flight geometry of Theorems B.2 and C.1.

For fixed representatives of bounded Borel functions put

$$\mathcal{S}_a(f) = \sup_{W\in\mathcal{W}_R^{\mathrm{s}}} |W|^{-a} \int_W |f|\, d\ell, \qquad 0 \le a < 1.$$

This absolute stable-test seminorm is specified here to avoid identifying it with a full anisotropic distribution norm. With respect to the invariant measure ν, the ordinary density transfer is $\mathcal{L}_R h = h\circ T_R^{-1}$. For the original record observable $\mathbf{g}_R = (\kappa_1, \kappa_2, \tau, \mathbf{1}_{Y_R^*})$ and real $\xi = (\xi_1, \xi_2, \xi_t, \xi_n)$, define

$$\mathcal{L}_{R,\xi} h(x) = e^{i\xi\cdot \mathbf{g}_R(T_R^{-1}x)} h(T_R^{-1}x).$$

The three discrete frequencies may be taken modulo 2π; the physical-roof frequency is real and unbounded. This definition does not assert boundedness of this operator on an unstated Banach space. On bounded functions its modulus is simply $|h| \circ T_R^{-1}$.

Theorem D.2 (A uniform future-propagated clearance gain)**.** *Under Input D.1, let $|h| \le H$ pointwise on the regular collision space, with $h = 0$ off E_δ^-. Then for every $l \ge 0$, real ξ, and $0 \le a < 1$,*

$$\int_W |\mathcal{L}_{R,\xi}^l h|\, d\ell \le H \min\{|W|, C_g C_{\mathrm{geo}}\delta\}, \tag{D.2}$$

$$\mathcal{S}_a(\mathcal{L}_{R,\xi}^l h) \le H(C_g C_{\mathrm{geo}}\delta)^{1-a}. \tag{D.3}$$

The constants do not depend on the entire future collision word or on any of the four real frequencies. The bounds also hold for any bounded multiplier of modulus at most one applied at arbitrary intermediate collisions, with the same support at the marked time.

Proof. On regular branches positivity and $|e^{i\xi\cdot\mathbf{g}_R}| = 1$ give $|\mathcal{L}_{R,\xi}^l h(x)| \le H\mathbf{1}_{E_\delta^-}(T_R^{-l}x)$. Change variables separately on the inverse branches of W. The full nonnegative sum, completed by monotone convergence, satisfies

$$\int_W |\mathcal{L}_{R,\xi}^l h|\, d\ell \le H \sum_{V\in\mathcal{G}_l(W)} (\sup_V J_V T_R^l)\, |V \cap E_\delta^-| \le H C_{\mathrm{geo}}\delta C_g.$$

The pointwise bound by H supplies the other term in the minimum. For $x > 0$ and $D > 0$, splitting into $x \le D$ and $x \ge D$ gives $x^{-a}\min(x, D) \le D^{1-a}$. Apply this with $x = |W|$ and $D = C_g C_{\mathrm{geo}}\delta$ to obtain (D.3). The modulus of a product of additional multipliers is at most one, so precisely the same positive upper bound applies. No stochastic independence, wordwise supremum in place of a physical integral, or Fourier inversion has been used. □

Corollary D.3 (The complete depth schedule in stable tests)**.** *Let $f_0 = \mathbf{1}_{Y_R^*}/c_*$. For $0 \le j < m$ set*

$$h_j = \mathbf{1}_{E^-_{\delta_j}} \mathcal{L}_{R,\xi}^{j+1} f_0, \qquad f_{m,j} = \mathcal{L}_{R,\xi}^{m-j-1} h_j.$$

Then

$$\mathcal{S}_a(f_{m,j}) \le c_*^{-1} (C_g C_{\mathrm{geo}} \delta_j)^{1-a}. \tag{D.4}$$

If $2\varepsilon \le \delta_*$ and $\delta_j = 2\varepsilon r^{d'_j}$, the terminal collision density restricted to the union of all original clearance witnesses has absolute stable-test seminorm at most

$$\frac{2(C_g C_{\mathrm{geo}})^{1-a} (2\varepsilon)^{1-a}}{c_*(1 - r^{1-a})}, \qquad 0 \le a < 1. \tag{D.5}$$

This holds uniformly over m, R and real record frequencies. Further original exact labels and terminal-section restrictions may be retained on the left by positive domination.

Proof. The initial density and all of its twisted iterates have modulus at most $1/c_*$. By (C.4), the past-flight event at collision $j+1$ is exactly the forward clearance mark on flight j. Thus the exponents $j+1$ and $m-j-1$ insert the mark at the actual collision time, including $j = 0$ and $j = m-1$ with zero remaining future steps. Theorem D.2 proves (D.4). The union indicator is bounded by the sum of its physical one-flight indicators. Subadditivity of $\mathcal{S}_a$ and at most two indices at each depth give

$$\sum_{j=0}^{m-1} \delta_j^{1-a} \le \frac{2(2\varepsilon)^{1-a}}{1 - r^{1-a}}.$$

This proves (D.5). Arbitrary exact-record selectors are bounded by one on the same trajectories, so they cannot increase the positive majorant. They are not replaced by independent weights, and no probability measure is renormalized. □

The estimate is genuinely uniform in the full number of physical collisions and in the full scheduled set of clearance marks. Its norm is nevertheless a stable-test norm on terminal collision space, not the essential height of a roof density. In particular, the absence of frequency growth in this absolute estimate is not decay as $|\xi_t| \to \infty$ and does not justify integration over that unbounded frequency. Such decay, or another rigorously justified pointwise bridge, is still required for the original raw target.

Appendix E. Exact-record disintegration at an unvisited mark

The flux coordinates also give a direct representation of the remaining roof problem which does not discard the exact return, displacement or collision information. We state it as an equality of measures first, because a conditional angular roof kernel need not have a density without proof.

For $0 \le j < m$ define the initial state

$$X_{R,j}(\vartheta, \sigma, u) = T_R^{-j} X_R(\vartheta, \sigma, u)$$

on its regular set, with a fixed null-set convention otherwise. The complete actual record from this state is used. Displacements are computed in the original lattice basis, relative to its initial lift; translating the marked disk to zero changes none of these relative displacements. For almost every u, define the positive Borel measure on the original roof line

$$\mathcal{K}_{m,R,n,k,j,u}(A) = \sum_{\sigma=\pm 1} \int_0^{2\pi} \mathbf{1}_{\Pi_{n,k,m,R}}(X_{R,j}) \mathbf{1}_{\{L_{m,R}(X_{R,j}) \in A\}} \, d\vartheta. \tag{E.1}$$

It has mass at most 4π. No density or angular derivative bound is included in this definition.

Theorem E.1 (Disintegration of a physically marked exact record)**.** *For every Borel set $A \subset \mathbb{R}$,*

$$\int \mathbf{1}_{\Pi} N_{\delta}(T_R^j x)\mathbf{1}_{\{L_{m,R}\in A\}}\, d\nu_R^* = \frac{1}{4\pi R c_*}\int_0^{\delta} \mathcal{K}_{m,R,n,k,j,u}(A)\, du. \tag{E.2}$$

The identity also holds against every nonnegative Borel roof test. All singular boundary sets in these integrals use the original collision-area null-set convention, not deletion of paired boundary coefficients.

Proof. Replace $d\nu_R^*$ by $c_*^{-1}d\nu$ inside the integral, since $\Pi \subset Y_R^*$. Change the initial collision coordinate to $y = T_R^j x$ using invariance of ν. Apply Theorem B.2 with

$$a(y,z) = \mathbf{1}_{\Pi}(T_R^{-j}y)\mathbf{1}_{\{L_{m,R}(T_R^{-j}y)\in A\}}.$$

This is a bounded nonnegative function on the original physical states, independent of which near-miss mark is selected once y is fixed. The resulting integral is exactly (E.2). Countable regular partitions and Tonelli's theorem justify the integration over u; their exceptional parameters have two-dimensional measure zero. For arbitrary nonnegative roof tests use monotone simple-function approximation. In particular the factor is $1/(4\pi R c_*)$, with no second section normalization and no new conditioning on the near miss. □

E.1. **A sufficient pointwise bridge, with its status explicit.** Fix M. The following measure domination is one concrete, stronger sufficient condition for paying all original clearance witnesses: there exists C_M such that, for all sufficiently large m, all R, n, k, j and almost every $0 < u < \delta_*$,

$$\mathcal{K}_{m,R,n,k,j,u}\big|_{\mathcal{C}_{m,R,n,k}(M)} \le C_M m^{-2}\, dt. \tag{E.3}$$

An almost-everywhere set may depend on the fixed R, m, n, k, j; C_M must be common. This is an *unproved sufficient condition*, not a newly assumed premise of the organizing raw local limit assertion. It asserts absence of singular conditional roof mass as well as an upper density bound. Neither Theorem B.2 nor Theorem D.2 establishes it.

Proposition E.2 (The exact normalization a pointwise bridge must pay)**.** *If* (E.3) *holds, the original roof measure restricted to any clearance witness, with $2\varepsilon \le \delta_*$, is dominated on the central target by a density d_{clr} with*

$$\sup_{R,n,k} m^2 \|d_{\mathrm{clr}}\|_{L^\infty(\mathcal{C}_{m,R,n,k}(M))} \le \frac{C_M \varepsilon}{\pi R_- c_*(1-r)}. \tag{E.4}$$

The same upper bound holds for any positive source already dominated by that original collision measure, restricted to these witnesses.

Proof. The witness-union indicator is at most $\sum_j N_{2\varepsilon r^{d'_j}}(T_R^j x)$ off the fixed null set. Apply (E.2) to every term, then the measure domination (E.3). The resulting density bound is $(4\pi R c_*)^{-1} C_M m^{-2} \sum_j 2\varepsilon r^{d'_j}$. The sum is at most $4\varepsilon/(1-r)$, proving the stated constant. Positive domination preserves the bound for the indicated source. At the original outer schedule this would tend to zero after the collision limsup and then $B \to \infty$; no window has been changed. □

On a regular physical branch, the new parametrization has the exact chain rule

$$DX_{R,j} = DT_R^{-j}(X_R)\, DX_R, \qquad D(L_{m,R}\circ X_{R,j}) = DL_{m,R}(X_{R,j})\, DX_{R,j}.$$

The one-flight lower bound for $|b'_z|$ does not lower-bound this long-time roof derivative. Inverse-iterate growth, cancellations and critical angular roofs must be controlled separately. A bound after integration in u can allow angular folds which a uniform bound on each $\mathcal{K}_{j,u}$ would exclude. No supremum is passed through that disintegration here.

A proof of the stronger condition (E.3) is not necessary if a suitably summed kernel bound or comparison-capacity argument can be proved instead. The proposition only makes explicit which normalization and which measure must be controlled. It is not counted as an unconditional closure or a second payment of a source already paid in revision 112.

E.2. **Compatibility with the complete original source.** All identities through (15.3) and the full pool $P_{r_B}^{\mathrm{ng}}$ are unchanged. We have not subtracted the clearance source from $\mathsf{U}_B^{\mathrm{low}}$ using only a weak norm. Consequently the exact remaining target is still (15.8), with the original order of limits. The new results treat the mechanism which is invisible when $J_\varepsilon = \varnothing$: genuine unvisited-disk near misses. They do not pay low-severity shallow incidence without clearance, prove abundance of the full nongrazing comparison pool, or settle positive critical-entrance and section-cut pairing traces.

For precision concerning a minor referee point, the inequality $\alpha f(N) < u_m$ already proved in Corollary 15.2 implies $N < 2\sqrt{u_m/\alpha}$ on the retained source. The independent quadratic count theorem is retained as a finite-count theorem, not presented as another missing payment for that already restricted source. The term "low" in all inherited source symbols means low *incidence* load only. In the monotone exhaustion of Proposition 10.4, every marked weight is explicitly assigned zero on the omitted collision-area null set; its extended integral is taken only over the stated regular charts. This convention never sets a positive one-sided pairing trace to zero.

Appendix F. Scope of the retained comparison statements

For clarity, we record the precise scope of the earlier allocation statements. None of their algebraic maximality assertions is used as a substitute for the physical estimate in Theorem 7.1.

Proposition F.1 (Fixed-window scalar maximality and old-family dominance)**.** *Fix a Borel comparison subspace, $0 < r < \Delta$, and $K > 0$. Consider nonnegative measurable weights w_+, w_- with $w_+ + w_- \le 1$, supported on the respective one-sided window conditions in* (6.2)*. Among scalar accepted heights $Z(t)$ satisfying*

$$0 \le Z(t) \le U(t), \qquad Z(t) \le K \int (w_+ + w_-) p_\vee \, d\theta,$$

the greatest is $V_{r,K}(t) = \min\{U(t), KP_r(t)\}$. The allocation is global at the roof and does not couple individual source sheets to comparison sheets.

For a compatible old fixed family with window width r_{old}, physical overlap N and block cost κ, the explicit conditions

$$0 < r \le r_{\mathrm{old}}, \qquad \ell_{\mathrm{comparison}} < \Delta, \qquad K = N\kappa$$

imply $V^{\mathrm{old}}(t) \le V_{r,N\kappa}(t)$ on the same original source and comparison space.

Proof. The support and one-copy conditions give $w_+ + w_- \le Q_r$. Hence every Z is bounded by the stated minimum. On $P_r > 0$ choose $w_\pm = V_{r,K} q_\pm / (KP_r)$; use zero weights otherwise. This is a measurable attaining certificate. The separate source multiplier is $\beta_{r,K}$ defined in Section 7. For the old family, retain only active source blocks at t. Their comparison supports lie in the r-pool by the displayed width condition. Summing the blockwise budget and using physical overlap at most N gives $V^{\mathrm{old}} \le N\kappa P_r$; disjoint original source blocks also give $V^{\mathrm{old}} \le U$. The minimum proves the assertion, with countable families obtained by monotone convergence on their common roof versions. □

Proposition F.2 (Strictness relative to roof-independent whole-interval blocks)**.** *In the interval-source model, the roofwise certificate class is strictly larger than the old class of roof-independent angular source blocks that requires a comparison interval to contain the translate of the* entire

source interval by $[0,r]$ *or* $[-r,0]$. *This statement compares certificate classes and does not assert a Lorentz realization.*

Proof. Let $s = 1/1000$ and $r = 1/10$. On two disjoint angular sets of measure one give both endpoint traces value one. Let their physical intervals be $(0,s)$ and $(3s/8, r+7s/8)$, and select only the first as source. Take $\Delta > r + s/2$ and $K = 1$. For $3s/8 < t < 7s/8$ the second interval contains $[t, t+r]$, so $P_r(t) \geq U(t) = 1$. Neither physical interval has length $r+s$, which is necessary for the old translated-whole-source inclusion. Angular subdivision does not alter these intervals. Thus the old class cannot allocate a nonzero whole-source block while the roofwise class accepts the source on the indicated roof interval. □

F.1. **Limit order and the remaining estimate.** The constructions in this article are evaluated first at finite m, R, n, k with B, r_B, h_B, K_B already fixed. Positive physical sums are completed before taking the essential roof supremum, the radius and exact-label suprema, and the collision limsup. Only then is B sent to infinity. The assertion that every positive-length segment contains a sufficiently small window at a fixed count supplies no uniform modulus for this order.

The remaining estimate is precisely

$$\lim_{B\to\infty} \limsup_{m\to\infty} \sup_{R,n,k} m^2 \operatorname*{ess\,sup}_{t\in\mathcal{C}_{m,R,n,k}(M)} \left(\mathsf{U}^{\mathrm{micro}}_{B,m}(t) - K_B P^{\mathrm{ng}}_{r_B,m}(t)\right)_+ = 0.$$

No inference to this estimate is made from bounded wordwise optical weights, local integrability, a finite diagnostic, or a positive finite-count incidence. Theorem 7.1 proves the entire visited-grazing payment in this order; all remaining positive classes in the displayed expression are still part of the original pointwise problem.

Appendix G. Optical cofactors, multiple marks and simultaneous endpoints

We supply explicit matrix details for the optical identity and isolate the one-sided endpoint statement used in the trace theorem. These calculations take place on the original finite itinerary. They do not assert a bound for a sum over itineraries.

Lemma G.1 (The signed corner cofactor)**.** *For* $m \geq 1$*, with the empty determinant convention when* $m = 1$*,*

$$G_{0m} = (-1)^{m-1} \frac{R^2 c_0 c_m}{(\prod_{i=0}^{m-1} \tau_i) \det J}. \tag{G.1}$$

In particular the absolute value, not the signed mixed entry, gives $\omega = |G_{0m}|/R$.

Proof. Suppose first that $m > 1$ and put $n = m-1$. The minor obtained by deleting the last row and first column of the tridiagonal matrix J is triangular, with diagonal entries $\tau_1^{-1}, \ldots, \tau_{n-1}^{-1}$. Cramer's rule therefore gives

$$(J^{-1})_{1n} = (-1)^{n+1} \frac{\prod_{i=1}^{n-1} \tau_i^{-1}}{\det J}.$$

For $n = 1$ its numerator is the empty product and the formula says $(J^{-1})_{11} = 1/J_{11}$. The only endpoint–interior couplings are $H_{01} = R^2 c_0 c_1/\tau_0$ and $H_{nm} = R^2 c_n c_m/\tau_{m-1}$, whereas $H_{0m} = 0$. Since $H_{II}^{-1} = R^{-2} C_I^{-1} J^{-1} C_I^{-1}$, the Schur complement gives

$$G_{0m} = -\frac{R^2 c_0 c_m}{\tau_0 \tau_{m-1}} (J^{-1})_{1n},$$

which proves (G.1). If $m = 1$, the mixed entry follows directly from the single square in (4.2) and has positive sign.

For clarity, the equality between the determinant denominator and $\mathfrak{b}_m$ can also be checked without a sign convention for optical coordinates. Both quantities have initial values $B_0 = 0$, $B_1 = \tau_0$ and satisfy

$$B_{l+1} = \left(1 + \tau_l\kappa_l + \frac{\tau_l}{\tau_{l-1}}\right) B_l - \frac{\tau_l}{\tau_{l-1}} B_{l-1}.$$

For the determinant this is the tridiagonal recurrence after multiplication by the flight lengths. For the optical product it follows by eliminating the lower entry of its second column. This verifies the denominator identity used in Lemma 4.3. □

Put $T_{a,b} = \sum_{i=a}^{b-1} \tau_i$ for $0 \le a < b \le m$. For $S = \{j_1 < \cdots < j_s\} \subset \{1, \dots, m-1\}$ set $j_0 = 0$ and $j_{s+1} = m$.

Proposition G.2 (Positive expansion and simultaneous optical marks)**.** *The exact denominator has the positive expansion*

$$\mathfrak{b}_m = \sum_{S \subset \{1,\dots,m-1\}} \left(\prod_{j\in S} \kappa_j\right) \prod_{\nu=0}^{|S|} T_{j_\nu, j_{\nu+1}}. \tag{G.2}$$

For every fixed set S of s distinct interior marks,

$$\frac{\omega}{\prod_{j\in S} c_j} \le \frac{R_+^{s+1}}{2^s \tau_-^{s+1}} \varrho^{m-s-1}. \tag{G.3}$$

For $S = \varnothing$ this is the first optical bound in (4.8)*; for $|S| = 1$ it is the second.*

Proof. Let $E = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$. Expand every factor $\mathsf{C}(\kappa) = I + \kappa E$ in the product (4.6). Between two selected marks the shears combine according to $\mathsf{S}(a)\mathsf{S}(b) = \mathsf{S}(a+b)$, and $E\mathsf{S}(T)E = TE$. The top-right entry of the term with marked set S is exactly the corresponding summand in (G.2). The empty term is $T_{0,m}$. Every coefficient is positive.

To obtain the stronger estimate retaining the unmarked reflections, expand only the factors at the prescribed marks. The coefficient of $\prod_{j\in S} \kappa_j$, with all other factors retained, is the product of the top-right entries of the $s+1$ intervening optical blocks. Indeed $E = e_2 e_1^{\mathsf{T}}$, so repeated multiplication factors that coefficient into these entries. The block with $j_{\nu+1} - j_\nu$ flights has top-right entry at least $\tau_- \varrho^{-(j_{\nu+1} - j_\nu - 1)}$, by the same positive-column argument as in (4.9). Keeping only this nonnegative term gives

$$\mathfrak{b}_m \ge \left(\prod_{j\in S} \frac{2}{R c_j}\right) \tau_-^{s+1} \varrho^{-(m-s-1)}.$$

Substitute this in $\omega = R c_0 c_m / \mathfrak{b}_m$, and use $c_0 c_m \le 1$ and $R \le R_+$, to prove (G.3). □

This estimate controls simultaneous vanishing incidences on a fixed finite trajectory. Its constant depends on the number of marks, not on their locations. No word-count factor has been suppressed, and no pressure conclusion is drawn by summing its right side.

Lemma G.3 (One-sided limits at simultaneous physical mechanisms)**.** *Fix a positive-length pairing segment and approach one of its ends from the regular interior. The collision points, velocities, flight lengths, visited incidences and third-disk clearances have limits along the entire approach, even if several of the following occur simultaneously: visited grazing, unvisited tangency, a section boundary and a critical incoming endpoint. The limiting reduced Hessian and coarea factor are those in* (5.3) *and* (5.4)*. No continuation of the exact event through its boundary is asserted.*

Proof. Lemma 5.1 gives limits for q and z along the entire segment, not only a subsequence. Hence the initial collision point and $p_0 = -z\cos\theta/R$ converge. The section cap gives the outgoing normal component $c_0 > 0$ at the limit, so the initial velocity converges as well.

Suppose inductively that the departure point x and unit velocity v of a designated flight converge. For its designated next disk with centre O, the entry time is

$$\tau = -(x-O)\cdot v - \sqrt{((x-O)\cdot v)^2 - |x-O|^2 + R^2}.$$

This is the entry root on the actual regular itinerary. Its radicand is nonnegative along the approach and has a nonnegative limit. The root is continuous also when the radicand tends to zero; $\tau \geq \tau_- > 0$ prevents a zero-length flight. The arrival point therefore converges. Reflection by $v \mapsto v - 2(v\cdot n)n$ is continuous even at tangency, so the next velocity converges. Induction applies a finite number of times. An earlier disk may become tangent only at the limiting flight; this does not alter the designated entry-root limit taken from the original event.

At fixed itinerary and radius, the bounded flight lengths restrict all possibly nearby third-disk centres to a finite set. Distance from a compact line segment to each such disk is continuous in the endpoints. Their finite minimum, and hence the original clearance, has a limit. A section decision may change at the limiting state, but its coordinate limit is still obtained from the same side. None of these continuity arguments requires the boundary mechanisms to occur separately. Formula (5.3), with Z^{-1} uniformly bounded, then gives the Hessian limit even with several vanishing incidences. The fixed-direction coarea formula gives the asserted limit for χ. □

Corollary G.4 (An explicit uniform trace modulus)**.** *A permissible constant in* (5.6) *is*

$$C_{\mathrm{H}} = \frac{500 C_{\mathrm{reg}}\sqrt{2\Lambda_\circ}}{4\pi c_* \lambda_\circ^2}. \tag{G.4}$$

It is common to all original sheets, collision counts, radii and exact labels. Its inherited dependence is precisely the regular variation constant C_{reg}; the remaining constants are the fixed geometry and section constants displayed in this article.

Proof. Corollary 4.4 and $\chi \geq \lambda_\circ$ give $AI_m \leq 500\varrho^{(m-2)_+}/(4\pi c_*\lambda_\circ)$. Integrate the regular derivative bound between u and v, and apply Lemma 5.1 to that subsegment:

$$\int_u^v z^{-1}\,ds \leq \lambda_\circ^{-1}\sqrt{2\Lambda_\circ(v-u)}.$$

Multiply the constants and pass to the already proved endpoints. □

Any nonempty collection of visited endpoint grazings belongs to Γ, and one vanishing marked incidence is already sufficient for its zero source statement. Proposition G.2 additionally controls their simultaneous product. This does not alter the distinct treatment of an unvisited third-disk tangency.

Appendix H. Dependency, reference and normalization audit

The direct clearance proofs use the original positive disk separation, the uniform finite-horizon hypothesis, circular curvature, elastic reflection and elementary change of variables. The arbitrary-time stable-test estimate additionally uses Input D.1. Neither uses uniform pressure, a twisted spectral gap, or the unproved localized-shortage estimate to prove its own premise.

H.1. **The precise imported curve estimate.** In the inspected 10 October 2013 author version of Demers–Zhang [6], Lemma 3.2(b), printed page 14, bounds the sum of suprema of the forward arclength Jacobians on homogeneous inverse-image pieces, uniformly for W and for maps in the stipulated class. The geometric-family statement is Theorem 2.7 in that version; the varying-arclength qualification is Remark 2.9(b). The inherited module 241's numbering must therefore be matched to a specified version rather than silently treated as verified. Its stronger finite-testing and pressure argument is not needed for Theorem D.2.

For this paper the coordinates, Euclidean arclength Jacobian, short-curve curvature bound, homogeneous partition, fixed subdivision length, and map-family constants must all agree with Input D.1. Changing boundary perimeter with R requires the coordinate transport and its uniform equivalence; changing the subdivision threshold changes C_g by a bounded factor. These interfaces remain explicit retained hypotheses here, rather than a claim of an independent specialist reconstruction.

In the inspected 19 September 2022 author version of Baladi–Demers [7], Lemma 2.2, printed pages 7–8, is a distortion statement across stable curves in the same homogeneous backward cell. Its closure convention excludes true singularity curves. It is not a four-record, roof-conditioned m^{-2} estimate. The present proofs make no such inference.

H.2. **Checks which enter the proofs.** The radius factor in (B.4) is R, not R^2 or $1/R$. The two impact signs and the full direction circle give the factor $4\pi\delta$ before collision normalization. Exactly one section factor c_*^{-1} occurs in (E.2). The foot has physical time coordinate t_z, so the sign in (C.3) is a minus for both the boundary motion and the direction motion. Increasing-slope curves, not forward stable curves, are used for the forward event.

The uniform lower bounds used above are strictly positive on the whole closed radius interval:

$$g_* - \delta_* = 9/200, \qquad \kappa_- - (1 - R_+)^{-1} = 600/2491, \qquad a_0 = 9/94.$$

The occurrence of N_* in the curve estimate, but not in the exact flux identity, is deliberate: the former is a one-flight union bound on a test curve, whereas the latter is a global one-copy change of section. Empty future blocks are included by $l = 0$. The denominator $1 - r^{1-a}$ is finite only for $a < 1$; no estimate at the endpoint $a = 1$ is claimed.

The source identities in the retained proof bodies are unchanged. All positive sums are completed before essential suprema and the collision limsup. Taking $B \to \infty$ in a stable-test estimate would not by itself permit taking that limit in the target-roof norm. Fixed-m simulations and symbolic identities check algebra and implementation only; they do not certify either the inherited uniform dynamical inputs or the remaining raw pointwise theorem.

H.3. **What is retained, and what has not been reconstructed.** The complete active revision-112 manuscript and its ten active core proof bodies are retained in this manuscript, and an unmodified copy of the downloaded active packet is supplied in `provenance/v112-active`. The original repository additionally contains the recursive revision-111 archive, with source tree `790e627efe3a89a4549123d7e9acacf24089c287`. That recursive archive was not exported by the completed CI artifact available for this revision. Its identity is preserved; it has not been represented as a locally reconstructed, independently audited set of 267 historical proofs. The included source manifest and preservation audit distinguish byte-verified local preservation from remote tree pinning. In particular the historical spectral, arithmetic, induced-return and four-coordinate inverse-problem inputs are not replaced by the new clearance lemmas.

Appendix I. Critical-roof and fold audit: normalization and ordered closure

The new identities have different kinds of uniformity. The clearance-to-endpoint determinant is an identity for each finite physical itinerary and every marked flight. The fold and finite-type

bounds are uniform over Borel selectors on a specified chart, with specified derivative constants. The critical coarea error is uniform over selectors but depends on the chart's third-derivative and weight-modulus constants. The critical coefficient sum is over all regular physical words at fixed collision count, relative to the retained uniform pressure input; it is unscaled. None of these qualifiers can be replaced by another.

I.1. **Two distinct uses of reversibility.** At a normal endpoint, the actual reflected ray reverses its incoming velocity. This proves the reversing-periodic-orbit assertion without adding a reflection at an unvisited mark. In collision coordinates, time reversal also gives the derivative identity $\mathcal{M} = JB^{-1}JB$. The first assertion concerns actual points and their section tests; the second concerns the tangent map on the regular domain. At an unvisited tangency, only the one-sided reduced action coefficient is used. A smooth extension of that action does not define the physical map through the forbidden side.

Even at a regular reversing orbit the original section need not be reversal invariant. The occupation identity is $A_{2m} = A_m + A_m^{\mathcal{I}}$ when both original endpoints are in the section, not automatically $2n$. The coefficient $\mathfrak{J}$ describes nearby collision area pushed to the roof. It is not a conditional law on a measure-zero periodic orbit.

I.2. **The exact small-scale error and its range.** The radius r_0 in Theorem 17.3 is a local endpoint-chart radius. The height increment is $s = t - t_x$, and its allowed range is $0 < s < r_0^2/3$. Neither r_0 nor s replaces the original microscopic observation window. In particular the explicit error constant

$$2\pi\sqrt{3}\{2L_w + Kw(0)\}$$

has not been shown summable after multiplying the full physical source by m^2. Decreasing r_0 separately for each orbit does not establish that summability. A general Borel selector retains the exact fraction $a_E(s)$; only a verified finite tangent-sector structure justifies a limiting angular fraction.

The local fold exponent $1/2$ is produced by integrating a square-root conditional singularity across a transverse clearance direction. A higher finite type has its stated exponent $1/\ell$ and derivative budget. Full critical roofs have a different two-dimensional local coarea law; the fold theorem is not applied there when its clearance transversality fails. A countable disjoint source-chart covering preserves positivity and exact multiplicity, but does not itself bound the sum of the chart constants.

I.3. **The unchanged closure statement.** All original positive coefficients and the complete nongrazing pool are still those of Section 15. The remaining statement is (15.8), with the full physical sum first, then the exact-label and radius suprema and the original roof essential supremum, then the collision limsup, and the outer limit last. A successful continuation must control clearance and critical charts, positive one-sided pairing traces, the low-incidence source, and the actual available comparison capacity in that topology. The new calculations neither subtract these unpaid classes nor introduce a new name for the shortage.

The accompanying programs check the signed generating derivative, stability determinant, normalization, fold constants, selector error and actual horizontal period-two physical rays. The latter are not asserted to lie in the large-collision central roof window. Exact rational checks of their optical Schur complements and finite ray-derivative comparisons test independent implementations; they are not substitutes for the written continuous proofs or the unresolved uniform dynamical estimates.

Appendix J. Cross-flight normalization and source-scope audit

The physical order used in Section 19 is

$$\text{first mark foot} \xrightarrow{a} q_1 \xrightarrow{\ell_1} \cdots \xrightarrow{\ell_{h-1}} q_h \xrightarrow{b} \text{second mark foot}.$$

Only $q_1, \dots, q_h$ are reflections. The incident collision before the first foot and the incident collision after the second foot are also present in the original full trajectory. Neither is deleted from the record; neither auxiliary marked circle changes the clock.

The finite-horizon quantity τ_+ denotes the fixed uniform bound on the length of every real free flight in the original radius family, as specified in the retained model. This revision does not certify a new numerical value for it. All new constants display that dependency. In particular, with

$$L_{\mathrm{pair}} = \tau_+ + 97/100, \qquad L_{\mathrm{end}} = \tau_+ + 191/200,$$

N_{pair} and $N_\circ$ are the numbers of nonzero lattice vectors of length at most L_{pair} and L_{end}, respectively. For either L, they can be evaluated exactly as

$$\#\{(a,b) \in \mathbb{Z}^2 \setminus \{(0,0)\} : a^2 + ab + b^2 \le L^2\}.$$

A conservative bound, without any enumeration assumption, is $\lfloor (2L+1)^2 \rfloor - 1$: the disjoint open radius-$1/2$ disks around these centers and the origin lie in the radius-$(L + 1/2)$ disk; area comparison proves the claim. This is an end-candidate count, never a bound on the number of full physical itineraries.

The two-mark same-flight identity counts *ordered* distinct centers. For chronological pairs the expansion $\mathscr{N}(\mathscr{N}-1) = \sum_j N_j(N_j-1) + 2\sum_{j<k} N_j N_k$ accounts explicitly for both orders. A repeated center at two times still supplies two chronological marks. The exact physical prefactor $1/(4\pi R)$ precedes every clearance substitution, while the original section restriction contributes $1/c_*$ only when passing to ν_R^*. Neither the factor two nor the section normalization is absorbed into a supposed independent probability.

In the inherited identity $E_\delta^- = \mathcal{I} E_\delta^+ = T_R E_\delta^+$, both expressions on the right denote *set images* on the regular invertible collision space. Thus $\mathbf{1}_{E_\delta^-}(x) = \mathbf{1}_{E_\delta^+}(T_R^{-1}x)$ almost everywhere; the displayed image is not $T_R^{-1}E_\delta^+$. The null-set convention is a bulk-measure convention, not a rule that positive one-sided endpoint traces vanish. Similarly, the critical coarea density in Section 17 is the contribution of its stated coordinate ball and selector. It is not automatically the complete global density near that roof value.

The new proof chain has three distinct levels. The foot standoff, Jacobi identity, tangential-end uniqueness, exact coarea and stable metric comparison are proved directly in Sections 19–20. The uniform all-separation density and complete factorial bounds additionally require the explicitly retained unconditioned physical pressure inequality $\sup_{R,l} P_l(R) \le Q_*$. The raw pointwise theorem still requires (15.8). No conclusion at the second level is substituted for the third. The native historical proof bodies 260–276 are preserved, and modules 277–278 alone are the new numbered proof modules in this author revision.

Appendix K. Angular-source dependency, metric and limit audit

The new geometric source formulas use only lattice separation, actual circle reflection, the inherited physical finite-horizon bound and the exact one-mark parametrization. The direct source norm in (21.8) is $\sum_C \|\rho_C\|_\infty$, not a sum of masses and not a global collision-space L^∞ norm. It contains both signed first marks. The second mark also has two signs, already paid in C_t; no additional sign or root factor is to be appended to H_*. The one-mark normalization is inserted only after the root sum, as $1/(4\pi R)$. The original source normalization appears once, as c_*^{-1}, inside a positive exact-event domination.

External curve growth and physical coordinates. The inspected author version of [6] dates from 10 October 2013. Its Lemma 3.2(b) is the homogeneous stable curve-Jacobian sum; its proof specifies (H2)–(H4). Theorem 2.7 states uniform properties for its geometric family, and Remark 2.9(b) discusses a small change of boundary arclength. We use none of these statements as an exact-label roof local limit theorem or as a sum over all physical cylinders. Input 22.1 states exactly the forward counterpart needed here; source-curve membership and a common curvature allowance are essential parts of its invocation. Theorem 21.1 verifies a concrete bound K_0 for that membership check. It does not independently re-prove expansion and distortion for the admissible class.

Time reversal is $\mathcal{I}(s,\varphi) = (s,-\varphi)$, an isometry of the original collision-coordinate arclength, and $\mathcal{I}T_R\mathcal{I} = T_R^{-1}$ on the regular set. Thus forward pieces of an unstable curve correspond to backward pieces of its reflected stable curve, with equal arclength Jacobians. This conversion does not reverse the original measured return record in a theorem statement; it only verifies the curve estimate.

For a local common boundary coordinate at reference radius R_0, write $A_R(s,\varphi) = (R_0 s/R, \varphi)$. If $m_A \le |A_R v|/|v|$ and $|A_R v|/|v| \le M_A$, then conjugating a curve map gives

$$J^{\mathrm{common}}T_R^l(x) = J^{\mathrm{physical}}T_R^l(x)\frac{|A_R v_l|/|v_l|}{|A_R v_0|/|v_0|}.$$

There is only an endpoint ratio, at most M_A/m_A, not a factor $(M_A/m_A)^l$. A compact finite local cover of $[R_-, R_+]$ gives common constants whenever the cited geometric curve hypotheses are verified in those local charts. This is a coordinate comparison, not an affine deformation of the triangular physical lattice. The input is kept explicit rather than asserting that compactness alone verifies (H2)–(H4) or the larger pressure and pairing chain.

Boundary and quantifier conventions. All regular first-hit intervals are genuinely physical. Future singular cuts are exhausted through the one-copy curve partition. This removes zero angular or curve measure in bulk integrals only. It says nothing about the positive one-sided trace of a paired source at a singular endpoint. Those traces and critical ball selectors remain as in the inherited optical proofs. Likewise the set-image convention is unchanged: $\mathcal{I}E_\delta^+$ and $T_R E_\delta^+$ mean images on the regular set, whereas $T_R^{-1}(E)$ denotes its preimage; inherited equalities are used with their specified null-set convention.

The curve-growth supremum ranges over all stated radii, all nonnegative iterate counts and the entire admissible curve class. The new kernel holds for every fixed positive first clearance and all positive integer separations. The root sum is countable and positive before any essential supremum. These facts do not commute a roof supremum with an integral over first clearance. The original central roof window and arithmetic labels remain fixed. The collision limsup is taken before the outer B limit. The explicit remaining proof obligation is (15.8), including first-order clearance, critical and boundary sums, incidence-only source and original nongrazing capacity. None is assigned zero mass or zero height merely because the angular-source theorem has been proved.

References

[1] Q. Qi, *Collision records, arithmetic, and raw local inversion in a triangular Lorentz gas*, A2-DYN clearance-roof author revision, October 15, 2026. Complete source and 47-page manuscript committed at `8a5a7c2be5630bf5919e9f74ee79de01af0dbb85`; active article Git tree `c2e9e83f39985a34690300433f4b40de4f1f88ac`.

[2] Q. Qi, *Collision records, arithmetic, and raw local inversion in a triangular Lorentz gas*, A2-DYN revision 111, October 14, 2026. Historical source archive pinned at Git tree `790e627efe3a89a4549123d7e9acacf24089c287`;

its scoped source objects are retained in the verified historical DAG, without expanding recursive copies. Author commit `8c617cafa5e5493e26817401e866b2415522f7c0`.

[3] Q. Qi, *Collision records, arithmetic, and raw local inversion in a triangular Lorentz gas*, A2-DYN revision 110, October 14, 2026. Historical source, with recovered anchors retained in the provenance packet. Author commit `0abce3fcfbc735ff5cc6d085279abea5f32a7a22`.

[4] Q. Qi, *Collision records, arithmetic, and raw local inversion in a triangular Lorentz gas*, A2-DYN revision 109, October 14, 2026. Historical source, with recovered anchors retained in the provenance packet. Author commit `ab0ad79b9c101b82c80e33511529588d101a45c0`.

[5] Q. Qi, *Collision records, arithmetic, and raw local inversion in a triangular Lorentz gas*, A2-DYN revision 108, October 14, 2026. Historical manuscript and supplement; retained scoped source objects are recoverable from the historical DAG. Author commit `2144665f89362ca80116f1999d99c80875833dcf`.

[6] M. F. Demers and H.-K. Zhang, *A functional analytic approach to perturbations of the Lorentz gas*, Communications in Mathematical Physics **324** (2013), 767–830. The prior audit inspected the author version dated 10 October 2013; the fresh version check is recorded in the literature audit.

[7] V. Baladi and M. F. Demers, *Thermodynamic formalism for dispersing billiards*, Journal of Modern Dynamics **18** (2022), 415–493. The prior audit inspected the author version dated 19 September 2022; the fresh primary-source check is recorded in the literature audit.

[8] D. Dolgopyat and P. Nándori, *On mixing and the local central limit theorem for hyperbolic flows*, Ergodic Theory and Dynamical Systems **40** (2020), 142–174, Definition 2.2 and Section 6.4, Proposition 6.3. The exact inspected version and scope are recorded in the literature audit.

[9] S. Boyd and L. Vandenberghe, *Convex Optimization: Duality*, official lecture notes, `https://web.stanford.edu/class/ee364a/lectures/duality.pdf`. Used for general background; the special disk-word duality is proved in full in the present article.